

Custom Compiler™ Electrical Checking and Reporting User Guide

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About This User Guide

This user guide explains how to use the various tools and features available in the Custom Compiler™ environment to perform electrical checks and reporting.

This user guide assumes that you are familiar with the Custom Compiler design flow, and the available tools such as the Schematic Editor and the Layout Editor.

This preface includes the following sections:

- [New in This Release](#)
- [Related Products, Publications, and Trademarks](#)
- [Conventions](#)
- [Customer Support](#)
- [Statement on Inclusivity and Diversity](#)

New in This Release

Information about new features, enhancements, and changes, known limitations, and resolved Synopsys Technical Action Requests (STARs) is available in the Custom Compiler Release Notes on the SolvNetPlus site.

Related Products, Publications, and Trademarks

For additional information about the Custom Compiler tool, see the documentation on the Synopsys SolvNetPlus support site at the following address:

<https://solvnetplus.synopsys.com>

You might also want to see the documentation for the following related Synopsys products:

Synopsys® Custom Compiler™
Synopsys PrimeWave™ Design Environment
Synopsys PrimeSim™ Continuum
Synopsys PrimeSim™
Synopsys PrimeSim™ EMIR
Synopsys IC Validator™

- Synopsys PrimeSim™ Pro
- Synopsys PrimeSim™ SPICE
- Synopsys PrimeSim™ HSPICE®
- Synopsys PrimeSim™ XA
- Synopsys StarRC™

Conventions

The following conventions are used in Synopsys documentation.

Convention	Description
Courier	Indicates syntax, such as <code>write_file</code> .
<i>Courier italic</i>	Indicates a user-defined value in syntax, such as <code>write_file design_list</code>
Courier bold	Indicates user input—text you type verbatim—in examples, such as <code>prompt> write_file top</code>
Purple	• Within an example, indicates information of special interest. • Within a command-syntax section, indicates a default, such as <code>include_enclosing = true false</code>
[]	Denotes optional arguments in syntax, such as <code>write_file [-format fmt]</code>
...	Indicates that arguments can be repeated as many times as needed, such as <code>pin1 pin2 ... pinN</code> .
	Indicates a choice among alternatives, such as <code>low medium high</code>
\	Indicates a continuation of a command line.
/	Indicates levels of directory structure.
Bold	Indicates a graphical user interface (GUI) element that has an action associated with it.
Edit > Copy	Indicates a path to a menu command, such as opening the Edit menu and choosing Copy .

Convention	Description
Ctrl+C	Indicates a keyboard combination, such as holding down the Ctrl key and pressing C.

Customer Support

Customer support is available through SolvNetPlus.

Accessing SolvNetPlus

The SolvNetPlus site includes a knowledge base of technical articles and answers to frequently asked questions about Synopsys tools. The SolvNetPlus site also gives you access to a wide range of Synopsys online services including software downloads, documentation, and technical support.

To access the SolvNetPlus site, go to the following address:

<https://solvnetplus.synopsys.com>

If prompted, enter your user name and password. If you do not have a Synopsys user name and password, follow the instructions to sign up for an account.

If you need help using the SolvNetPlus site, click REGISTRATION HELP in the top-right menu bar.

Contacting Customer Support

To contact Customer Support, go to <https://solvnetplus.synopsys.com>.

Statement on Inclusivity and Diversity

Synopsys is committed to creating an inclusive environment where every employee, customer, and partner feels welcomed. We are reviewing and removing exclusionary language from our products and supporting customer-facing collateral. Our effort also includes internal initiatives to remove biased language from our engineering and working environment, including terms that are embedded in our software and IPs. At the same time, we are working to ensure that our web content and software applications are usable to people of varying abilities. You may still find examples of non-inclusive language in our software or documentation as our IPs implement industry-standard specifications that are currently under review to remove exclusionary language.

1

About Electrical Checking and Reporting

Electrical checking and reporting is essential to robust power distribution in integrated circuit designs, thus ensuring high performance and reliability. The Custom Compiler environment offers comprehensive tools that you can use for electrical checking and reporting.

This chapter describes the following:

- [In-Design Electrical Checking and Reporting](#)
- [Tools to Use for Electrical Checking and Reporting](#)

In-Design Electrical Checking and Reporting

The performance and reliability of integrated circuits are affected by the following:

- Electromigration, the erosion of wire material from high current density, can result in short circuitry or circuit failure. As design size decreases, electromigration effects increase.
- Resistance in wires can result in voltage drops across the power network, impacting signal strength. To control the impact of voltage drops, you can add capacitors to the power network.
- Signal noise and crosstalk can reduce signal integrity. To reduce signal noise and interference, you can add shielding to isolate certain circuits from other parts of the design.

To address these factors and to prevent failure, you can run electrical checks and reports. Traditionally, these checks and reports are performed after the design is finished. The Custom Compiler tools permit you to run the checks and reports throughout the design process so that you can analyze, monitor, and fix potential issues earlier and reduce the overall design timeline.

Tools to Use for Electrical Checking and Reporting

The Custom Compiler environment provides the following tools you can use for electrical checking and reporting:

Task	Tools	Description
Electromigration	PrimeWave Design Environment Electrical Tools Custom Compiler Electromigration Checker EM Calculator EM Reporter	Analyzes net-based electrical current in routed designs. Available features in the tool annotates currents and reports violations. You can also monitor current flows in wire segments and vias. For more information, see Reporting and Checking Electromigration (EM) . Estimates compliant wire widths or current values without design layouts and simulations. See Calculating EM Without Layouts and Simulations .
Resistance	Parasitic Reporter	Allows you to control the resistance in your design using specified constraints. For more information, see Reporting and Checking Resistance .
Capacitance	Parasitic Reporter	Reports capacitance values between partially or fully routed nets in a design. For more information, see Reporting and Checking Capacitance .
Voltage	PrimeWave Design Environment Voltage Annotation Tool VDRC Assistant	Annotates the design schematic or layout with simulated voltage data. See Annotating Voltage Data for VDRC .
Shielding	Shield Reporter Shield Checker	Reports the shield coverage ratio for each layer. For more information, see Reporting and Checking Shield Coverage .

2

Calculating EM Without Layouts and Simulations

The EM Calculator provides quick current or wire width estimates without using design layouts and simulations.

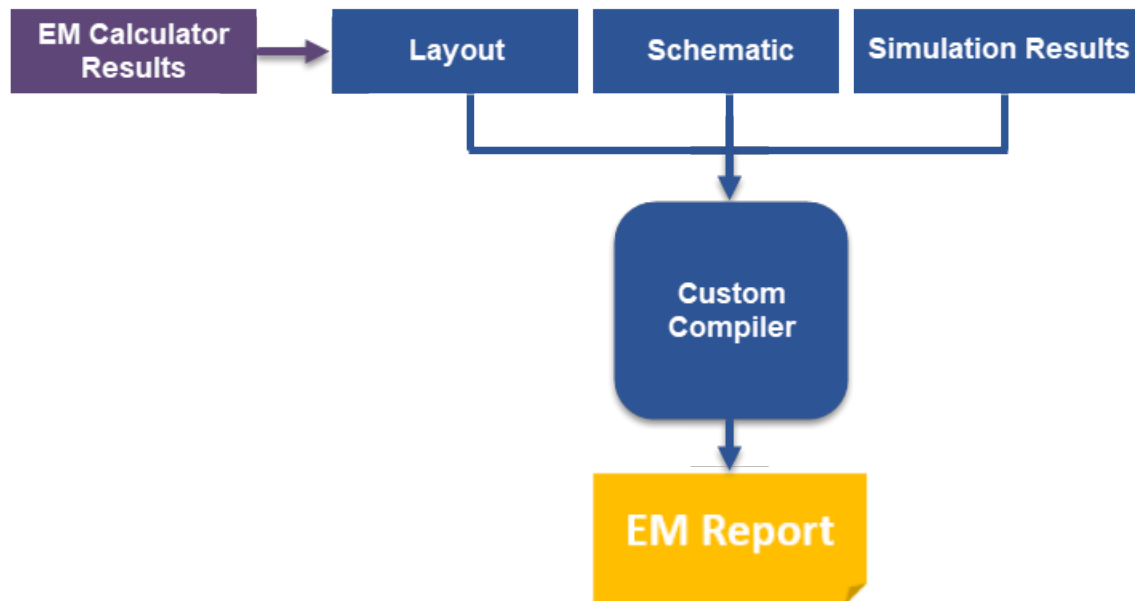
This topic describes the following:

- [About the EM Calculator](#)
- [Opening the EM Calculator Dialog Box](#)
- [Specifying the EM Constraints and Layer Mapping File](#)
- [Specifying Inputs for EM Calculations](#)
- [Calculating and Viewing the EM Results](#)
- [EM Calculator Dialog Box Reference](#)

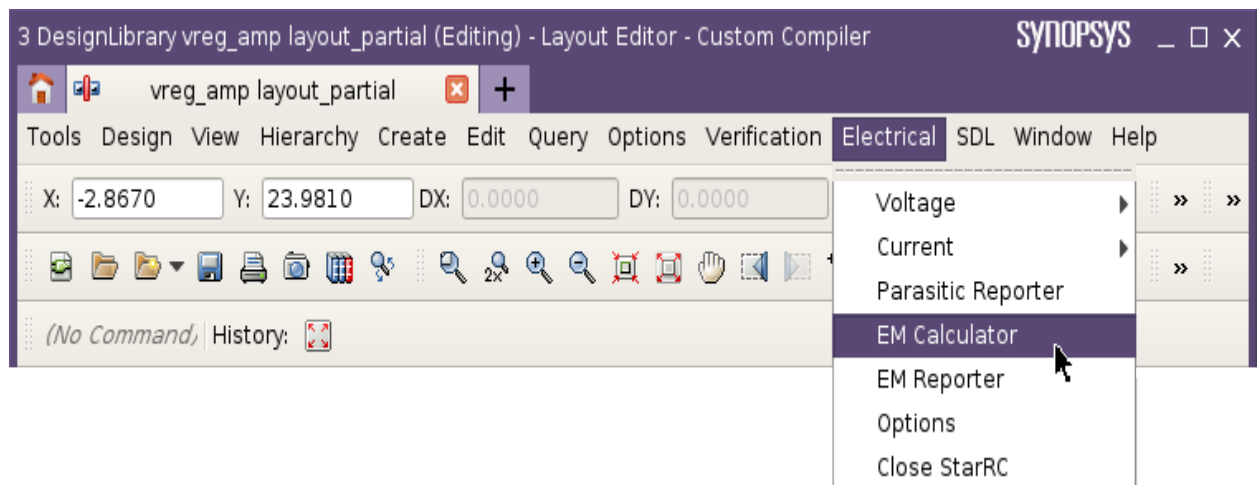
About the EM Calculator

The EM Calculator calculates EM information for a given metal or via layers by directly using foundry EM Rules. Layouts and simulations are not required. Use the EM Calculator to anticipate or fix wire failures before running the EM Reporter.

The in-design EM Reporter lets you catch causes of electromigration (EM) early in the design process, thus reducing the number of costly fixes downstream. However, the EM Reporter requires a partially or fully completed design layout and simulation runs to determine the wire failures. To work around the requirements, you can run the EM Calculator to quickly estimate compliant values, thus further reducing the number of design iterations.



You access the EM Calculator from the Layout Editor **Electrical** menu.



In the **EM Calculator** dialog box, you can supply a number of inputs, including the following:

- Type of EM analysis.
- Temperature to consider in the calculation.
- Wire width (for calculating the resulting current values).
- Current value (for calculating the compliant wire widths).

- Layers (name and color) to consider in the calculation.
- Via characteristics to consider in the calculation.

The calculated output depends on the information you supplied as input. The output is displayed in the **EM Calculator** dialog box.

Opening the EM Calculator Dialog Box

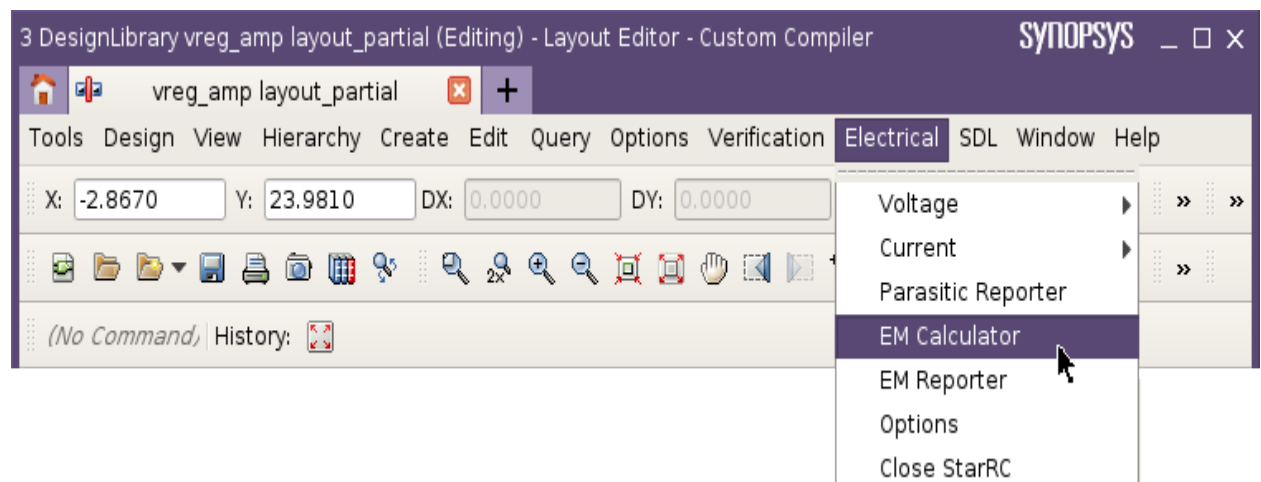
The **EM Calculator** dialog box lets you enter inputs for quickly calculating compliant wire width or current values without design layouts and simulation runs.

To open the **EM Calculator** dialog box:

1. With the design layout opened in the Layout Editor, choose **Tools > Electrical**.

The **Electrical** menu appears on the Layout Editor menu bar.

2. Choose **Electrical > EM Calculator**.



The **EM Calculator** dialog box opens.

EM Calculator

EM Type: ☒ Avg ☐ Peak ☐ RMS ☐ DC ☐ All

EM Temperature: 110 ▼

Metals Vias

☒ Calculate Width ☐ Calculate Current

Layer: All ▼ Color: NoColor ▼

EM Length (um): 0

Width

Current (mA): 0

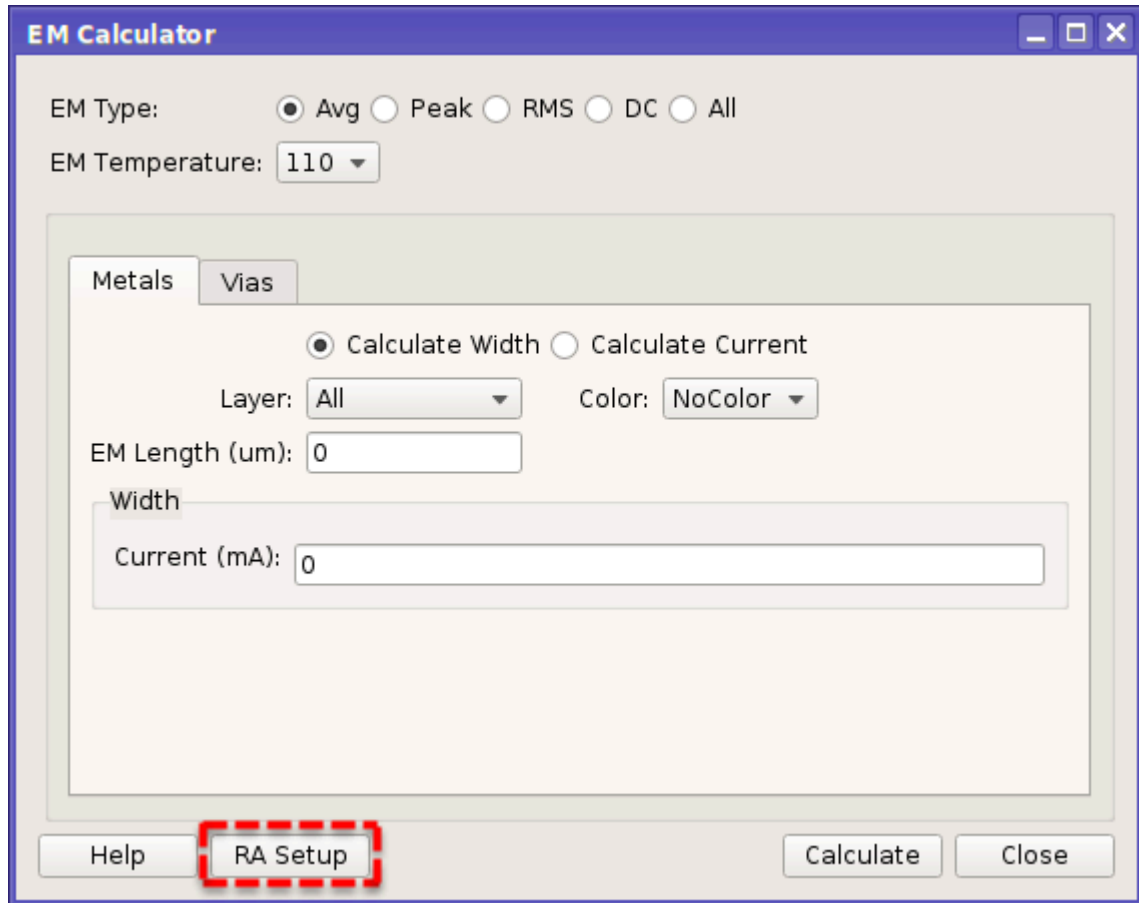
Help RA Setup Calculate Close

Specifying the EM Constraints and Layer Mapping File

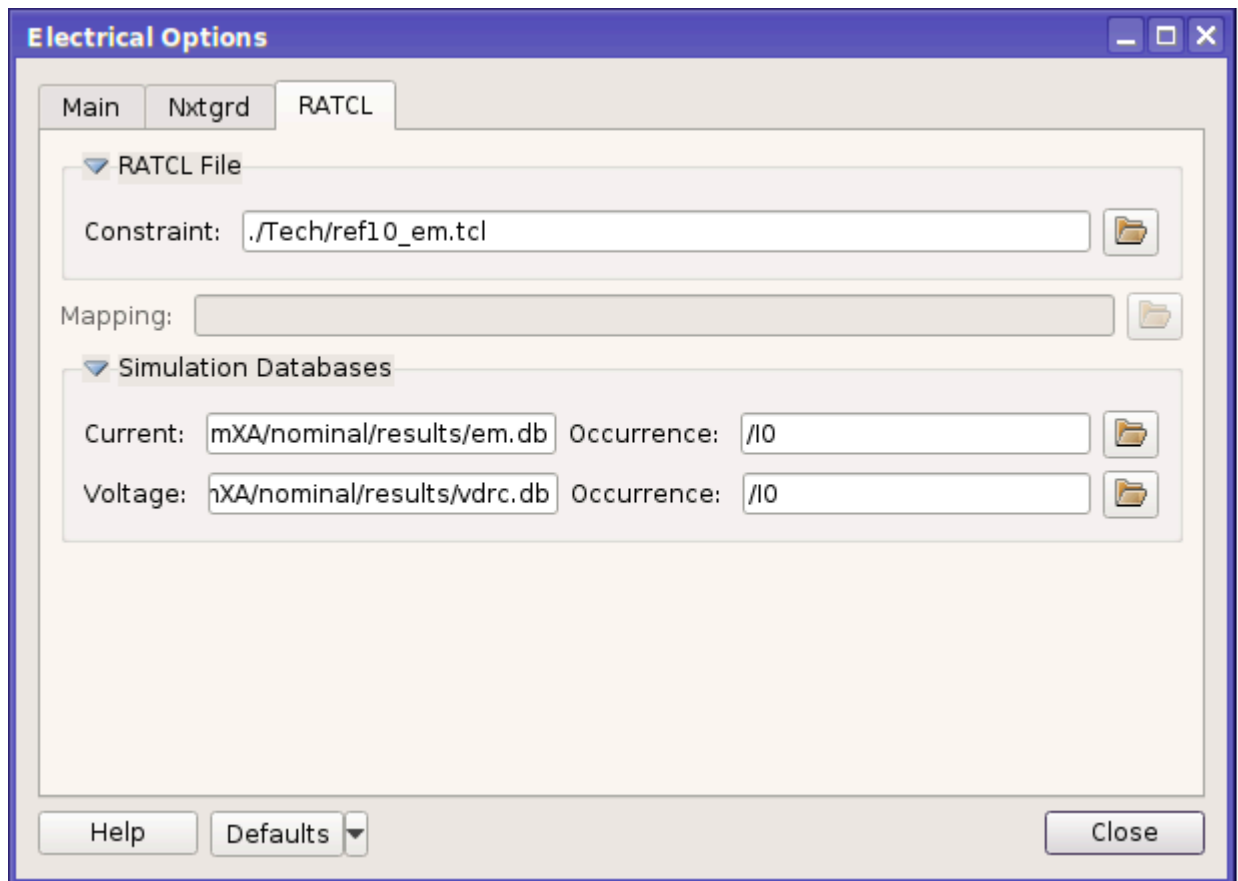
Before you can calculate the compliant wire widths or current values, you must specify the EM constraints and the layer mapping file you want to use.

To specify the EM constraint file and the layer mapping file:

1. In the **EM Calculator** dialog box, click **RA Setup**.



The **Electrical Options** dialog box opens with the **RATCL** tab displayed.



2. In the **RATCL File** section, select the **Constraint** file (the file containing the EM constraints).
3. Select the layer **Mapping** file that maps the OA layer names to the corresponding RATCL layer names.
4. Because you are not running a simulation, skip the settings in the **Simulation Databases** section.
5. Click **Close** to return to the **EM Calculator** dialog box.

Specifying Inputs for EM Calculations

In the **EM Calculation** dialog box, you can provide various inputs for the wire width or current calculation.

This topic describes the inputs for the following:

- [Selecting Current Type and Temperature for EM Calculations](#)
- [Specifying Inputs for Calculating Wire Widths or Current Values](#)
- [Specifying Inputs for Calculating Via Number or Via Current](#)

Selecting Current Type and Temperature for EM Calculations

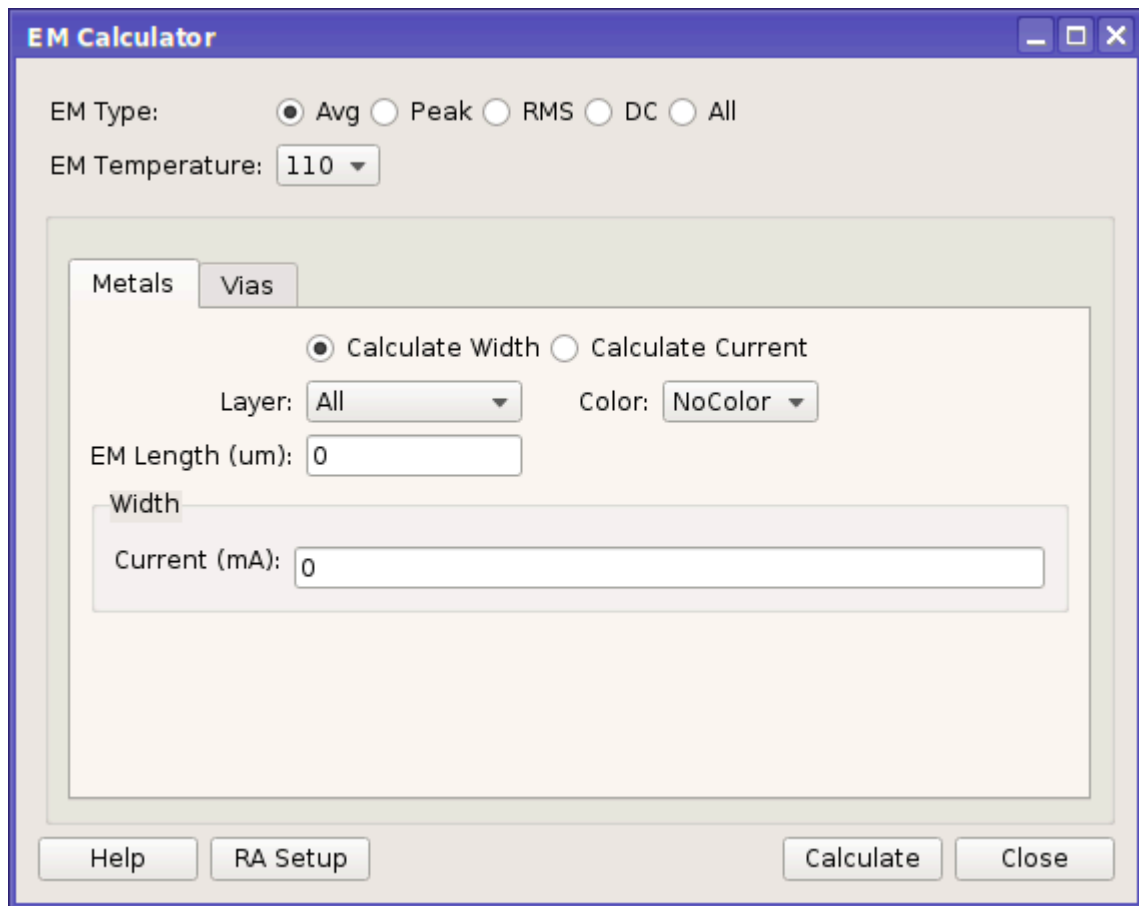
In the **EM Calculator** dialog box, you can specify the current type (average, peak, RMS, or all) and the EM temperature. The two settings are used to calculate the wire width and current value.

To specify the current type and the EM temperature:

1. In the **EM Calculator** dialog box, select the **EM Type** to use during the calculation.

EM Type is the set of EM constraints for the specific current type. The constraints are defined in the RATCL file. See [Specifying the EM Constraints and Layer Mapping File](#).

Argument	Description
Avg	EM rules for the AC average current
Peak	EM rules for the AC peak current
RMS	EM rules for the Root-mean-square current
All	All EM rules in the RATCL file



The image shows a software dialog box titled "EM Calculator". It has a standard Windows-style title bar with minimize, maximize, and close buttons. The dialog is divided into several sections. At the top, there's a section for "EM Type" with radio buttons for "Avg" (selected), "Peak", "RMS", "DC", and "All". Below this is a dropdown menu for "EM Temperature" set to "110". The main area of the dialog is a large rectangle containing two tabs: "Metals" (selected) and "Vias". Inside the "Metals" tab, there are two radio buttons: "Calculate Width" (selected) and "Calculate Current". Below these are two dropdown menus: "Layer" set to "All" and "Color" set to "NoColor". There are two text input fields: "EM Length (um):" with the value "0" and "Width" (highlighted in a light grey box) with the value "0". At the bottom of the dialog, there are four buttons: "Help", "RA Setup", "Calculate", and "Close".

2. Select the **EM Temperature** to use in the calculation.

See Also

- [Opening the EM Calculator Dialog Box](#)
- [Specifying the EM Constraints and Layer Mapping File](#)
- [Specifying Inputs for Calculating Wire Widths or Current Values](#)
- [Specifying Inputs for Calculating Via Number or Via Current](#)
- [Calculating and Viewing the EM Results](#)

Specifying Inputs for Calculating Wire Widths or Current Values

In the **EM Calculator** dialog box, you can specify the wire width and calculate the resulting current values. Alternatively, you can specify the current value and calculate the compliant wire width.

To specify the wire width or current values for the EM calculations:

1. In the **EM Calculator** dialog box, click the **Metals** tab.

The screenshot shows the **EM Calculator** dialog box with the following settings:

- EM Type:** ☒ Avg ☐ Peak ☐ RMS ☐ DC ☐ All
- EM Temperature:** 110
- Metals** tab is selected (Vias tab is also visible).
- ☒ Calculate Width ☐ Calculate Current
- Layer:** All
- Color:** NoColor
- EM Length (um):** 0
- Width** section is expanded, showing **Current (mA):** 0
- Buttons at the bottom: Help, RA Setup, Calculate, Close

2. Select the type of calculation you want to run:

Argument	Description
Calculate Width	Calculates the wire width based on the current value you provide.
Calculate Current	Calculates the current value based on the wire width you provide.

3. If you want to run the calculation for specific layers, select those **Layers**.
4. If you want to run the calculation for specific colored layers, select the **Color**.
5. Type the **EM Length** value, which is the wire length value used during the EM calculation.

If you do not provide a value, the wire length value is determined from the design layout during the calculation

6. If you want to calculate the wire width, specify the **Current Value** (mA).
7. If you want to calculate the current value, specify the **Width Value** (μm).

See Also

- [Opening the EM Calculator Dialog Box](#)
- [Calculating and Viewing the EM Results](#)
- [Specifying the EM Constraints and Layer Mapping File](#)
- [Selecting Current Type and Temperature for EM Calculations](#)
- [Specifying Inputs for Calculating Via Number or Via Current](#)

Specifying Inputs for Calculating Via Number or Via Current

In the **EM Calculator** dialog box, you can specify the via information and calculate the resulting current values. Alternatively, you can specify the current value and calculate the via details.

To specify the via information or via current values for the EM calculations:

1. In the **EM Calculator** dialog box, click the **Vias** tab.

The screenshot shows the 'EM Calculator' dialog box with the 'Vias' tab selected. At the top, 'EM Type' has radio buttons for 'Avg' (selected), 'Peak', 'RMS', 'DC', and 'All'. Below it, 'EM Temperature' is set to '110'. The 'Vias' tab contains two radio buttons: 'Calculate Via Number' and 'Calculate Via Current' (selected). Below these are dropdowns for 'Layer' (set to 'All') and 'Current Direction' (set to 'UP'), followed by a 'Cut Class' dropdown. There are input fields for 'Upper Layer Width' (0), 'Upper Layer Length' (0), 'Lower Layer Width' (0), and 'Lower Layer Length' (0). A 'Current Calculation' section has input fields for 'Column Number' (0) and 'Row Number' (0). At the bottom are buttons for 'Help', 'RA Setup', 'Calculate', and 'Close'.

2. Select the type of calculation you want to run:

Argument	Description
Calculate Via Number	Calculates the number of cuts based on the current value you provide.
Calculate Via Current	Calculates the via current value based on the number of cuts you specify.

3. If you want to run the calculation for specific layers, select those **Layers**.

4. Choose the **Current Direction**:

Argument	Description
Up	The current is flowing upward through the via.
Down	The current is flowing downward through the via.

5. If you want to run the calculation for specific via cut classes, choose the **Cut Class**.

6. For the via top layer, specify the following:

- **Upper Layer Width**
- **Upper Layer Length**

7. For the via bottom layer, specify the following:

- **Lower Layer Width**
- **Lower Layer Length**

8. If you want to calculate the number of vias, specify the **Via Current Value (mA)**.

9. If you want to calculate the via current value, specify values for the following:

- **Column Number**
- **Row Number**

See Also

- [Opening the EM Calculator Dialog Box](#)
- [Specifying the EM Constraints and Layer Mapping File](#)
- [Selecting Current Type and Temperature for EM Calculations](#)
- [Specifying Inputs for Calculating Wire Widths or Current Values](#)
- [Calculating and Viewing the EM Results](#)

Calculating and Viewing the EM Results

After you specify the inputs for either the wire width, current, or vias, you can run the EM calculations and view the results.

To calculate and view the results:

1. In the **EM Calculator** dialog box, click **Calculate**.

EM Calculator

EM Type: ☒ Avg ☐ Peak ☐ RMS ☐ DC ☐ All

EM Temperature: 110 ▼

Metals Vias

☒ Calculate Width ☐ Calculate Current

Layer: All ▼ Color: NoColor ▼

EM Length (um): 70

Width

Current (mA): 5

Help RA Setup Calculate Close

The results appear in the dialog box.

2. Review the results that appear in the dialog box.

The following example shows the results when EM Type: All was selected, and a current value was specified. The wire width results are listed by layer.

Calculate Width				
	Avg	Peak	RMS	DC
m1	0.07um	0.07um	0.07um	0.07um
m2	0.07um	0.07um	0.07um	0.07um
m3	0.07um	0.07um	0.07um	0.07um
m4	0.07um	0.07um	0.07um	0.07um
m5	0.07um	0.07um	0.07um	0.07um
m6	0.07um	0.07um	0.07um	0.07um

The following example shows the results when EM Type: All was selected, and the Cut Column and Cut Row values were specified. The via current results are listed by layer.

Calculate Via Current				
	Avg	Peak	RMS	DC
VIA0				
VIA1	1.333mA	0mA	0mA	1.333mA
VIA2	1.333mA	0mA	0mA	1.333mA
VIA3	1.458mA	0mA	0mA	1.458mA
VIA4	0.714mA	0mA	0mA	0.714mA
VIA5	0.965mA	0mA	0mA	0.965mA
VIA6	0.965mA	0mA	0mA	0.965mA
VIA7	0.965mA	0mA	0mA	0.965mA

EM Calculator Dialog Box Reference

This topic provides a quick reference of the **EM Calculator** dialog box. For instructions, see [Calculating EM Without Layouts and Simulations](#).

The **EM Calculator** dialog box consists of the following:

- [General Options](#)
- [Metals Tab](#)
- [Vias Tab](#)

EM Calculator

EM Type: ☒ Avg ☐ Peak ☐ RMS ☐ DC ☐ All

EM Temperature: 110 ▼

Metals **Vias**

☒ Calculate Width ☐ Calculate Current

Layer: All ▼ Color: NoColor ▼

EM Length (um): 0

Width

Current (mA): 0

Help RA Setup Calculate Close

General Options

EM Type:

Argument	Description
Avg	EM rules for the AC average current
Peak	EM rules for the AC peak current
RMS	EM rules for the Root-mean-square current
All	All EM rules in the RATCL file

EM Temperature: The temperature to use in the EM calculation.

Metals Tab

The screenshot shows the 'EM Calculator' dialog box with the 'Metals' tab selected. The 'EM Type' section has radio buttons for 'Avg' (selected), 'Peak', 'RMS', 'DC', and 'All'. The 'EM Temperature' is set to '110' in a dropdown menu. The 'Metals' tab is active, showing options to 'Calculate Width' (selected) or 'Calculate Current'. Below these are dropdowns for 'Layer' (set to 'All') and 'Color' (set to 'NoColor'). There is a text input for 'EM Length (um)' with the value '0'. A sub-section labeled 'Width' contains a text input for 'Current (mA)' with the value '0'. At the bottom are buttons for 'Help', 'RA Setup', 'Calculate', and 'Close'.

Calculation:

Argument	Description
Calculate Width	Calculates the wire width based on the current value you provide.
Calculate Current	Calculates the current value based on the wire width you provide.

Input:

Option	Description
Layer	Layers to be considered for the EM calculation.
Color	Colors of the layers to be considered for the EM calculation.

Option	Description
EM Length	Wire length value used during the EM calculation.
Width: Current (mA)	Current value, in mA, to be used when calculating the wire width.
Current: Width (μm)	Width value, in μm, to be used when calculating the current.

Vias Tab

Calculation:

Argument	Description
Calculate Via Number	Calculates the number of cuts based on the current value you provide.

Argument	Description
Calculate Via Current	Calculates the via current value based on the number of cuts you specify.

Input:

Option	Description
Layer	Layers to be considered for the EM calculation.
Current Direction	Direction of the current through the via: Up , or Down .
Cut Class	Via cut class to be considered for the EM calculation.
Upper Layer Width	Width of the top via layer.
Upper Layer Length	Length of the top via layer.
Lower Layer Width	Width of the bottom via layer.
Lower Layer Length	Length of the bottom via layer.
Number: Current (mA)	Current value, in mA, to be used when calculating the number of vias.
Current Calculation: • Column Number • Row Number	Number of columns and rows of vias to be used when calculating the via current.

3

Reporting and Checking Electromigration (EM)

This chapter explains how to generate interactive electromigration reports for partially completed designs.

This chapter contains the following topics:

- [About Electromigration \(EM\)](#)
- [Defining Current Density Constraints](#)
- [Preparing the Layer Mapping File](#)
- [Measuring Terminal Current and Voltage Data Using the PrimeWave Design Environment](#)
- [Measuring Terminal Current Data Using the Schematic Editor](#)
- [Adding cdsTerm Labels in Design Schematics](#)
- [Annotating Schematics With Current Data](#)
- [Annotating Layouts with Current Data](#)
- [Viewing Current Data in the Layout Editor](#)
- [Editing Current Annotations](#)
- [Reporting Electromigration](#)
- [Checking Electromigration](#)

Reporting and checking electromigration is part of in-design electrical checking, which includes [Reporting and Checking Resistance](#) and [Reporting and Checking Shield Coverage](#).

About Electromigration (EM)

Electromigration (EM) is the gradual erosion of wire material caused by long-term exposure to high current density. It can cause reliability issues, especially as the number of design components increases, and the clearance between the components decreases.

To assess electromigration risks, you can run EM analysis. Instead of waiting until you complete the layout, the Custom Compiler tools allow you to run the analysis during the design layout process.

This topic describes the following:

- [Types of EM Analysis](#)
- [Tools for Reporting and Checking EM](#)
- [Workflow for Static EM Analysis](#)
- [Workflow for Dynamic EM Analysis](#)

To include self-heating effect reporting, see [Reporting Self-Heating Effects](#).

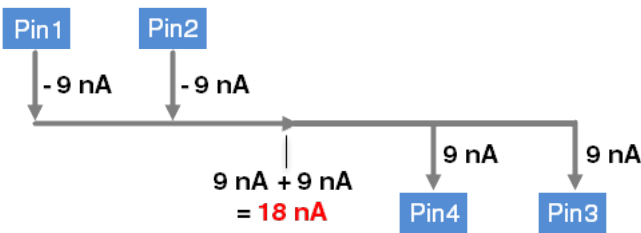
Types of EM Analysis

The type of in-design electromigration (EM) analysis flow you use depends on the source current information you have and the level of details you want in the results.

The following table lists the types of in-design EM analysis available in the Custom Compiler environment.

Type of EM Analysis	Analyzes...
Static	AC average current data from instance terminals. You can obtain the terminal current data from pre-layout simulations. Alternatively, you can manually enter the values in the Property Editor. Analysis is limited to the 0x level only. Use this method if you want faster but rough current value estimates.

Average Current

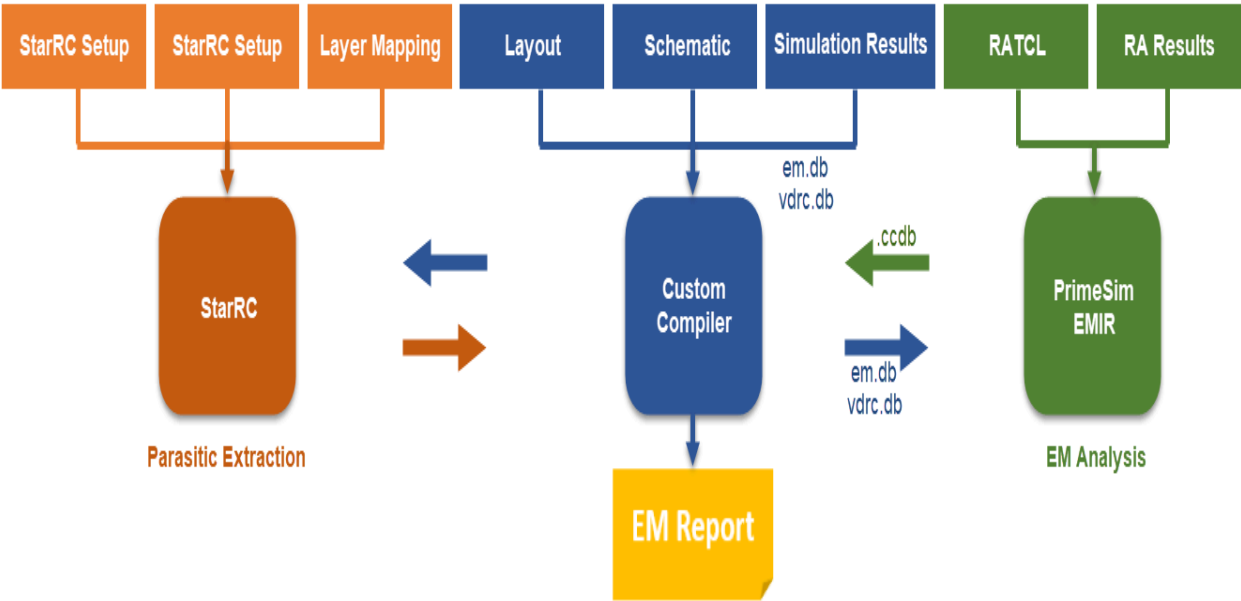


Type of EM Analysis	Analyzes...
Dynamic	AC average, peak, and root-mean square (RMS) data from instance terminals. You can obtain the current and voltage data from prelayout simulations. Because waveform data are time-based, the estimated current values are more accurate for peak and RMS reporting. The EM analysis extends below the 0x level, to the first instance terminals that have current data from the prelayout simulations. Analysis can be run across all hierarchy levels. Use this method if you want more accurate current values, and you do not mind longer analysis time. Peak Current

The in-design dynamic EM flow uses:

- The StarRC tool for Parasitic RC netlist extraction.
- The PrimeSim EMIR tool for EM analysis.

After obtaining the instance terminal current and voltage data from the simulation results, the Custom Compiler tool passes the data to the PrimeSim EMIR tool using the piecewise linear (PWL) file.



Tools for Reporting and Checking EM

Reporting and checking electromigration involves the analysis of the current flow through the resistance network of physically routed nets. After a net is routed, EM analysis can proceed immediately, even though other nets are not yet routed and the layout is not yet completed.

You can use the following Custom Compiler tools when estimating and addressing electromigration in your designs:

Tool	Description
PrimeWave Design Environment	Saves terminal current information from design simulations.
Electrical > Current Tool	Uses the current measurements generated in the PrimeWave Design Environment to annotate the schematic and layout.

Tool	Description
EM Reporter	Reports various current data and violations in the EM Reporter dialog box. The EM Reporter compares calculated current values (based on path widths, vias, and pin current) with the maximum allowed current values in the technology database, an ITF-EM file, or an iRCX file. You can view and monitor current flows in wire segments and vias. Based on the results, you can modify components in your design, such as increasing the wire width, to reduce electromigration and avoid violations. The EM Reporter reports results for the following current types: Average, peak, and RMS.
EM Checker	Uses the same input data as the EM Reporter, adds error markers in the design, and reports violations in the Error Viewer.

Workflow for Static EM Analysis

The static EM analysis flow processes terminal current data and generates AC average current data. Analysis is limited to the 0x level only. Use this method if you want faster but rough current value estimates.

The following table shows the high-level workflow for reporting electromigration (EM) using static EM analysis, which requires a design layout and simulation runs. The workflow is an iterative process that involves reporting, checking, viewing violations, fixing violations, and reporting again until measurements achieve compliance.

To run quick calculations and estimate wire widths and current values without using a design layout and simulations runs, see [Calculating EM Without Layouts and Simulations](#).

Step	Description
1	Define current density constraints. Make sure process-level current density information is defined. You can define current density process constraints by defining OpenAccess MaxACCurrentDensity constraints in the technology database, or by referencing an iRCX or ITF-EM constraint file. See Defining Current Density Constraints.
2	Prepare the layer mapping file. If you plan to reference an iRCX or ITF-EM constraint file, you have the option of specifying a layer mapping file. See Preparing the Layer Mapping File.

Step	Description
3	Obtain current measurements. Use one of the following methods: • Measuring Terminal Current and Voltage Data Using the PrimeWave Design Environment • Measuring Terminal Current Data Using the Schematic Editor (if you have existing waveform files) The task produces the EM file that you need to annotate the schematic or layout in steps 5 and 6. If you are not using the PrimeWave Design Environment, you can use a Tcl API to post process the simulation results outside of the PrimeWave Design Environment.
4	(Optional) Add cdsTerm Labels. If you are missing current annotations and find that cdsTerm labels are missing from the symbol view, you can add them using the Add/Delete cdsTerm Labels command. See Adding cdsTerm Labels in Design Schematics.
5	(Optional) Annotate terminal currents in the schematic. Using the electrical tools, annotate the schematic with current measurements obtained from step 3. See Annotating Schematics With Current Data. Perform this step only if you want to visualize the currents in the schematic.
6	Annotate current measurements in the layout. Using the electrical tools, annotate the layout with current information you selected and saved in the EM current database. See Annotating Layouts with Current Data.
7	(Optional) View the current data and estimate the minimum EM-compliant wire widths for nets. View the current data in the layout. Using the current annotations in the layout and the maximum current density constraints, determine the minimum required wire widths. See Viewing Current Data in the Layout Editor.
8	(Optional) Edit current values. Edit the current values in the layout, if necessary. See Editing Current Annotations.
9	Route the nets. Route each net fully. To use the Custom Router, see About the Custom Router. To add routes manually using the Create > Interconnect command or other features in the Layout Editor, see Creating and Editing Interconnects.
10	Report electromigration. Generate electromigration reports to view and monitor the current density, margins, and failures in detail. See Reporting Electromigration.
11	Check the current density. Check for current density compliance. See Checking Electromigration.

Step	Description
12	Adjust the design. Adjust design components to fix violations. For example, you can stretch the wire width to decrease the effects of electromigration. For instructions, see Stretching the Wire Width. You can use the EM Calculator to determine the compliant current or wire width to use. See Calculating EM Without Layouts and Simulations.

Workflow for Dynamic EM Analysis

The dynamic EM analysis flow processes terminal current and voltage data and generates AC average, peak, and root-mean square (RMS) data. Analysis can be run across all hierarchy levels. Use this method if you want more accurate current values, and you do not mind longer analysis time.

The following table shows the high-level workflow for reporting electromigration using dynamic EM analysis, which requires a design layout and simulation runs. Note that the workflow is an iterative process that involves reporting, checking, viewing violations, fixing violations, and reporting again until measurements achieve compliance.

To run quick calculations and estimate wire widths and current values without using a design layout and simulations runs, see [Calculating EM Without Layouts and Simulations](#).

Step	Description
1	Define current density constraints. Make sure process-level current density information is defined. You can define current density process constraints by defining <code>OpenAccess MaxACCurrentDensity</code> constraints in the technology database, or by referencing an iRCX or ITF-EM constraint file. See Defining Current Density Constraints.
2	Prepare the layer mapping file. If you plan to reference an iRCX or ITF-EM constraint file, you have the option of specifying a layer mapping file. For details, see Preparing the Layer Mapping File.
3	Obtain current and voltage measurements. Use one of the following methods: • Measuring Terminal Current and Voltage Data Using the PrimeWave Design Environment • Measuring Terminal Current Data Using the Schematic Editor (if you have existing waveform files) The task produces the EM and VDRC files that you need as input for the PrimeSim EMIR tool, which generates the CCDB file required for results analysis. If you are not using the PrimeWave Design Environment, you can use a Tcl API to post process the simulation results outside of the PrimeWave Design Environment.

Step	Description
4	Route the nets. Route each net fully. To use the Custom Router, see About the Custom Router. To add routes manually using the Create > Interconnect command or other features in the Layout Editor, see Creating and Editing Interconnects.
5	Report electromigration. Generate electromigration reports to view and monitor the current density, margins, and failures in detail. See Reporting Electromigration .
6	View the violations in the results viewer. View the violation data in the PrimeSim EMIR results viewer. The PrimeSim EMIR results viewer displays automatically only if you selected Load Report in Results Browser in the Electrical Options dialog box.
7	Check the current density. Check for current density compliance. For information, see Checking Electromigration .
8	Adjust the design. Adjust design components to fix violations. For example, you can stretch the wire width to decrease the effects of electromigration. For instructions, see Stretching the Wire Width. You can use the EM Calculator to determine the compliant current or wire width to use. See Calculating EM Without Layouts and Simulations .

Defining Current Density Constraints

Before you start electromigration reporting and checks, you must make sure process-level current density information is defined in one of the following:

- OpenAccess `maxACCurrentDensity` constraints in the technology database
- ITF-EM file
- iRCX file
- RA-TCL file*

By default, the EM Reporter and Checker will use the OpenAccess `maxACCurrentDensity` constraints in the technology database.

ITF-EM, iRCX, and RA-TCL are rule files that provide the current density requirements. The Custom Compiler environment supports the reading of these formats for obtaining current density information.

You can choose one of these rule sources when you set the electromigration reporting and checking options. For information, see [Generating EM Reports](#).

* The PrimeSim XA tool uses RA-TCL rule files for electrical signoff verification. For consistency and to facilitate the design process, you can use the same RA-TCL file for in-design electrical checks.

Preparing the Layer Mapping File

When you generate electromigration, resistance, and capacitance reports using an external rule source, you have the option of referencing a layer mapping file that maps OA or Milkyway layer names to corresponding layer names in the external rule source:

Electrical Report	External Rule Sources
Resistance	TLUPlus
Capacitance	nxtgrd
Electromigration	iRCX ITF RATCL

The layer mapping files must be in the ASCII format. Each file should consist of two sections: `conducting_layers` and `via_layers`. The syntax for the sections are as follows:

```
conducting_layers
<db_metallayer_name><externalsource_metallayer_name>

via_layers
<db_vialayer_name><externalsource_vialayer_name>
```

Field	Description
db_metallayer_name	The name of the conducting (metal) layer in the OA or Milkyway database.
externalsource_metallayer_name	The name of the conducting (metal) layer in the external rule source.
db_vialayer_name	The name of the via layer in the OA or Milkyway database.
externalsource_vialayer_name	The name of the via layer in the external rule source.

For example:

```
conducting_layers
PO          PO
met1        M1
met2        M2
```

met3	M3
met4	M4
met5	M5
met6	M6
via	layers
v1	VIA1
v2	VIA2
v3	VIA3
v4	VIA4
v5	VIA5

Measuring Terminal Current and Voltage Data Using the PrimeWave Design Environment

In both the static and dynamic in-design EM analysis, you must first capture current data and save the information to an EM (`em.db`) file. In the dynamic in-design EM analysis, you must also obtain the voltage data and save the information to a VDRC (`vdrc.db`) file.

You use one of the following methods to obtain the necessary current and voltage data:

Method	Description
PrimeWave Design Environment	Runs simulations to: • (Static and dynamic EM analyses) Measure the current, and then saves the terminal current data to the EM files. • (Dynamic EM analysis only) Measure the voltage, and then saves the voltage data to the VDRC files.
Schematic Editor	(Static EM analysis only) Processes current data from existing waveform files, and then saves the terminal current data to EM files. Use this method if you have an existing waveform files that contain all the current data for the design.
Tcl API	Runs commands, such as the <code>sa::createEMDB</code> and <code>sa::createVDRCDB</code> Tcl commands, to obtain terminal current and voltage data from simulations and saves the data to an EM file and a VDRC file.

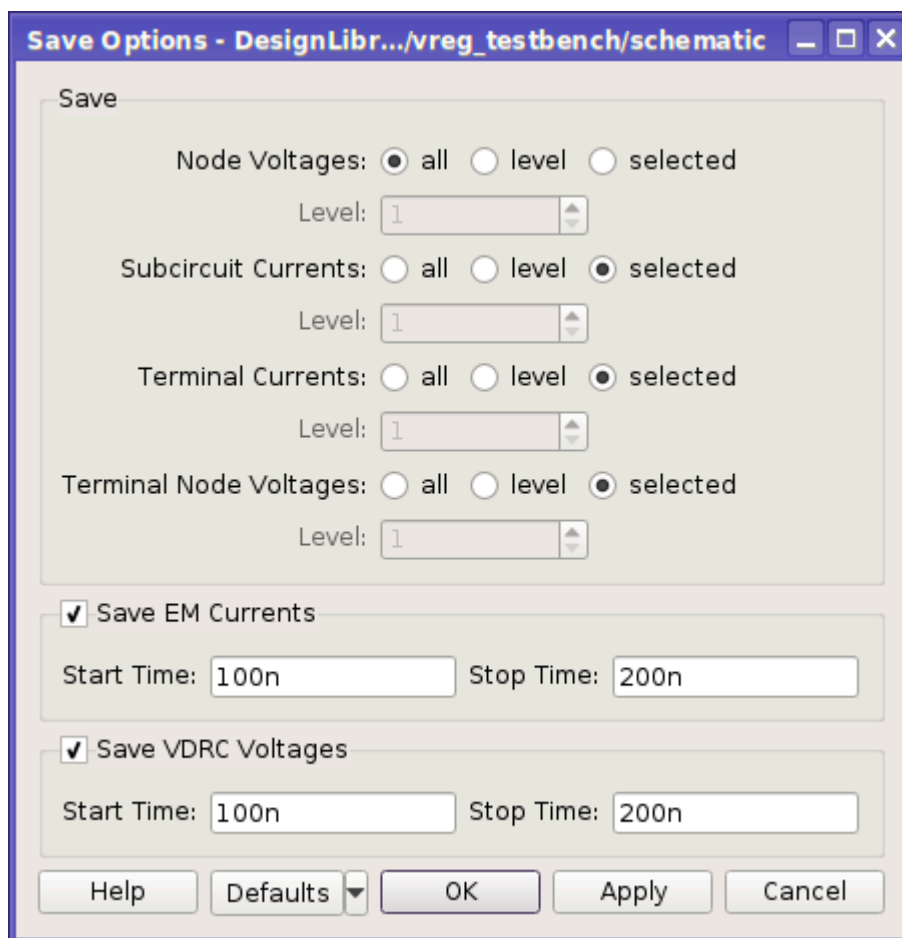
This topic describes how to run the PrimeWave Design Environment to obtain current and voltage measurements only. For instructions on how to run the simulators for other uses, see [Setting Up Testbenches](#).

For information on using the Schematic Editor to obtain the current and voltage data, see [Measuring Terminal Current Data Using the Schematic Editor](#).

For information about the `sa::createEMDB` Tcl command, see the *Custom Compiler Tcl Reference Manual*.

To run PrimeWave Design Environment and save the current and voltage data:

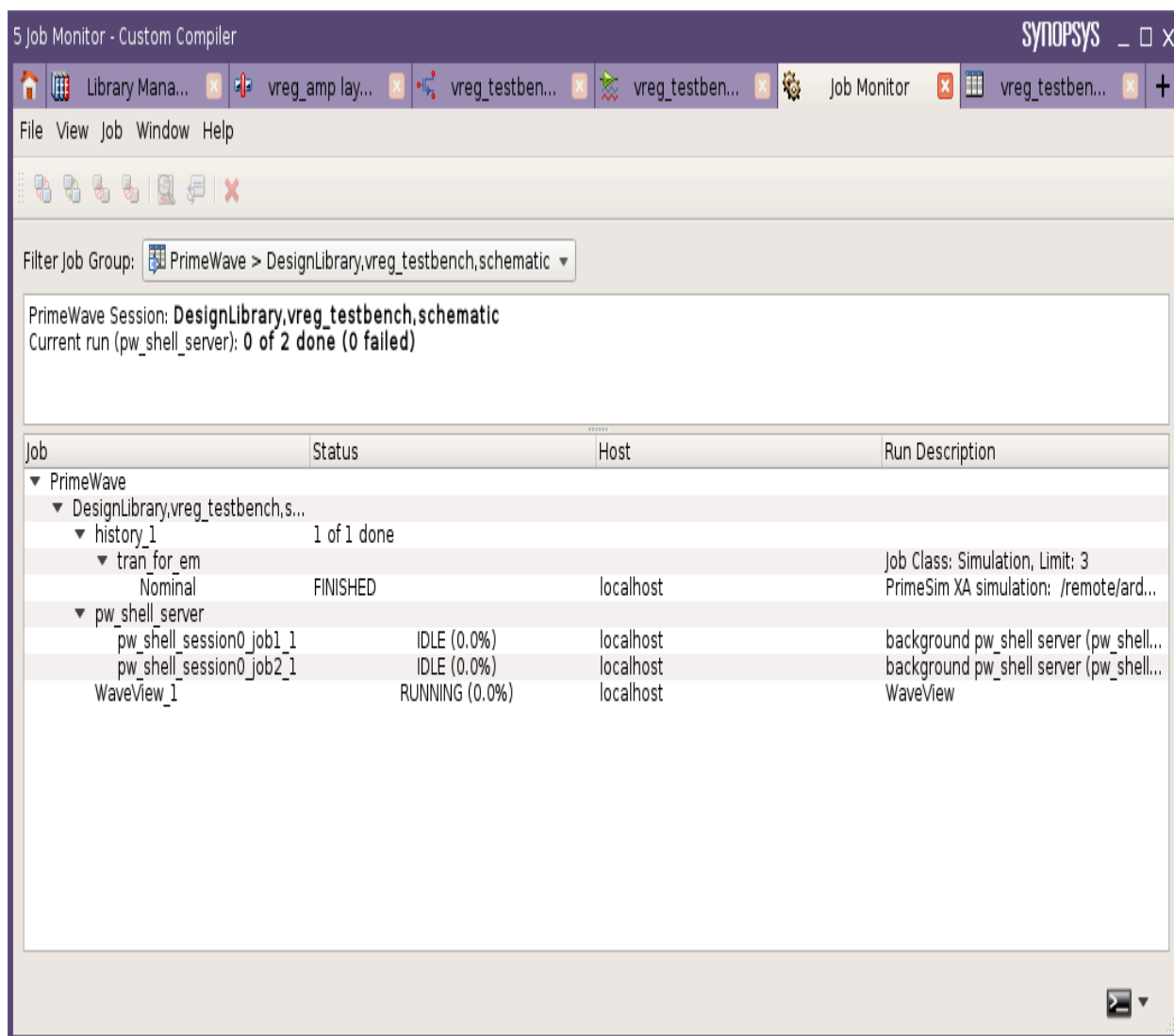
1. Open the schematic design. See Open Designs.
2. In the Schematic Editor, choose **Tools > PrimeWave** to open the PrimeWave Design Environment. The design testbench displays.
3. In the PrimeWave Design Environment, set up a new or load an existing simulation state as you would normally.
4. In the PrimeWave Design Environment, choose **Outputs > Save Options**. In the **Save Options** dialog box:



- a. Select **Save EM Currents**.
- b. In the **Start Time** and **Stop Time** boxes, specify the time window within which the PrimeWave Design Environment will obtain the current data while post-processing

the waveforms. Make sure you specify the time-unit abbreviations with the values (for example, 1n).

- c. (Dynamic EM analysis only) Select **Save VDRC Voltages**.
 - d. (Dynamic EM analysis only) In the **Start Time** and **Stop Time** boxes, specify the time window within which the PrimeWave Design Environment will obtain the minimum and maximum voltage values while post-processing the waveforms. Make sure you specify the abbreviated unit of time with the values (for example, 1n).
 - e. Select the remaining settings.
 - f. When you are finished, click **OK**. The current data settings are saved.
5. In the PrimeWave Design Environment, choose **Simulation > Netlist and Run**. The design simulation starts. The Job Monitor opens and displays the simulation status.
- When the simulation is finished, the WaveView window opens and displays plotted waveforms. The Job Monitor shows that the simulation has ended and displays run messages.



- Verify that the terminal current data is saved to an `em.db` file. You can locate the file in the design's simulation results folder.

`em.db` is a file that contains all the terminal current measurements for each instance terminal and port in the design hierarchy from the simulation.

- Verify that the voltage data is saved to a `vdrc.db` file, which contains the voltage measurements, the minimum and maximum values for the time interval you specified, and the waveform information. You can locate the file in the design's simulation results folder.

Measuring Terminal Current Data Using the Schematic Editor

For static in-design EM analysis, if you have existing waveform files that contain current data, you can save the current data to an EM (`em.db`) file while in the schematic. The `em.db` file is then used for adding current annotations in the schematic or layout.

You use one of the following methods to obtain the necessary current data:

Method	Description
PrimeWave Design Environment	Runs simulations to: • (Static and dynamic EM analyses) Measure the current, and then saves the terminal current data to the EM files. • (Dynamic EM analysis only) Measure the voltage, and then saves the voltage data to the VDRC files.
Schematic Editor	(Static EM analysis only) Processes current data from existing waveform files, and then saves the terminal current data to EM files. Use this method if you have an existing waveform files that contain all the current data for the design.
Tcl API	Runs commands, such as the <code>sa::createEMDB</code> and <code>sa::createVDRCDB</code> Tcl commands, to obtain terminal current and voltage data from simulations and saves the data to an EM file and a VDRC file.

This topic describes how to use the Schematic Editor to obtain current data. For instructions on using other features of the Schematic Editor, see [Schematic Editor Description](#).

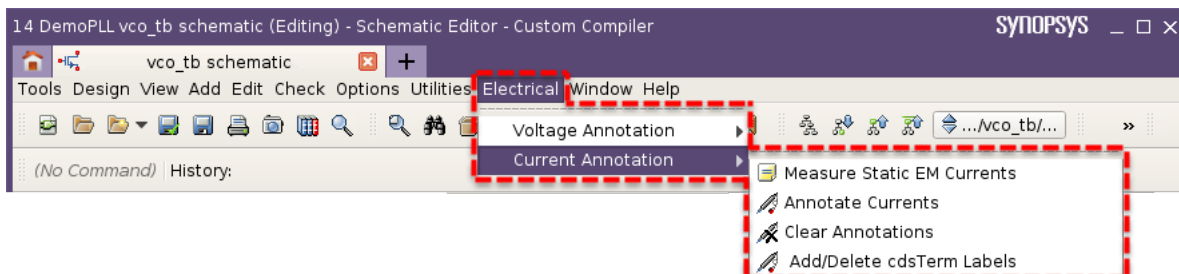
For information on using the PrimeWave Design Environment to obtain the current data, see [Measuring Terminal Current and Voltage Data Using the PrimeWave Design Environment](#).

For information about the `sa::createEMDB` Tcl commands, see the *Custom Compiler Tcl Reference Manual*.

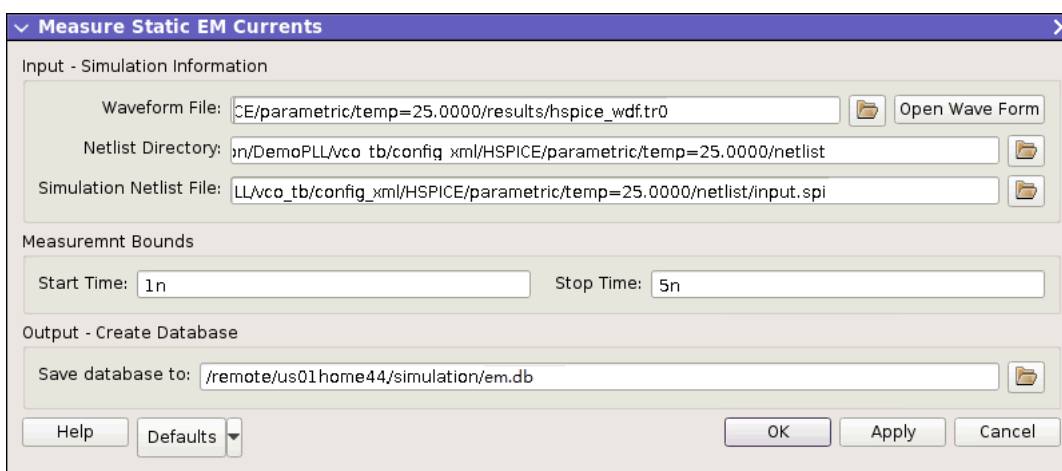
Before you start, make sure you have the waveform file that contains all the current data for the design. In addition, you must have the corresponding simulation netlist file and netlist directory ready. The netlist directory should contain name mapping information and the simulation input file (SPICE netlist file). If you do not have these files, run a simulation according to the instructions in [Measuring Terminal Current and Voltage Data Using the PrimeWave Design Environment](#).

To use the Schematic Editor to obtain the current data:

1. Open the schematic design. For instructions, see Open Designs.
2. In the Schematic Editor, choose **Tools > Electrical**. The **Electrical** menu appears on the menu bar.



3. Choose **Electrical > Current Annotation > Measure Static EM Currents**. The **Measure Static EM Currents** dialog box opens.



4. In the **Input** area:
 - a. Select the **Waveform File**, an HSPICE WDF, PSF, or FSDB file.
 - b. Select the **Netlist Directory**.
 - c. Select the **Simulation Netlist File**, a .spi or .sp file in the netlist folder.
5. In the **Measurement Bounds** area Start Time and Stop Time boxes, specify the time window within which the PrimeWave Design Environment will obtain the minimum and maximum current values while post-processing the waveforms. Make sure you specify the abbreviated unit of time with the values (for example, 1n).

6. In the **Output - Create Database** area, specify the location for the `em.db` file.
7. When you are finished, click **OK**. The Schematic Editor creates the `em.db` file and saves it in the folder you specified.

Adding cdsTerm Labels in Design Schematics

Terminal current data can be derived from cdsTerm labels that are stored in symbol views. If the cdsTerm labels are not added, the pins will not have current data annotations.

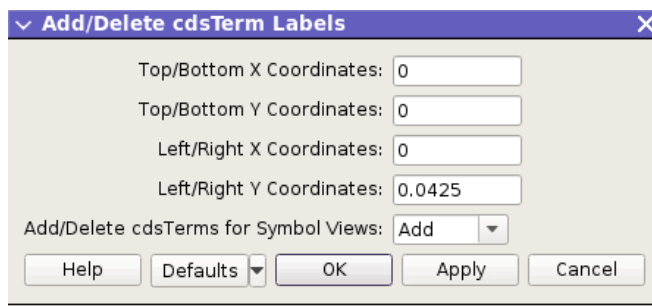
When you create symbol views in the Schematic Editor using the **Design > New CellView > From CellView** command, a cdsTerm label is automatically added to each pin in the symbol view. Creating symbol views manually or copying existing symbols might be prone to errors, such as missing cdsTerm labels.

If you are missing cdsTerm labels in the symbol view, you can automatically add them. This topic describes how you can automatically add cdsTerm labels in design schematics. To create symbol views, see [Creating Symbols From Schematics](#).

Before you start, make sure you have write permission for the symbol view. In addition, with the design open in the Schematic Editor, choose **Tools > Electrical**, so that the **Electrical** menu is displayed.

To automatically add cdsTerm labels in symbol views that have instances in the current design schematic:

1. In the design schematic, choose **Electrical > Current Annotation > Add/Delete cdsTerm Labels**. The **Add/Delete cdsTerm Labels** dialog box opens.



2. In the **Add/Delete cdsTerm Labels** dialog box, set the following options to position the cdsTerm labels relative to the pins:

Option	Description
Top/Bottom X Coordinates	X distance from the pin for labels that are placed above or below the pin

Option	Description
Top/Bottom Y Coordinates	Y distance from the pin for labels that are placed above or below the pin
Left/Right X Coordinates	X distance from the pin for labels that are placed to the left or right of the pin
Left/Right Y Coordinates	Y distance from the pin for labels that are placed to the left or right of the pin

3. In the **Add/Delete cdsTerms for Symbol Views** list, choose **Add**.
4. When you are finished, click **Apply** or **OK**.
5. Open the symbol view and check that the cdsTerm labels are added.
6. If the labels are not positioned correctly, return to the design schematic. In the **Add/Delete cdsTerm Labels** dialog box, choose **Delete**, and then repeat [Step 1](#) through [Step 5](#) to change the distances between the cdsTerm labels and the pins.

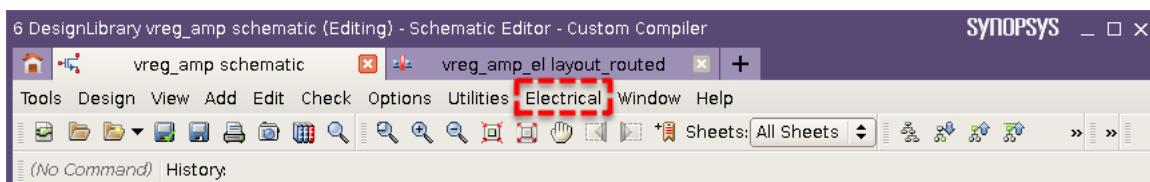
Annotating Schematics With Current Data

(Static EM analysis only) Perform this step only if you want to visualize the currents in the schematic. Otherwise, you can proceed to annotate the layout with current data. See [Annotating Layouts with Current Data](#).

You use one of the electrical tools to annotate the schematic design. The tool uses information in the EM current database that was created in [Measuring Terminal Current and Voltage Data Using the PrimeWave Design Environment](#).

To annotate the schematic design with current data:

1. Open the schematic design. For instructions, see Open Designs.
2. In the Schematic Editor, choose **Tools > Electrical**. The **Electrical** menu appears on the menu bar.



3. Choose **Electrical > Annotate Currents**. The **Annotate Currents** dialog box opens.

4. If you have saved a previous annotation setup, click **Load** to populate the data in the dialog box, and then skip to [Step 8](#).
5. In the **Root Directory** box, type the directory in which the `em.db` files are stored. In the **Get Current Data** menu, select whether to get the data **From .db Files Only** or **From Any Database File**, and then click **Get Current Data**. The list of datasets that are relevant to the opened design appears in the dialog box.
6. In the **Annotation Type** list, choose one of the following:

Annotation Type	Description
Annotate One Result (Select One Row)	Only the dataset is used for annotation. You will be permitted to select only one dataset in the list in Step 8 .
Worst Case (Across All)	Only the maximum current (worst-case) value from across all available current data sets for the design under test. You will not be permitted to select any dataset in the list in Step 8 .
Worst Case (Across Selected Rows)	Only the maximum current (worst-case) value from across the selected datasets is used for annotation. You will be permitted to select the datasets in the list in Step 8 .

7. In the **Current Type** list, choose the current values you want to use in the annotation:

Current Type	Description
Average	Average current values.
RMS	Root mean square current values.
Peak	Peak current values.
Average & RMS	Average and root mean square current values.
Average & Peak	Average and peak current values.
RMS & Peak	Root mean square and peak current values.
Average & RMS & Peak	Average, root mean square, and peak values.

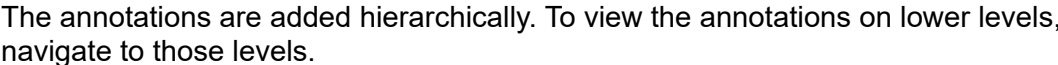
8. Select the current datasets you want to use for the annotation process.

To view a terminal current dataset, right-click the dataset in the list, and choose **Open Database File**. A window opens and displays the dataset contents. When you are finished, close the window in which the database is displayed.

	File Name	Testbench	SimInputFile	Occurrence
1	...ominal/results/em.db	...testbench/schematic	...minal/netlist/input.spi	/10
2	...ominal/results/em.db	...ench_vdd2/schematic	...n	/112
3	...ominal/results/em.db	...ench_vdd2/schematic	...minal/netlist/input.spi	/113

Your ability to select datasets depends on the Annotation Type you chose in [Step 6](#). If you selected Worst Case (across all), you will not be permitted to select any dataset.

9. When you are finished, click **Apply**. The schematic design is annotated with current data. You can zoom in the design to verify that the annotations are added.



The current annotations are saved as an `em_config` cellview. The Layout Editor uses this cell view to annotate the design layout.

10. To save the annotation setup, click **Save**. An `em_config` cellview is created. You can reload ([Step 4](#)) and use the same settings later.

Saving the settings also allows layout designers to reload the saved state and proceed without having to re-enter the necessary information.

11. To clear the existing annotation and re-annotate using different data selections, click **Clear**, and then repeat steps 4 through 10. You can re-annotate as many times as desired.

(Static EM analysis only) You can use the current data saved in the EM current database to annotate the design layout. In the design layout, current values are saved as instance terminal and pin attributes.

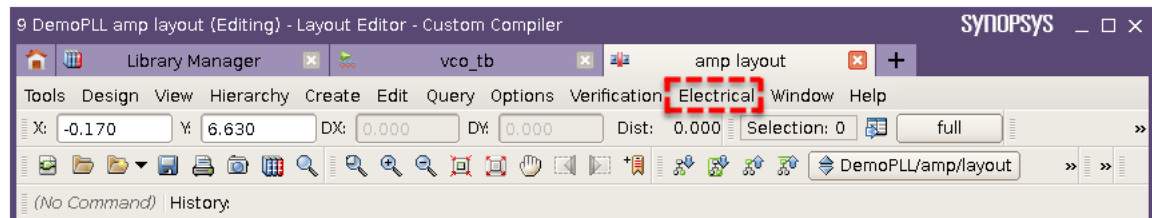
This topic describes how to automate the annotation process. If you want to manually annotate the layout, or if you want to override an automatically annotated value after this step, you can do so in the Connection section of the Property Editor.

Before you start, make sure you:

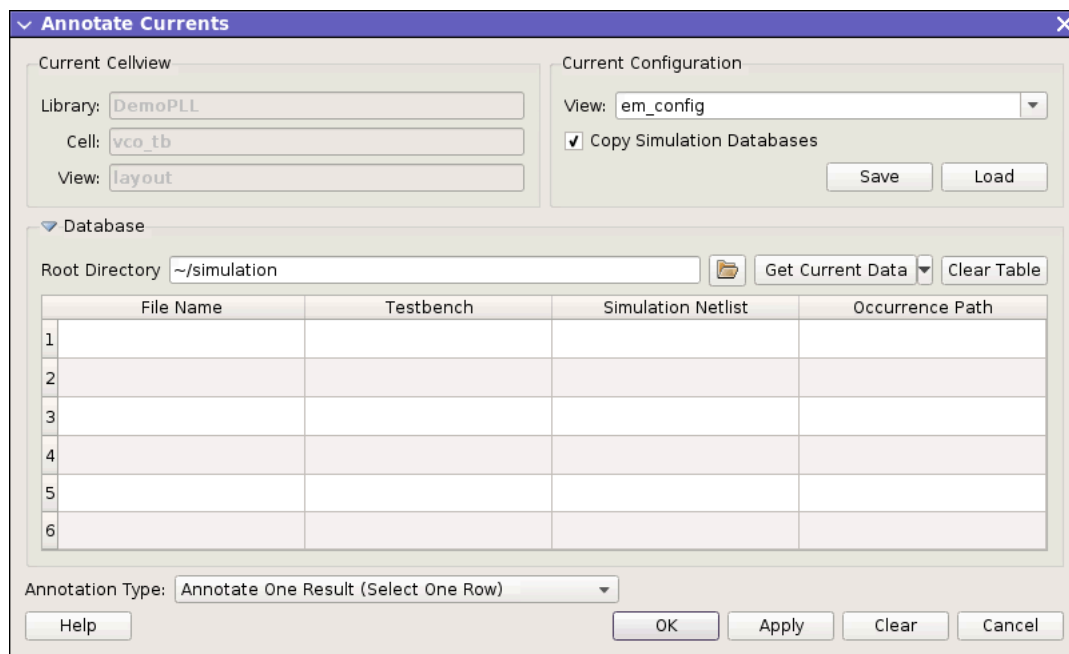
- Open the design in the SDL mode.
- Realize the instances in the layout with correct connectivity so that you can run automate the current annotation in the Layout Editor.

To annotate the design layout with current data:

1. In the Layout Editor, choose **Tools > Electrical**. The **Electrical** menu appears in the menu bar.



2. Choose **Electrical > Current Annotation > Annotate Currents**. The **Annotate Currents** dialog box opens.



3. To load the annotation setup you created previously, click **Load**.

4. To set up a new annotation configuration:

- a. In the **Root Directory** box, type the directory in which the `em.db` files are stored. In the **Get Current Data** menu, select whether to get the data **From .db Files Only** or **From Any Database File**, and then click **Get Current Data**. The list of datasets that are relevant to the opened design appears in the dialog box.
- b. In the **Annotation Type** list, choose one of the following:

Annotation Type	Description
Annotate One Result (Select One Row)	Only the data dataset is used for annotation. You will be permitted to select only one dataset in the list in Step 4.c .
Worst Case (Across All)	Only the maximum current (worst-case) value from across all available current data sets for the design under test. You will not be permitted to select any dataset in the list in Step 4.c .
Worst Case (Across Selected Rows)	Only the maximum current (worst-case) value from across the selected datasets is used for annotation. You will be permitted to select the datasets from the list in Step 4.c .

- c. Select the current datasets you want to use for the annotation process.

To view a terminal current dataset, right-click the dataset in the list, and choose **Open Database File**. A window opens and displays the dataset contents. When you are finished, close the window in which the database is displayed.

File Name	Testbench	SimInputFile	Occurrence
1 ...ominal/results/em.db	...testbench/schematic	...minal/netlist/input.spi	/10
2 ...ominal/results/em.db	...ench_vdd2/schematic	...n	/11 2
3 ...ominal/results/em.db	...ench_vdd2/schematic	...minal/netlist/input.spi	/11 3

Your ability to select datasets depends on the Annotation Type you chose in [Step 6](#). If you selected Worst Case (across all), you will not be permitted to select any dataset.

5. In the **Annotate Currents** dialog box, click **OK**. The Layout Editor starts to load the current data into the design layout. During this process, the current measurements are automatically mapped from the schematic terminals to the corresponding pins in the design layout.

6. To save the annotation setup, click **Save**. An `em_config` cellview is created. You can reload ([Step 3](#)) and use the same configuration later.
7. To clear the existing annotation and re-annotate using different data selections, click **Clear**, and then repeat steps 3 through 6. You can re-annotate as many times as desired.

Viewing Current Data in the Layout Editor

After you annotate the layout with current data, you can use various ways to view the current information in the layout. Use the procedures in this topic for static EM analysis only.

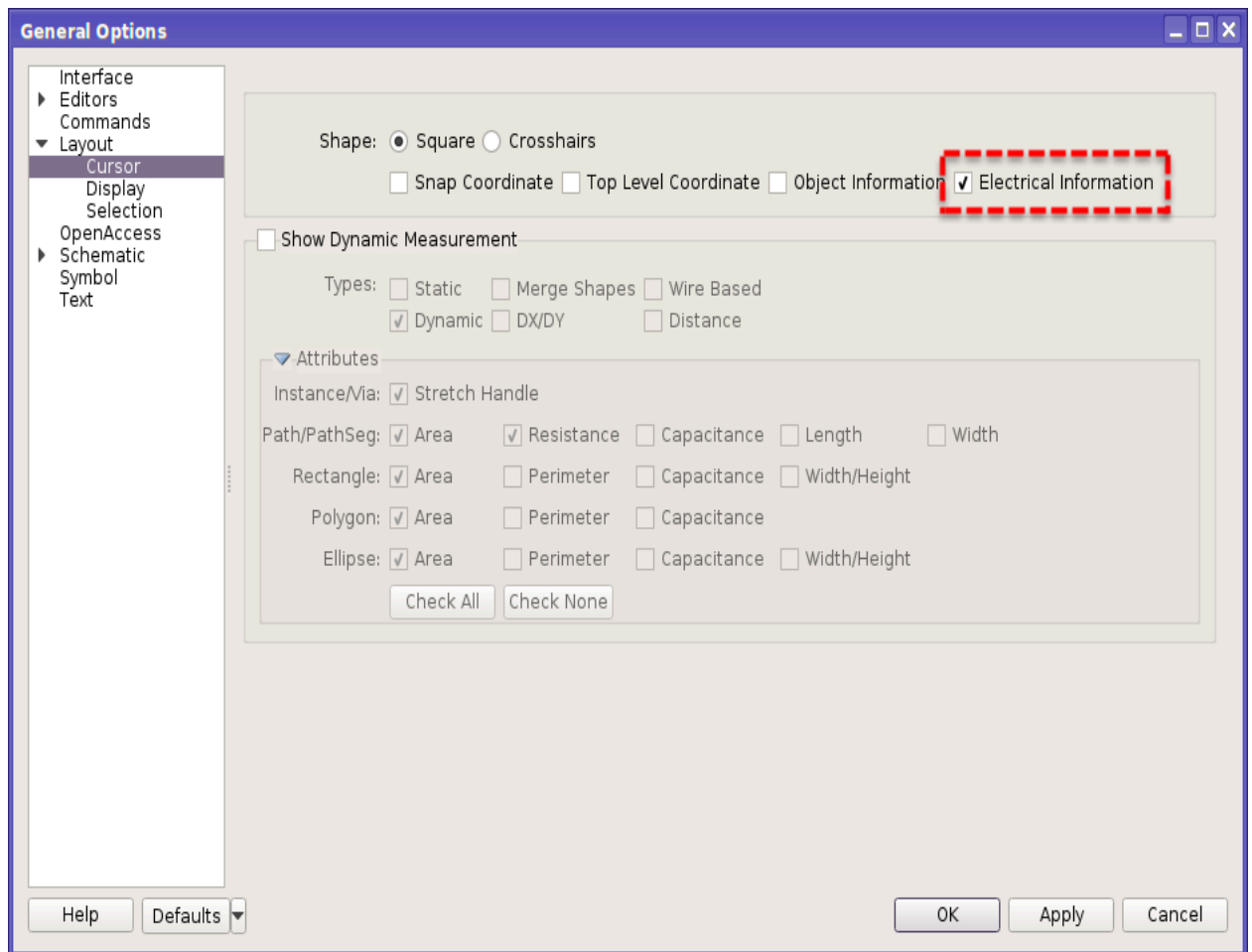
This topic describes the ways you can view the current data:

- [Viewing Current Data in Tooltips](#)
- [Viewing Recommended Width of a Single Wire](#)
- [Viewing All Recommended Wire Widths](#)

Viewing Current Data in Tooltips

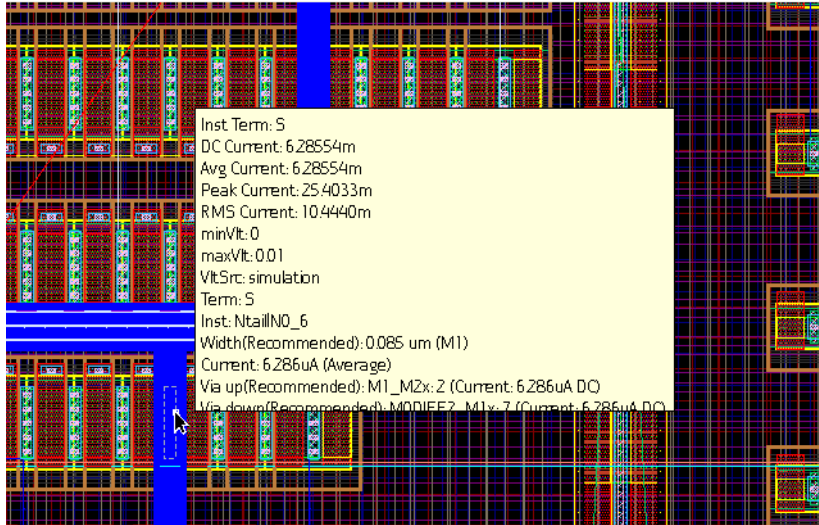
To display the current data in tooltips:

1. In the Layout Editor, display the **Layout Display Options** dialog box. See [Setting Layout Display, Cursor, and Selection Options](#).
2. Click the **Cursor** tab and select **Show: Electrical Information**.



3. Click **OK** to apply the setting.

4. On the canvas, locate the pin of interest and rest the pointer on it. The tooltip displays the electrical information, including the recommended EM-compliant wire width for the pin.



Viewing Recommended Width of a Single Wire

You can view the recommended width of a single wire using one of the following methods:

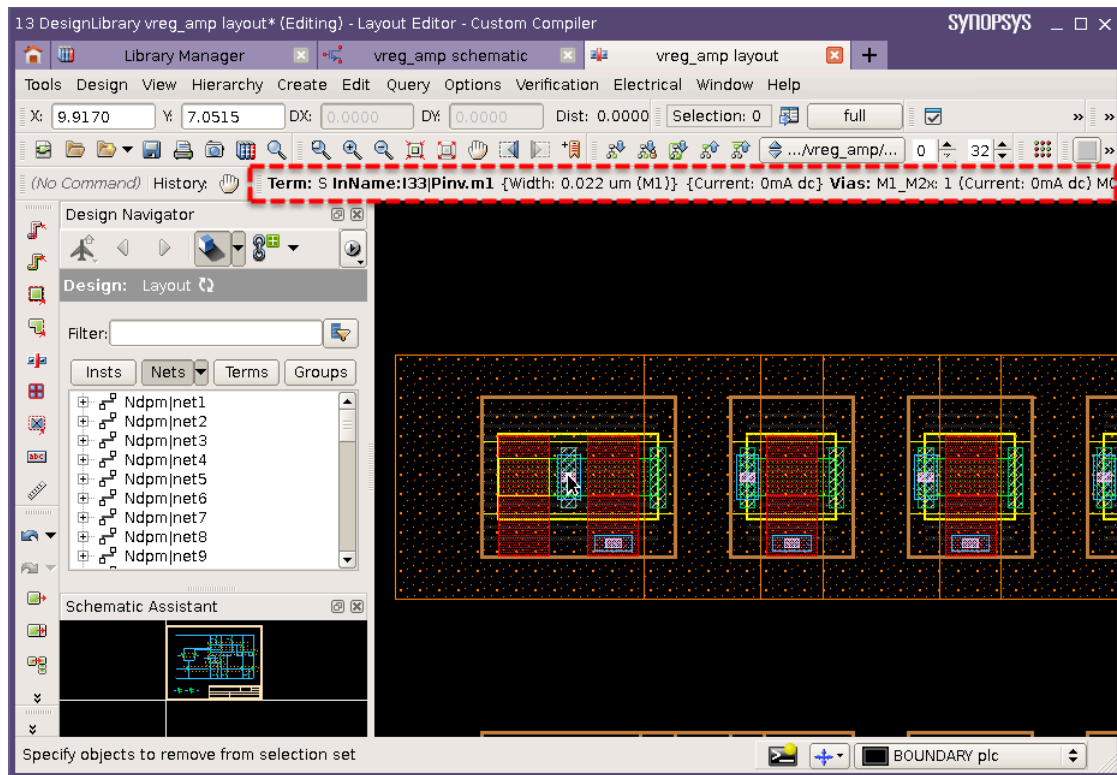
- In a tooltip. See [Viewing Current Data in Tooltips](#).
- On the Object Info toolbar.

This topic describes how to view the estimated EM-compliant width of a single wire on the Object Info toolbar. To view all the recommended wire widths in the design, see [Viewing All Recommended Wire Widths](#).

To view the recommended width of a single wire on the Object Info toolbar:

1. If you have not already done so, in the Layout Editor, choose **Tools > Electrical**. The **Electrical** menu appears on the menu bar.
2. Choose **Electrical > Current Annotation > Show EM Width in Toolbar**.

3. On the canvas, locate and rest the pointer on the pin of interest. The Object Info toolbar displays the recommended EM-compliant wire width for the pin.

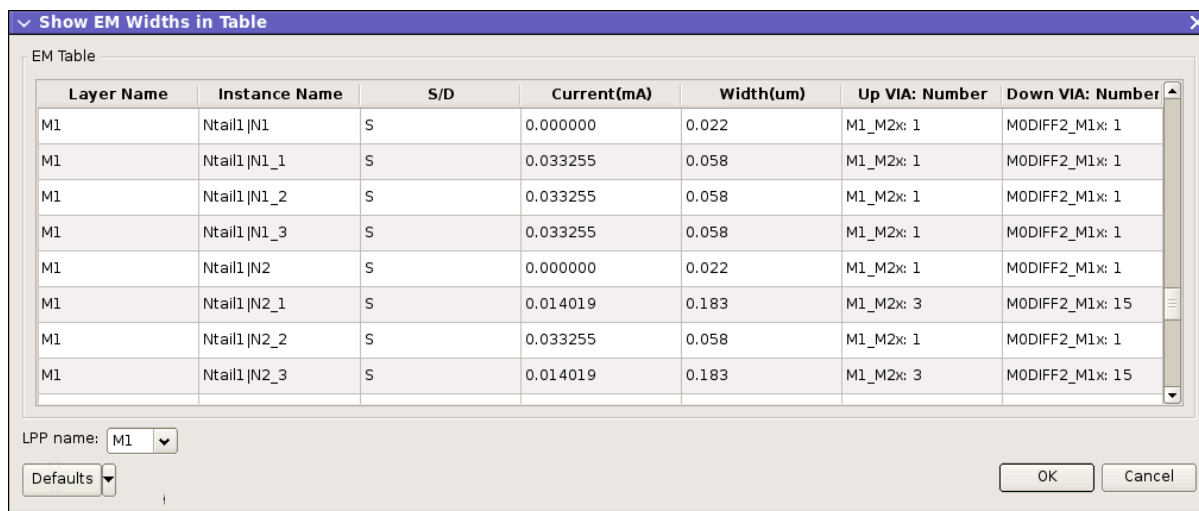


Viewing All Recommended Wire Widths

This topic describes how to view the estimated EM-compliant widths across all layers in the design. To view the recommended width of a single wire, see [Viewing Recommended Width of a Single Wire](#).

To view all the estimated EM-compliant widths:

1. In the Layout Editor, choose **Electrical > Current > Show EM Widths in Table**. The **Show EM Widths in Table** dialog box opens.



2. Review the information in the **EM Table**.

Column Heading	Description
Layer Name	Name of the layer on which the wire is located
Instance Name	Name of the instance
S/D	Type of terminal: Source or Drain
Current(mA)	Current flowing through the wire
Width(um)	Recommended width of the wire
Up VIA: Number	Via that connects the adjacent layer above the wire
Down VIA: Number	Via that connects the adjacent layer below the wire

3. (Optional) To display only the information of a specific layer, choose the **LPP name**.
4. When you are finished, click **OK** to close the dialog box and return to the layout.

Editing Current Annotations

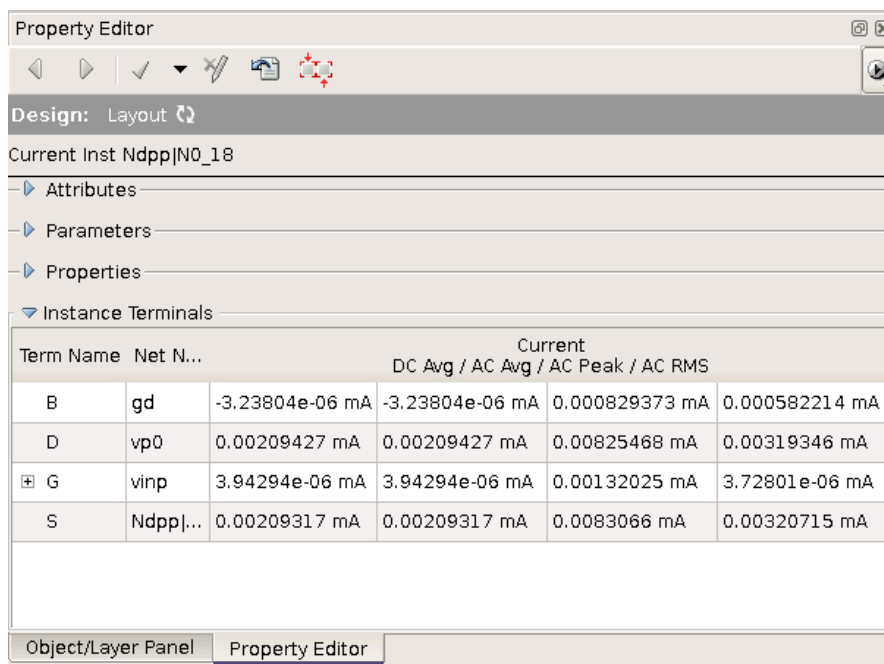
After you annotate the layout with current data, you can view and edit the current information in the layout. Use the procedures in this topic for static EM analysis only.

You can edit simulated current annotations to provide more realistic values. You can also use editing procedures to add current values manually instead of using the Schematic Editor **Electrical > Current Annotation > Annotate Current** to automate the annotation process.

In design layouts, current values are saved as instance terminal and pin attributes. You use the Property Editor to view and edit these attributes.

To view the current data in design layouts:

1. Display the Property Editor Assistant.
2. In the design, select the pin whose current value you want to view.
3. In the Property Editor, expand the **Instance Terminals** section. A table lists the terminal name, net name, and the available current information.



4. Modify one or more current values.
5. When you are finished, click **Apply Changes** .

Reporting Electromigration

Caution:

A net must be fully routed before you can report electromigration. For routing instructions, see the Custom Router Description.

You can use the EM Reporter to analyze wire dimensions and electrical current to determine the reliability of wire segments. The EM Reporter:

- Compares the wire widths and vias against current density rules.
- Generates reports that lets you locate wire segments or vias that might cause reliability problems.

The following sections describe how to create electromigration reports:

- [Specifying Input Information](#)
- [Generating EM Reports](#)
- [Viewing EM Reports in the EM Reporter Dialog Box](#)
- [Clearing Report History in the EM Reporter Dialog Box](#)
- [Viewing EM Violations in the PrimeSim EMIR Results Viewer](#)
- [EM Reporter Dialog Box Reference](#)

Specifying Input Information

Electromigration reports are based on input information that you supply, such as the rule file and extraction settings. You can supply the information using one of the following methods:

Method	Description
Preload the input information during environment setup	You can set preferences by running a Tcl script during environment setup. In this case, you do not need to specify the settings in the EM Reporter or the Electrical Options dialog box. You only need to display the dialog box and click Report. See Generating EM Reports. To find out whether the input information is preloaded with a Tcl script, check with your administrator or CAD manager. To see the preferences you can set, see Setting EM, Resistance, and Capacitance Options and EM Reporter Dialog Box Reference.
Preload the input information by setting the Tcl preferences in the Console	Instead of running a Tcl script, you can type Tcl commands in the Console to set the preferences. To see the preferences you can set, see Setting EM, Resistance, and Capacitance Options and EM Reporter Dialog Box Reference.

Method	Description
Supply the input in the EM Reporter dialog box	You can enter the input information in the dialog box every time you generate the report. To enter the input information, see Generating EM Reports .

Generating EM Reports

You use the Design Navigator to generate net-based electromigration reports.

This section describes how to generate electromigration reports using the Design Navigator. You can also use the Tcl `file::reportEM` command, see the *Custom Compiler Tcl Reference Manual*.

 [Show me](#) an example of how to report electromigration violations.

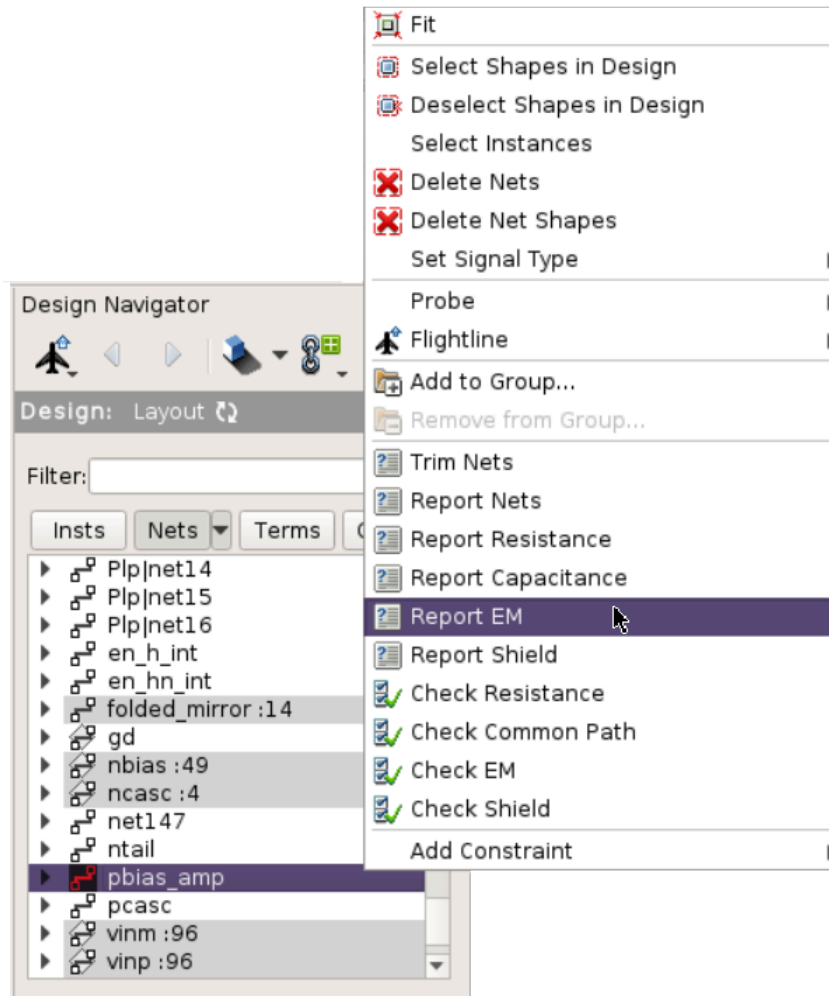
Before you start, if you want to save the report for future viewing, select the **Save History** setting.

To include self-heating effect data, see [Reporting Self-Heating Effects](#).

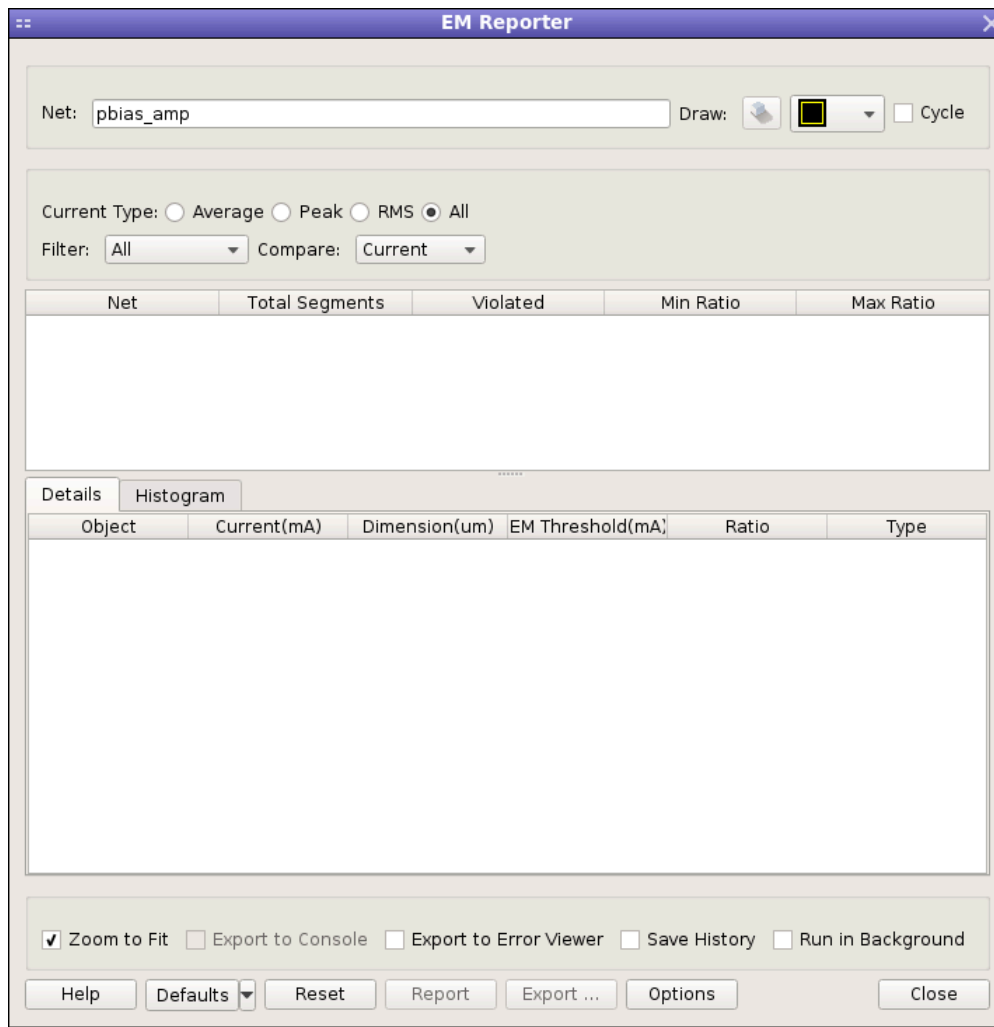
To report electromigration calculations for nets:

1. (Optional) Select one or more nets:
 - a. In the Layout Editor, open the Design Navigator.
 - b. Click the **Nets** tab and select the nets. You can do this from either the **Design: Layout** mode or the **Design: Schematic** mode.

You can also select multiple nets and choose **Report EM** to extract parasitics in a single run. If you select more than 10 nets, the **Reporter - Interactive StarRC** dialog box appears. Click **OK** to continue or **Cancel**. If you select **Cancel**, you should select less than 10 nets and continue extraction in the interactive mode.
2. Right-click the selected nets, and then choose **Report EM**.



The **EM Reporter** dialog box opens. The names of the nets you selected appear in the **Net** box.



The EM Reporter dialog box is shown with the following settings:

- Net: pbias_amp
- Draw: [Icon] [Color Selection] [Cycle]
- Current Type: ☐ Average ☐ Peak ☐ RMS ☒ All
- Filter: All
- Compare: Current

Net	Total Segments	Violated	Min Ratio	Max Ratio
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Details | Histogram

Object	Current(mA)	Dimension(um)	EM Threshold(mA)	Ratio	Type
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☒ Zoom to Fit ☐ Export to Console ☐ Export to Error Viewer ☐ Save History ☐ Run in Background

Help Defaults Reset Report Export ... Options Close

3. If the information in the dialog box was preloaded as part of the environment setup:
 - a. Review the settings in the **EM Reporter** dialog box.
 - b. Click **Options** (at the bottom of the dialog box), and review the settings in the **Electrical Options** dialog box. See [Setting EM, Resistance, and Capacitance Options](#) and [Specifying Input Information](#).
 - c. Skip to step 5.

4. If you did not preload settings in step 3, set the options in the following:
 - **EM Reporter** dialog box. See [EM Reporter Dialog Box Reference](#).
 - **Electrical Options** dialog box. See [Setting EM, Resistance, and Capacitance Options](#).

If you want to view the results in the PrimeSim EMIR results browser, select **Load Report in Results Browser** in the **Electrical Options** dialog box.

Note:

Values you enter in the dialog boxes override preloaded values.

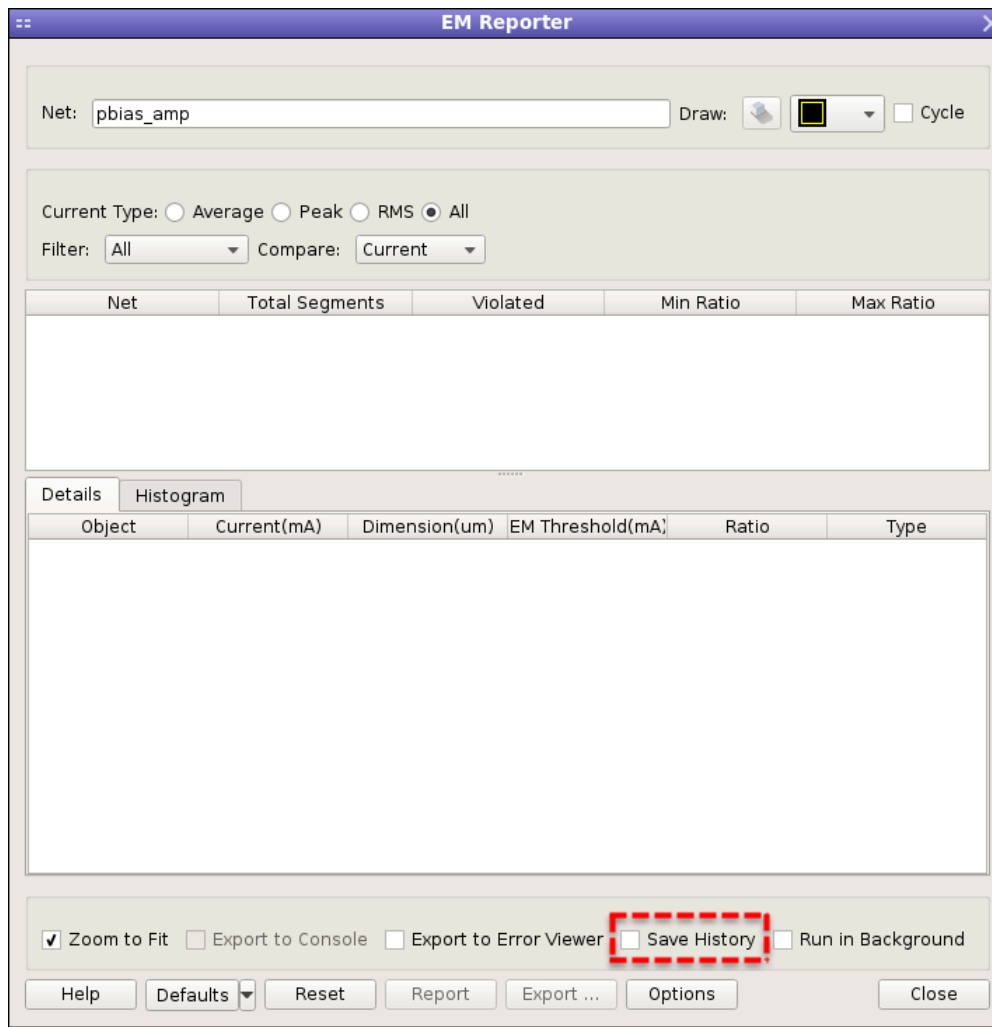
5. To automatically save the generated electromigration report so that you can recall it in future design sessions, select **Save History**. (Alternatively, set the `leEMReporterAutoSave` Tcl preference to `true` before you generate the report.) The **History** section appears.

All newly generated reports will be automatically saved in the following directory and file:

```
$env(SYNOPSYS_CUSTOM_LOCAL)/em_reporter/<cellview>_em_result.db
```

If a reported net already exists in the `.db` file, it will be overwritten by the latest data.

To recall the saved report, see [Viewing EM Reports in the EM Reporter Dialog Box](#).

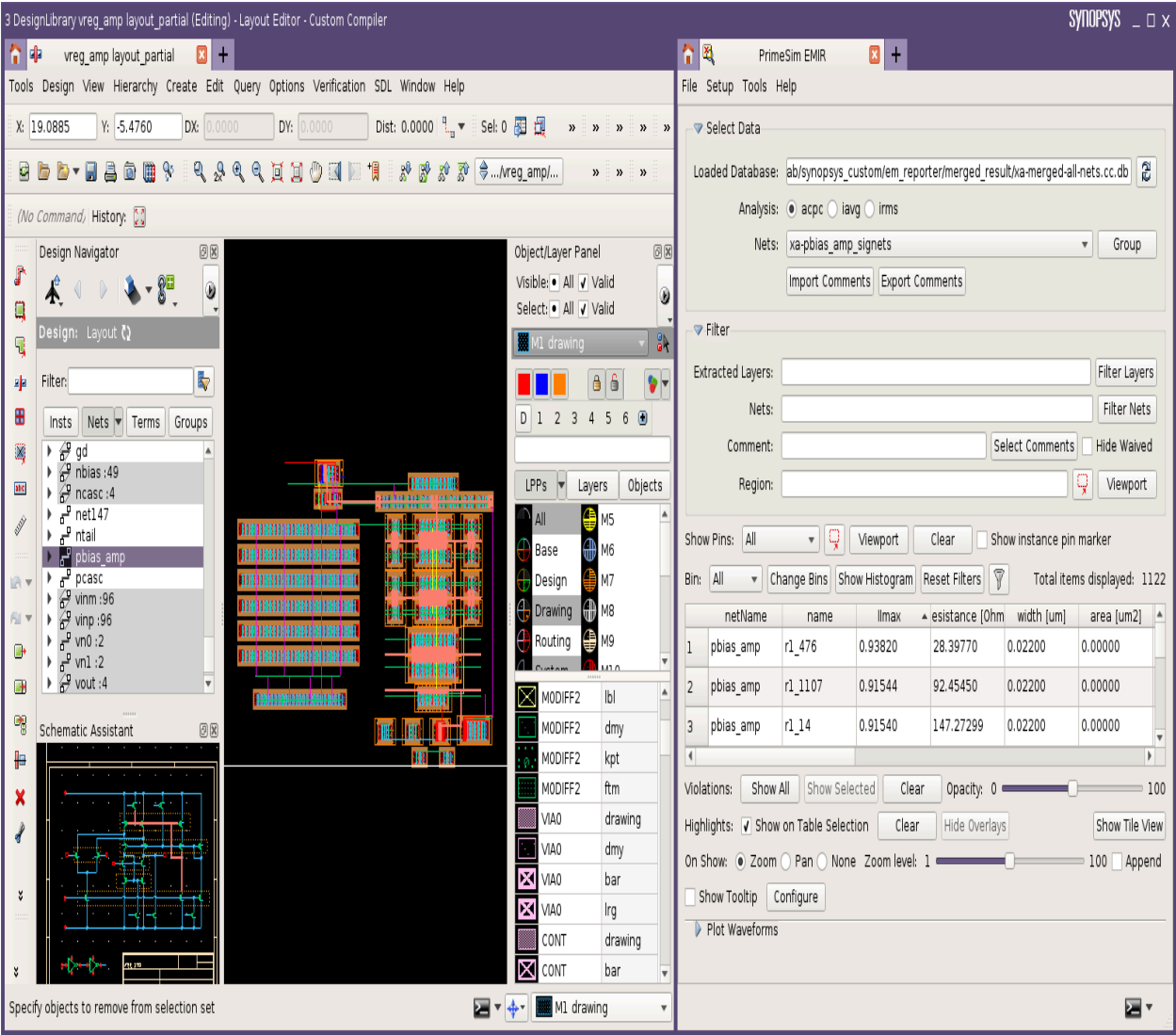


6. To export and list the violations in the Error Viewer, select **Export to Error Viewer**.
7. If you want the EM analysis in the background so that you can continue to work in the Layout Editor, select **Run in Background**.

This option is only available if you selected **Electromigration: StarRC / PrimeSim EMIR** in the **Electrical Options** dialog box. See [Setting EM, Resistance, and Capacitance Options](#).

8. Click **Report**.

If you selected **Load Report in Results Browser** in the **Electrical Options** dialog box, the EM Reporter automatically displays the results in the PrimeSim EMIR results viewer.



If you cleared the **Load Report in Results Browser** check box, the EM Reporter displays the results in the **EM Reporter** dialog box.

The EM Reporter dialog box displays the following information:

Net: Draw: ☐ Cycle

Current Type: ☐ Average ☒ Peak ☐ RMS ☐ All

Filter: Compare:

Net	Total Segments	Violated	Min Ratio	Max Ratio
pbias_amp	504	0	0	0.9382

Details Histogram

Object	Current(mA)	Dimension(um)	EM Threshold(mA)	Ratio
Wire on M3	-2.0696	0.022	2.2059	0.9382
Wire on M4	2.0696	0.022	2.2607	0.91544
Wire on M4	2.0695	0.022	2.2607	0.9154
Wire on M4	2.0694	0.022	2.2607	0.91538
Wire on M1	-2.0694	0.028	2.6227	0.78906
Wire on M3	2.0696	0.028	2.8075	0.73716
Wire on M3	-2.0694	0.028	2.8075	0.7371
Wire on M3	2.0694	0.028	2.8075	0.7371

History

Net	Total Segments	Violated
pbias_amp	726	0
vph	878	7

☒ Zoom to Fit ☐ Export to Console ☐ Export to Error Viewer ☒ Save History ☐ Run in Background

See Also

- [Viewing EM Reports in the EM Reporter Dialog Box](#)
- [Viewing EM Violations in the PrimeSim EMIR Results Viewer](#)
- [EM Reporter Dialog Box Reference](#)
- [Electrical Options Dialog Box Reference](#)

Viewing EM Reports in the EM Reporter Dialog Box

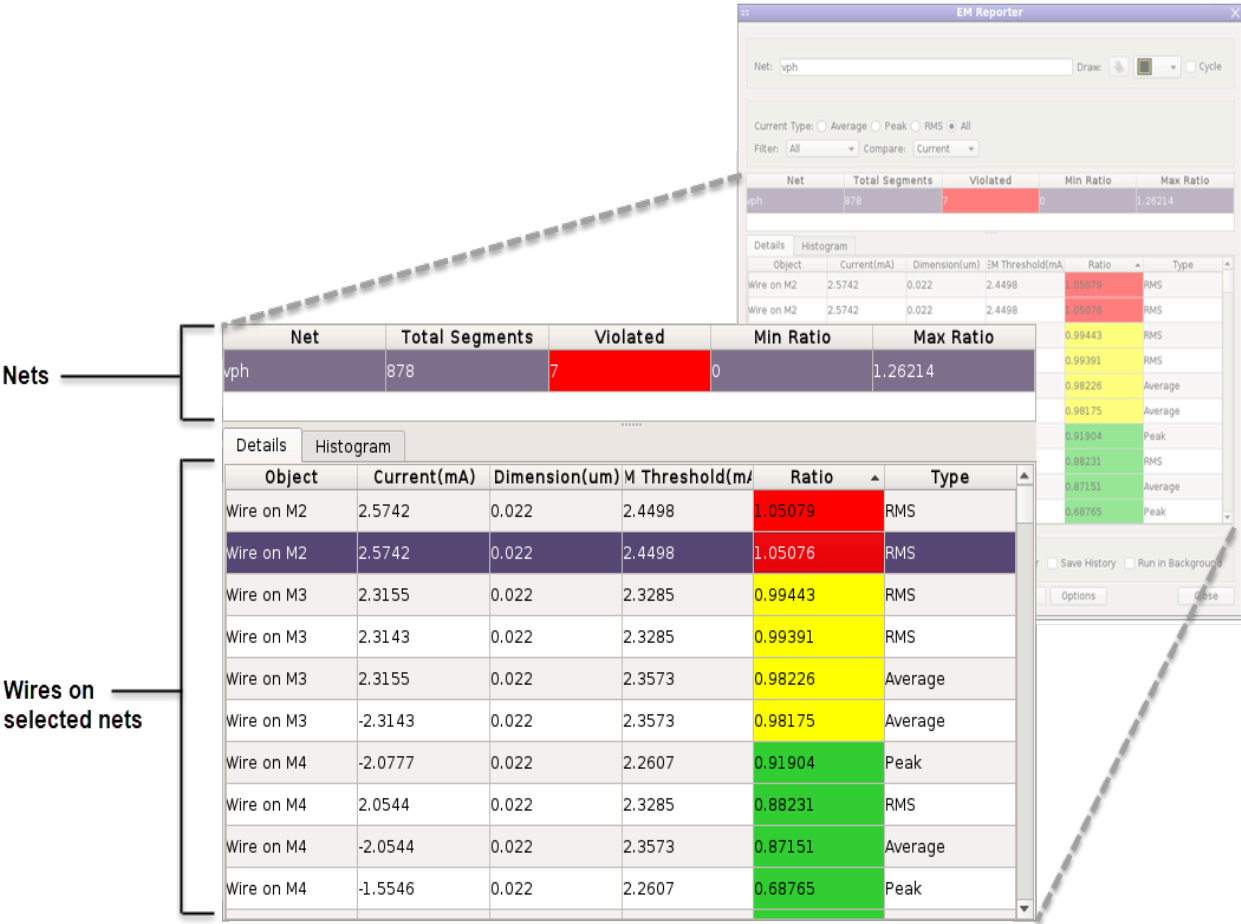
All newly generated reports are automatically displayed in the **EM Reporter** dialog box. This topic describes how to view the report in the **EM Reporter** dialog box. For dynamic EM analysis, if you want to display the results in the PrimeSim EMIR results viewer, see [Generating EM Reports](#).

If you selected **Save History** or set the `leEMReporterAutoSave` Tcl preference to `true`, reports are automatically saved to the following directory and file so that you can load and view them in future sessions:

```
$env(SYNOPSYS_CUSTOM_LOCAL)/em_reporter/<cellview>_em_result.db
```

The electromigration report in the **EM Reporter** dialog box consists of the following:

- **Nets** table: Lists the nets you selected for reporting.
- **Details** tab: Shows the individual wire segments of the net you select in the Nets table.
- **Histogram** tab: Shows the distribution of the various data in the EM report.



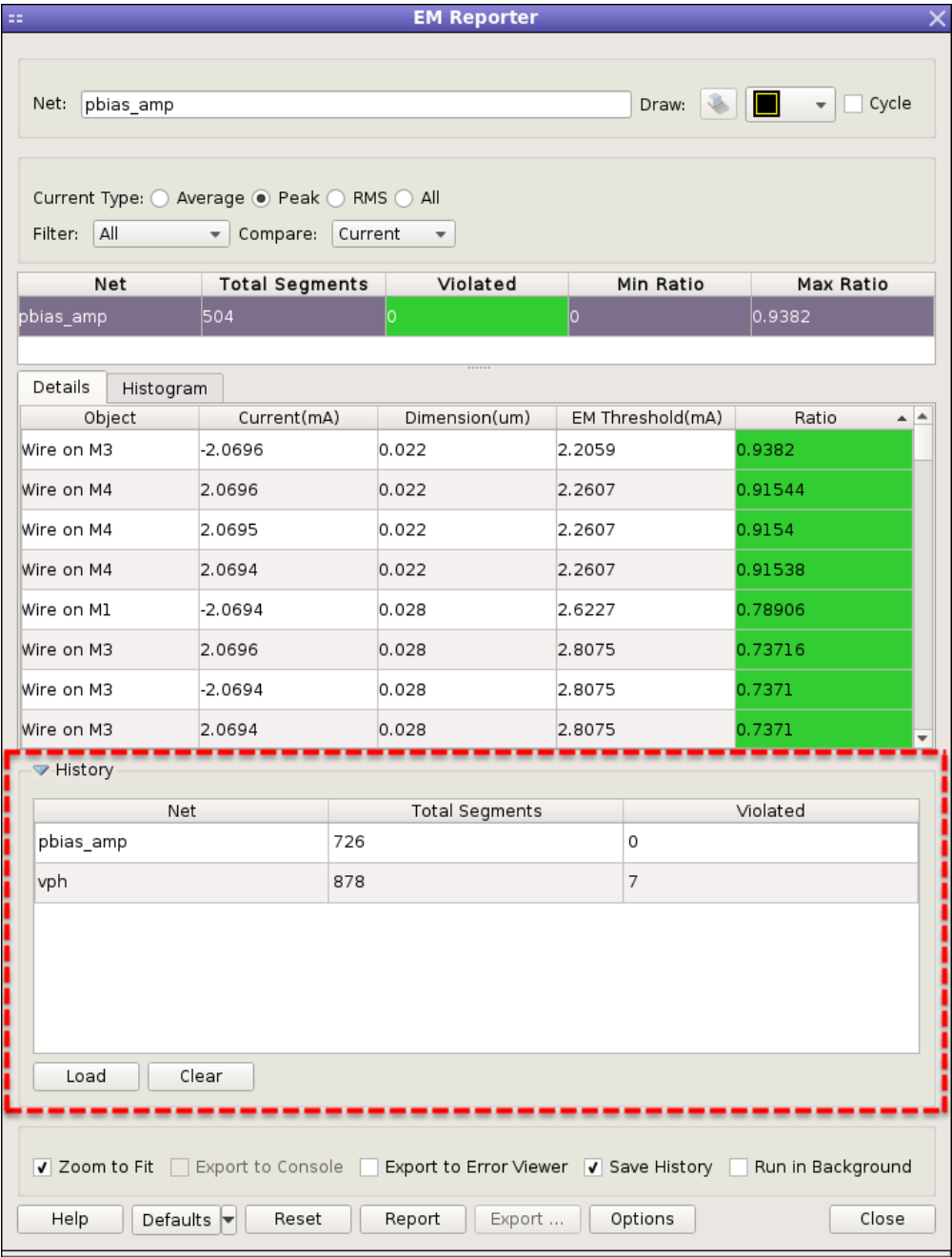
To view the electromigration report:

1. In the **EM** dialog box, adjust the report contents by selecting the option in the **Filter** and **Compare** lists.

Option	Description
Filter	Displays the selected category in the report: ◦ Violation: Wire segments that have electromigration violations ◦ Non-optimal: Wire segments that do not have electromigration violations, but can potentially decrease wire dimensions ◦ All: All wire segments

Option	Description
Compare	Specifies the method to use when determining potential electromigration: ◦ Current: Compares the current the wire can transport against its given width. The Electromigration Reporter calculates and displays the ratio between the computed current for the wire (Current column) and the maximum current allowed (Max Current column). The ratio will be displayed in the Ratio column. ◦ Dimension: Compares the wire width against the minimum wire dimension that avoids a violation. The Electromigration Reporter calculates and displays the difference between the actual wire width (the first Dimension column) and the minimum wire dimension (the second Dimension column). The difference will be displayed in the Delta column.

2. If you want to view a saved report:
 - a. Expand the **History** section.
 - b. In the **History** table, select the nets of interest.
 - c. Click **Load**. The report for the selected nets appears in the tables above the History section.



3. Review the information in the **Nets** table:

Column Name	Description
Net	The net name.

Column Name	Description
Total Segments	The number of path segments on the net.
Violated	The number of path segments that have electromigration violations.
Min Delta	(If you selected Compare: Dimension) The minimum difference between wire widths on the net.
Max Delta	(If you selected Compare: Dimension) The maximum difference between wire widths on the net.
Min Ratio	(If you selected Compare: Current) The minimum ratio value on the net. The ratio is Current/Max Current.
Max Ratio	(If you selected Compare: Current) The maximum ratio value on the net. The ratio is Current/Max Current.

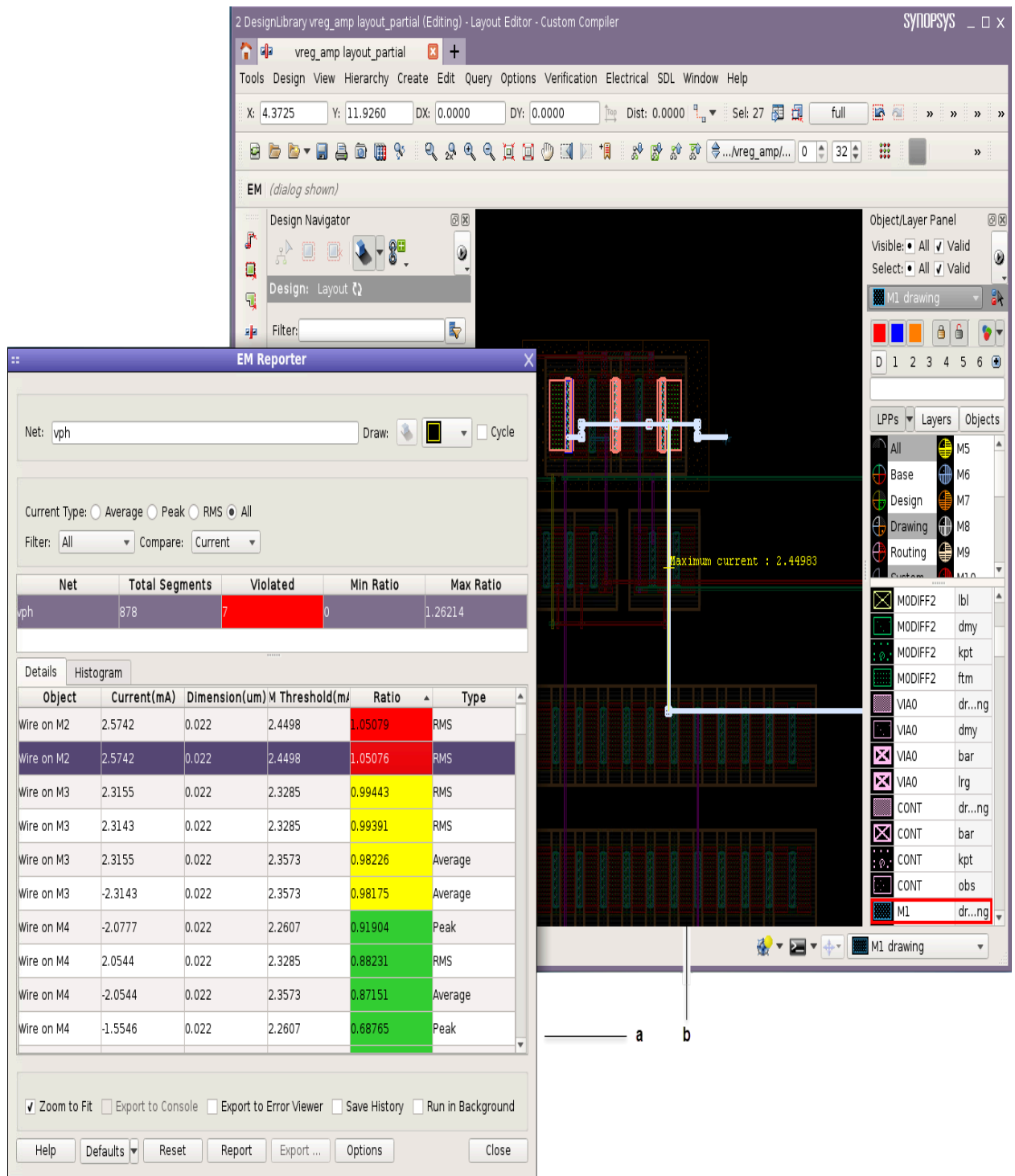
- Select a net in the **Nets** table to display the detailed electromigration report in the **Details** tab. The data is listed by wire segment.
- View the detailed report in the **Details** tab. The report contains the following information per wire segment:

Option	Description
Object	The layer on which the wire segment is located.
Current	The calculated current of the wire segment.
Dimension	The width of the wire segment.
EM Threshold	(If you selected Compare: Current) The maximum current allowed for the wire segment.
Min Dimension	(If you selected Compare: Dimension) The minimum wire width required to avoid violation.
Ratio	(If you selected Compare: Current) The ratio of the calculated current for the wire (Current column) and the maximum current allowed (Max Current column): Current/Max Current. The Ratio is color-coded as follows: - Red: Ratio > leEMViolationThreshold - Yellow: Ratio <= leEMViolationThreshold, and >= leEMWarningThreshold - Green: Ratio < leEMWarningThreshold
Type	The type of current: AC Average (average alternating current).

6. To view a segment or via on the canvas, select **Zoom** if you want to zoom in when viewing the segment or via, and then select a segment in the table (a). The Electromigration Reporter zooms in and displays the segment on the canvas (b).

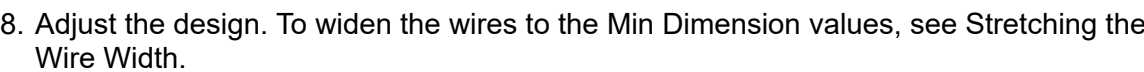
Chapter 3: Reporting and Checking Electromigration (EM)

Reporting Electromigration



7. To view the histograms of the report data, click the **Histogram** tab, and then:
 - a. Choose the **Type** of data you want to view.
For the descriptions of the data, see steps 3 and 5.
 - b. Set the **Color Level**, the number of colors you want to use when displaying the histogram and heat map on the canvas.
The data between the **Min** and **Max** values are binned into the number of colors you specify.
 - c. Specify the **Min** and **Max** values for the data type you chose in step a.
 - d. View the bar chart that shows the distribution of the data type you selected.
 - e. To view the corresponding heat map on the canvas, click **Show heat map** . To hide the heat map, click **Hide heat map** .
 - f. Repeat steps a through e to view other data types and the corresponding distribution. To refresh the histogram and the heat map on the canvas, click **Refresh** .

If you want to automatically refresh the histogram and heat map display, select the check box below .



9. After you make the adjustments:
 - a. Return to the **EM Reporter** dialog box.
 - b. Click **Reset** to clear any old data.
 - c. Click **Report** to generate a new electromigration report. Verify that the adjustments you made corrected the violations.
10. Continue to adjust the design and generate new reports until the electromigration calculations are satisfactory.

Note:

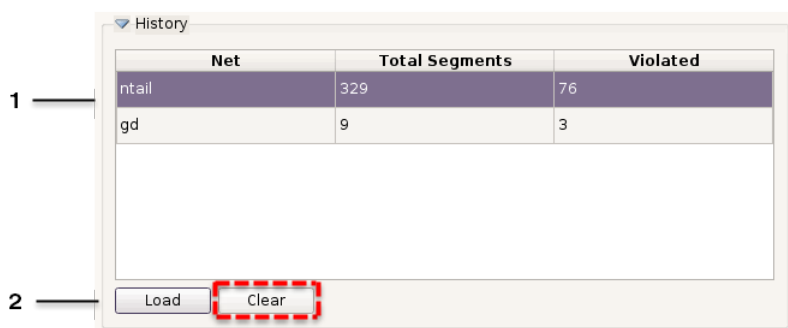
The Export feature is available for Resistance reporting only.

Clearing Report History in the EM Reporter Dialog Box

In the **EM Reporter** dialog box, you can remove EM report history to clean up the report section before generating new reports.

To remove EM reports from EM Reporter dialog box:

1. At the bottom of the **EM Reporter** dialog box, select **Save History** to display the **History** section.
2. In the **History** section, select the net. If you want to remove all reports, do not select any net.
3. Click **Clear**. The selected net (report) is removed. If you did not select any net, all net reports are removed.



Viewing EM Violations in the PrimeSim EMIR Results Viewer

For dynamic EM analysis, you have the option of displaying the results in the PrimeSim EMIR results viewer. This topic describes how to view the report in the PrimeSim EMIR results viewer.

If you selected **Load Report in Results Browser** in the **Electrical Options** dialog box, the EM Reporter automatically displays the results in the PrimeSim EMIR results viewer. See [Generating EM Reports](#) and [Electrical Options Dialog Box Reference](#).

The screenshot shows the PrimeSim EMIR results viewer window. The interface is divided into three main sections indicated by brackets on the left:

- Select Data:** This section includes the 'Select Data' panel with a 'Loaded Database' field (ab/synopsys_custom/em_reporter/merged_result/xa-merged-all-nets.cc.db), 'Analysis' radio buttons (acpc, iavg, irms), a 'Nets' dropdown (xa-pbias_amp_signets), and buttons for 'Import Comments' and 'Export Comments'. Below this is the 'Filter' panel with input fields for 'Extracted Layers', 'Nets', 'Comment', and 'Region', along with buttons for 'Filter Layers', 'Filter Nets', 'Select Comments', 'Hide Waived', and 'Viewport'.
- View Data:** This section contains a table of results. The table has columns: netName, name, lmax, esistance [Ohm], width [um], and area [um2]. The data shows three entries for 'pbias_amp' with different 'name' values (r1_476, r1_1107, r1_14) and their corresponding 'lmax' and 'esistance' values.
- Set Up Canvas:** This section includes controls for 'Show Pins' (All, Viewport, Clear), 'Bin' (All, Change Bins, Show Histogram, Reset Filters), 'Violations' (Show All, Show Selected, Clear), 'Highlights' (Show on Table Selection, Clear, Hide Overlays, Show Tile View), 'On Show' (Zoom, Pan, None, Zoom level: 1, Append), 'Show Tooltip' (Configure), and a 'Plot Waveforms' button.

	netName	name	lmax	esistance [Ohm]	width [um]	area [um2]
1	pbias_amp	r1_476	0.93820	28.39770	0.02200	0.00000
2	pbias_amp	r1_1107	0.91544	92.45450	0.02200	0.00000
3	pbias_amp	r1_14	0.91540	147.27299	0.02200	0.00000

To view the report in the PrimeSim EMIR results viewer:

1. Select the data you want to view. See [Selecting the Analysis Data for Viewing](#).
2. View the report data in the table. See [Viewing the Violations in the Data Table](#).
3. Set up the canvas and view the violations on the layout canvas. See [Viewing the Violations on the Layout Canvas](#).

For example, you can configure the tooltips to show resistor name, layer information, and current value when you rest the pointer on a resistor shape.

For all other features in the PrimeSim EMIR results viewer, see [Workflow for Performing EMIR Analysis](#).

4. After viewing the report, adjust the design to correct violations or other issues.

To widen the wires to the Min Dimension values, select the violating wire and modify its width using the Property Editor. See also [Stretching the Wire Width](#).

To address EM issues without widening or lengthening the wires, you can stack the routes. See [Setting the Layer and Width Attributes when pattern routing](#).

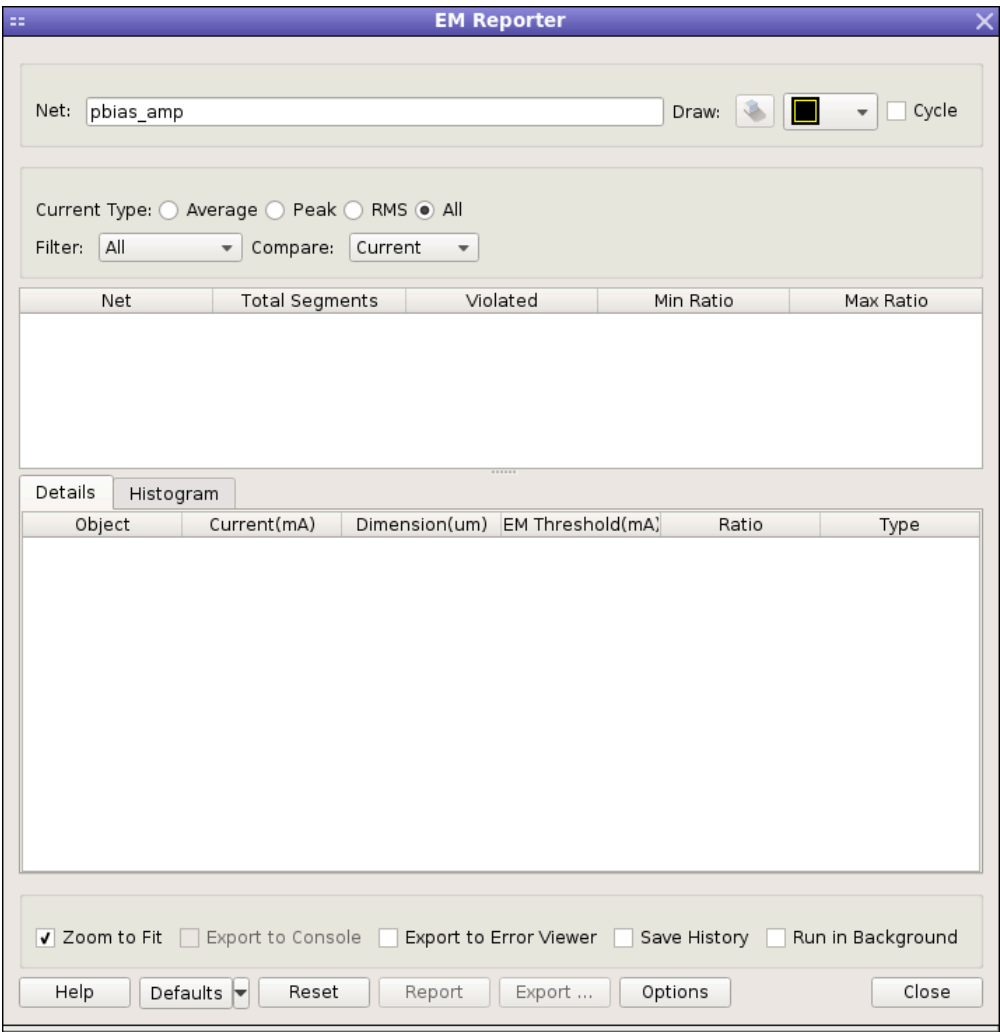
5. After you make the adjustments:
 - a. Return to the **EM Reporter** dialog box.
 - b. Click **Reset** to clear any old data.
 - c. Click **Report** to generate a new electromigration report. Verify that the adjustments you made corrected the violations.
6. Continue to adjust the design and generate new reports until the electromigration calculations are satisfactory.

EM Reporter Dialog Box Reference

The **EM Reporter** dialog box lets you generate and view EM reports. A link opens the **Electrical Options** dialog box to let you set options for report generation.

This topic provides a quick reference of the **EM Reporter** dialog box. To generate EM reports, see [Generating EM Reports](#). To set the required reporting options, see [About the Electrical Options Dialog Box](#).

The **EM Reporter** dialog box contains the following options:



Net Options

Option	Description
Net	Generates a report for the specified net. By default, the net you selected in the Design Navigator or in the layout appears in the box.
Draw	Applies the selected color to the flight line as you select the source and destination points.
Cycle	Cycles packet colors as objects on the canvas change. Disabling this option uses the current color shown.

Report Data Options

Option	Description
Current Type	Lists the available current types for EM reporting. The available current types displayed depends on the analysis types in the .ccdb file. Select the type of interest: - If you selected the Built-in electromigration reporting engine in the Electrical Options dialog box, the only current type available is Average (average alternating current). - If you selected the StarRC/PrimeSim EMIR electromigration reporting engine in the Electrical Options dialog box, the list of available current types from the .ccdb file are displayed. The .ccdb file is the output from EMIR simulations. See also About the Electrical Options Dialog Box.
Filter	Displays the selected category in the report: Violation: Wire segments that have electromigration violations Non-optimal: Wire segments that do not have electromigration violations, but can potentially decrease wire dimensions All: All wire segments
Compare	Specifies the method to use when determining potential electromigration: Current: Compares the current the wire can transport against its given width. The Electromigration Reporter calculates and displays the ratio between the computed current for the wire (Current column) and the maximum current allowed (Max Current column). The ratio is displayed in the Ratio column. Dimension: Compares the wire width against the minimum wire dimension that avoids a violation. The Electromigration Reporter calculates and displays the difference between the actual wire width (the first Dimension column) and the minimum wire dimension (the second Dimension column). The difference is displayed in the Delta column.

Summary Report Table

Column Name	Description
Net	The net that is queried for the electromigration report. Default: The net you selected in the Design Navigator in Generating EM Reports
Total Segments	The number of path segments on the net.
Violated	The number of path segments that have electromigration violations.

Column Name	Description
Min Ratio	The minimum ratio value on the net. The ratio is Current/Max Current.
Max Ratio	The maximum ratio value on the net. The ratio is Current/Max Current.

Detailed Report Tab

Column Name	Description
Object	The layer on which the wire segment is located.
Current	The calculated current of the wire segment.
Dimension	The width of the wire segment.
EM Threshold	(If you selected Compare: Current) The maximum current allowed for the wire segment.
Ratio	(If you selected Compare: Current) The ratio of the calculated current for the wire (Current column) and the maximum current allowed: Current/Max Current. - Ratio > 1 is a violation, and the table cell is red. - Ratio < 1 is non-optimal, and the table cell is green.
Min Dimension	(If you selected Compare: Dimension) The minimum wire width required to avoid violation.
Delta	(If you selected Compare: Dimension) The difference between the actual wire width (Dimension column) and the minimum wire dimension (Min Dimension column): Dimension – Min Dimension. - Difference < 0 is a violation, and the table cell is red. - Difference > 0 is non-optimal, and the table cell is green.

Histogram Tab

Column Name	Description
Type	Type of data you want to display. See the data descriptions in the Summary Report Table and the Detailed Report Tab sections.
Color Level	Number of colors to use when displaying the histogram and heat map on the canvas. The data between the Min and Max values are binned into the number of colors you specify.
Min	Minimum value of the data.

Column Name	Description
Max	Maximum value of the data.
	Displays the heat map based on the histogram on the canvas.
Show heatmap	
	Hides the heat map.
Hide heatmap	
	Updates the histogram based on updated settings in the Histogram tab.
Refresh histogram	
☒	Automatically updates the histogram as soon as you modify a setting in the Histogram tab.
Automatically refresh histogram	

History Section

History Column Name	Description
Net	The net name.
Total Segments	The total number of wire segments on the net.
Violated	The number of violations on the net.
Load	Loads the selected report and displays the data in the report tables.
Clear	Removes the selected report from the History section.

Global Options

Option	Description
Zoom to Fit	Zooms in to the selected path segment on the canvas.
Export to Console	Exports the EM report to the Console.
Save History	Retains the EM reports you are running in the History section of the dialog box. You can later reload the saved reports.
Export to Error Viewer	Exports and lists the violations in the Error Viewer.

Option	Description
Run in Background	Runs the EM analysis in the background so that you can continue to work in the Layout Editor. This option is only available if you selected Electromigration: StarRC / PrimeSim EMIR in the Electrical Options dialog box. See Setting EM, Resistance, and Capacitance Options . Tcl preference: <code>leReportEMRCBackgroundMode</code>
Report	Generates the point-to-point resistance report based on the settings you specified.
Options	Opens the Electrical Options dialog box. See About the Electrical Options Dialog Box .

Checking Electromigration

The EM Checker and the EM Reporter use the same data to generate reports. The difference is the EM Checker adds error markers in the design, and you can view the violations using the Error Viewer. The markers are saved with the design so that they are available for viewing in later sessions.

If you want a detailed electromigration report that lists current values of every wire in the design, see [Reporting Electromigration](#).

Before you start, make sure you have supplied the correct input data. See [Generating EM Reports](#).

To check electromigration:

1. In the Layout Editor, open the design.
2. Open the Design Navigator.
3. In the Design Navigator, select the net.
4. Do one of the following:
 - In the Design Navigator, right-click the net name, and then choose **Check EM**.

You can also select multiple nets and choose **Check EM** to extract parasitics in a single run. If you select more than 10 nets, the **Reporter - Interactive StarRC** dialog box appears. Click **OK** to continue or **Cancel**. If you select **Cancel**, you should select less than 10 nets and continue extraction in the interactive mode.

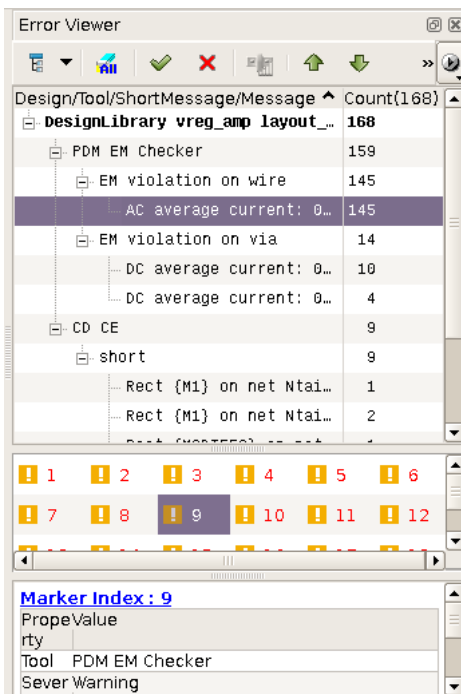
- On the canvas, right-click the path segment, and then choose **Check EM**.
- Enter the following Tcl command in the Console:
`le::checkEMResistanceConstraints`

The tool calculates equivalent current for each edge based on the current distribution, checks edge current for each edge against the current constraint for the net, and reports any violations to the Error Viewer.

If there are no violations, you do not have to proceed to step 5.

5. In the Error Viewer, expand the list to show violations per wire segment and via.

To display the full message of each violation, select the wire segment or via. The detailed violation information appears in the area below the table.



Where a measurement is in violation, you have two options:

- Change the measurement using the EM/R Fixer.
- Change the measurement using the Property Editor.

4

Reporting Self-Heating Effects

Heat generation and dissipation in devices can cause current reduction and other reliability issues. The Custom Compiler in-design self-heating electromigration (EM) flow allows you to assess and correct self-heating effects early in the design process.

This chapter describes the following:

- [About the In-Design Self-Heating EM Flow](#)
- [Setting Up for In-Design SHE EM Flow](#)
- [Viewing SHE EM Results](#)

About the In-Design Self-Heating EM Flow

Device self-heating effect (SHE) in transistors from advanced process technologies such as FinFET and FDSOI has reliability impact and must be considered during the design process. The Custom Compiler in-design EM reporting flow lets you include SHE reporting and analysis.

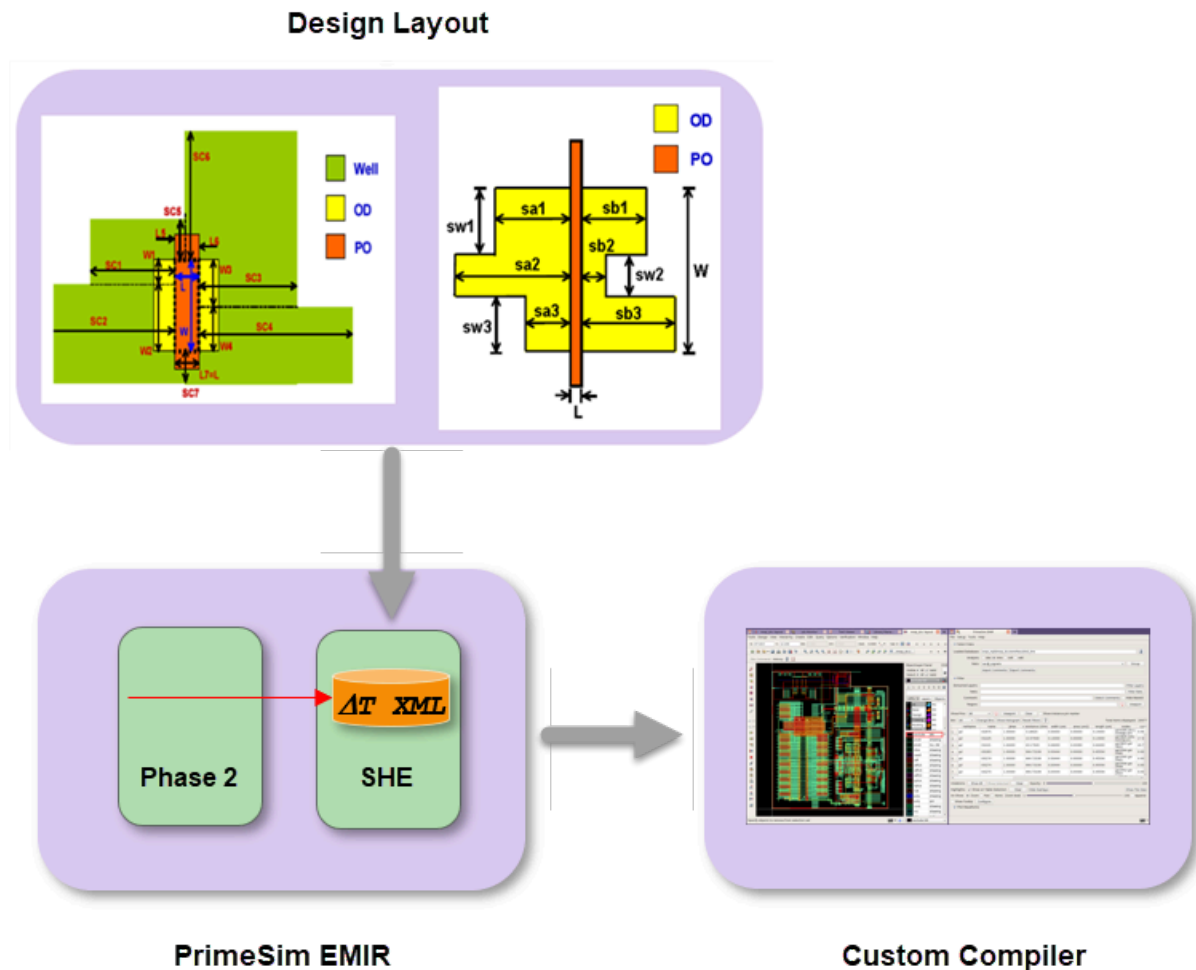
In the Custom Compiler environment, you can generate EM reports that include self-heating effect data. When setting up for EM reporting, you:

- Select the StarRC / PrimeSim EMIR engine for EM reporting.
- Select the Self-Heating Flow option.
- Specify the delta temperature file and other options as input.

During the EM reporting operation, the following are processed to provide accurate results for SHE analysis:

- Self-heating device parameters in SPF files generated from original foundry runsets
- Delta temperature for every device
- OD GDS data from special runsets

When the EM reporting operation is finished, the EM report results, including the SHE data, appear in the PrimeSim EMIR results viewer. You can adjust different settings to view different aspects of the results.



See Also

- [Opening the EMIR Tool](#)

Setting Up for In-Design SHE EM Flow

Self-heating effect (SHE) analysis is run when you report electromigration (EM). Before you report EM, you must select the option in the **Electrical Options** dialog box.

To set up for in-design SHE EM flow:

1. In the Layout Editor, choose **Electrical > Options**.
2. In the **Electrical Options** dialog box, select the following:
 - **Electromigration: StarRC / PrimeSim EMIR**
 - **Enable Self-Heating Flow**

Electrical Options

Main

Nxtgrd

RATCL

Self-Heating Options

Flow Settings

Resistance:

☐ Built-in
 ☒ StarRC

Nxtgrd

Capacitance:

☒ StarRC

Nxtgrd

Electromigration:

☐ Built-in
 ☒ StarRC / PrimeSim EMIR

RATCL

StarRC Settings

☒ Device Layers

Command File:

Dynamic_EM_Reporting/Labs/PDK10NM/starrrc/star_indesign_cmd

Halo Width:

10

um

EM Settings

Current Scaling Factor:

100

%

PrimeSim XA Command File:

PrimeSim EMIR Command File:

☐ Load Report in Results Viewer

☒ Enable Self-Heating Flow

Analysis Temperature (C)

R/C:

105

EM (Max):

105

EM (Delta):

5

Layout Extraction Settings

Top Pin Extract Rule:

Technology Library

Global:

☒ From layout
 ☐ Density:

0.35

Local:

☒ From layout
 ☐ Spacing:

1

times of Smin

Width:

1

times of Wmin

Inter Layer Separation:

1 layers

Inter Layer Orientation:

Perpend

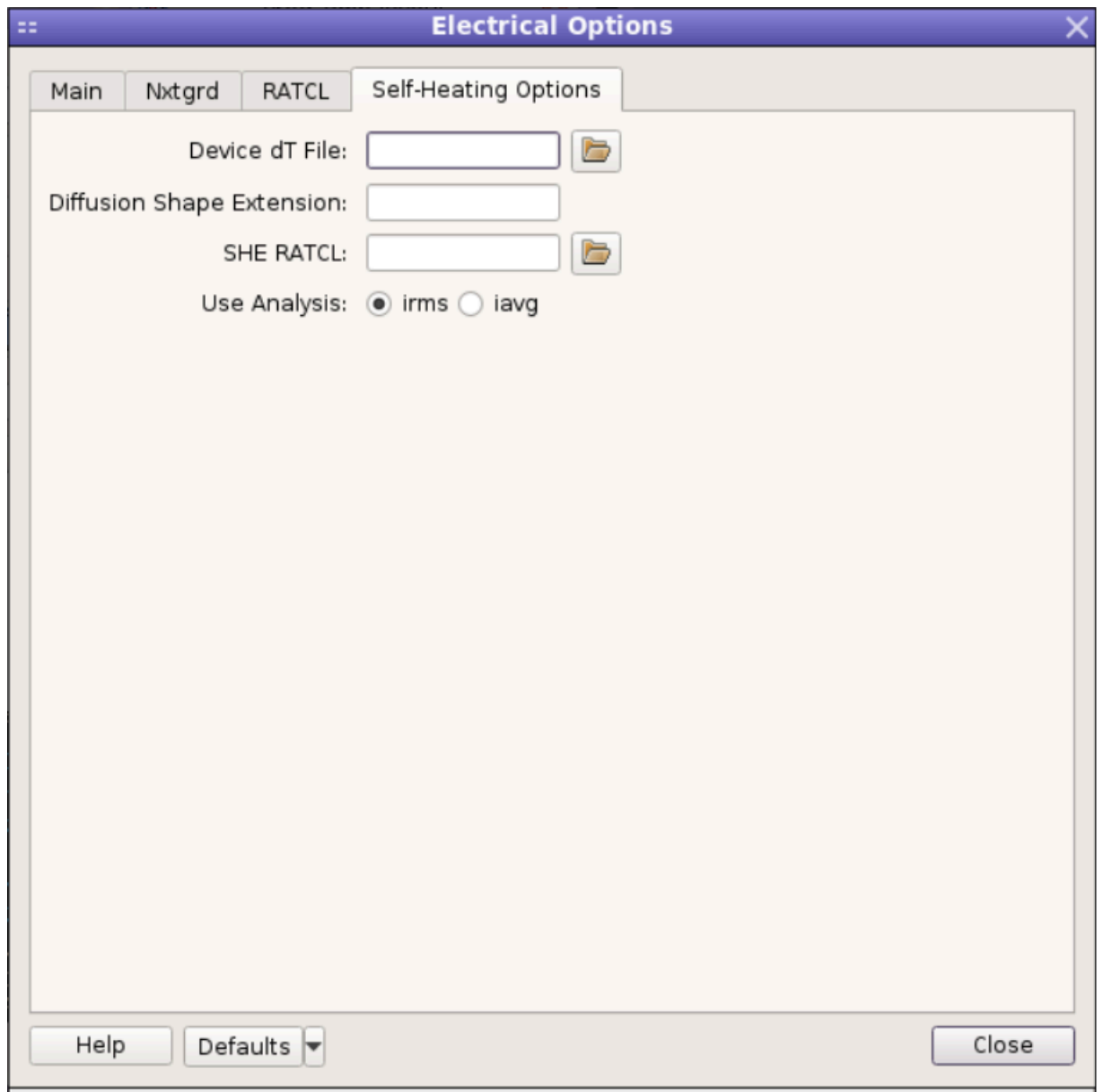
Help

Defaults

Close

3. In the **Self-Heating Options** tab, set the following:

Option	Description
Device dT File	The XML file that contains the device delta temperature information. The file is generated from the prelayout simulation.
Diffusion Shape Extension	The amount to extend the diffusion shapes, in μm .
SHE RATCL	Additional RA Tcl file containing the constraints.
Use Analysis	The type of analysis to run: irms or iavg .



4. Complete the remaining settings for EM reporting. See [Reporting Electromigration](#).
5. When you are finished, click **Close**.
6. Run the EM report. See [Reporting and Checking Electromigration \(EM\)](#).

See Also

- [Setting EM, Resistance, and Capacitance Options](#)
- [Reporting and Checking Electromigration \(EM\)](#)

Viewing SHE EM Results

When the EM report operation finishes, the PrimeSim EMIR results viewer opens automatically.

To set up the viewer and examine the SHE results, see [Opening the EMIR Tool](#).

5

Reporting and Checking Resistance

This chapter explains how to generate interactive wire resistance reports for partially completed designs.

This chapter contains the following topics:

- [About Wire Resistance Reporting and Checking](#)
- [Workflow for Checking and Reporting Resistance](#)
- [Preparing for Resistance Reporting and Checking](#)
- [Setting Resistance Constraints](#)
- [Specifying Input Information](#)
- [Using the Resistance Reporter](#)
- [Checking Resistance](#)
- [Checking Common Path Resistance](#)

Checking and reporting resistance is part of in-design electrical checking, which includes [Reporting and Checking Electromigration \(EM\)](#) and [Reporting and Checking Shield Coverage](#).

For information about reliability analysis, see Reliability Verification.

About Wire Resistance Reporting and Checking

Resistance in wires can result in voltage drops across the power network, impacting signal strength. To monitor potential voltage drops, you can check and report resistance during the design process. Reporting and fixing resistance issues during design development can save time and costly iterations downstream.

This topic describes the following:

- [In-Design Resistance Reporting](#)
- [Point-to-Point Resistance Calculations: Locations and Pin Patterns](#)

- [Parasitic Reporter](#)
- [StarRC Tool Integration](#)

In-Design Resistance Reporting

Checking resistance involves detecting the level of opposition to the passage of electrical current through wires and other elements. You can control the resistance in your design by [Setting Resistance Constraints](#) and then comparing the calculated resistance against the constraints. You can check point-to-point or individual net resistance. Based on the results (reporting of wires and vias that violate the constraints), you can modify components in your design, such as increasing the wire width, to adjust the resistance.

The in-design Resistance Reporter and Checker need to resolve current direction to correctly calculate the resistance values. The tools support most of the rectangles, polygons, wires (path and path segments), and vias; and uses the following formulas to perform net-based resistance calculations throughout the design hierarchy:

$$\text{Resistance}_{\text{wire}} = \frac{\text{Length}_{\text{wire}}}{\text{Width}_{\text{wire}}} \times \text{Sheet Resistance}$$

$$\text{Resistance}_{\text{via}} = \frac{\text{Resistance per cut}}{\text{Number of cuts}}$$

The wire and via resistance are calculated using TLUPlus (or common nxtgrd) files, or Sheet Resistance. You must supply the resistance constraints before checking and reporting resistance. For details, see [Workflow for Checking and Reporting Resistance](#) and [Calculating Resistance Using Sheet Resistance](#).

Point-to-Point Resistance Calculations: Locations and Pin Patterns

Point-to-point resistance measurements provide fast and accurate resistance information between source and destination locations on an interconnect segment. You can check the measured resistance against the resistance constraints you set.

For point-to-point resistance reporting, you can select one of the following as the start point (source) and end point (destination):

- A pin or terminal (the Resistance Reporter uses the intersection of the pin and wire)



- A coordinate on the pin or terminal (for more accurate calculations)



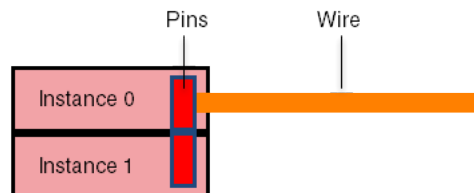
- A coordinate anywhere along the wire



The Resistance Reporter supports the following pin patterns:

- *Pins made of one rectangle:* These pins may be used as feedthroughs connecting one side of a cell to another side. There could be an internal connection along that pin that you intend to use as a destination point for the resistance calculation.
- *Pins made of multiple rectangles abutting against each other:* These are typically standard cell pins. There could be an internal connection to that pin within the cell that you intend to use as a destination point for the resistance calculation.

The following example shows two pin instances that are abutted against each other. The wire is connected to the top-level (Instance 0) pin.



To improve accuracy, you can select a pin instance to focus on a specific branch of a net, differentiating the resistance of the pins. In addition, the Parasitic Reporter decomposes large via arrays into smaller via cuts, covering the entire overlap between the two paths that the via array is connecting.

Parasitic Reporter

You use the following to set up and report resistance:

Feature	Description
Parasitic Reporter dialog box	Contains the following tabs: • Resistance: Contains settings for reporting resistance. • Capacitance: Contains settings for reporting total capacitance. See Reporting and Checking Capacitance.
Electrical Options dialog box	Contains common optional settings for reporting electromigration, resistance, and total capacitance. See Setting EM, Resistance, and Capacitance Options .

You can use the Parasitic Reporter to calculate and report the resistance between the following:

- Pins on a net
- Terminals on a net
- Specific points (net assignment is not necessary)
- Pins (or terminals) and points on a net

Because the Parasitic Reporter supports hierarchical connections, you can select points inside subcells. In addition, you can use the resistance report to determine the physical objects that are causing the resistance.

StarRC Tool Integration

You have the option of using the Synopsys StarRC tool to extract resistance information from the layout. The integration facilitates the reporting and troubleshooting processes.

After the extraction process is finished, instead of opening the output file, you can view the results while in the Resistance Reporter. In addition, when you select the resistance values of interest, the reporter automatically zooms in and locates the corresponding nets on the canvas for dynamic viewing. See [Viewing Point-to-Point Resistance Reports](#).

See also [Managing StarRC Version and License Use](#).

Workflow for Checking and Reporting Resistance

[Table 1](#) shows the high-level workflow for checking, reporting, and adjusting wire resistance in your design. Checking resistance is an iterative process. Typically, steps 3 through 5 are repeated until the wire resistance is optimized.

Table 1 Checking and Reporting Resistance Workflow

Step	Description
1	Prepare for resistance reporting requirements. Determine the method you want to use for resistance reporting, and then prepare for the requirements. See Preparing for Resistance Reporting and Checking .
2	Set resistance constraints. Set resistance constraints using the Constraint Editor. When setting common path resistance constraints, you have the option of excluding specific instance terminals. For instructions on setting resistance constraints, see Setting Resistance Constraints .
3	(Optional) Check wire resistance in your design. Compare the calculated wire resistance with the constraints to check for violations. For information, see Checking Resistance .
4	Report the resistance. To obtain resistance information between specific points on a wire, generate a point-to-point resistance report. See: • Reporting Resistance for Selected Path (Pin-to-Pin, Terminal-to-Terminal, Instance-to-Instance, Point-to-Point) • Reporting Resistance for All Pins, All Terminals, or All Instances on a Net • Reporting Resistance for All Constraints • Reporting Resistance From Selected Queries See also Managing Resistance Report Queries .
5	Adjust the design. Adjust design components to fix violations. For example, you can stretch the wire width to decrease resistance. For instructions, see Stretching the Wire Width .

Preparing for Resistance Reporting and Checking

This topic describes the requirements for resistance reporting:

- [Before You Start](#)
- [Custom Compiler Resistance Reporting With TLUPlus Files](#)
- [StarRC Resistance Reporting With nxtgrd Files](#)
- [Calculating Resistance Using Sheet Resistance](#)

Before You Start

Before running the Resistance Reporter, determine the method you will use so that you can prepare for the requirements:

Method	Requirement
Custom Compiler tool (Built-in Flow Setting)	• If the input is OA technology, you do not need to prepare the data. Proceed to Setting Resistance Constraints. • If the input is TLUPlus file, see Custom Compiler Resistance Reporting With TLUPlus Files.
StarRC tool (StarRC Flow Setting)	See StarRC Resistance Reporting With nxtgrd Files .

Custom Compiler Resistance Reporting With TLUPlus Files

When using TLUPlus files for resistance reporting, make sure you have the following ready:

- TLUPlus file
- Layer name mapping file
- Layer thickness file (optional)

When you generate point-to-point resistance reports using a TLUPlus file, you have the option of specifying a layer mapping file that maps OA or Milkyway layer names to corresponding ITF layer names. The files must be in the ASCII format. Each file should consist of two sections: `conducting_layers` and `via_layers`. The syntax for the sections are as follows:

```
conducting_layers
<db_metallayer_name><tluplus_metallayer_name>
```



```
via_layers
<db_vialayer_name><tluplus_vialayer_name>
```

Field	Description
db_metallayer_name	The name of the conducting (metal) layer in the OA or Milkyway database.
tluplus_metallayer_name	The ITF name of the conducting (metal) layer.
db_vialayer_name	The name of the via layer in the OA or Milkyway database.
tluplus_vialayer_name	The ITF name of the via layer.

For example:

```
conducting_layers
PO PO
met1 M1
met2 M2
met3 M3
met4 M4
met5 M5
met6 M6
```

```
via_layers
v1 VIA1
v2 VIA2
v3 VIA3
v4 VIA4
v5 VIA5
```

Example Layer Name Mapping File:

```
conducting_layers

m1 metal1
m2 metal2
m3 metal3
poly poly
v1 via1
v2 via2
v3 via3
```

You can optionally prepare a layer thickness file. The layer thickness file is an ASCII file that contains the list of layers and the corresponding thickness values, modeling the thickness variations in the TLUPlus file. A thickness value can be a floating point value in user units, or it is assigned one of the following keywords: Minimum, Nominal, or Maximum. Minimum and maximum are the smallest and largest values in the TLUPlus file, respectively. Nominal is default thickness value. The import process uses the keywords to

look up the values in the TLUPlus file, and then uses the values when creating OA sheet resistance constraints.

Example Layer Thickness File:

```
layer_thickness

metal1    nominal
metal2    minimum
metal3    maximum
metal4    0.41
metal5    0.45
```

StarRC Resistance Reporting With nxtgrd Files

When you generate resistance and total capacitance reports using external rule sources, you have the option of referencing a layer mapping file that maps OA, IC Compiler II, or IC Compiler layer names to corresponding StarRC layer names in the nxtgrd files. The same mapping file is used for both resistance and total capacitance reporting. See [Preparing the Layer Mapping File](#) in the capacitance reporting topic.

Calculating Resistance Using Sheet Resistance

The wire and via resistance can also be calculated using Sheet Resistance, which is obtained from sheet resistance constraints. You can set sheet resistance constraints as follows:

Table 2 *Resistance Constraints*

Sheet resistance constraints set in the ...	Are applicable to...	The constraint information is stored in...
Technology	All designs that reference the technology	The technology library. You can supply the constraint data in the technology library using one of the following methods: • Import a TLUPlus file directly into the technology library. • Export the technology data to a technology file, add the constraint in the file, and then import the file into the technology library.

Table 2 Resistance Constraints (Continued)

Sheet resistance constraints set in the ...	Are applicable to...	The constraint information is stored in...
TLUPlus or common-nxtgrd file	Specific designs	TLUPlus or common-nxtgrd files. You can specify one or more TLUPlus or nxtgrd files that provide the following information: • Minimum operating conditions • Maximum operating conditions In addition, you can specify the layer name mapping file.

Any color-related rules defined in TLUPlus, ITF, and nxtgrd files are automatically taken into consideration for resistance calculations.

For information about adding sheet resistance constraints in technology files, see the *Custom Compiler Technology and Display Resource Reference Manual*. For information about adding sheet resistance constraints using the Constraint Manager, see the *Custom Compiler Environment User Guide*.

Note:

You can use Tcl scripts to add sheet resistance constraints to the Fusion Compiler and IC Compiler II library opened in a Custom Compiler session. For instructions, contact Synopsys Technical Support Center.

Setting Resistance Constraints

Before you check and report resistance, you should set design-specific resistance constraints. You use the Constraint Editor to set these constraints.

This section describes the following:

- [Defining Terminal-to-Terminal Resistance Constraints](#)
- [Defining Pin-to-Pin Resistance Constraints](#)
- [Defining Net-Level Resistance Constraints](#)
- [Specifying Max Common Resistance](#)

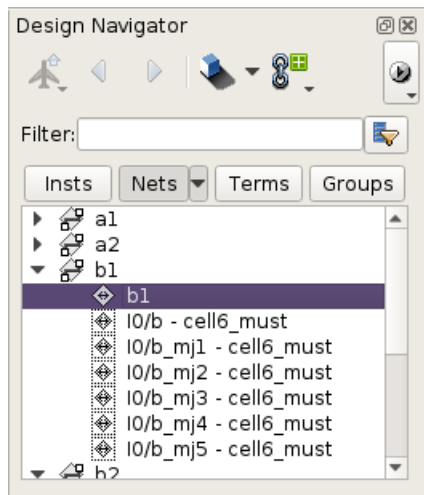
Defining Terminal-to-Terminal Resistance Constraints

To control the resistance in your design, you define terminal-to-terminal resistance constraints. You can specify the minimum and maximum permissible resistance values. You can also specify the terminals to exclude from the constraints.

You can define Resistance constraints in either the Schematic Editor or the Layout Editor. The resistance constraint data defined for a design schematic are automatically transferred to the layout during the Schematic-Driven Layout (SDL) operation.

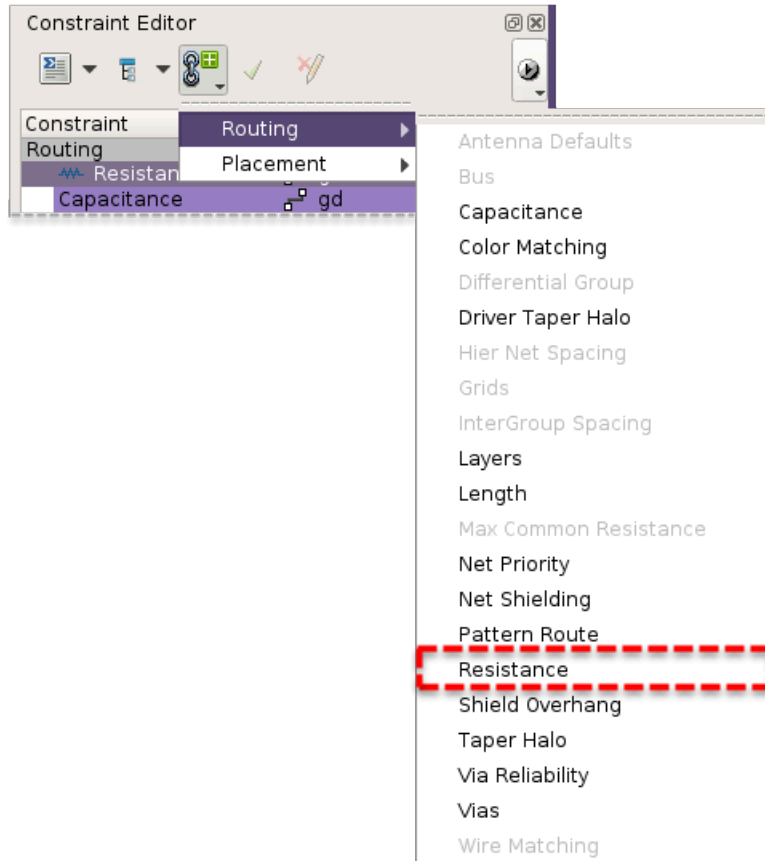
To define a terminal-to-terminal resistance constraint in the Schematic Editor or the Layout Editor:

1. In the Design Navigator, click the **Nets** tab or the **Terms** tab, and select an instance terminal.

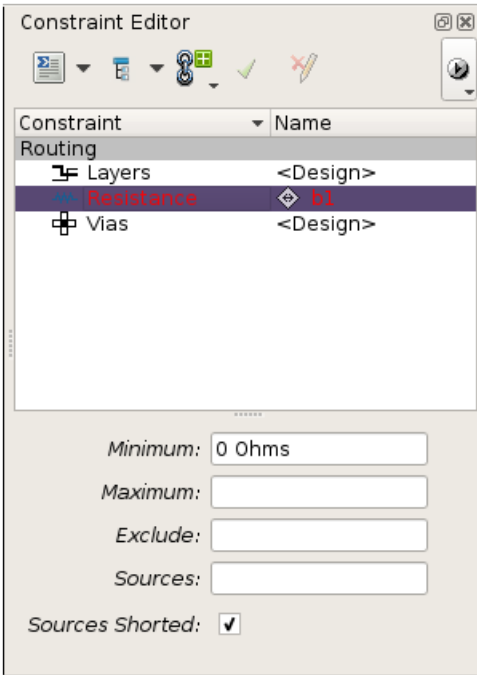


2. Display the Constraint Editor.

3. In the Constraint Editor, click **Create Constraint** , and then choose **Routing > Resistance**.



4. In the Constraint Editor, locate the **Routing** section, and then select the newly created Resistance constraint. The parameters appear in the Constraint Editor.



5. Set the following parameters:

Parameter	Description
Minimum	The minimum resistance allowed on the selected nets. Specify a floating-point value ≥ 0 .
Maximum	The maximum resistance allowed on the selected nets. Specify a floating point value > 0 .
Exclude	The names of terminals and instance terminals that you want to exclude from the constraint. Separate consecutive names with a character space. Note: If a terminal or instance terminal has a resistance constraint, and the selected net also has a resistance constraint, the terminal or instance terminal resistance constraint will override the net resistance.
Sources	The names of same-net terminals and instance terminals to be considered for the resistance calculations. Separate consecutive names with a character space. The supported formats are term, inst/term, and inst/term/pin. If you do not specify any source, all connections on the net will be considered in the resistance calculation.
Sources Shorted	Treats multiple sources as a single source.

6. Notice that after you change a parameter value, a  button appears next to the changed value. To reset a parameter back to its default value, click .
7. When you are finished modifying a constraint:
 - Click **Apply Changes** () to save the modifications.
 - Click **Discard Changes** () to revert back to the previously saved settings.

See Also

- [Defining Pin-to-Pin Resistance Constraints](#)
- [Defining Net-Level Resistance Constraints](#)
- [Specifying Max Common Resistance](#)

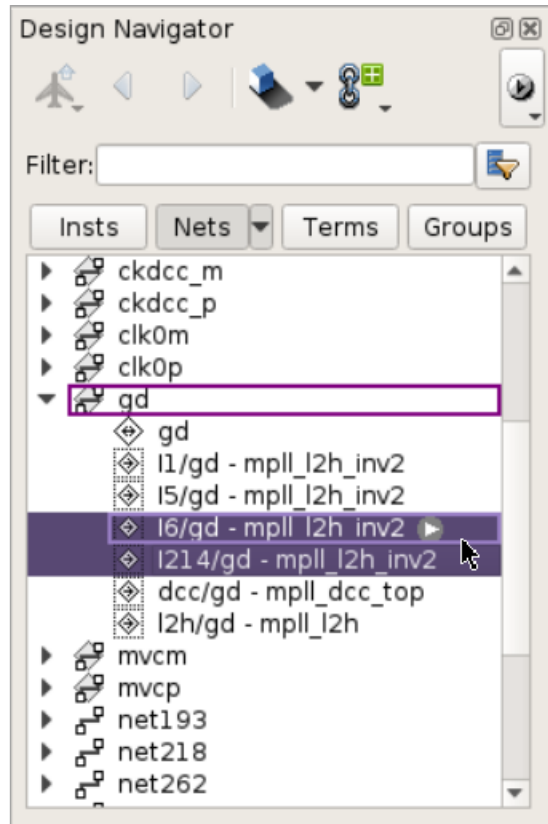
Defining Pin-to-Pin Resistance Constraints

To control the resistance in your design, you can define pin-to-pin resistance constraints. You can specify the source and destination objects. You can also set the minimum and maximum permissible resistance values.

You can define Resistance constraints in either the Schematic Editor or the Layout Editor. The resistance constraint data defined for a design schematic are automatically transferred to the layout during the Schematic-Driven Layout (SDL) operation.

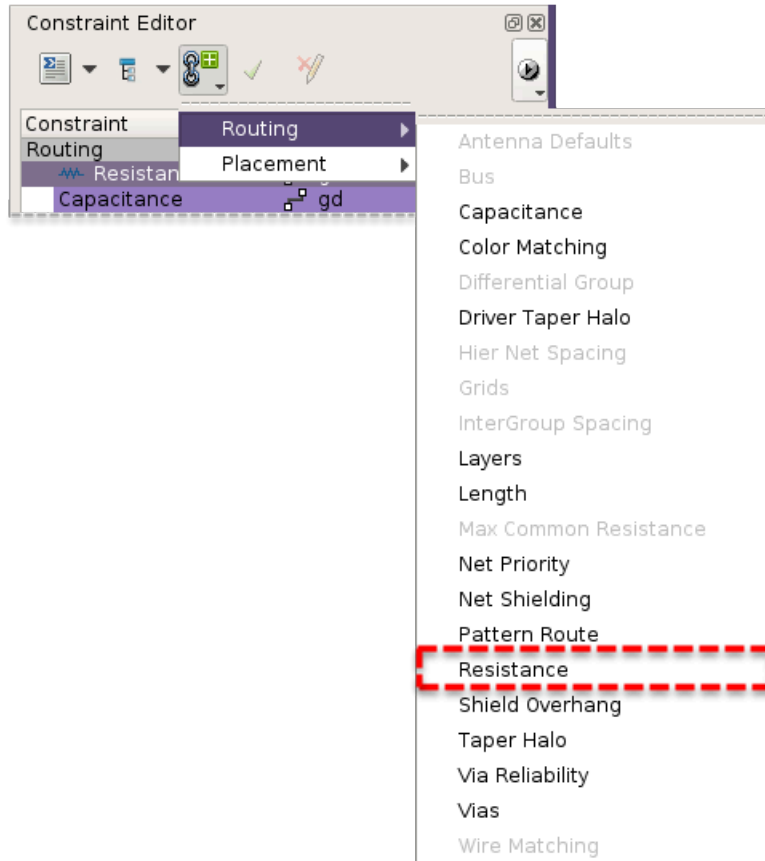
To define a pin-to-pin resistance constraint in the Schematic Editor or the Layout Editor:

1. In the Design Navigator, click the **Nets** tab, and select one or more pins.



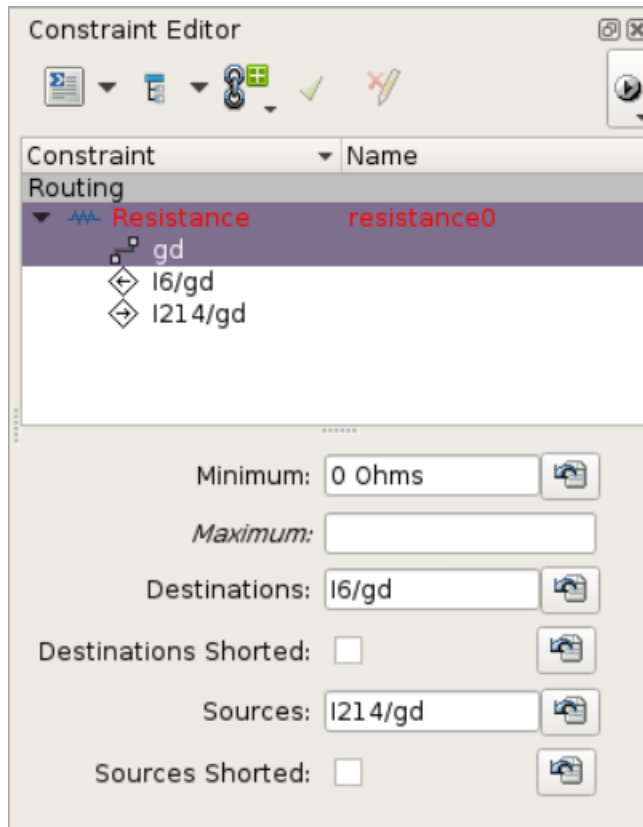
2. Display the Constraint Editor.

3. In the Constraint Editor, click **Create Constraint** , and then choose **Routing > Resistance**.



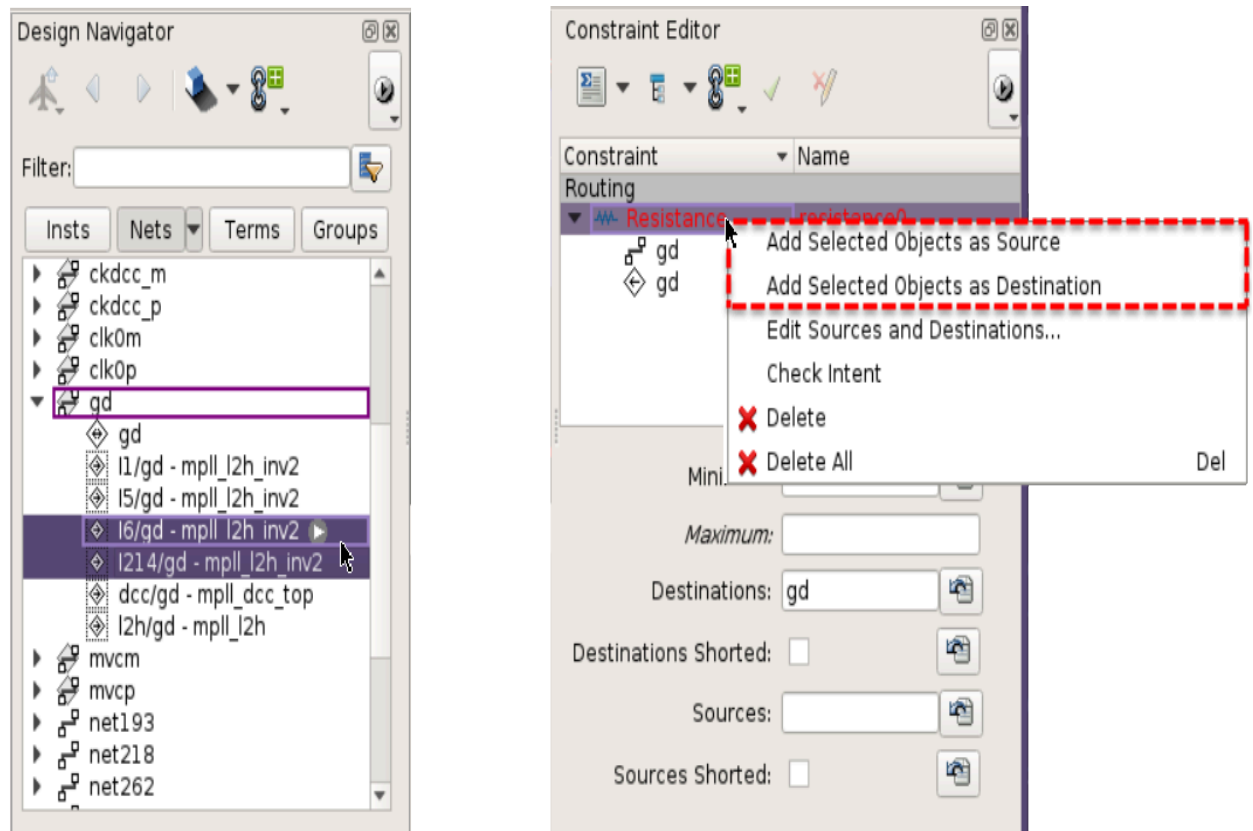
4. In the Constraint Editor, locate the **Routing** section, and then select the newly created Resistance constraint.

The parameters appear in the Constraint Editor. By default, if you selected more than one instance pin, the first instance pin appears as a Destination and the remaining pins are Sources.

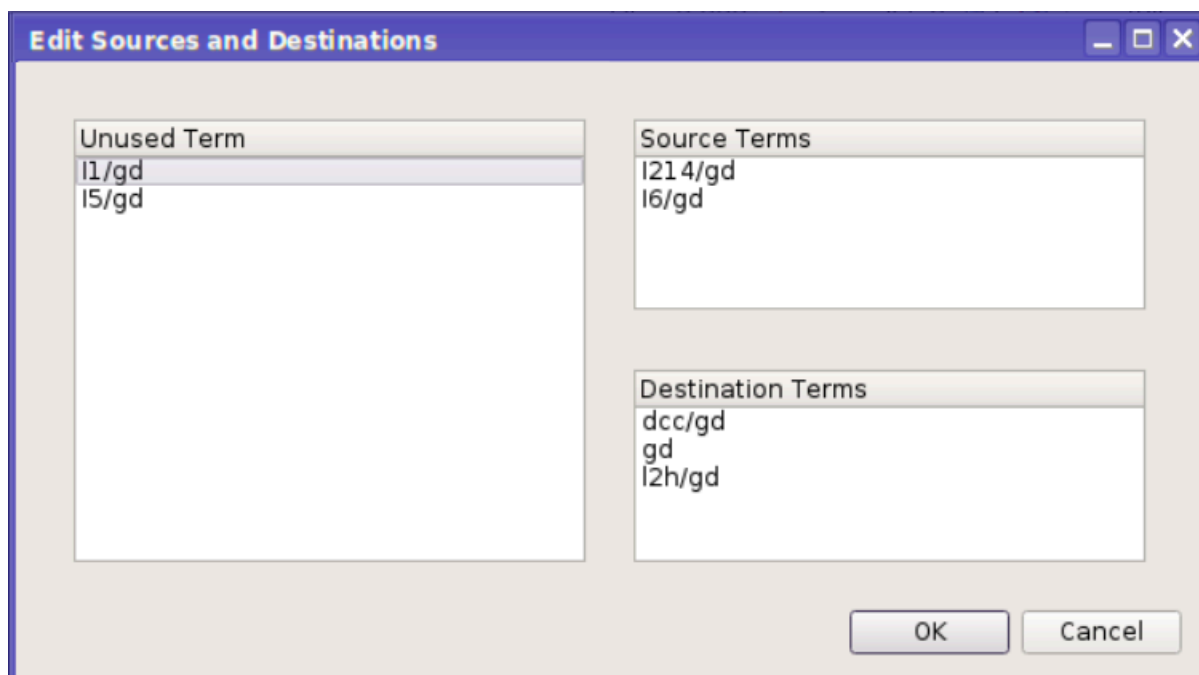


5. To add source and destination pins:
 - a. In the Design Navigator, select the source pins.
 - b. In the Constraint Editor, right-click the newly created constraint, and choose **Add Selected Objects as Source**.
 - c. In the Design Navigator, select the destination pins.
 - d. In the Constraint Editor, right-click the newly created constraint, and choose **Add Selected Objects as Destination**.

In the Constraint Editor, the selected source and destination pin names populate the **Sources** and **Destinations** boxes.



6. To edit the existing sources and destinations:
 - a. In the Constraint Editor, right-click the resistance constraint, and choose **Edit Sources and Destinations**.
 - b. In the **Edit Sources and Destinations** dialog box, select and drag the source and destination term names to the target boxes.
 - c. Click **OK**.



In the Constraint Editor, confirm that the changes you made are displayed in the **Sources** and **Destinations** boxes.

7. In the Constraint Editor, set the **Minimum** and the **Maximum** resistance values allowed for the nets of the selected pins.

For Minimum, specify a floating-point value ≥ 0 . For Maximum, specify a floating point value > 0 .

8. Select **Sources Shorted** and **Destination Shorted** to treat multiple sources and multiple destinations as a single source, respectively.
9. Notice that after you change a parameter value, a  button appears next to the changed value. To reset a parameter back to its default value, click .
10. When you are finished modifying a constraint:
 - Click **Apply Changes** () to save the modifications.
 - Click **Discard Changes** () to revert back to the previously saved settings.

See Also

- [Defining Terminal-to-Terminal Resistance Constraints](#)
- [Defining Net-Level Resistance Constraints](#)
- [Specifying Max Common Resistance](#)

Defining Net-Level Resistance Constraints

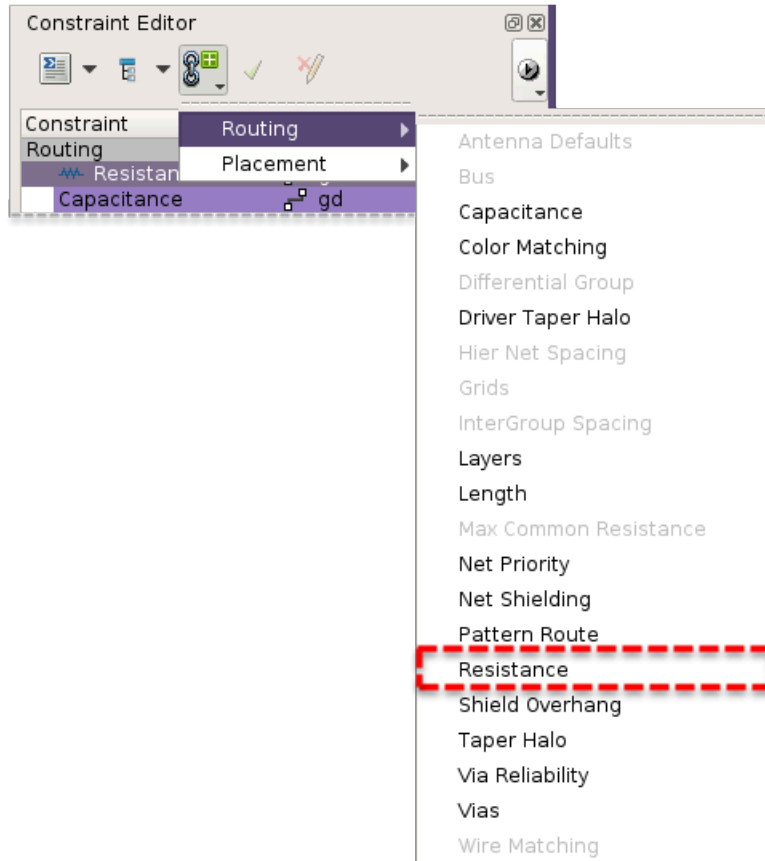
To control the resistance along wires, you set net-level Resistance constraints. You can specify the minimum and maximum permissible resistance values. You can also specify the terminals to exclude from the constraint.

You can define Resistance constraints in either the Schematic Editor or the Layout Editor. The resistance constraint data defined for a design schematic are automatically transferred to the layout during the Schematic-Driven Layout (SDL) operation.

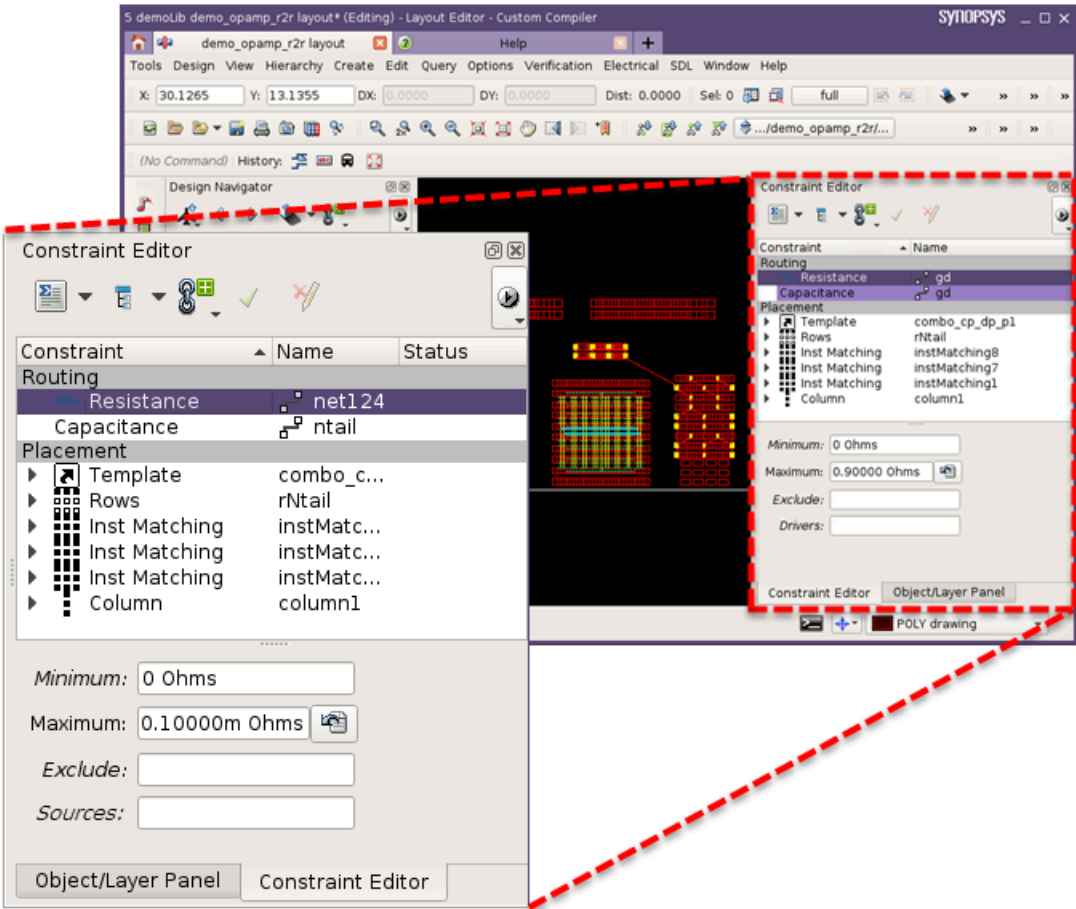
This section describes how to set the Resistance constraint using the Constraint Editor. For information on using the `cm::createResistance` Tcl command, see the *Custom Compiler Tcl Reference Manual*.

To view or modify the Resistance constraint parameters in the Schematic Editor or the Layout Editor:

1. In the Design Navigator, select one or more nets.
2. Display the Constraint Editor.
3. To create a new Resistance constraint, click **Create Constraint** , and then choose **Routing > Resistance**.



4. To view the existing Resistance constraint, in the Constraint Editor, locate the **Routing** section, and then choose **Resistance**. The parameters appear in the Constraint Editor.



5. Modify the following parameters:

Parameter	Description
Minimum	The minimum resistance allowed on the selected nets. Specify a floating-point value ≥ 0 .
Maximum	The maximum resistance allowed on the selected nets. Specify a floating point value > 0 .
Exclude	The names of terminals and instance terminals that you want to exclude from the constraint. Separate consecutive names with a character space. Note: If a terminal or instance terminal has a resistance constraint, and the selected net also has a resistance constraint, the terminal or instance terminal resistance constraint will override the net resistance.

Parameter	Description
Sources	The names of same-net terminals and instance terminals to be considered for the resistance calculations. Separate consecutive names with a character space. The supported formats are term, inst/term, and inst/term/pin. If you do not specify any source, all connections on the net will be considered in the resistance calculation.
Sources Shorted	Treats multiple sources as a single source.

6. Notice that after you change a parameter value, a  button appears next to the changed value. To reset a parameter back to its default value, click .
7. When you are finished modifying a constraint:
 - Click **Apply Changes** () to save the modifications.
 - Click **Discard Changes** () to revert back to the previously saved settings.

Specifying Max Common Resistance

Common path routes feed electrical current to multiple terminals. To ensure that sufficient signal reaches the terminals, you can decrease the resistance along the common path route.

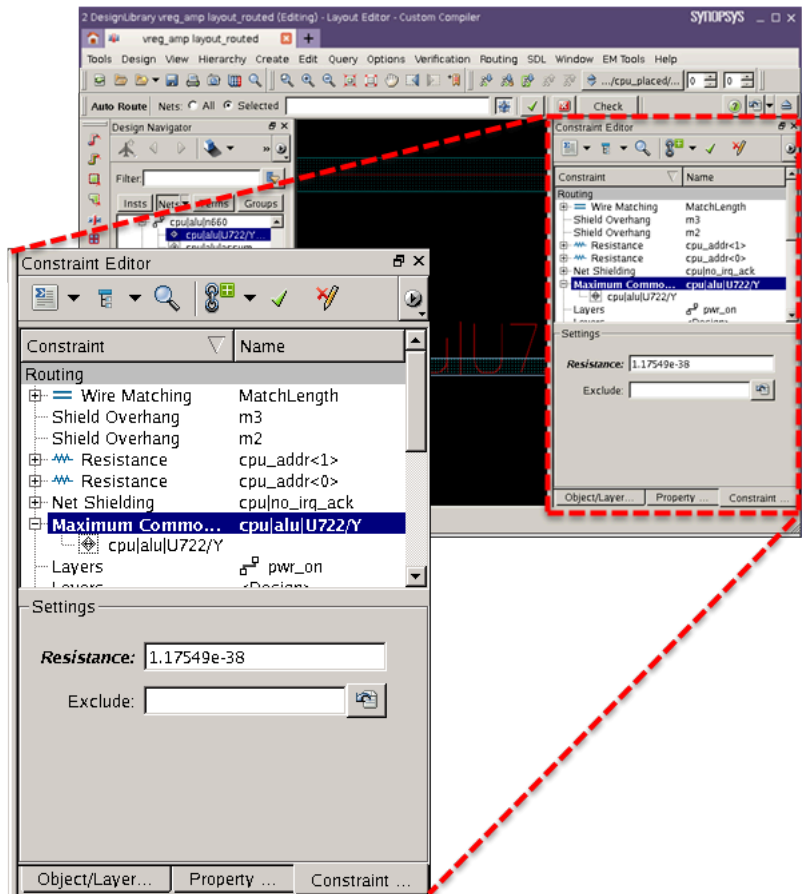
You set the Max Common Resistance constraint to lower the resistance along the common path route. The common path resistance should be less than or equal to the maximum resistance allowed elsewhere along the route.

This section describes how to set the Max Common Resistance constraint using the Constraint Editor. For information on using the `cm::createMaxCommonResistance` Tcl command, see the *Custom Compiler Tcl Reference Manual*. For information about setting the Resistance constraint, see [Defining Net-Level Resistance Constraints](#).

To create, view, or modify the Max Common Resistance constraint parameters:

1. In the Design Navigator, select the terminals.
2. In the Layout Editor, display the Constraint Editor.
3. To create a new Max Common Resistance constraint, click **Create Constraint** , and then choose **Routing > Max Common Resistance**.

4. To view the existing Max Common Resistance constraint, in the Constraint Editor, locate the **Routing** section, and then choose **Max Common Resistance**. The parameters appear in the Constraint Editor.



5. Modify the following parameters:

Parameter	Description
Resistance	The maximum resistance allowed along the common path route.
Exclude	The names of terminals and instance terminals that you want to exclude from the constraint. Type a character space after each name.

6. Notice that after you change a parameter value, a  button appears next to the changed value. To reset a parameter back to its default value, click .

7. When you are finished modifying a constraint:

- Click **Apply Changes** () to save the modifications.
- Click **Discard Changes** () to revert back to the previously saved settings.

Specifying Input Information

Resistance reports are based on input information that you supply, such as the resistance analysis temperature, rule file, and extraction settings. You can supply the information using one of the following methods:

Method	Description
Preload the input information during environment setup	You can set preferences by running a Tcl script during environment setup. In this case, you do not need to specify the settings in the Parasitic Reporter or the Electrical Options dialog box. You only need to display the dialog box and click Report. See Reporting Resistance for Selected Path (Pin-to-Pin, Terminal-to-Terminal, Instance-to-Instance, Point-to-Point). To find out whether the input information is preloaded with a Tcl script, check with your administrator or CAD manager. To create the script, see Introduction to Custom Compiler Tcl.
Preload the input information by setting the Tcl preferences in the Console	Instead of running a Tcl script, you can type Tcl commands in the Console to set the preferences. To see the preferences you can set, see About the Electrical Options Dialog Box and Parasitic Reporter - Resistance Tab Reference.
Supply the input in the Electrical Options dialog box	You can enter the input information in the dialog box every time you generate the report. To enter the input information, see: • Reporting Resistance for Selected Path (Pin-to-Pin, Terminal-to-Terminal, Instance-to-Instance, Point-to-Point) • Reporting Resistance for All Pins, All Terminals, or All Instances on a Net • Reporting Resistance for All Constraints • Reporting Resistance From Selected Queries

Using the Resistance Reporter

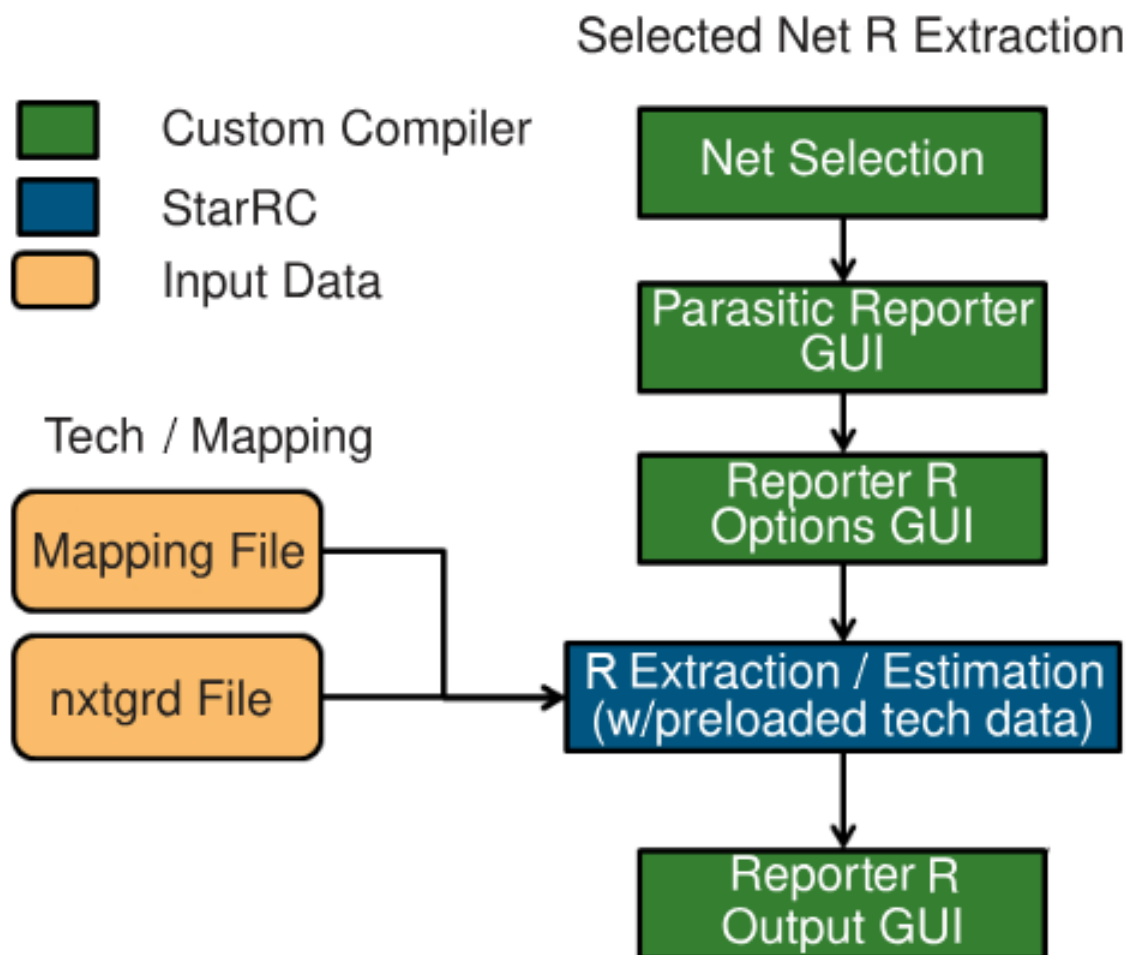
The Resistance Reporter allows you to probe and analyze resistance dynamically as you create the design. Reporting and fixing resistance issues during design development can save time and costly iterations downstream.

This topic describes the following:

- [Resistance Extraction Workflow](#)
- [Reporting Resistance for Selected Path \(Pin-to-Pin, Terminal-to-Terminal, Instance-to-Instance, Point-to-Point\)](#)
- [Reporting Resistance for All Pins, All Terminals, or All Instances on a Net](#)
- [Reporting Resistance for All Constraints](#)
- [Reporting Resistance From Selected Queries](#)
- [Managing Resistance Report Queries](#)
- [Reporting Resistance and Capacitance](#)
- [Viewing Point-to-Point Resistance Reports](#)
- [Exporting Point-to-Point Resistance Reports](#)
- [Parasitic Reporter - Resistance Tab Reference](#)

Resistance Extraction Workflow

The following diagram shows the high-level in-design resistance extraction workflow:



Reporting Resistance for Selected Path (Pin-to-Pin, Terminal-to-Terminal, Instance-to-Instance, Point-to-Point)

Resistance measurements between points, pins, terminals, and instances provide fast and accurate resistance information between source and destination locations on an interconnect segment. You can check the measured resistance against the resistance constraints you set.

Before you start, make sure the resistance source and destination points are connected at the extraction level.

To report resistance between points, pins, terminals, and instances:

1. (Optional) Select the net:
 - a. In the Layout Editor, open the Design Navigator.
 - b. Click the **Nets** tab and select the nets. You can do this from either the **Design: Layout** mode or the **Design: Schematic** mode.

You can skip this step if you want to report non-net point-to-point resistance.

2. Right-click the selected net, and then choose **Report Resistance**.

The **Parasitic Reporter** dialog box opens. The net you selected in [Step 1](#) appears in the **Net** box in the **Resistance** tab.

The image shows the Parasitic Reporter dialog box with the Resistance tab selected. The settings are as follows:

- Net:** ntail
- Draw:** [Icon] [Color: Black] [Cycle: ☐] [Auto Highlight: ☐]
- Depth:** ☒ 0x ☐ View ☐ Full ☐ Include Parent Designs
- Area:** ☒ Design ☐ Region
- Report using:** Selected Path [Manage Query] ☒ Elmore Delay
- Path Definitions:**
 - Select using: pin [Add Query] ☐ Auto Add
 - ☒ Source [Field] ☐ Short
 - ☐ Destination [Field] ☐ Short
- Tables:**

Net	Path Reported	Min Path R (Ohms)	Max Path R (Ohms)

Source	Destination	Path R (Ohms)	Length (um)	Elmore Delay (ps)
- Details:**

Description	R (Ohms)	Wire Width (um) Via Column/Rows	Wire Length (um) Via Cut Class
- History:** [Field]
- Options:**
☒ Zoom to Fit ☒ Export to Console ☐ Export to Error Viewer ☒ Save History ☐ Run in Background ☐ Report RC

Buttons at the bottom: Help, Defaults, Reset, Report, Export ..., Options, Close.

3. If the information in the dialog box was preloaded as part of the environment setup:
 - a. Review the settings in the **Resistance** tab.
 - b. Click **Options** (at the bottom of the dialog box), and review the settings in the **Electrical Options** dialog box. See [Setting EM, Resistance, and Capacitance Options](#) and [Specifying Input Information](#).
 - c. Skip to [Step 5](#).

4. If you did not preload settings in [Step 3](#), on the **Resistance** tab of the **Parasitic Reporter** dialog box:
 - Set the **Depth** and **Area** options. See [Parasitic Reporter - Resistance Tab Reference](#).
 - Click **Options** (at the bottom of the dialog box), and set the options in the **Electrical Options** dialog box. See [Setting EM, Resistance, and Capacitance Options](#).

Caution:

Values you enter in the dialog boxes override preloaded values.

5. In the **Report using** list, choose **Selected Path**. The **Path Definition** section displays.
6. In the **Select using** list, choose one of the following:
 - **pin**: Lets you choose one or more individual pins for the Source or Destination.
 - **terminal**: Lets you choose one or more terminals for the Source or Destination. When you choose a terminal, all pins within the terminal are selected.
 - **instance**: Lets you choose one or more individual instances for the Source or Destination.

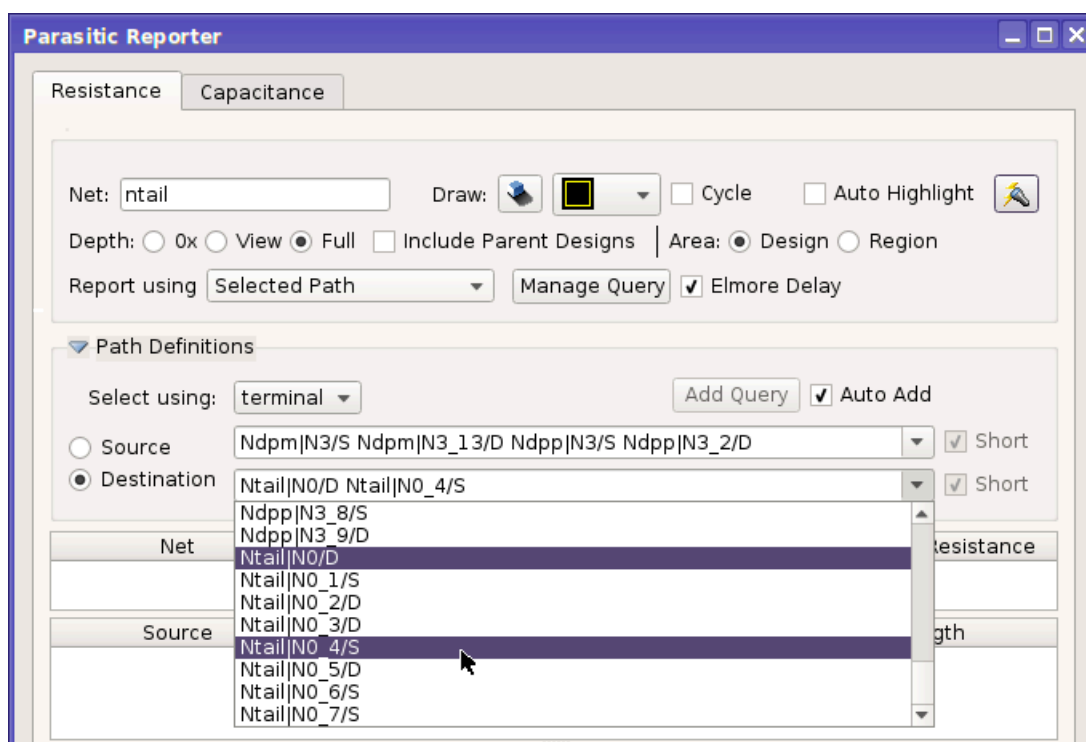
If you plan to use points, you can ignore this setting.

7. To apply Elmore Delay analysis during the calculations, select **Elmore Delay**.

The option is only available when you select the StarRC tool for resistance reporting in the **Electrical Options** dialog box. See [Setting EM, Resistance, and Capacitance Options](#).

8. Select the **Source** and **Destination** pins, instances, terminals, or points.

If you are using pins, terminals, and instances, choose one or more pins, terminals, or instances from the list, or type the pin, terminal or instance names in the box.



If you are using points, on the layout canvas:

- Click a location on a shape to select one source or destination point.
- Shift+click multiple locations on different shapes to select multiple source or destination points.

In cases where multiple shapes are stacked, press 1 to cycle to the shape you want to select.

The coordinates and layer name of the selected points appear in the **Source** and **Destination** boxes.

9. To save the query for future reporting:

- Click **Add Query**. You can recall and generate a resistance report later.
- Select **Auto Add**. The query and resistance report results are saved when you click **Report**.

See [Managing Resistance Report Queries](#) and [Reporting Resistance From Selected Queries](#).

10. If you want to display the results in the Custom Compiler Console, select **Export To Console**.

11. To automatically save the generated resistance report so that you can recall it in future design sessions, select **Save History**. (Alternatively, set the `leEMRCAutoSaveHistory` Tcl preference to `true` before you generate the report.) The **History** section appears.

All newly generated reports will be automatically saved in the following directory and file:

```
$env(SYNOPSYS_CUSTOM_LOCAL)/r_reporter/<LCV>_R_result.db
```

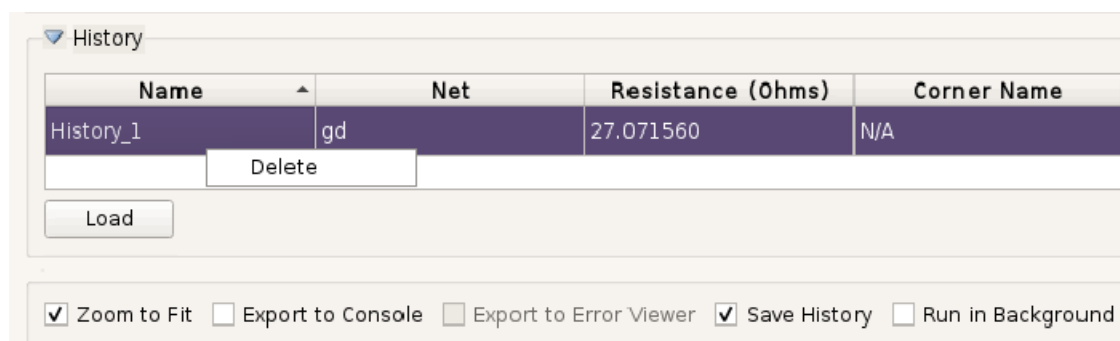
If a reported net already exists in the `.db` file, it will be overwritten by the latest data.

To recall the saved report, see [Viewing Point-to-Point Resistance Reports](#).

By default, the maximum number of reports retained is 30. To change the number, set the `leEMRCMaxHistoryNum` Tcl preference.

To remove the history of a report, right-click on the entry in the **History** section and select **Delete**.

testing

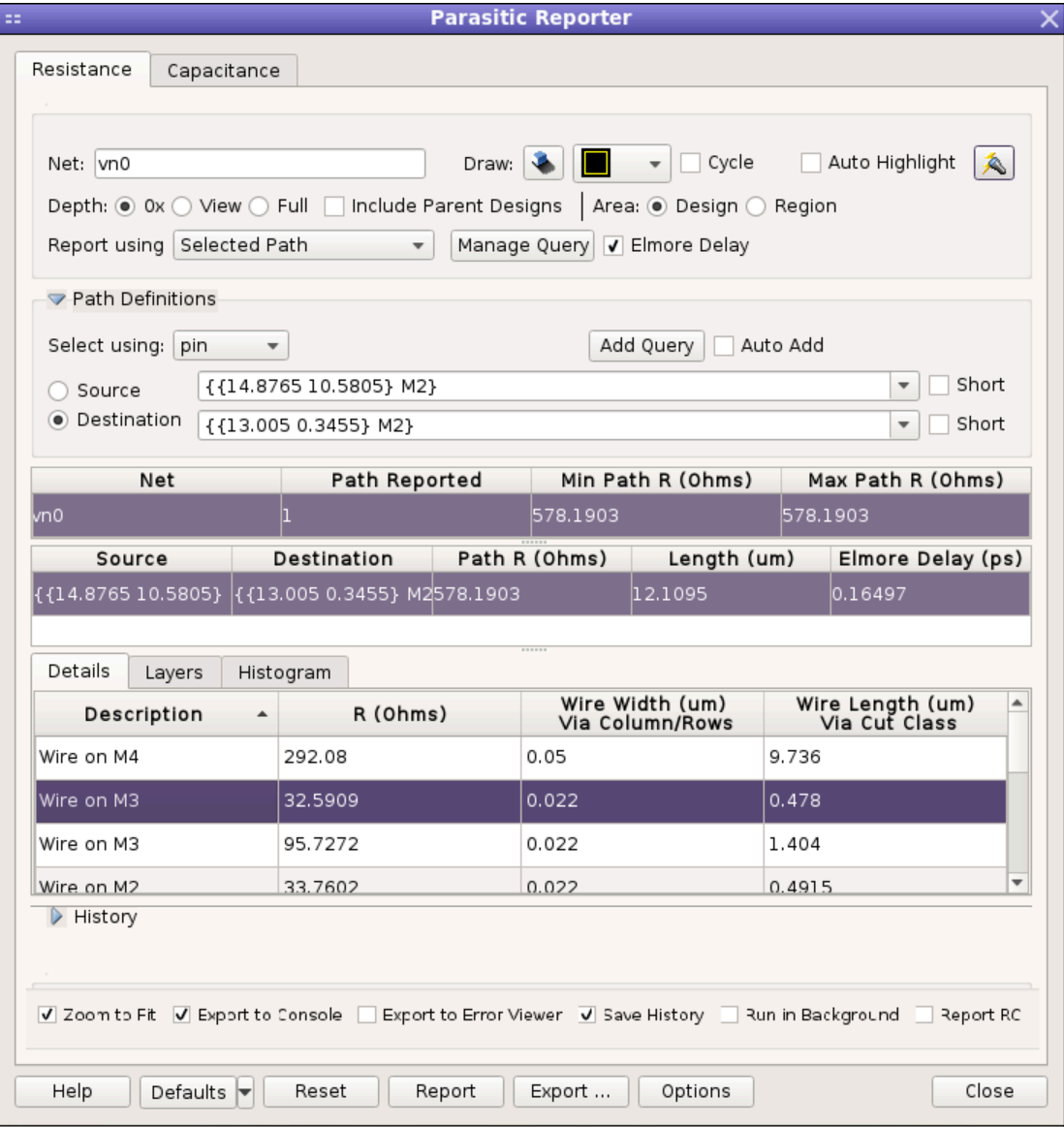


12. If you want the resistance analysis in the background so that you can continue to work in the Layout Editor, select **Run in Background**.

Background mode supports both StarRC shared and non-shared modes. When the report is generated, you can locate the results in the **History** section of the dialog box.

13. When you are ready to generate and view the report, click **Report**.

The Parasitic Reporter displays the results in the tables at the bottom half of the **Resistance** tab. See [Viewing Point-to-Point Resistance Reports](#).



Reporting Resistance for All Pins, All Terminals, or All Instances on a Net

Instead of reporting resistance for specific source and destination pairs, you can report resistance for all unique pin-, instance-, or terminal-pairs on one or more selected nets.

Before you start, make sure the resistance source and destination points are connected at the extraction level.

To report resistance for all unique pin-, instance-, or terminal-pairs on selected nets:

1. (Optional) Select one or more nets:
 - a. In the Layout Editor, open the Design Navigator.
 - b. Click the **Nets** tab and select the nets. You can do this from either the **Design: Layout** mode or the **Design: Schematic** mode.

You can also select multiple nets and choose **Report Resistance** to extract parasitics in a single run. .

Note:

- If you select more than 10 nets, the **Reporter - Interactive StarRC** dialog box appears. Click **OK** to continue.
- If you select **Cancel**, you must select less than 10 nets and continue extraction in the interactive mode.

2. Right-click the selected nets, and then choose **Report Resistance**.

The **Parasitic Reporter** dialog box opens. The nets you selected in [Step 1](#) appear in the **Net** box in the **Resistance** tab.

The screenshot shows the 'Parasitic Reporter' dialog box with the 'Resistance' tab selected. The 'Net' field is set to 'vn0 ntail'. The 'Draw' section has a blue icon selected. The 'Depth' section has '0x' selected. The 'Area' section has 'Design' selected. The 'Report using' dropdown is set to 'All Pins'. The 'Elmore Delay' checkbox is checked. Below these settings are two empty tables. The first table has columns: Net, Path Reported, Min Path R (Ohms), and Max Path R (Ohms). The second table has columns: Source, Destination, Path R (Ohms), Length (um), and Elmore Delay (ps). Below the tables are three tabs: 'Details', 'Layers', and 'Histogram'. The 'Details' tab is selected, showing a table with columns: Description, R (Ohms), Wire Width (um) Via Column/Rows, and Wire Length (um) Via Cut Class. Below the table is a 'History' section. At the bottom, there are checkboxes for 'Zoom to Fit', 'Export to Console', 'Export to Error Viewer', 'Save History', 'Run in Background', and 'Report RC'. The 'Zoom to Fit' and 'Save History' checkboxes are checked. At the very bottom are buttons for 'Help', 'Defaults', 'Reset', 'Report', 'Export ...', 'Options', and 'Close'.

3. If the information in the dialog box was preloaded as part of the environment setup:
 - a. Review the settings in the **Resistance** tab.
 - b. Click **Options** (at the bottom of the dialog box), and review the settings in the **Electrical Options** dialog box. See [Setting EM, Resistance, and Capacitance Options](#) and [Specifying Input Information](#).
 - c. Skip to [Step 5](#).

4. If you did not preload settings in [Step 3](#), on the **Resistance** tab of the **Parasitic Reporter** dialog box:
 - Set the **Depth** and **Area** options. See [Parasitic Reporter - Resistance Tab Reference](#).
 - Click **Options** (at the bottom of the dialog box), and set the options in the **Electrical Options** dialog box. See [Setting EM, Resistance, and Capacitance Options](#).

Caution:

Values you enter in the dialog boxes override preloaded values.

5. In the **Report using** list, choose **All Pins**, **All Terminals**, or **All Instances**.
6. To apply Elmore Delay analysis during the calculations, select **Elmore Delay**.

The option is only available when you select the StarRC tool for resistance reporting in the **Electrical Options** dialog box. See [Setting EM, Resistance, and Capacitance Options](#).

7. If you want to display the results in the Custom Compiler Console, select **Export To Console**.
8. To automatically save the generated resistance report so that you can recall it in future design sessions, select **Save History**. (Alternatively, set the `leEMRCAutoSaveHistory` Tcl preference to `true` before you generate the report.) The **History** section appears.

All newly generated reports will be automatically saved in the following directory and file:

```
$env(SYNOPSYS_CUSTOM_LOCAL)/r_reporter/<LCV>_R_result.db
```

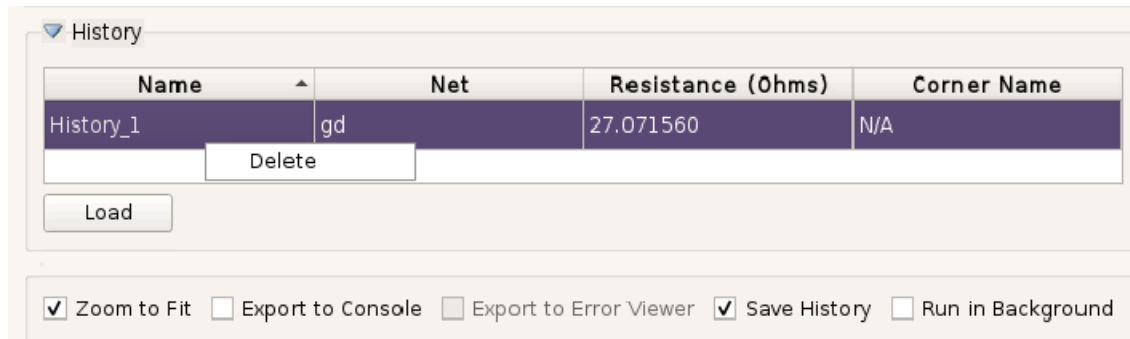
If a reported net already exists in the `.db` file, it will be overwritten by the latest data.

To recall the saved report, see [Viewing Point-to-Point Resistance Reports](#).

By default, the maximum number of reports retained is 30. To change the number, set the `leEMRCMaxHistoryNum` Tcl preference.

To remove the history of a report, right-click on the entry in the **History** section and select **Delete**.

testing



9. If you want the resistance analysis in the background so that you can continue to work in the Layout Editor, select **Run in Background**.

Background mode supports both StarRC shared and non-shared modes. When the report is generated, you can locate the results in the **History** section of the dialog box.

- When you are ready to generate and view the report, click **Report**. The Parasitic Reporter displays the results in the tables at the bottom half of the **Resistance** tab. See [Viewing Point-to-Point Resistance Reports](#).

Parasitic Reporter

Resistance **Capacitance**

Net: Draw: ☐ Cycle ☐ Auto Highlight

Depth: ☒ 0x ☐ View ☐ Full ☐ Include Parent Designs | Area: ☒ Design ☐ Region

Report using ☒ Elmore Delay

Net	Path Reported	Min Path R (Ohms)	Max Path R (Ohms)
vn0	3	164.7729	627.6293
ntail	7	717.70011	848.37831

Source	Destination	Path R (Ohms)	Length (um)	Elmore Delay (ps)
Ncasc0.m1/S/S	Ntail4 N0_0/S/S	627.6293	12.298	0.39716
Ncasc0.m2/D/D	Ntail4 N0_0/S/S	595.0384	11.82	0.38865

Details **Layers** **Histogram**

Description	R (Ohms)	Wire Width (um) Via Column/Rows	Wire Length (um) Via Cut Class
Wire on M4	292.08	0.05	9.736
Wire on M3	32.5909	0.022	0.478
Wire on M3	95.7272	0.022	1.404
Wire on M2	34.0227	0.022	0.499
Wire on M1	1.75	0.06	0.07
Wire on M1	1.75	0.06	0.07
Wire on M1	0.45	0.06	0.018

History

☒ Zoom to Fit ☒ Export to Console ☐ Export to Error Viewer ☒ Save History ☐ Run in Background ☐ Report RC

Reporting Resistance for All Constraints

Instead of reporting resistance for specific source and destination pairs, you can report resistance for every unique pair of instance terminals and ports across multiple nets with defined resistance constraints.

Before you start, make sure you create Resistance constraints for the nets. See [Setting Resistance Constraints](#).

To report resistance for multiple nets with defined resistance constraints:

1. Select the nets:
 - a. In the Layout Editor, open the Design Navigator.
 - b. Click the **Nets** tab and select the nets. You can do this from either the **Design: Layout** mode or the **Design: Schematic** mode.

You can also select multiple nets and choose **Report Resistance** to extract parasitics in a single run.

Note:

- If you select more than 10 nets, the **Reporter - Interactive StarRC** dialog box appears. Click **OK** to continue.
- If you select **Cancel**, you should select less than 10 nets and continue extraction in the interactive mode.

2. Right-click the selected nets, and then choose **Report Resistance**.

The **Parasitic Reporter** dialog box opens. The nets you selected in [Step 1](#) appear in the **Net** box in the **Resistance** tab.

The screenshot shows the 'Parasitic Reporter' dialog box with the 'Resistance' tab selected. The 'Capacitance' tab is also visible. The 'Net' field contains 'ntail'. The 'Draw' section has a blue square icon selected. The 'Depth' section has '0x' selected. The 'Area' section has 'Design' selected. The 'Report using' dropdown is set to 'All Constraints'. The 'Elmore Delay' checkbox is checked. Below these settings are two empty tables. The first table has columns: Net, Path Reported, Min Path R (Ohms), and Max Path R (Ohms). The second table has columns: Source, Destination, Path R (Ohms), Length (um), and Elmore Delay (ps). Below the tables are three tabs: 'Details', 'Layers', and 'Histogram'. The 'Details' tab is selected, showing a table with columns: Description, R (Ohms), Wire Width (um) Via Column/Rows, and Wire Length (um) Via Cut Class. Below the table is a 'History' section. At the bottom, there are checkboxes for 'Zoom to Fit', 'Export to Console', 'Export to Error Viewer', 'Save History', 'Run in Background', and 'Report RC'. The 'Save History' checkbox is checked. At the very bottom are buttons for 'Help', 'Defaults', 'Reset', 'Report', 'Export ...', 'Options', and 'Close'.

3. If the information in the dialog box was preloaded as part of the environment setup:
 - a. Review the settings in the **Resistance** tab.
 - b. Click **Options** (at the bottom of the dialog box), and review the settings in the **Electrical Options** dialog box. See [Setting EM, Resistance, and Capacitance Options](#) and [Specifying Input Information](#).
 - c. Skip to [Step 5](#).

4. If you did not preload settings in [Step 3](#), on the **Resistance** tab of the **Parasitic Reporter** dialog box:
 - Set the **Depth** and **Area** options. See [Parasitic Reporter - Resistance Tab Reference](#).
 - Click **Options** (at the bottom of the dialog box), and set the options in the **Electrical Options** dialog box. See [Setting EM, Resistance, and Capacitance Options](#).

Caution:

Values you enter in the dialog boxes override preloaded values.

5. In the **Report using** list, choose **All Constraints**.
6. To apply Elmore Delay analysis during the calculations, select **Elmore Delay**.

The option is only available when you select the StarRC tool for resistance reporting in the **Electrical Options** dialog box. See [Setting EM, Resistance, and Capacitance Options](#).
7. If you want to display the results in the Custom Compiler Console, select **Export To Console**.
8. To automatically save the generated resistance report so that you can recall it in future design sessions, select **Save History**. (Alternatively, set the `leEMRCAutoSaveHistory` Tcl preference to `true` before you generate the report.) The **History** section appears.

All newly generated reports will be automatically saved in the following directory and file:

```
$env (SYNOPSYS_CUSTOM_LOCAL) /r_reporter/<LCV>_R_result.db
```

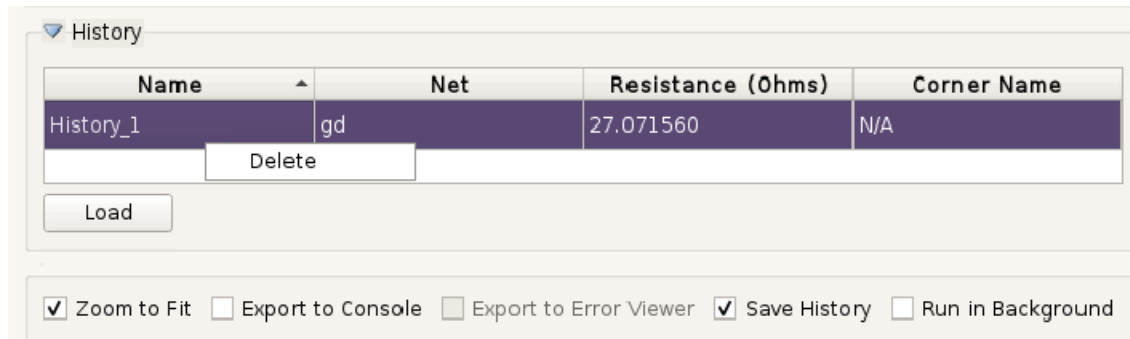
If a reported net already exists in the `.db` file, it will be overwritten by the latest data.

To recall the saved report, see [Viewing Point-to-Point Resistance Reports](#).

By default, the maximum number of reports retained is 30. To change the number, set the `leEMRCMaxHistoryNum` Tcl preference.

To remove the history of a report, right-click on the entry in the **History** section and select **Delete**.

testing



Name	Net	Resistance (Ohms)	Corner Name
History_1	gd	27.071560	N/A

Delete

Load

☒ Zoom to Fit ☐ Export to Console ☐ Export to Error Viewer ☒ Save History ☐ Run in Background

9. If you want to view the violations in the Error Viewer, select **Export To Error Viewer**.
10. If you want the resistance analysis in the background so that you can continue to work in the Layout Editor, select **Run in Background**.

Background mode supports both StarRC shared and non-shared modes. When the report is generated, you can locate the results in the **History** section of the dialog box.

11. When you are ready to generate and view the report, click **Report**. The Parasitic Reporter displays the results in the tables at the bottom half of the **Resistance** tab. See [Viewing Point-to-Point Resistance Reports](#).

Parasitic Reporter

Resistance ☒ Capacitance

Net: Draw: ☐ Cycle ☐ Auto Highlight

Depth: ☐ 0x ☐ View ☒ Full ☐ Include Parent Designs | Area: ☒ Design ☐ Region

Report using: ☒ Elmore Delay

Net	Path Reported	Path Violated	Min Path R (Ohms)	Max Path R (Ohms)
ntail	3	3	294.1006	483.28557

Source	Destination	Path R (Ohms)	Length (um)	Constraint (Ohm)	Elmore Delay (ps)
Ndpp N3_1/S	Ndpm N3_0/S	294.1006	3.198	100 ~ 250	0.49266
Ndpm N3_20/S	Ndpm N3_0/S	299.12187	4.82	100 ~ 250	0.05995
Ndpm N3_18/S	Ntail N0_6/S	483.28557	6.6115	100 ~ 250	0.45898

Details ☒ Layers ☐ Histogram

Description	R (Ohms)	Wire Width (um) Via Column/Rows	Wire Length (um) Via Cut Class
Wire on M3	75.102200	0.022	1.1015
Wire on M2	6.136360	0.022	0.09
Wire on M2	6.136360	0.022	0.09
Wire on M2	6.136360	0.022	0.09
Wire on M2	47.045400	0.022	0.69
Wire on M2	6.136360	0.022	0.09
Wire on M2	6.136360	0.022	0.09
Wire on M2	6.136360	0.022	0.09
Wire on M2	47.045400	0.022	0.69

History

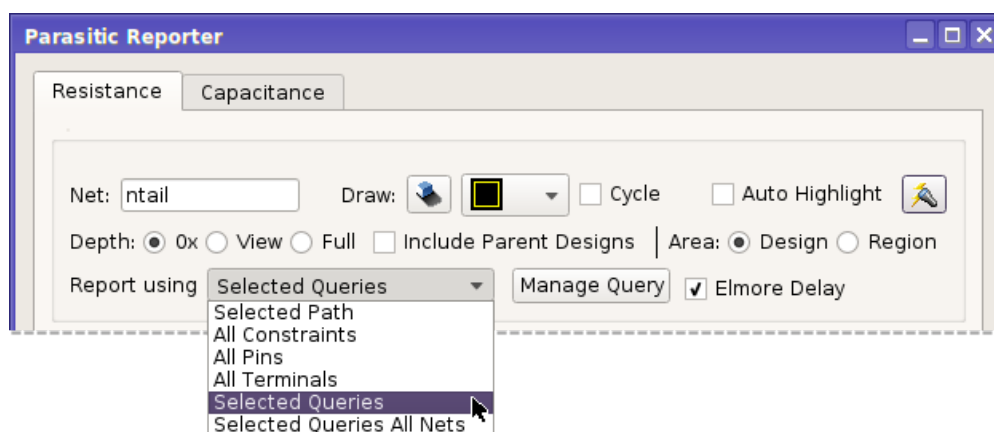
☒ Zoom to Fit ☒ Export to Console ☐ Export to Error Viewer ☒ Save History ☐ Run in Background ☐ Report RC

Reporting Resistance From Selected Queries

After you set up and save your resistance report queries, you can select the queries and create new reports.

On the **Resistance** tab, you can choose the query reporting type from the **Report using** list:

- [Reporting Selected Queries](#)
- [Reporting Selected Queries for All Nets](#)
- [Using Tcl command to report existing queries](#)



See also [Managing Resistance Report Queries](#).

Reporting Selected Queries

The **Reporting using: Selected Queries** option reports all selected queries of selected nets.

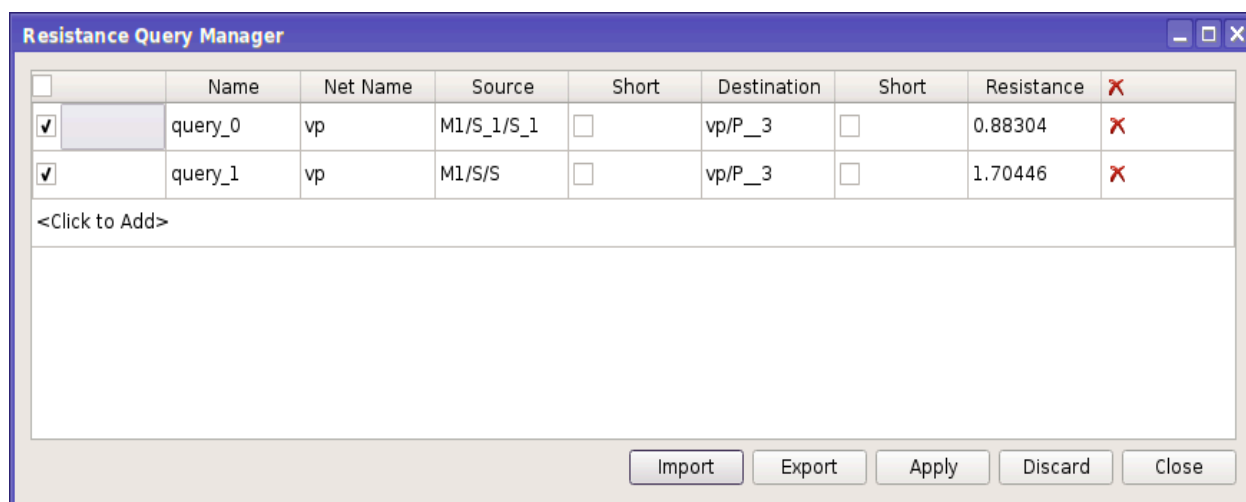
To generate resistance report for selected queries of selected nets:

1. Select the nets:
 - a. In the Layout Editor, open the Design Navigator.
 - b. Click the **Nets** tab and select the nets.

You can also select multiple nets and choose **Report Resistance** to extract parasitics in a single run. If you select more than 10 nets, the **Reporter - Interactive StarRC** dialog box appears. Click **OK** to continue or **Cancel**. If you select **Cancel**, you should select less than 10 nets and continue extraction in the interactive mode.

2. On the **Resistance** tab in the **Parasitic Reporter** dialog box, click **Manage Query**.

The **Resistance Query Manager** dialog box opens.



3. Select the queries.

4. On the **Resistance** tab:

a. In the **Report Using** list, choose **Selected Queries**.

b. To apply Elmore Delay analysis during the calculations, select **Elmore Delay**.

The option is only available when you select the StarRC tool for resistance reporting in the **Electrical Options** dialog box. See [Setting EM, Resistance, and Capacitance Options](#).

c. To automatically save the generated resistance report so that you can recall it in future design sessions, select **Save History**. (Alternatively, set the `leEMRCAutoSaveHistory` Tcl preference to `true` before you generate the report.) The **History** section appears.

All newly generated reports will be automatically saved in the following directory and file:

```
$env(SYNOPSYS_CUSTOM_LOCAL)/r_reporter/<LCV>_R_result.db
```

If the report data for a net already exists in the `.db` file, it will be overwritten by the latest data.

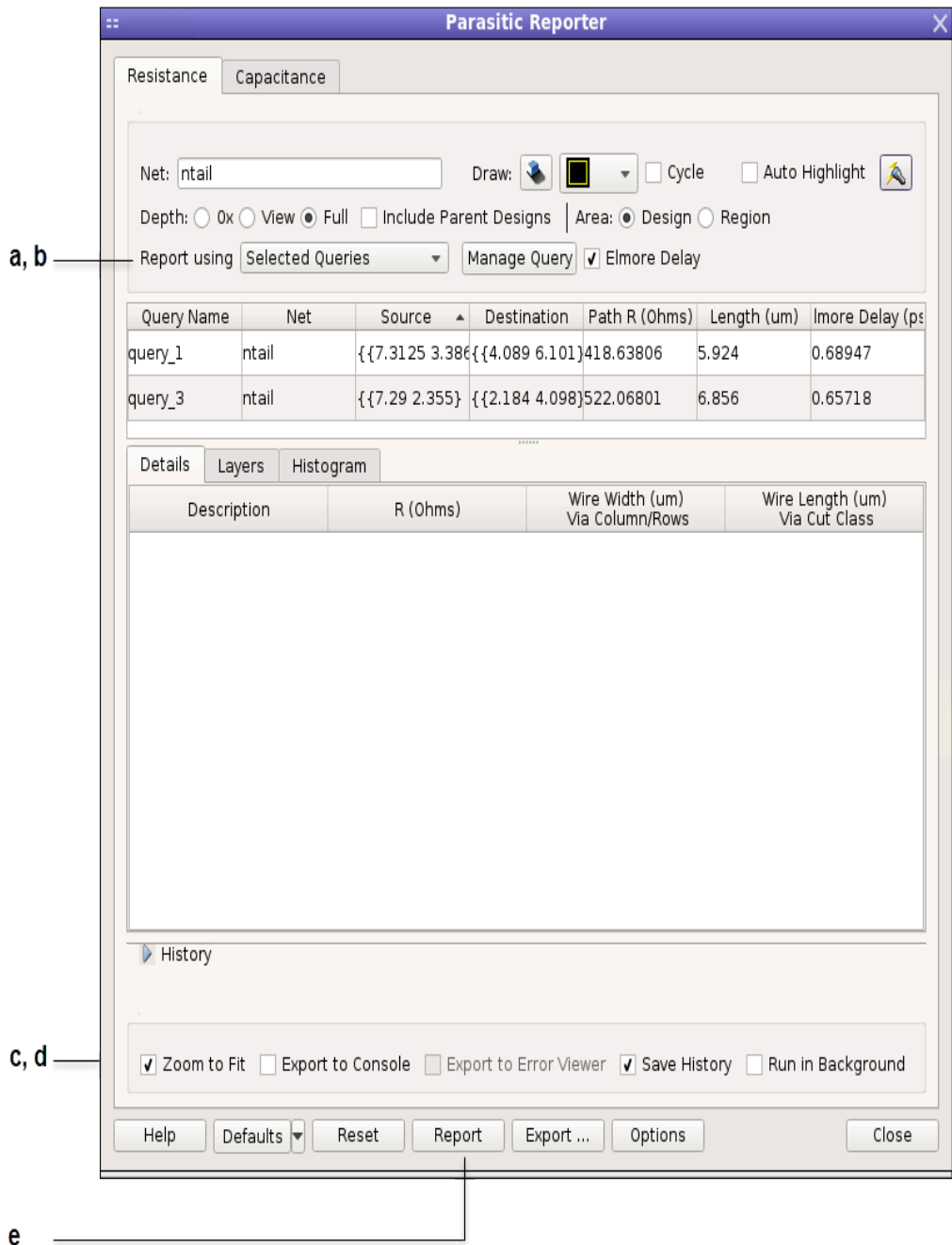
To recall the saved report, see [Viewing Point-to-Point Resistance Reports](#).

To remove the history of reports, click **Clear**.

d. If you want the resistance analysis in the background so that you can continue to work in the Layout Editor, select **Run in Background**.

Background mode supports both StarRC shared and non-shared modes. When the report is generated, you can locate the results in the **History** section of the dialog box.

- e. Click **Report**.



The **Resistance** tab tables refresh to display the new reports. See [Viewing Point-to-Point Resistance Reports](#).

Reporting Selected Queries for All Nets

The **Reporting using: Selected Queries All Nets** option reports all selected queries of all nets (net selection is ignored).

To generate resistance report for all selected queries of all nets:

► On the **Resistance** tab in the **Parasitic Reporter** dialog box:

- a. In the **Report Using** list, choose **Selected Queries All Nets**.
- b. To apply Elmore Delay analysis during the calculations, select **Elmore Delay**.

The option is only available when you select the StarRC tool for resistance reporting in the **Electrical Options** dialog box. See [Setting EM, Resistance, and Capacitance Options](#).

- c. To automatically save the generated resistance report so that you can recall it in future design sessions, select **Save History**. (Alternatively, set the `leEMRCAutoSaveHistory` Tcl preference to `true` before you generate the report.) The **History** section appears.

All newly generated reports will be automatically saved in the following directory and file:

```
$env(SYNOPSISYS_CUSTOM_LOCAL)/r_reporter/<LCV>_R_result.db
```

If the report data for a net already exists in the `.db` file, it will be overwritten by the latest data.

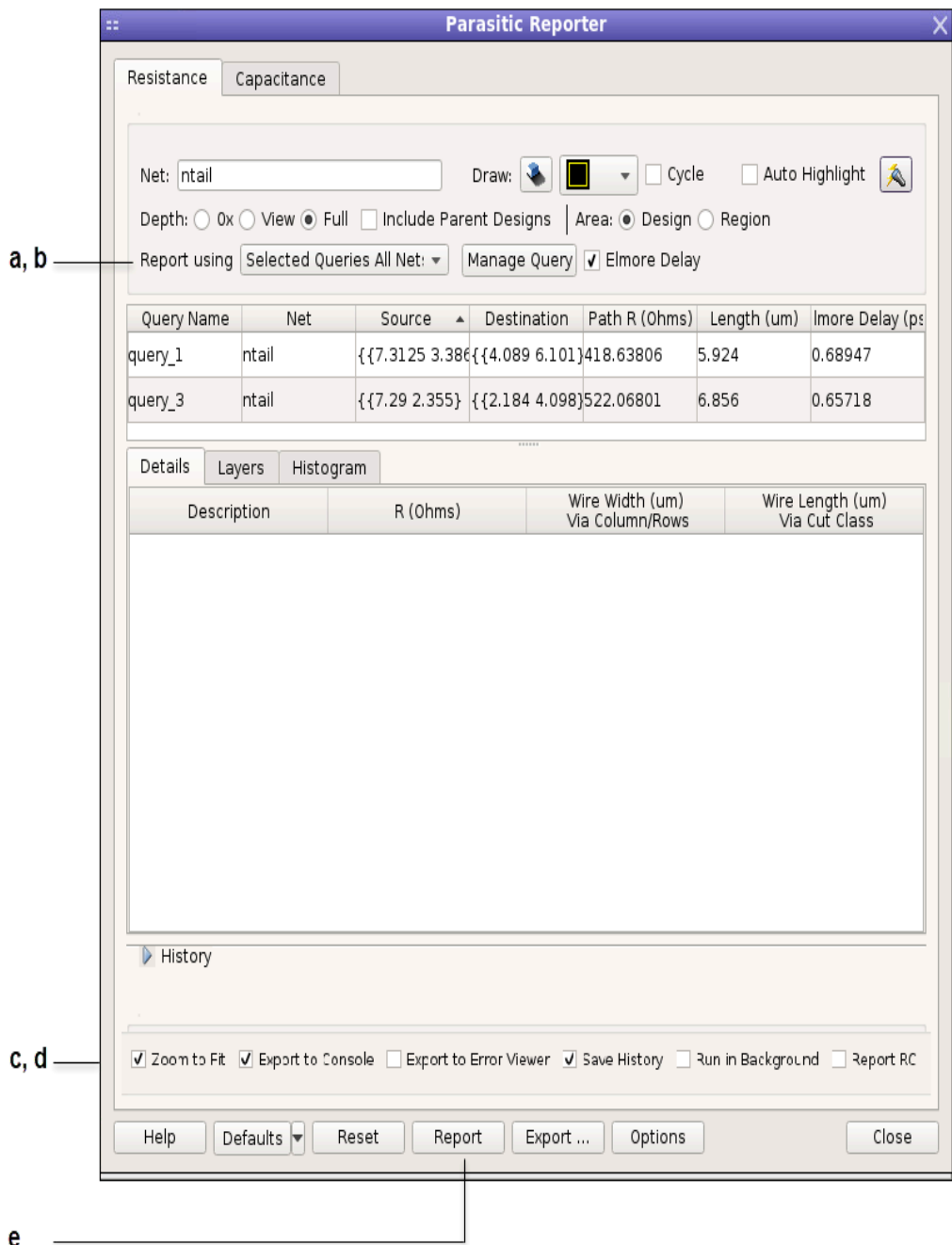
To recall the saved report, see [Viewing Point-to-Point Resistance Reports](#).

To remove the history of reports, click **Clear**.

- d. If you want the resistance analysis in the background so that you can continue to work in the Layout Editor, select **Run in Background**.

Background mode supports both StarRC shared and non-shared modes. When the report is generated, you can locate the results in the **History** section of the dialog box.

- e. Click **Report**.



The **Resistance** tab tables refresh to display the new reports. See [Viewing Point-to-Point Resistance Reports](#).

Using Tcl command to report existing queries

Instead of using the query features in the **Parasitic Reporter** dialog box, you can use the `leReportResistanceSet` Tcl preference with the `le::reportResistance` Tcl command to recall one or more query sets in a startup Tcl script.

For example, to save queries, you can use the following Tcl code:

```
db::setPrefValue leReportResistanceSet -value "<queryName>"
```

To report resistance for a specific query, you can use the following Tcl code:

```
le::reportResistance [ed] -query <queryName>
```

You can also use Tcl code to report resistance for all queries:

```
le::reportResistance [ed] -queryAll
```

Managing Resistance Report Queries

You can save Selected Path resistance report queries and recall them later to generate new reports. With the Resistance Query Manager, you can also edit existing queries, delete individual queries, and import or export the queries.

This topic describes the following:

- [Saving Queries Automatically](#)
- [Saving Queries Manually](#)
- [Modifying Queries](#)
- [Deleting Queries](#)
- [Importing and Exporting Queries](#)

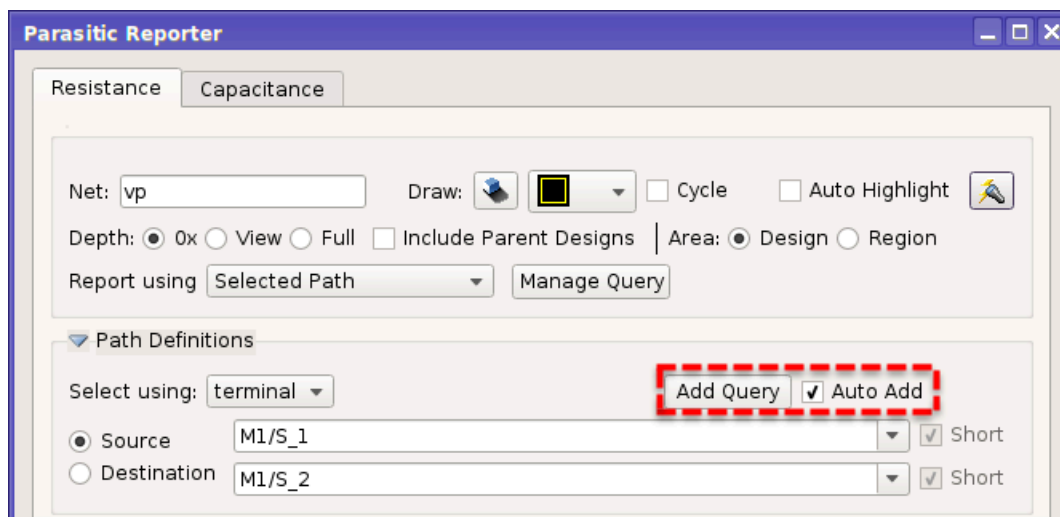
To generate new resistance reports from queries, see [Reporting Resistance From Selected Queries](#).

Saving Queries Automatically

After you set up a Selected Path query on the **Resistance** tab, you can save the query automatically.

1. On the **Resistance** tab, after you select the sources and destinations, click **Add Query**. The Parasitic Reporter saves the query information to the Resistance Query Manager. Use this method if you want to save the query without having to generate the report first.

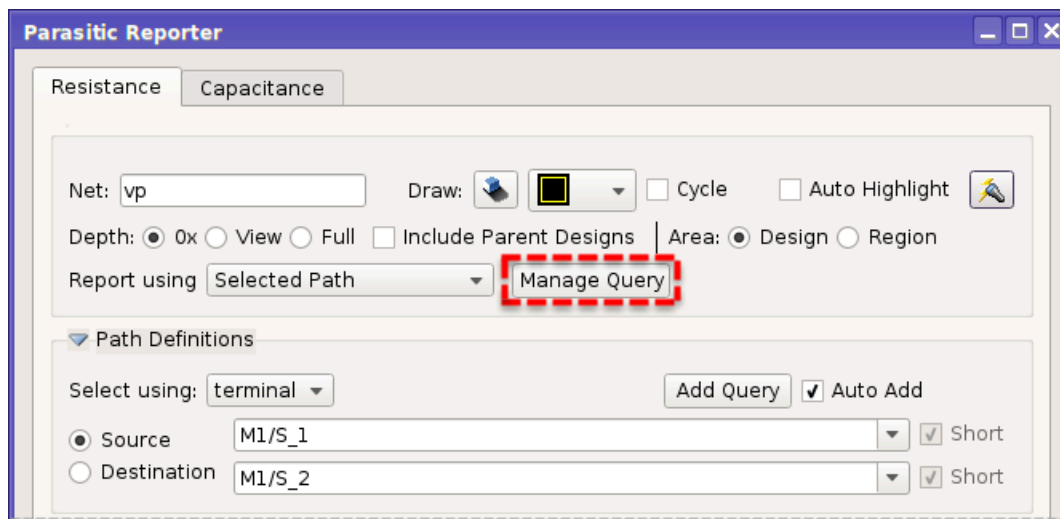
2. (Optional) Select **Auto Add** to add the query when you click **Report**. See [Reporting Resistance for Selected Path \(Pin-to-Pin, Terminal-to-Terminal, Instance-to-Instance, Point-to-Point\)](#).



Saving Queries Manually

You have the option of adding a Selected Path query manually.

1. On the **Resistance** tab, click **Manage Query**.



2. In the **Resistance Query Manager** dialog box, click **Click to Add**. A new row appears in the query table.

The Resistance Query Manager dialog box contains a table with the following data:

	Name	Net Name	Source	Short	Destination	Short	Resistance	
☐								✗
☒	query_0	gd	M1/S_1/S_1	☐	vp/P_3	☐		✗
☒	query_1	vp	M1/S/S	☐	vp/P_3	☐	1.70446	✗
☒	query_2	vp	M1/S_1/S_1	☐	M1/S_2/S_2	☐	1.61	✗
☒	query_3	gd		☐		☐		✗

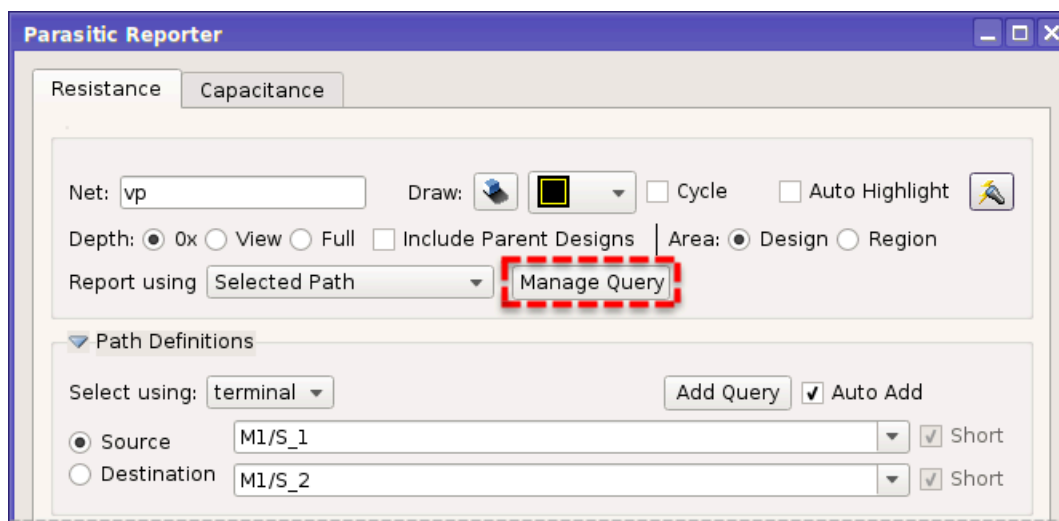
Below the table is a text area with the placeholder text "<Click to Add>". At the bottom right of the dialog are five buttons: Import, Export, Apply, Discard, and Close.

3. In the **Name** column, type a name for the query you just created.
4. Specify the **Net Name**, if applicable.
5. Specify the **Source** and **Destination**. The Source and Destination can be a pin name, terminal name, or coordinates on the layout canvas. For details, see [Reporting Resistance for Selected Path \(Pin-to-Pin, Terminal-to-Terminal, Instance-to-Instance, Point-to-Point\)](#).
6. When you are finished, click **Apply** to save the changes.

Modifying Queries

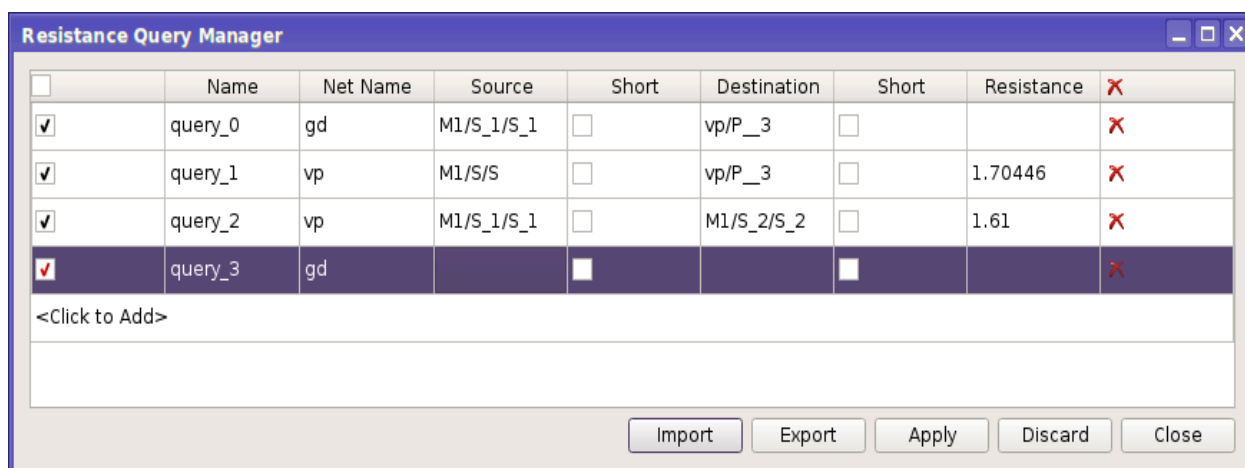
You have the option of modifying existing queries. Doing so can save you from having to re-enter all the query information just for minor changes.

1. On the **Resistance** tab, click **Manage Query**.



2. In the **Resistance Query Manager** dialog box, locate the query and edit the columns. For details about the columns, see [Reporting Resistance for Selected Path \(Pin-to-Pin, Terminal-to-Terminal, Instance-to-Instance, Point-to-Point\)](#).

Note that you cannot add or edit the **Resistance** column.

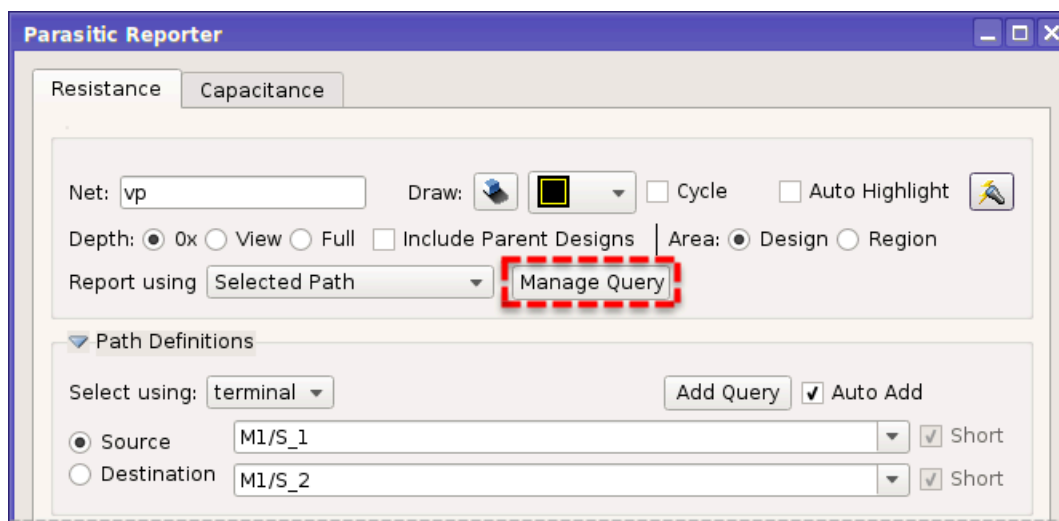


3. If you made a mistake, click **Discard** to undo the modifications. This must be done before you click **Apply** and commit the changes.

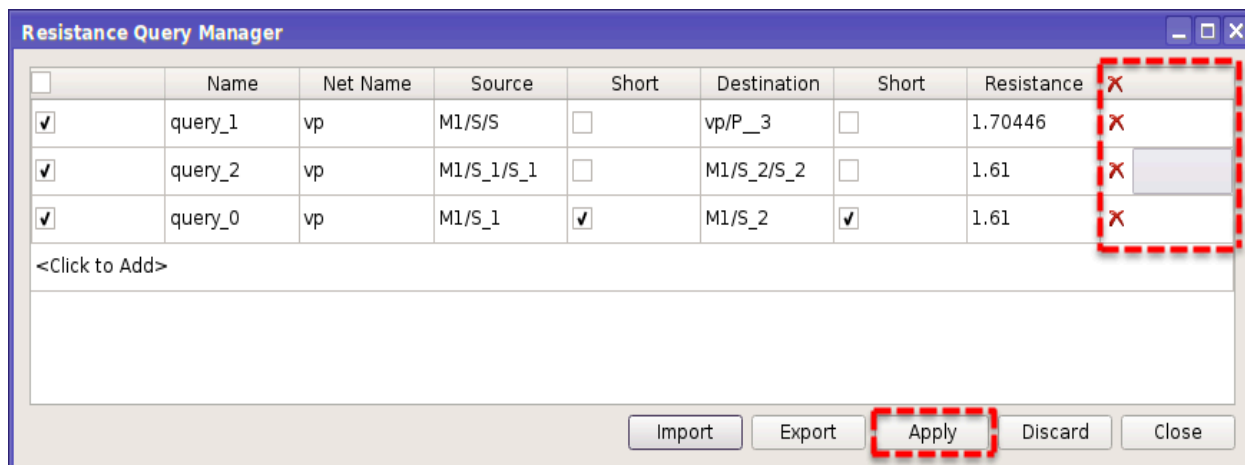
Deleting Queries

You can delete queries that are obsolete, thus preventing the list from becoming cluttered.

1. On the **Resistance** tab, click **Manage Query**.



2. In the **Resistance Query Manager** dialog box, do one of the following:
 - To delete one or more queries, locate the queries and click  in the corresponding rows.
 - To delete all the queries, click  in the table heading.
3. If you made a mistake, click **Discard** to undo the deletions. This must be done before you click **Apply** and commit the deletions.
4. When you are finished, click **Apply** to commit the deletions.

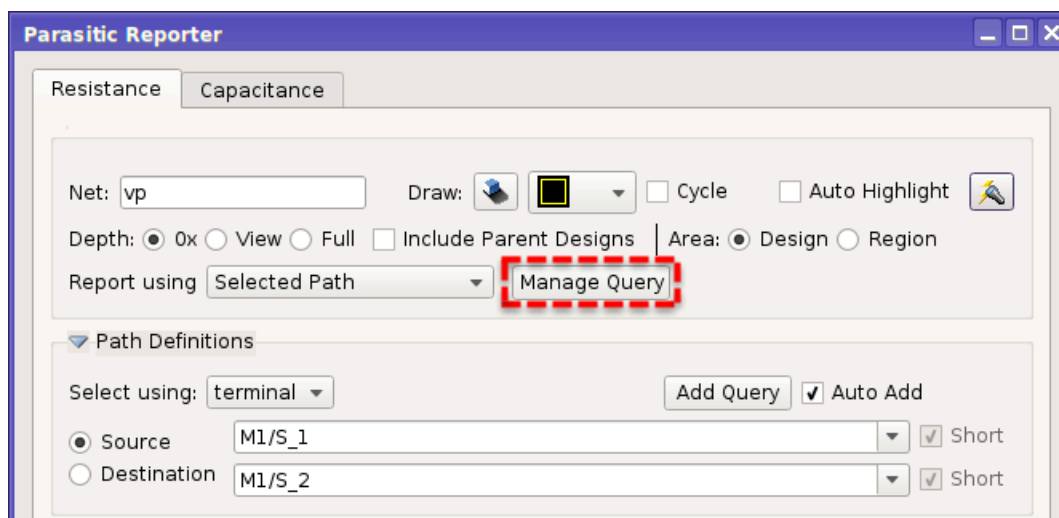


Importing and Exporting Queries

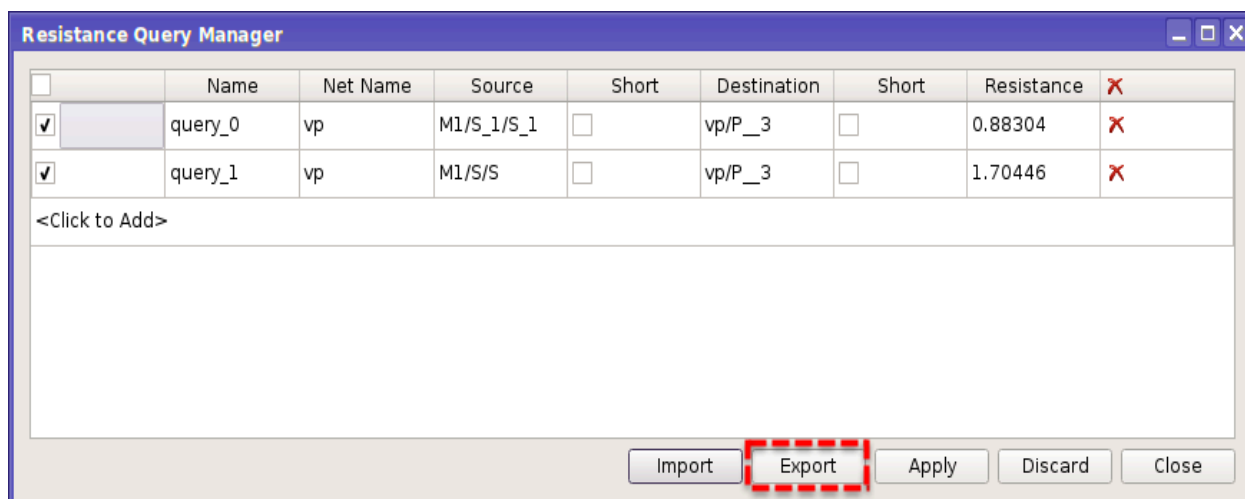
You can export the queries to an XML file, modify it to create a new query XML file, and then import the new XML file to generate new reports.

To export the queries:

1. On the **Resistance** tab, click **Manage Query**.



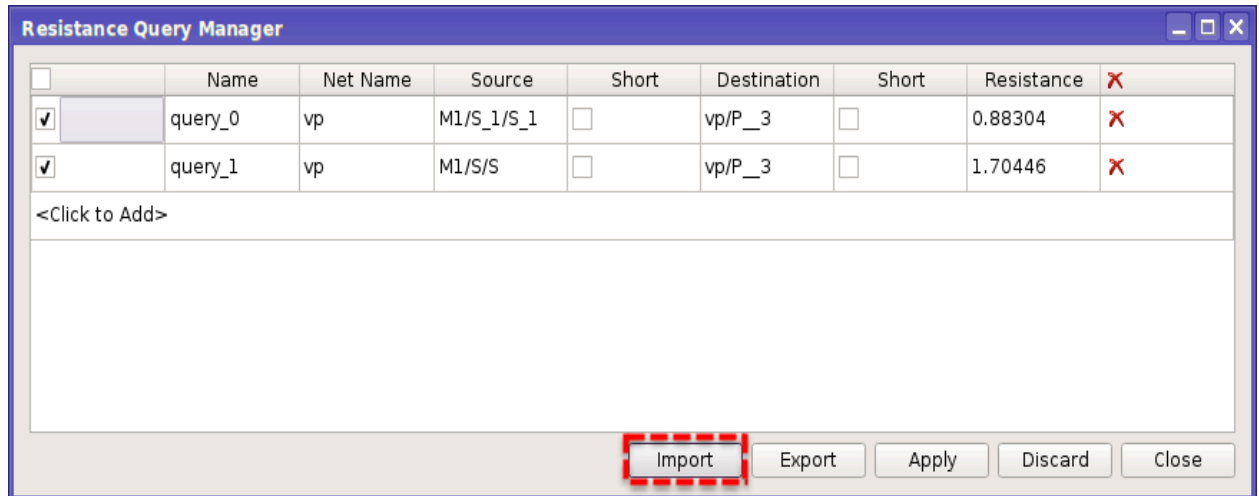
2. If you modified the queries, click **Apply** to save the changes so that the modifications will be exported.
3. In the **Resistance Query Manager** dialog box, click **Export**. All selected and non-selected queries in the Resistance Query Manager are exported.



To import the queries:

1. In the **Resistance Query Manager** dialog box, click **Import**.

You can import resistance queries in XML files to **Resistance Query Manager** using the `le::importResistanceQuery` command.



2. In the **Open File for Query** dialog box, select the query (XML) file you want to import, and click **Open**.

If the import process encounters existing queries, you have the option of overwriting the queries or appending the queries. During the append process, the Query Manager lets you know of query name conflicts so that you can resolve them.

The imported queries appear in the **Resistance Query Manager** dialog box.

3. Review the queries, and do one of the following:
 - To keep the imported queries as they are, click **Apply**.
 - To modify the queries, see [Modifying Queries](#) and [Deleting Queries](#). When you are finished, click **Apply**.
 - If you do not want to keep any of the imported queries, click **Discard**, and then click **Apply**.

Reporting Resistance and Capacitance

The **Report RC** option reports resistance and capacitance simultaneously.

To generate reports of resistance and capacitance:

1. Set the `leRCReporterByOneExtraction` preference to `true` to enable the **Report RC** option.

Note:

Since **Report RC** runs with StarRC only, make sure that you choose **Options > Resistance > StarRC**.

2. Select **Report**. The results are displayed in the **Details** section of the **Resistance** tab.
3. To view the history of RC reports, select the **Save History** option and run the RC report. The results are displayed in the **History** section.

Viewing Point-to-Point Resistance Reports

The resistance report in the **Resistance** tab consists of the following:

- **Nets** table: Lists the nets you selected, the number of pin-to-pin or terminal-to-terminal paths reported on the net, and the calculated minimum and maximum resistance. If you selected two points that are not on any net, nothing appears in the net table.
- **Summary report** table: Contains the source and destination used, the resistance calculated between the source and destination, and the length of the wire between the source and destination.
- **Details** tab: Displays segment-level resistance information.
- **Histogram** tab: Shows the distribution of the various data in the resistance report.

Parasitic Reporter

Resistance | Capacitance

Net: Draw: ☐ Cycle ☐ Auto Highlight

Depth: ☐ 0x ☐ View ☒ Full ☐ Include Parent Designs | Area: ☒ Design ☐ Region

Report using ☒ Elmore Delay

Net	Path Reported	Path Violated	Min Path R (Ohms)	Max Path R (Ohms)
ntail	3	2	209.92167	390.4799

Source	Destination	Path R (Ohms)	Length (um)	Constraint (Ohm)	Elmore Delay (ps)
Ndpm N3_1/S	Ndpm N3/S	390.4799	12.409	100 ~ 250	0.80735
Ndpp N3/S	Ndpm N3_20/S	280.01757	7.661	100 ~ 250	0.52619
Ndpp N3_18/S	Ntail N0_6/S	209.92167	5.219	100 ~ 250	0.53001

Details | Layers | Histogram

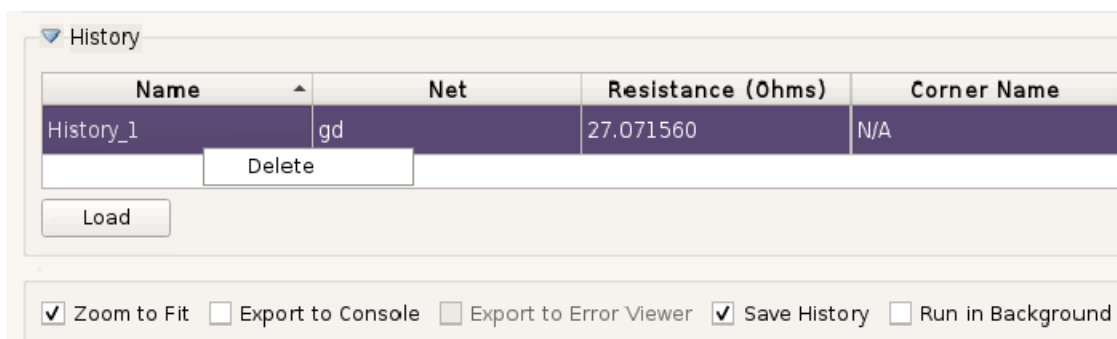
Description	R (Ohms)	Wire Width (um) Via Column/Rows	Wire Length (um) Via Cut Class
Wire on M3	17.925	0.06	0.717
Wire on M3	47.8	0.06	1.912
Wire on M3	29.875	0.06	1.195
Wire on M2	1.55	0.06	0.062
Wire on M2	1.55	0.06	0.062
Wire on M2	1.55	0.06	0.062
Wire on M2	1.55	0.06	0.062
Wire on M2	1.55	0.06	0.062
Wire on M2	1.55	0.06	0.062
Wire on M2	1.55	0.06	0.062
Wire on M2	10.25	0.06	0.41

History

☒ Zoom to Fit ☒ Export to Console ☐ Export to Error Viewer ☒ Save History ☐ Run in Background ☐ Report RC

Before you start, if you want to view a saved report, in the **Parasitic Reporter** dialog box:

1. Select **Save History**.
2. In the **History** section that appears, select the report (row) you want to view.
3. Click **Load**. The report displays in the tab and displaces any existing report in the tab.



4. To preview details about a saved report, rest the pointer on the row. A tooltip displays the additional information.

To view the resistance report:

1. Review the data in the net table.

Column Name	Description
Net	Name of the net.
Path Reported	Number of pin-to-pin, terminal-to-terminal or instance-to-instance paths reported for the net.
Path Violated	Number of pin-to-pin, terminal-to-terminal, or instance-to-instance paths that have resistance values that violate the defined resistance constraint. The value is green if it meets the resistance constraint value. The value is red if it does not meet the constraint value.
Min Path Resistance	Minimum resistance calculated between the source and destination.
Max Path Resistance	Maximum resistance calculated between the source and destination.

2. In the nets table, select a net and review the data in the summary report table.

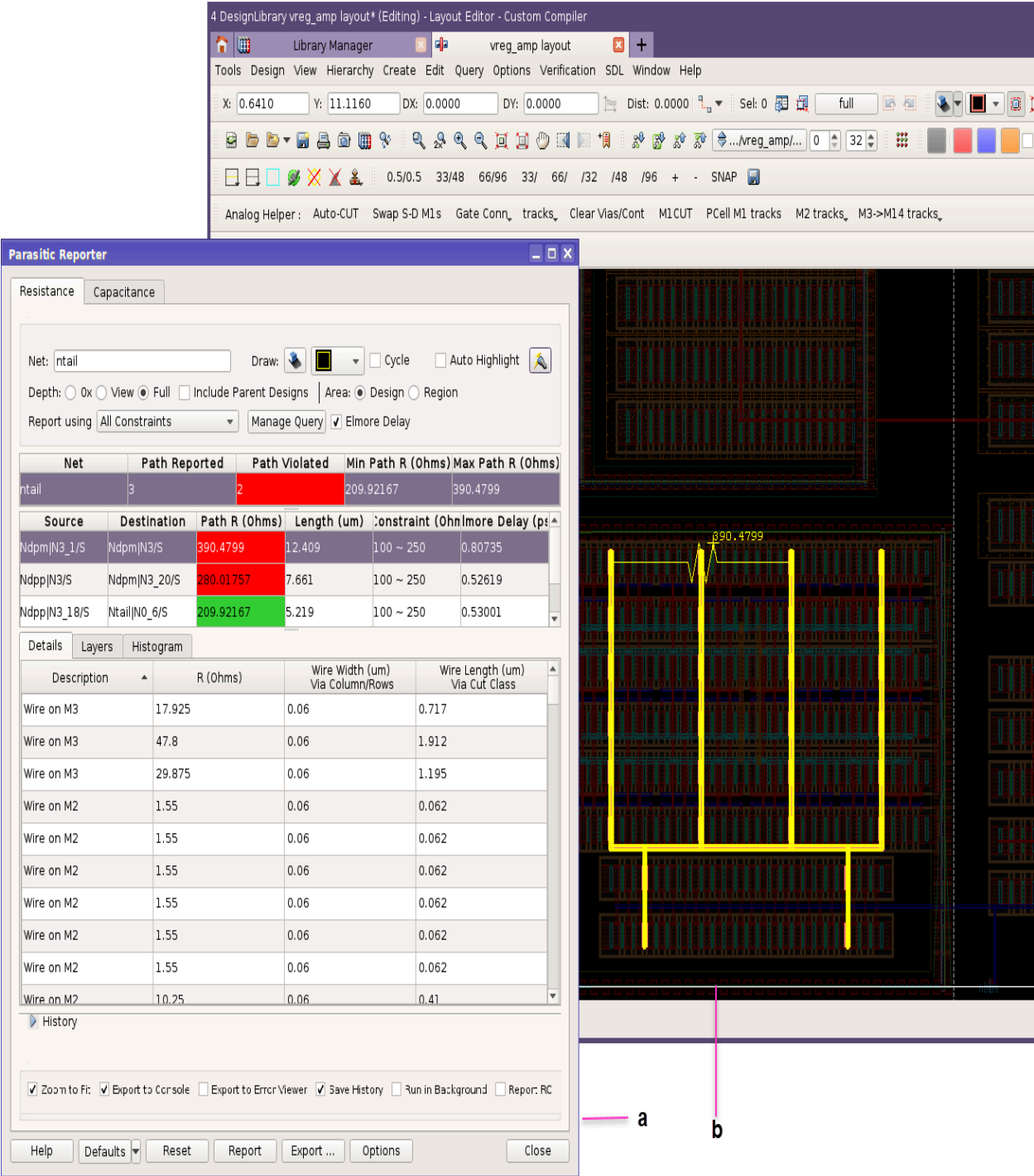
Column Name	Description
Source	Name of the source pin, terminal, instance, or coordinates of the source point.
Destination	Name of the destination pin, terminal, instance, or coordinates of the destination point.

Column Name	Description
Path Resistance	Resistance calculated between the source and destination. The value is green if it meets the resistance constraint value. The value is red if it does not meet the constraint value. If the value is in violation (red), check the Res Constraint column.
Length	Length of the wire between the source and destination.
Res Constraint	Minimum and maximum resistance values defined by the resistance constraint.
Elmore Delay	Estimated time delay (in picoseconds) from the source. This column only appears if you selected Elmore Delay in Net and Display Options .

3. In the summary table, select a row to display segment-level information in the detailed report table. The detailed report table populates with the following:

Column Name	Description
Description	The layer on which the wire segment is located.
Resistance	The calculated resistance (ohms) of the segment.
Wire Width Via Columns/Rows	The width (μm) of the wire, or the number of columns and rows of cuts in the via.
Wire Length Via Cut Class	(Wires only) The length (μm) of the wire segment, or the via cut class.

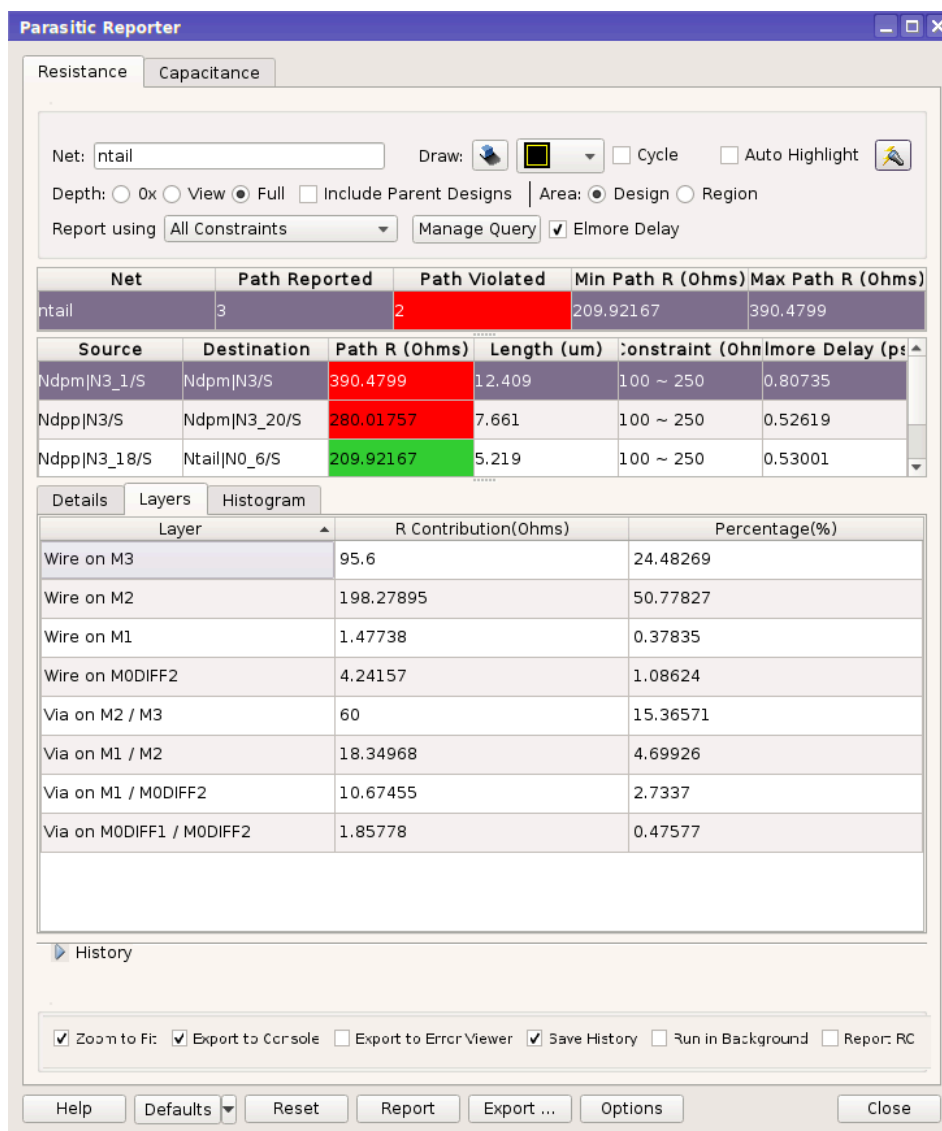
4. To view the flight line on the canvas, select a row in the summary report table. The canvas displays the area of the design that contains the flight line.
5. To view a wire segment or via on the canvas, select the segment in the detailed report table (a). The canvas zooms in and displays a magnified view of the segment (b). Alternatively, you can click a segment of interest on the canvas (b), and the Parasitic Reporter highlights the corresponding row in the report (a).



6. To view the resistance contribution from a specific layer, click the **Layers** tab. The contents of the **Layers** tab depends on the selections you make in the **Summary report** table .

The **Layers** tab provides the following information:

Column	Description
Layer	Layer name.
R Contribution	The resistance contributed by the layer.
Percentage	The percent of resistance contribution by the layer compared to the resistance of the entire path: $((R_{\text{layer}} / R_{\text{total_path}}) * 100)$.

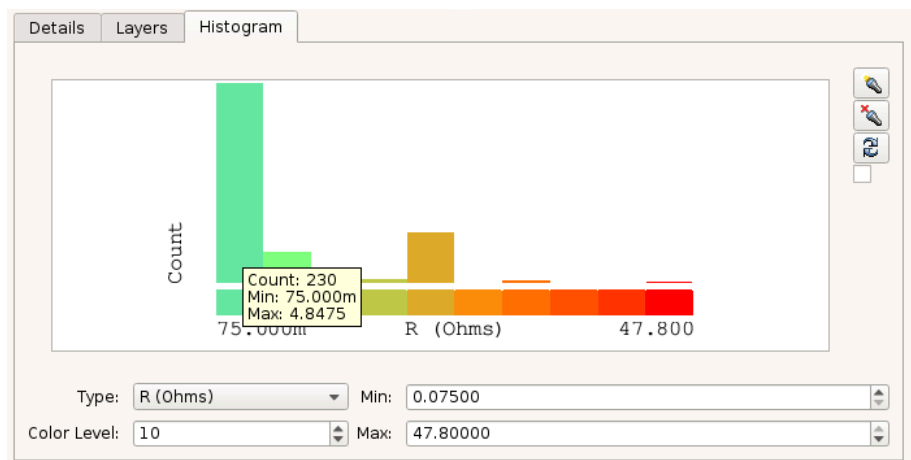


7. To view the histograms of report data, click the **Histogram** tab, and then:
 - a. Choose the **Type** of data you want to view. Only Min Path R (Ohms), Max Path R (Ohms), R (Ohms) are available in heat map views.
For the descriptions of the data, see steps 1, 2, and 3.
 - b. Set the **Color Level**, the number of colors you want to use when displaying the histogram and heat map on the canvas.

The data between the **Min** and **Max** values are binned into the number of colors you specify.

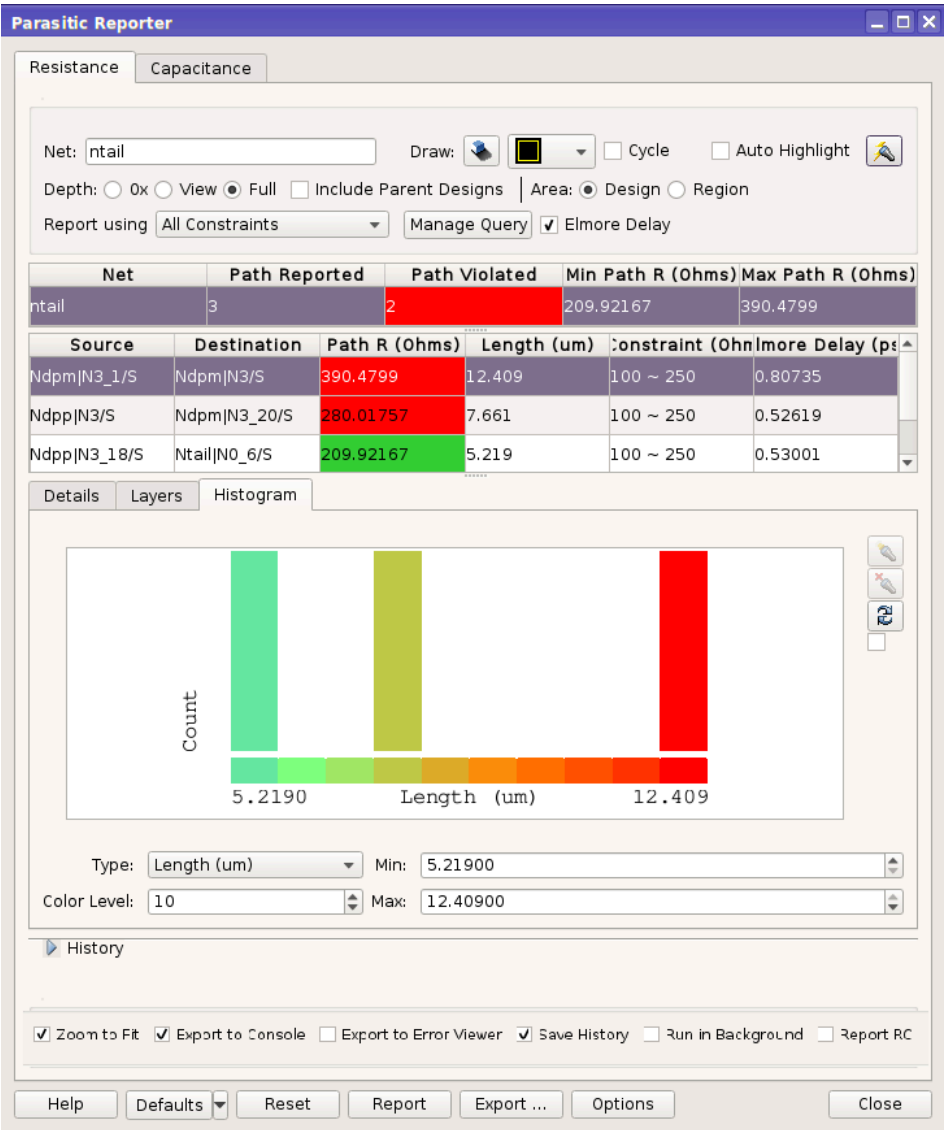
- c. Specify the **Min** and **Max** values for the data type you chose in step a.
- d. View the bar chart that shows the distribution of the data type you selected.
- e. To view the corresponding heat map on the canvas, click **Show heat map** . To hide the heat map, click **Hide heat map** .

To view the number of values within a bin range, rest the pointer on a colored bar in the histogram.



- f. Repeat steps a through e to view other data types and the corresponding distribution. To refresh the histogram, click **Refresh** .

If you want to automatically refresh the histogram, select the check box below .



8. Adjust the design and generate new reports until the resistance calculations are satisfactory.

To adjust wire widths, see [Stretching the Wire Width](#).

Exporting Point-to-Point Resistance Reports

If you want to export the resistance report to the Console or a CSV file, you must make the selection before you generate the report. This topic describes how to specify the export settings.

You can export point-to-point resistance reports to the following:

- Console
- Comma-separated values (CSV) files, in which the comma character separates each field of text. You can open CSV files in software applications such as Microsoft Excel.

To export point-to-point resistance reports to the console:

1. In the **Resistance** tab, select **Export to Console**.
2. If you want to view the violations in the Error Viewer, select **Export To Error Viewer**.

3. Click **Report**. The Parasitic Reporter exports the report to the Console.

The Parasitic Reporter dialog box is shown with the **Resistance** tab selected. The configuration options are as follows:

- Net:** ntail
- Draw:** [Icon] [Color: Black] [Cycle: ☐] [Auto Highlight: ☐
- Depth:** ☐ 0x ☐ View ☒ Full ☐ Include Parent Designs
- Area:** ☒ Design ☐ Region
- Report using:** All Constraints [Manage Query]

The main table displays path violations for the selected net:

Net	Path Reported	Path Violated	Min Path R (Ohms)	Max Path R (Ohms)
ntail	3	2	196.86392	376.67512

Below this, a detailed table shows the source, destination, path resistance, length, and constraint for each violation:

Source	Destination	Path R (Ohms)	Length (um)	Constraint (Ohms)
Ndpm N3_1/S	Ndpm N3/S	376.67512	12.387	100 ~ 250
Ndpp N3/S	Ndpm N3_20/S	266.19266	7.639	100 ~ 250
Ndpp N3_18/S	Ntail N0_6/S	196.86392	5.215	100 ~ 250

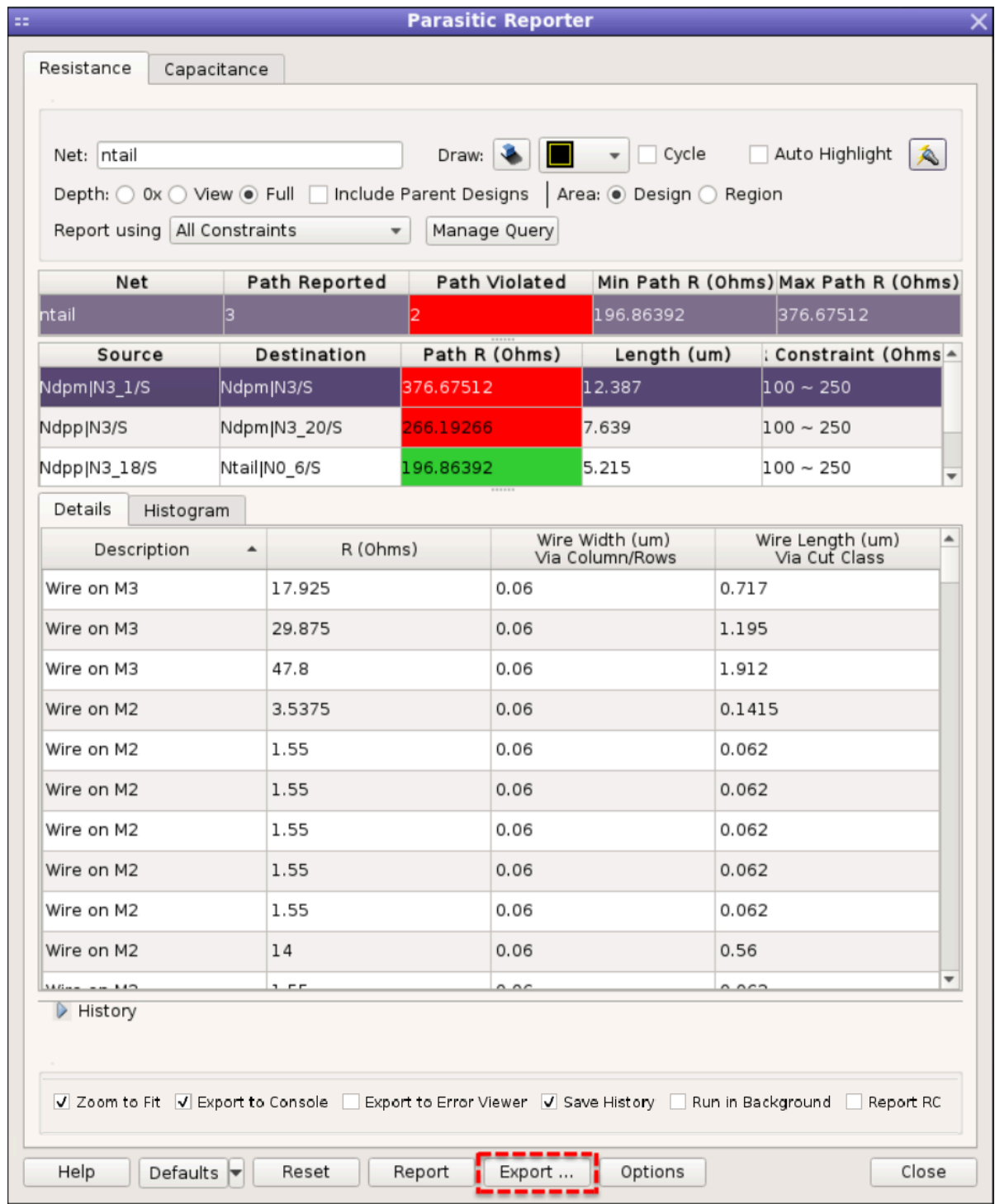
The **Details** tab is also visible, showing a table of wire segments:

Description	R (Ohms)	Wire Width (um) Via Column/Rows	Wire Length (um) Via Cut Class
Wire on M3	17.925	0.06	0.717
Wire on M3	29.875	0.06	1.195
Wire on M3	47.8	0.06	1.912
Wire on M2	3.5375	0.06	0.1415
Wire on M2	1.55	0.06	0.062
Wire on M2	1.55	0.06	0.062
Wire on M2	1.55	0.06	0.062
Wire on M2	1.55	0.06	0.062
Wire on M2	1.55	0.06	0.062
Wire on M2	14	0.06	0.56

At the bottom, there are checkboxes for **Zoom to Fit**, **Export to Console**, **Export to Error Viewer**, **Save History**, **Run in Background**, and **Report RC**. The **Report** button is highlighted.

To export point-to-point resistance reports to CSV files:

1. In the **Resistance** tab, click **Export**. The **Export Parasitic** dialog box appears.



2. Select the location for the file, type a name in the **File name** box, and click **Save**. The Parasitic Reporter exports the report to a CSV file.

Parasitic Reporter - Resistance Tab Reference

This section provides a quick reference of the **Resistance** tab in the **Parasitic Reporter** dialog box. To generate point-to-point resistance reports, see [Reporting Resistance for Selected Path \(Pin-to-Pin, Terminal-to-Terminal, Instance-to-Instance, Point-to-Point\)](#), [Reporting Resistance for All Pins, All Terminals, or All Instances on a Net](#), and [Reporting Resistance for All Constraints](#).

For the quick reference of the **Capacitance** tab, see [Parasitic Reporter - Capacitance Tab Reference](#).

The **Resistance** tab in the **Parasitic Reporter** dialog box contains the following options:

- [Net and Display Options](#)
- [Path Definition](#)
- [Net Table](#)
- [Summary Report Table](#)
- [Query Summary Report](#)
- [Details Tab](#)
- [Layers Tab](#)
- [Histogram Tab](#)
- [Global Options](#)

Net

Summary Report

Detailed Report

Parasitic Reporter

ResistanceCapacitance

Net: ntailDraw: ☐ Cycle ☐ Auto Highlight

Depth: ☐ 0x ☐ View ☒ Full ☐ Include Parent Designs | Area: ☒ Design ☐ Region

Report using All Constraints Manage Query ☒ Elmore Delay

Net	Path Reported	Path Violated	Min Path R (Ohms)	Max Path R (Ohms)
ntail	3	2	209.92167	390.4799

Source	Destination	Path R (Ohms)	Length (um)	Constraint (Ohm)	Elmore Delay (ps)
Ndpm N3_1/S	Ndpm N3/S	390.4799	12.409	100 ~ 250	0.80735
Ndpp N3/S	Ndpm N3_20/S	280.01757	7.661	100 ~ 250	0.52619
Ndpp N3_18/S	Ntail N0_6/S	209.92167	5.219	100 ~ 250	0.53001

DetailsLayersHistogram

Description	R (Ohms)	Wire Width (um) Via Column/Rows	Wire Length (um) Via Cut Class
Wire on M3	17.925	0.06	0.717
Wire on M3	47.8	0.06	1.912
Wire on M3	29.875	0.06	1.195
Wire on M2	1.55	0.06	0.062
Wire on M2	1.55	0.06	0.062
Wire on M2	1.55	0.06	0.062
Wire on M2	1.55	0.06	0.062
Wire on M2	1.55	0.06	0.062
Wire on M2	1.55	0.06	0.062
Wire on M2	10.25	0.06	0.41

History

☒ Zoom to Fit ☒ Export to Console ☐ Export to Error Viewer ☒ Save History ☐ Run in Background ☐ Report RC

HelpDefaultsResetReportExport ...OptionsClose

Net and Display Options

Option	Description
Net	The nets that will be queried for the resistance report. Default: The nets you selected in the Design Navigator in Reporting Resistance for Selected Path (Pin-to-Pin, Terminal-to-Terminal, Instance-to-Instance, Point-to-Point)

Option	Description
Draw	Specifies the following display options: : Dims non-highlighted nets to increase the visual contrast between highlighted and non-highlighted nets. Default: Off : Applies the selected color to the shapes that correspond to the source and destination selections in the report table. Default: Yellow Cycle: changes the color packet each time you select a different row in the report table. Clear the Cycle check box to use the currently selected color for all selections in reports. Default: Off
Auto Highlight	Automatically highlights the net as you soon as you report resistance from the Design Navigator, or when you change the net, depth, source, or destination. If you clear the Auto Highlight check box, you can use  to manually highlight selected nets. Default: Manual
Depth	Specifies the stop level of the extraction: • 0x: Extracts the top-level connecting shapes only. • View: Extracts all connecting shapes that are at the top level (0x) and the visible levels of the hierarchy. • Full: Extracts connecting shapes across the entire hierarchy in the design. Select Full if you want to select pins or points in subcells. Default: 0x
Include Parent Designs	Traces connected shapes through the parent hierarchy when you are in the edit-in-place mode at a low-level instance. Default: Off
Area	Specifies the area within which resistance reporting is performed: • Design: The entire design. • Region: The region you define. After you select Region, draw a rectangle on the canvas to define the reporting area.

Option	Description
Report Using	Specifies the type of resistance reporting: • Selected Path: Single-net or no-net pin-to-pin, terminal-to-terminal, instance-to-instance or point-to-point resistance reporting. See Reporting Resistance for Selected Path (Point-to-Point). • All Constraints: Every unique pair of instance terminals and ports on the multiple nets with constraints. See Reporting Resistance for All Constraints. • All Pins: All pin-pairs on multiple nets. See Reporting Resistance for All Pins or All Terminals on a Net. • All Terminals: All terminal-pairs on multiple nets. See Reporting Resistance for All Pins or All Terminals on a Net. • All Instances: All the possible combinations of the instances that belong to the target net. • Selected Queries: All selected queries of the selected nets. See Reporting Resistance From Selected Queries. • Selected Queries All Nets: All selected queries of all nets (ignores net selection). See Reporting Resistance From Selected Queries.
Manage Query	Opens the Resistance Query Manager dialog box and lets you: • Select queries for resistance reporting. • Edit query information. • Delete queries. • Import and export queries. See Managing Resistance Report Queries.
Elmore Delay	Applies Elmore Delay analysis during the calculations. The option is only available when you select the StarRC tool for resistance reporting in the Electrical Options dialog box. See Setting EM, Resistance, and Capacitance Options.

Path Definition

The **Path Definition** section is only available when you are setting up **Selected Path** reporting. See [Reporting Resistance for Selected Path \(Pin-to-Pin, Terminal-to-Terminal, Instance-to-Instance, Point-to-Point\)](#).

Parasitic Reporter

ResistanceCapacitance

Net: itail itailn

Draw:   

☐ Cycle☐ Auto Highlight 

Depth: ☒ 0x☐ View☐ Full☐ Include Parent Designs

Area: ☒ Design☐ Region

Report using All Instances

Manage Query

☐ Elmore Delay

Net	Path Reported	Path Violated	Min Path R (Ohms)	Max Path R (Ohms)
-----	---------------	---------------	-------------------	-------------------

Source	Destination	Path R (Ohms)	Length (um)	R Constraint (Ohms)
--------	-------------	---------------	-------------	---------------------

DetailsLayersHistogram

Description	R (Ohms)	Wire Width (um) Via Column/Rows	Wire Length (um) Via Cut Class
-------------	----------	------------------------------------	-----------------------------------

☒ Zoom to Fit☒ Export to Console☐ Export to Error Viewer☒ Save History☐ Run in Background☐ Report RC

HelpDefaultsResetReportExport ...Options

Close

Option	Description
Select Using	Specifies whether you want to use pin, terminal or instance when you select the Source and Destination for calculating resistance. • pin: Lets you select one or more pins. • terminal: Lets you select one or more terminals. All the pins within a terminal are selected. • instance: Lets you select one or more instances and report many to many. Note: ◦ The instance list is displayed in Source and Destination, and the list depends on the depth and the check status of Auto Highlight. ◦ You can select either a single instance or multiple instances in Source and Destination. If you select a single instance the Short button is disabled. When you select pins or terminals, the net name associated with the pins and terminals appears in the Net box.
Add Query	Saves the query settings in the dialog box so that you can recall and reuse the settings later. Does not require that you generate the report to save the query.
Auto Add	Automatically adds the query settings when you click Report .
Source	Specifies the source pins, terminals, instance, or coordinates of the source points. You can select the pin or terminal names from the Source list, or you can click on the layout canvas to add the coordinates to the Source box. To specify a coordinate on a pin, select the pin name, and then click on the canvas to add the coordinate. Select a pin name if you want the source to be the intersection of the selected pin and wire. Only the pins that are connected to the selected net appear in the list. Specify a coordinate if you want the source to be the specified coordinate on the pin or somewhere along the wire. If you selected Area: Region, the sources must be within the reporting area.
Short	Select Short if you want to short the source for the resistance calculation. Shorting the source would treat the multiple selections as a single source. For example, select Short if you have sources (such as VSS) in external designs.

Option	Description
Destination	Specifies the second pin, terminal, instance, or coordinate to use for the route path. You can select a pin or terminal name from the Destination list, or you can click on the layout canvas to add the coordinate to the Destination box. To specify a coordinate on a pin, select the pin name, and then click on the canvas to add the coordinate. Select a pin name if you want the destination to be the intersection of the selected pin and wire. Specify a coordinate if you want the destination to be a location on the pin or somewhere along the wire. If you selected Area: Region, the destinations must be within the reporting area.
Short	Select Short if you want to short the destination for the resistance calculation. Shorting the destination would treat the multiple selections as a single destination. For example, select Short if you have destinations (such as VDD) in external designs.

Net Table

Column Name	Description
Net	Name of the net.
Path Reported	Number of pin-to-pin, terminal-to-terminal or instance-to-instance paths reported for the net.
Path Violated	Number of pin-to-pin, terminal-to-terminal, or instance-to-instance paths that have resistance values that violate the defined resistance constraint. The value is green if it meets the resistance constraint value. The value is red if it does not meet the constraint value.
Min Path Resistance	Minimum resistance calculated between the source and destination.
Max Path Resistance	Maximum resistance calculated between the source and destination.

Summary Report Table

Column Name	Description
Source	Name of the source pin, terminal, instance, or coordinates of the source point.

Column Name	Description
Destination	Name of the destination pin, terminal, instance, or coordinates of the destination point.
Path Resistance	Resistance calculated between the source and destination. The value is green if it meets the resistance constraint value. The value is red if it does not meet the constraint value. If the value is in violation (red), check the Res Constraint column.
Length	Length of the wire between the source and destination.
Res Constraint	Minimum and maximum resistance values defined by the resistance constraint.
Elmore Delay	Estimated time delay (in picoseconds) from the source. This column only appears if you selected Elmore Delay in Net and Display Options .

Query Summary Report

The Query Summary Report appears only when you generate a resistance report based on queries you saved.

Query Summary Report

Parasitic Reporter

ResistanceCapacitance

Net: ntailDraw: ☐ Cycle ☐ Auto Highlight

Depth: ☐ 0x ☐ View ☒ Full ☐ Include Parent DesignsArea: ☒ Design ☐ Region

Report using Selected QueriesManage Query ☒ Elmore Delay

Query Name	Net	Source	Destination	Path R (Ohms)	Length (um)	Constraint (Ohms)	Elmore Delay (ps)
query_1	ntail	Ndpp N3_1/S/S	Ntail N0_6/S/S	141.80615	2.373	N/A	0.33317
query_0	ntail	Ndpm N3/S/S	Ntail N0_1/S/S	219.46847	5.629	N/A	0.64258

DetailsLayersHistogram

Column Name	Description
Query Name	Name of the query you saved.
Net	Name of the net.
Source	Name of the source pin, terminal, instance, or coordinates of the source points.
Destination	Name of the destination pin, terminal, instance, or coordinates of the destination points.

Column Name	Description
Path Resistance	Resistance calculated between the source and destination. The value is green if it meets the resistance constraint value. The value is red if it does not meet the constraint value. If the value is in violation (red), check the Res Constraint column.
Length	Length of the wire between the source and destination.
Res Constraint	Minimum and maximum resistance values defined by the resistance constraint.

Details Tab

The **Details** tab lists wire segments that connect the source and destination points you selected in the summary report table.

Column Name	Description
Description	The layer on which the wire segment is located.
Resistance	The calculated resistance (ohms) of the segment.
Wire Width Via Columns/Rows	The width (µm) of the wire, or the number of columns and rows of cuts in the via.
Wire Length Via Cut Class	(Wires only) The length (µm) of the wire segment, or the via cut class.

Layers Tab

The **Layers** tab shows the resistance contributions by layer. The layer information is available for both built-in and StarRC resistance reporting.

Column	Description
Layer	Layer name.
R Contribution	The resistance contributed by the layer.
Percentage	The percent of resistance contribution by the layer compared to the resistance of the entire path: $((R_{\text{layer}} / R_{\text{total_path}}) * 100)$.

Histogram Tab

Column Name	Description
Type	Type of data you want to display. See the data descriptions in the Summary Report Table and the Detailed Report Tab sections.
Color Level	Number of colors to use when displaying the histogram and heat map on the canvas. The data between the Min and Max values are binned into the number of colors you specify.
Min	Minimum value of the data.
Max	Maximum value of the data.
 Show heatmap	Displays the heat map based on the histogram on the canvas.
 Hide heatmap	Hides the heat map.
 Refresh histogram	Updates the histogram based on updated settings in the Histogram tab.
☒ Automatically refresh histogram	Automatically updates the histogram as soon as you modify a setting in the Histogram tab.

Global Options

Option	Description
Zoom to Fit	(For viewing the report only) Displays the area of the design that contains the specified wire segment in the viewport. Default: On
Export to Console	Exports the resistance report to the Console. See Exporting Point-to-Point Resistance Reports .
Export to Error Viewer	Exports the violations to the Error Viewer. This option is only available if you choose Report using: All Constraints .
Save History	Saves the generated resistance report in the History section automatically, so that you can recall it in future design sessions.

Option	Description
Run In Background	Runs the resistance analysis in the background so that you can continue to work in the Layout Editor.
Report RC	Reports resistance and capacitance simultaneously. See: Reporting Resistance and Capacitance
Report	Generates the point-to-point resistance report based on the settings you specified. See Reporting Resistance for Selected Path (Pin-to-Pin, Terminal-to-Terminal, Instance-to-Instance, Point-to-Point) .
Export	Exports the point-to-point resistance report to a CSV file. See Exporting Point-to-Point Resistance Reports .
Options	Opens the Electrical Options dialog box. See About the Electrical Options Dialog Box .

Checking Resistance

The Resistance Checker compares the calculated wire resistance with the point-to-point resistance constraints to check for violations. You can use the Resistance Checker to run high-level violation checks. For detailed reporting, use the [Resistance Reporter](#).

Before you Start

Ensure that the following settings are enabled:

- The net is connected to all I/O pads and top-level (0x) pins.
- The instance type is pad.
- All I/O pads are ground pads.
- All top-level (0x) pins are ground pins.

If you want to run the checker in the background so that you can continue to work in the Layout Editor:

- ▶ Set the `leReportEMRCBackgroundMode` Tcl preference to true. Alternatively, choose **Electrical > Parasitic Reporter**, and select **Run in Background**.

Background mode supports both StarRC shared and non-shared modes.

Checking Resistance From the Design Navigator

This section describes how to check resistance using the Design Navigator. To use the `le:checkEMResistanceConstraints` Tcl command, see the *Custom Compiler Tcl Reference Manual*.

To check resistance:

1. Open the design.
2. Set the resistance analysis temperature, specify the TLUPlus file, and set other electrical options using one of the following methods:
 - Open the **Electrical Options** dialog box and set the options. See [Opening the Electrical Options Dialog Box](#).
 - Open the Console, and type preferences that set the options. For information about the Tcl preferences, see the *Custom Compiler Tcl Reference Manual*.
3. In the Design Navigator, select the net:
 - a. In the Layout Editor, open the Design Navigator.
 - b. Click the **Nets** tab and select the nets. You can do this from either the **Design: Layout** mode or the **Design: Schematic** mode.

You can also select multiple nets and choose **Check Resistance** to extract parasitics in a single run. If you select more than 10 nets, the **Reporter - Interactive StarRC** dialog box appears. Click **OK** to continue or **Cancel**. If you select **Cancel**, you should select less than 10 nets and continue extraction in the interactive mode.

4. Do one of the following:

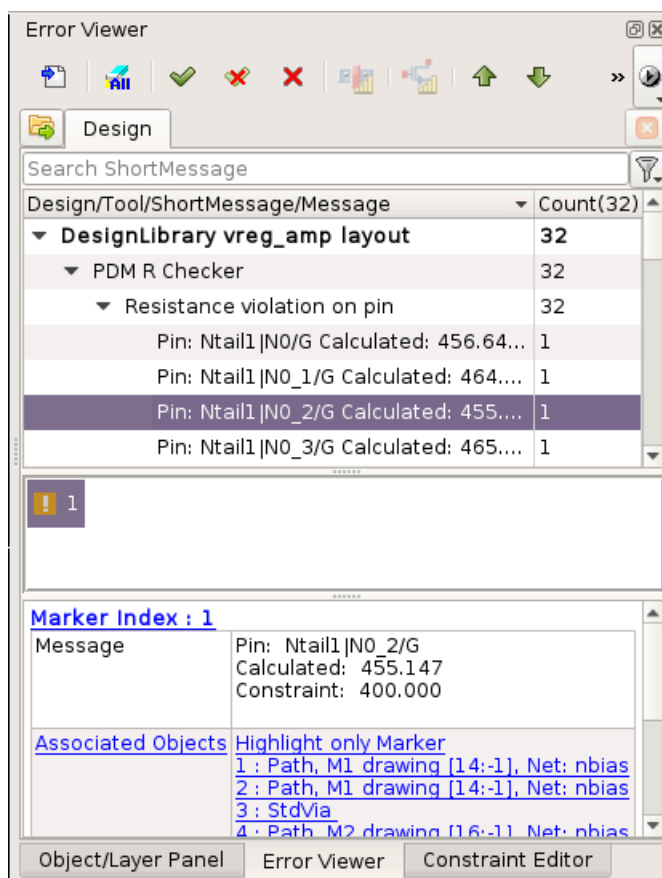
- In the Design Navigator, right-click the net, and then choose **Check Resistance**.
- On the canvas, right-click a path segment, and then choose **Check Resistance**.
- Enter the following Tcl command in the Console:

```
le::checkEMResistanceConstraints
```

The tool calculates equivalent current for each edge based on the current distribution, checks edge current for each edge against the current constraint for the net, and reports any violations to the Error Viewer. If the tool finds violations, the Error Viewer displays automatically and lists the violations.

If there are no violations, you do not need to proceed to the next step.

5. In the Error Viewer, expand the list to show the constraint violations. For a given constraint that failed, the assistant displays details about the error in the lower portion of the assistant.



Where a measurement is in violation, you have two options:

- Change the measurement using the EM/R Fixer.
- Change the measurement using the Property Editor

You can also view the pass or fail constraint status in the Constraint Editor and the Parasitic Reporter dialog box. See [Viewing Point-to-Point Resistance Reports](#).

Checking Common Path Resistance

After setting up Max Common Resistance constraints, you can use the Common Path Checker to report common-path resistance constraint violations. This topic describes how to use the checker from the Design Navigator graphical user interface.

Before you Start

- The net is connected to all I/O pads and top-level (0x) pins.
- The instance type is pad.
- All I/O pads are ground pads.
- All top-level (0x) pins are ground pins.

If you want to run the checker in the background so that you can continue to work in the Layout Editor:

- ▶ Set the `leReportEMRCBackgroundMode` Tcl preference to true. Alternatively, choose **Electrical > Parasitic Reporter**, and select **Run in Background**.

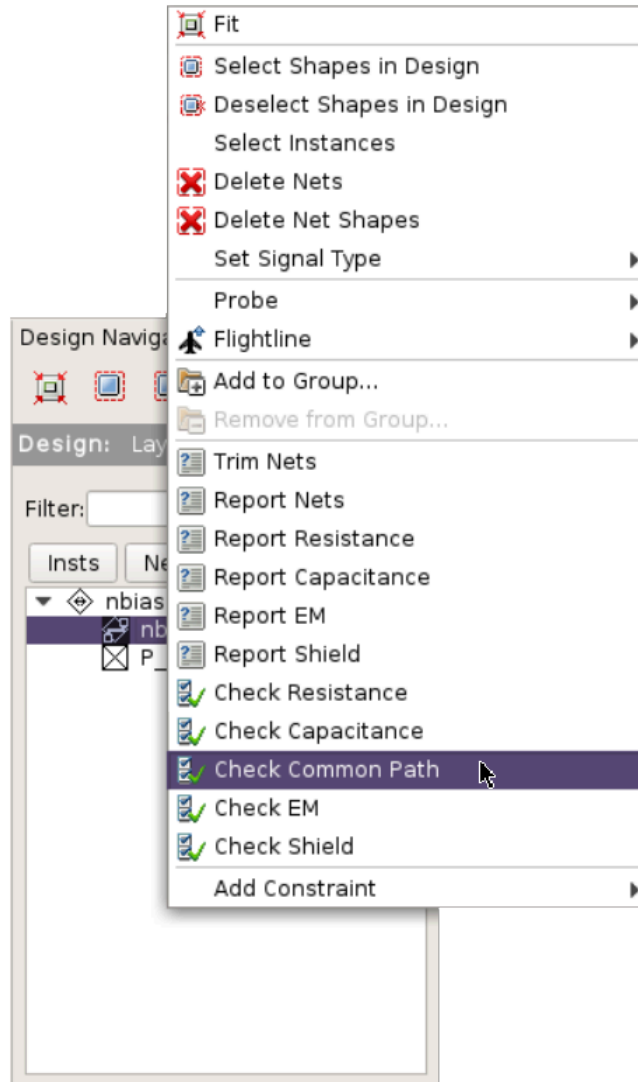
Background mode supports both StarRC shared and non-shared modes.

Instead of using the graphical user interface, you can use the `le::checkEMResConstraints` Tcl command in Tcl scripts and for batch processing.

Checking Common Path Resistance From the Design Navigator

To check common-path resistance constraint violations:

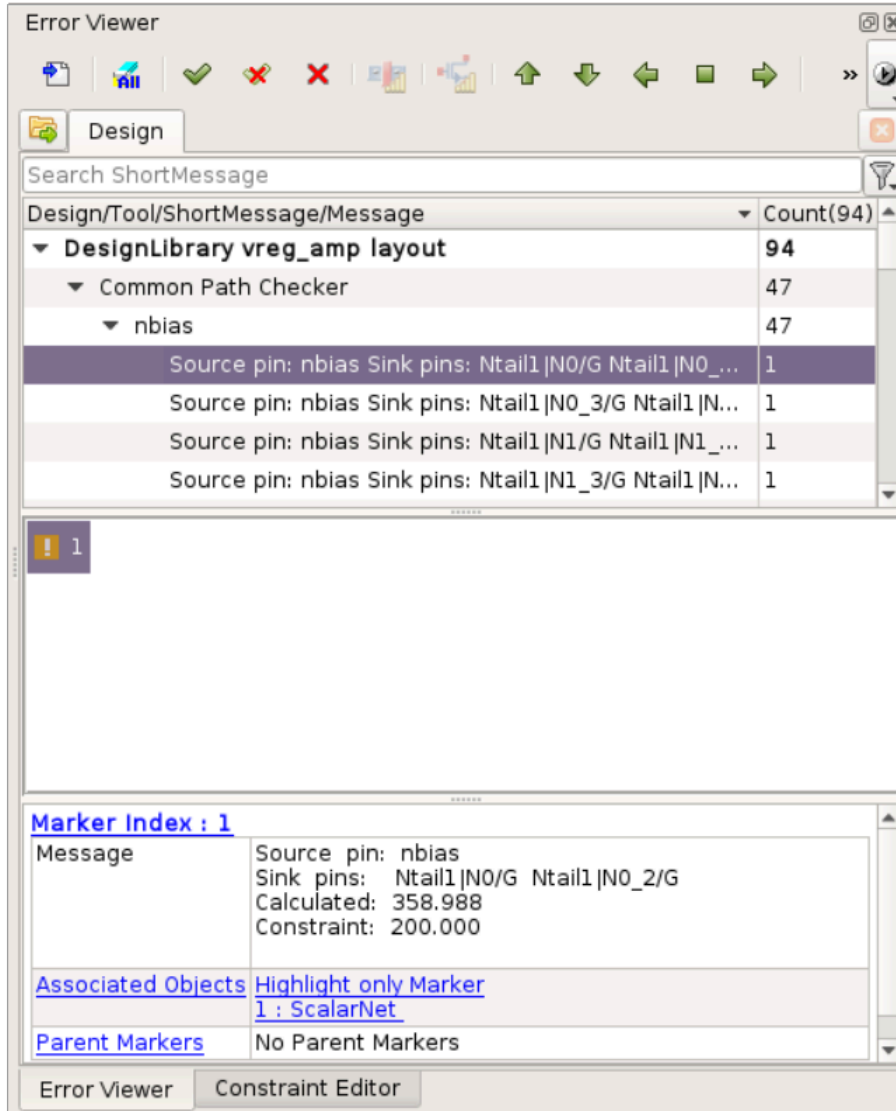
1. Open the design.
2. Set the resistance analysis temperature, specify the TLUPlus file, and set other electrical options using one of the following methods:
 - Open the **Electrical Options** dialog box and set the options. See [Opening the Electrical Options Dialog Box](#).
 - Open the Console, and type preferences that set the options. For information about the Tcl preferences, see the *Custom Compiler Tcl Reference Manual*.
3. In the Design Navigator, do one of the following:
 - In either the **Design: Schematic** or **Design: Layout** mode:
 - a. Click the **Nets** tab.
 - b. On the **Nets** tab, right-click the net, and then choose **Check Common Path**.
 - Make sure you are in the **Design: Layout** mode, and:
 - a. Click the **Terms** tab.
 - b. Right-click the term, and then choose **Check Common Path**.



The tool calculates the common-path resistance from the driver pin to all corresponding load pins, and compares the calculated resistance values against the constraint value.

- If the tool finds violations, the Error Viewer displays automatically and lists the violations.
- If there are no violations, you do not need to proceed to the next step (showing violations).

4. In the Error Viewer, expand the list to show the constraint violations. For a given constraint that failed, the assistant displays details about the error in the lower portion of the assistant.



See Also

- [Specifying Max Common Resistance](#)

6

Reporting and Checking Capacitance

This chapter explains how to use the interactive capacitance reporter to analyze coupling capacitance in a design.

This chapter describes the following:

- [About the In-Design Capacitance Reporter](#)
- [Workflow for Using the In-Design Capacitance Reporter](#)
- [Preparing the Design and Technology](#)
- [Setting Capacitance Constraints](#)
- [Specifying Input Information](#)
- [Using the Capacitance Reporter](#)
- [Checking Capacitance](#)

About the In-Design Capacitance Reporter

This topic describes the following:

- [In-Design Capacitance Reporting](#)
- [StarRC Tool Integration](#)
- [About the Halo Width Option](#)
- [Runtime Reduction](#)

In-Design Capacitance Reporting

Unlike traditional design flows, where capacitance reporting must be performed on completed designs that are LVS-clean, the Custom Compiler Capacitance Reporter allows you to probe and analyze capacitance dynamically as you create the design. You can modify components in your design and then re-run the interactive capacitance reporter

to view the changes in capacitance. Doing so catches errors during layout, thus reduces costly design iterations and accelerates the overall design process.

Note:

If you select **Device Layers** in the **Electrical Options** dialog box, routes on metal, device, base, and contactless layers are considered for capacitance extraction. If you disable **Device Layers**, only metal layers are considered for extraction. See [Electrical Options Dialog Box Reference](#).

StarRC Tool Integration

The Capacitance Reporter uses the Synopsys StarRC tool to extract capacitance information from the layout. The integration facilitates the reporting and troubleshooting processes.

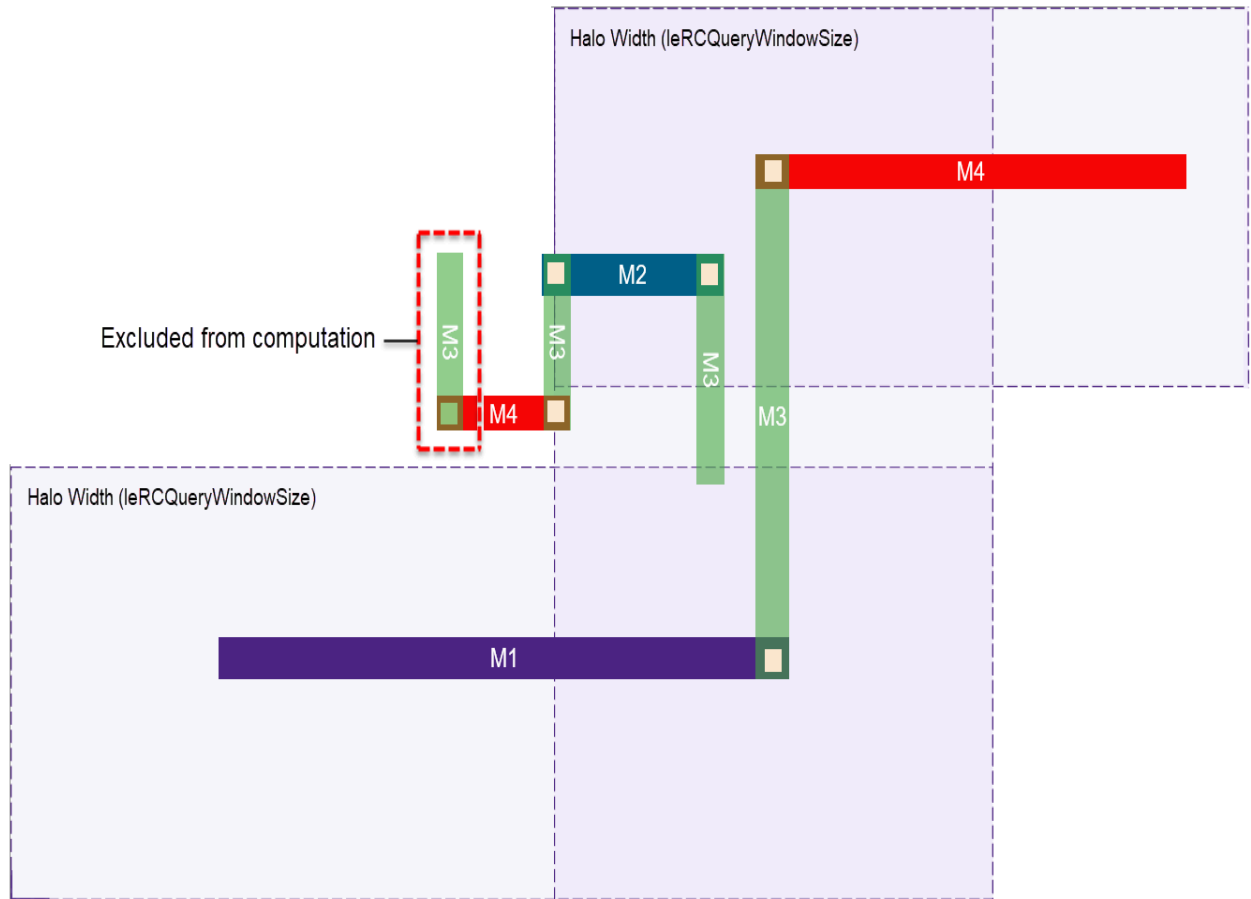
After the extraction process is finished, instead of opening the output file, you can view the results while in the Capacitance Reporter. In addition, when you select the capacitance values of interest, the reporter automatically zooms in and locates the corresponding nets on the canvas for dynamic viewing. See [Viewing Capacitance Reports](#).

See also [Managing StarRC Version and License Use](#).

About the Halo Width Option

When setting up for capacitance reporting, you have the option of specifying the **Halo Width** option in the **Electrical Options** dialog box. See [Using the Capacitance Reporter](#) and [Electrical Options Dialog Box Reference](#).

The **Halo Width** option lets you specify the non-default width of the enclosure around the net shapes of interest. The Capacitance Reporter includes shapes within and touching the enclosure of the target net for extraction.



Runtime Reduction

You can use the `leNetQueryWindowSize` Tcl preference to specify a second halo by which net assignments are determined. Use the preference to consider some shapes as ground net shapes if you anticipate heavy computations, especially when reporting total capacitance.

The `leNetQueryWindowSize` Tcl preference works with the **Halo Width** setting in the **Electrical Options** dialog box. `leNetQueryWindowSize` defines a second halo region

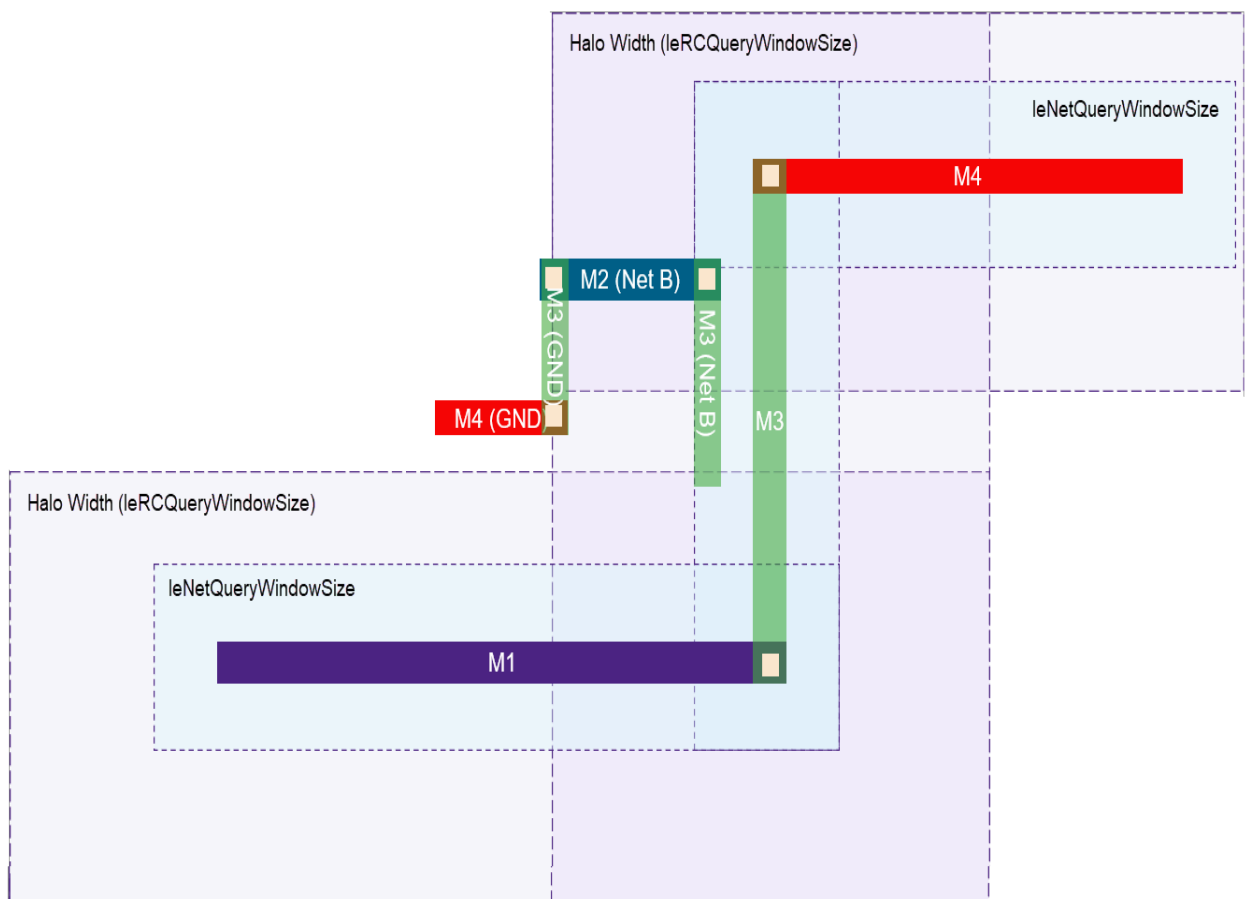
around the net of interest. Net assignments are then determined based on a shape's location:

- Shapes that are within and touching the `leNetQueryWindowSize` (second halo) enclosure are assigned actual net IDs.

Note:

If `leNetQueryWindowSize` is 0, neighboring shapes in the vertical direction can have coupling capacitance with the target net.

- Shapes that are outside the `leNetQueryWindowSize` region and within the **Halo Width** region are assigned to the ground (GND) net.



For example:

```
le::setPrefValue leNetQueryWindowSize -value 10 -scope  
[de::getActiveContext]
```

The default value of `leNetQueryWindowSize` is -1, which disables the effect of the Tcl preference.

`leNetQueryWindowSize` is ignored if its value is $\geq$ **Halo Width**.

Workflow for Using the In-Design Capacitance Reporter

The following table shows the high-level workflow for checking and reporting total electrical capacitance in your design. Analyzing total capacitance is an iterative process. Typically, steps 5 through 7 are repeated until the wire capacitance is optimized.

Analyzing total capacitance is an iterative process. You can use a high-level workflow such as the following:

- Run the interactive capacitance reporter to view capacitance values.
- Visualize needed changes in the design.
- Cross probe between the capacitors in the report and the layout. This will show a flight line with a C symbol between the two wire segment objects on the two nets, helping you visualize exactly where the capacitors occur in physical space.
- Modify the layout and re-run the interactive capacitance reporter.

Step		Description
1	Prepare the design and constraint data.	To report total capacitance successfully, you must make sure the design is set up properly. In addition, you must ensure the correct StarRC options are set. See Preparing the Design and Technology .
2	Prepare the layer mapping file.	To use external rule sources, you must reference a layer mapping file to map OA layer names to StarRC layer names in nxtgrd files. See Preparing the Layer Mapping File .
3	(Optional) Include additional StarRC options.	To include additional StarRC options that are not provided by the Capacitance Reporter user interface, see Including Additional StarRC Commands .
4	(Optional) Setting capacitance constraints.	View, modify, or set total capacitance constraints using the Constraint Editor. See Setting Capacitance Constraints .
5	(Optional) Check capacitance violations.	Run a check to see if the selected nets meet the capacitance constraints. See Checking Capacitance .
6	Report the capacitance.	Run the Capacitance Reporter. See Using the Capacitance Reporter .
7	Adjust the design.	Adjust design components to fix violations.

Preparing the Design and Technology

This topic describes the following:

- [Preparing the Design for Reporting Capacitance](#)
- [Setting Up the Technology for Capacitance Reporting](#)

Preparing the Design for Reporting Capacitance

To report total capacitance successfully, make sure:

- The net shapes in the design must have logical connectivity assignments; that is, they have oaNets assigned to them.
- The design has at least two nets (otherwise net-to-net coupling would contain no information).
- Un-netted shapes will be considered grounded or connected to the reference node.
- The technology data is set up in `nxtgrd` and mapping files. See [Setting Up the Technology for Capacitance Reporting](#).
- The StarRC tool is installed with a valid license and set up in the environment. Your Linux path should include StarXtract.

Setting Up the Technology for Capacitance Reporting

Before you can start StarRC in the Custom Compiler environment, you must define the following options in the **Electrical Options** dialog box:

- In the **Capacitance** section, select **Nxtgrd** or **CommonNxtgrd** to indicate the file you intend to use. The `nxtgrd` file contains capacitance data.
- In the **Nxtgrd** tab, set the path to a `nxtgrd` file for the **Minimum** field, the **Maximum** field, or both. (The field that is used is the one that is selected.)
- In the **Nxtgrd** tab, set the path to a **Layer Map** file.

See [About the Electrical Options Dialog Box](#).

Ideally, you should preset this information by setting the following preferences as part of your PDK/environment setup:

- `leCapacitanceModel`
- `leNxtgrdOperatingPoint`
- `leTCadGRDMinFile`

Chapter 6: Reporting and Checking Capacitance

Preparing the Design and Technology

- `leTCadGRDMaxFile`
- `leNXTGRDMetalMapFile`
- `leNXTGRDDeviceMapFile`
- `leNXTGRDSpecifyGeometryMapFile`

Note:

The Parasitic Reporter will create all necessary internal input files for StarRC.

Setting Capacitance Constraints

This topic describes the following:

- [When to Set Capacitance Constraints](#)
- [Creating Capacitance Constraints](#)
- [Creating Wire Matching Capacitance Constraints](#)
- [Viewing and Editing Capacitance Constraints](#)

When to Set Capacitance Constraints

In the typical design workflow, the schematic designer sets the capacitance constraint. The design retains the constraint information when you generate or open the layout using SDL. (See [Generating Layout](#).)

Before checking and reporting total capacitance in the Layout Editor, you have the option of viewing the existing capacitance constraint values that were set in the schematic. If necessary, you can modify the values while adjusting the layout. If the capacitance constraint was not set in the Schematic Editor, you can define it in the Layout Editor.

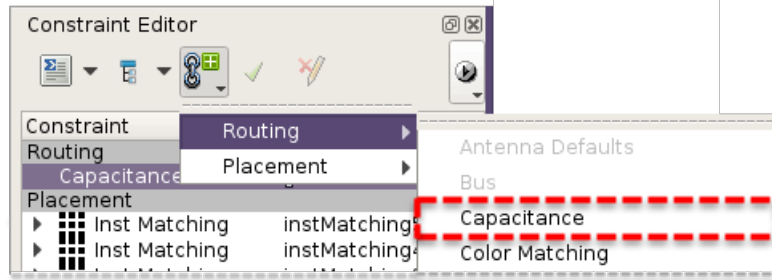
The capacitance constraint values are used when you check or report capacitance. See [Checking Capacitance](#) and [Using the Capacitance Reporter](#).

Creating Capacitance Constraints

This section describes how to create Capacitance constraints using the Constraint Editor in the Schematic Editor and the Layout Editor. For information on using the `cm::createCapacitance` Tcl command, see the *Custom Compiler Tcl Reference Manual*.

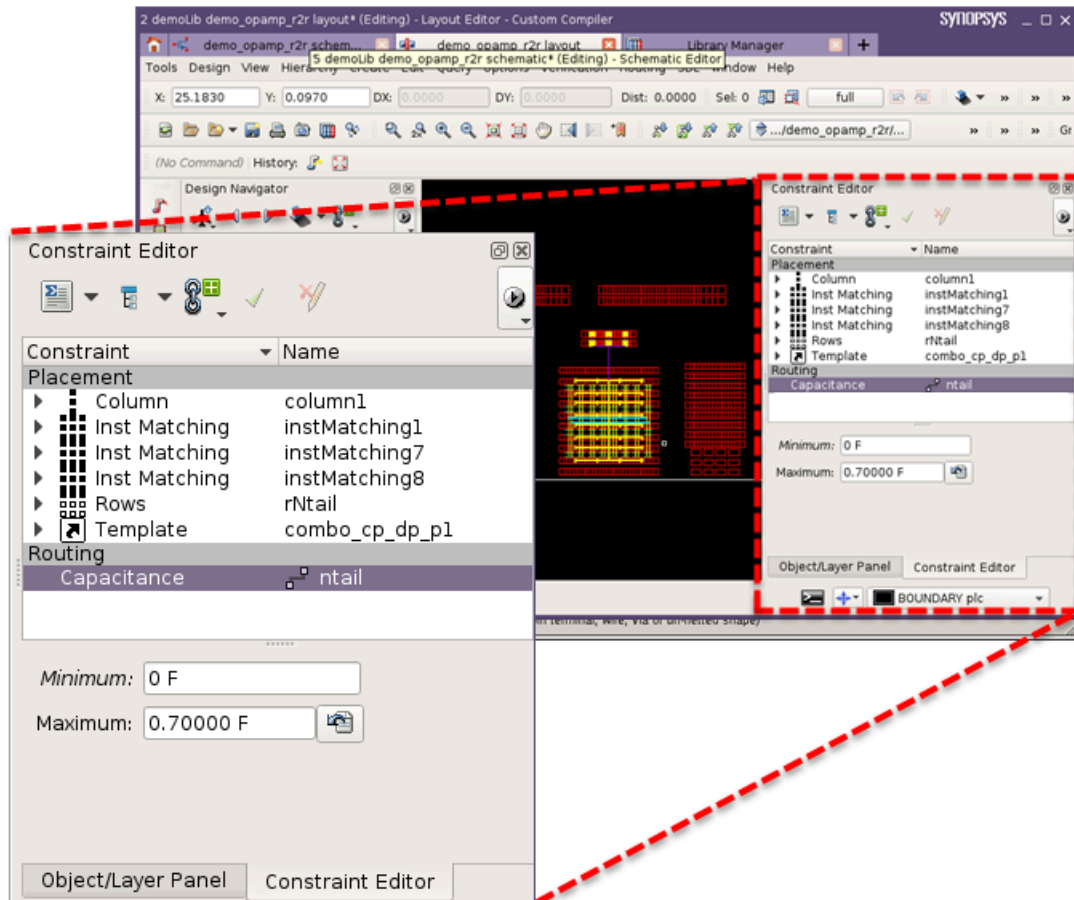
To create a Capacitance constraint:

1. In the Schematic Editor or the Layout Editor, display the Design Navigator and select the nets of interest in the **Nets** tab.
2. In the Schematic Editor or the Layout Editor, display the Constraint Editor.
3. In the Constraint Editor, click **Create Constraint** , and choose **Routing > Capacitance**.



A new **Capacitance** constraint appears below the **Routing** section. The nets associated with the constraint are shown in the **Name** column.

- Set the following parameters:



Parameter	Description
Minimum	The minimum capacitance, in farads (F), allowed for the selected nets. Specify a floating point value ≥ 0 . You can specify the order of magnitude for the minimum capacitance. For example, to specify 7 fF, type 7f. Default: 0
Maximum	The maximum capacitance, in farads (F), allowed for the selected nets. Specify a floating point value > 0 . You can specify the order of magnitude for the maximum capacitance. For example, to specify 9 nF, type 9n.

To define a threshold for the total maximum capacitance, specify only the **Maximum** value. To define the acceptable range for the total capacitance, specify the **Minimum** and **Maximum** values.

- Notice that after you change a parameter value, a  button appears next to the changed value. To reset a parameter back to its default value, click .
- When you are finished:
 - Click **Apply Changes** () to save the modifications.
 - Click **Discard Changes** () to revert back to the previously saved settings.

Creating Wire Matching Capacitance Constraints

To create a Wire Matching Capacitance constraint:

- In the Layout Editor, display the Design Navigator and select at least two nets of interest in the **Nets** tab.
- In the Layout Editor, display the Constraint Editor.
- In the Constraint Editor, click **Create Constraint** , and choose **Routing > Wire Matching**.
- A new **Wire Matching** constraint appears below the **Routing** section. The nets associated with the wire matching constraint are shown in the **Name** column.
- Set the **Style** as **Capacitance**.
- The **Absolute** constraint option is set as the default. Select the **Relative** to change the constraint option.

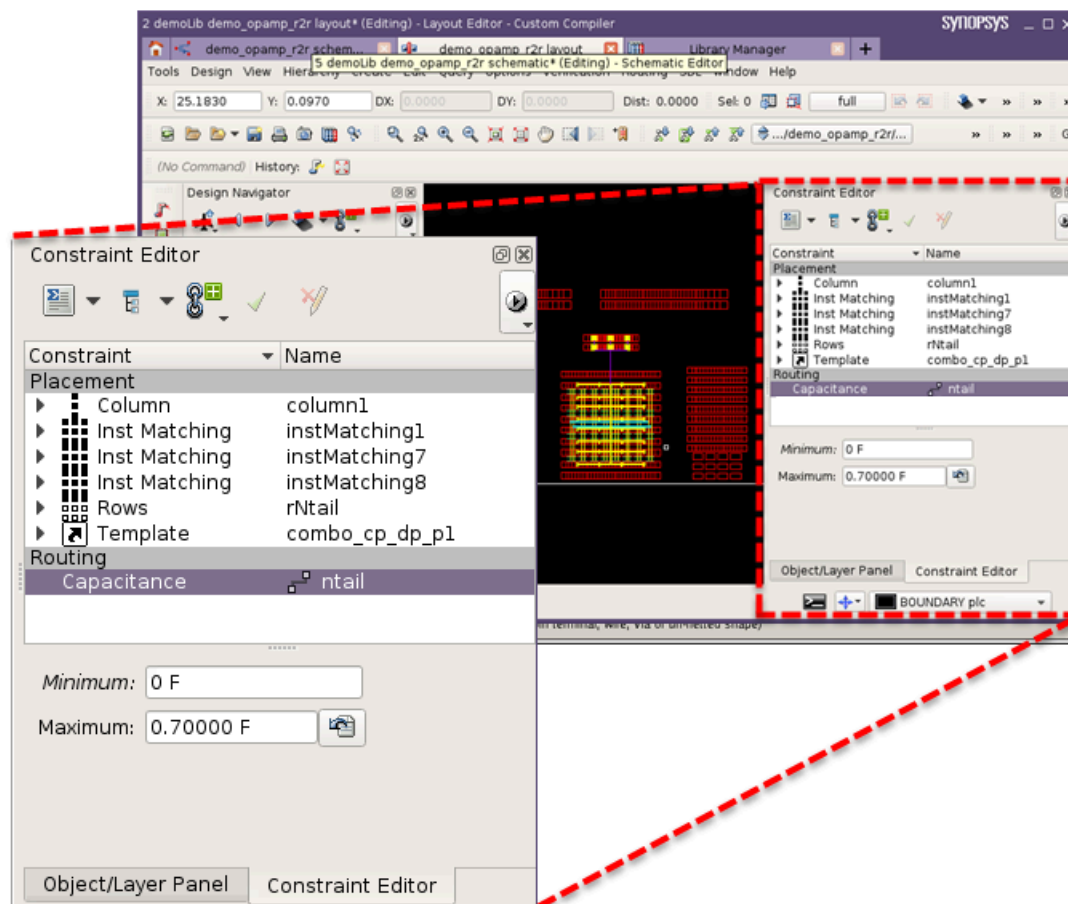
7. ◦ Click **Apply Changes** (✓) to save the modifications.
- Click **Discard Changes** (✗) to revert back to the previously saved settings.

Viewing and Editing Capacitance Constraints

This section describes how to view and edit existing Capacitance constraints using the Constraint Editor in the Schematic Editor and the Layout Editor. For information on using the `cm::createCapacitance` Tcl command, see the *Custom Compiler Tcl Reference Manual*.

To view or edit a Capacitance constraint:

1. In the Constraint Editor, locate the **Routing** section, and then select the **Capacitance** constraint for the nets of interest. The parameters appear in the Constraint Editor.



2. Modify the **Minimum** and **Maximum** parameters. For the description of the parameters, see [Creating Capacitance Constraints](#).

3. Notice that after you change a parameter value, a  button appears next to the changed value. To reset a parameter back to its default, click .
4. When you are finished modifying a constraint:
 - Click **Apply Changes** () to save the modifications.
 - Click **Discard Changes** () to revert back to the previously saved settings.

Specifying Input Information

Capacitance reports are based on input information that you supply, such as the rule file and extraction settings. You can supply the information using one of the following methods:

Method	Description
Preload the input information during environment setup	You can set preferences by running a Tcl script during environment setup. In this case, you do not need to specify the settings in the Parasitic Reporter or the Electrical Options dialog box. You only need to display the dialog box and click Report . See Generating Capacitance Reports . To find out whether the input information is preloaded with a Tcl script, check with your administrator or CAD manager. To see the preferences you can set, see About the Electrical Options Dialog Box and Parasitic Reporter - Capacitance Tab Reference .
Preload the input information by setting the Tcl preferences in the Console	Instead of running a Tcl script, you can type Tcl commands in the Console to set the preferences. To see the preferences you can set, see About the Electrical Options Dialog Box and Parasitic Reporter - Capacitance Tab Reference .
Supply the input in the Parasitic Reporter dialog box	You can enter the input information in the dialog box every time you generate the report. To enter the input information, see Generating EM Reports .

Using the Capacitance Reporter

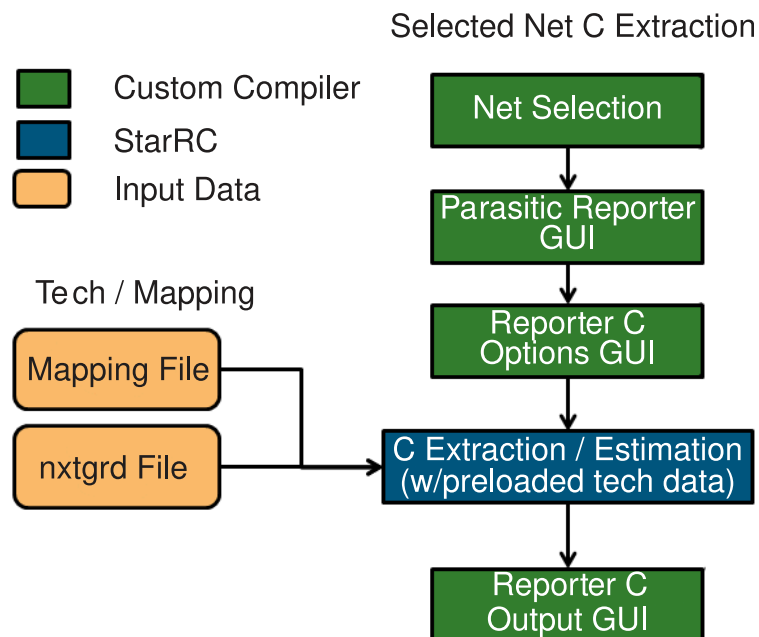
The Capacitance Reporter allows you to probe and analyze capacitance dynamically as you create the design. Reporting and fixing capacitance issues during design development can save time and costly iterations downstream.

This topic describes the following:

- [Capacitance Extraction Workflow](#)
- [Generating Capacitance Reports](#)
- [Viewing Capacitance Reports](#)
- [Exporting Capacitance Reports](#)
- [Parasitic Reporter - Capacitance Tab Reference](#)

Capacitance Extraction Workflow

The following diagram shows the high-level in-design capacitance extraction workflow:



Generating Capacitance Reports

You use the Design Navigator to generate total capacitance reports. Initially, the report displays a listing of the total coupling capacitance. You can then optionally select one of the entries in the initial report, and generate a detailed breakdown of all of the individual capacitance values.

This section describes how to generate total capacitance reports using the Design Navigator. To use the `le::reportCapacitance` Tcl command, see the *Custom Compiler Tcl Reference Manual*.

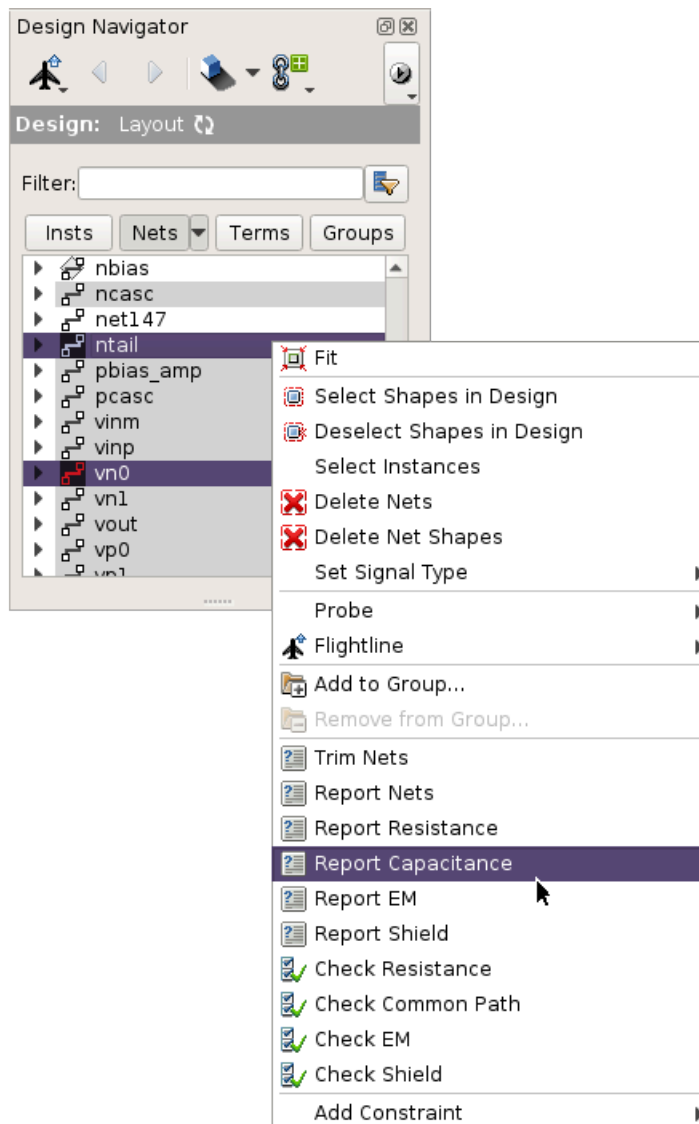
To check whether selected nets meet the capacitance constraint, see [Checking Capacitance](#).

To report total capacitance:

1. (Optional) Select one or more nets:
 - a. In the Layout Editor, open the Design Navigator.
 - b. Click the **Nets** tab and select the nets. You can do this from either the **Design: Layout** mode or the **Design: Schematic** mode.

You can also select multiple nets and choose **Report Capacitance** to extract parasitics in a single run. If you select more than 10 nets, the **Reporter - Interactive StarRC** dialog box appears. Click **OK** to continue or **Cancel**. If you select **Cancel**, you should select less than 10 nets and continue extraction in the interactive mode.

2. Right-click the selected nets, and then choose **Report Capacitance**.



The **Parasitic Reporter** dialog box opens and displays the **Capacitance** tab. Notice that the names of selected nets appear in the **Net** box.

Parasitic Reporter

Resistance **Capacitance**

Net: Draw:   ☐ Cycle

Net	Total C (fF)	Total Coupling C (fF)	Total Ground C (fF)	Constraint (fF)
-----	--------------	-----------------------	---------------------	-----------------

Group Results By Filter Display By %

From	To	Coupling C (fF)	Coupling/Total (%)
------	----	-----------------	--------------------

Details **Histogram**

From	To	Coupling C (fF)	Coupling/Total (%)
------	----	-----------------	--------------------

 History

☒ Zoom to Fit ☒ Export to Console ☐ Export to Error Viewer ☒ Save History ☐ Run in Background ☐ Report RC

3. If the information in the dialog box was preloaded as part of the environment setup:
 - a. Review the settings in the **Capacitance** tab.
 - b. Click **Options** (at the bottom of the dialog box), and review the settings in the **Electrical Options** dialog box. See [Setting EM, Resistance, and Capacitance Options](#) and [Specifying Input Information](#).
 - c. Skip to [Step 6](#).
4. If you did not preload settings in [Step 3](#), set the options in the following:
 - **Capacitance** tab in the **Parasitic Reporter** dialog box. See [Parasitic Reporter - Capacitance Tab Reference](#).
 - **Electrical Options** dialog box. See [Setting EM, Resistance, and Capacitance Options](#).

Caution:

The values you enter in the **Parasitic Reporter** and **Electrical Options** dialog boxes override preloaded values.

5. To automatically save the generated capacitance report so that you can recall it in future design sessions, select **Save History**. (Alternatively, set the `leEMRCAutoSaveHistory` Tcl preference to `true` before you generate the report.) The **History** section appears.

All newly generated reports are automatically saved in the following directory and file:

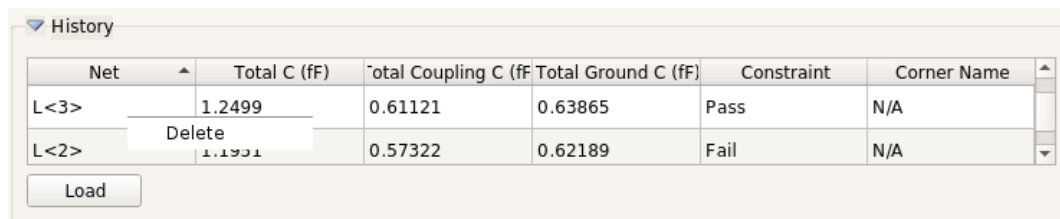
```
$env (SYNOPSYS_CUSTOM_LOCAL) /c_reporter/<LCV>_C_result.db
```

If a reported net already exists in the `.db` file, it is overwritten by the latest data.

To recall the saved report, see [Viewing Point-to-Point Resistance Reports](#).

By default, the maximum number of reports retained is 30. To change the number, set the `leEMRCMaxHistoryNum` Tcl preference.

To remove the history of a report, right-click on the entry in the **History** section and select **Delete**.



The screenshot shows a window titled "History" containing a table with the following data:

Net	Total C (fF)	Total Coupling C (fF)	Total Ground C (fF)	Constraint	Corner Name
L<3>	1.2499	0.61121	0.63865	Pass	N/A
L<2>	1.1991	0.57322	0.62189	Fail	N/A

Below the table is a "Load" button. A context menu with a "Delete" option is visible over the first row.

6. If you want to view the violations in the Error Viewer, select **Export To Error Viewer**.

7. If you want to run the capacitance reporter in the background so that you can continue to work in the Layout Editor, select **Run in Background**.

Background mode supports both StarRC shared and non-shared modes. When the report is generated, you can locate the results in the **History** section of the dialog box.

8. When you are ready to generate and view the report, click **Report**.

The Parasitic Reporter displays the results in the tables on the **Capacitance** tab. Note the net total capacitance values are displayed, and the values are green to indicate that they comply with the Capacitance constraints. See [Setting Capacitance Constraints](#).

Parasitic Reporter

Resistance Capacitance

Net: L<1> L<2> L<3> Draw: ☐ Cycle

Net	Total C (fF)	Total Coupling C (fF)	Total Ground C (fF)	Constraint
L<1>	1.0213	0.28321	0.73808	Fail
L<2>	1.1951	0.57322	0.62189	Fail
L<3>	1.2499	0.61121	0.63865	Pass

Group Results by Net Filter Display by Percentage >= 0 %

From	To	Coupling C (fF)	Coupling/Total (%)
L<2>	Ground	0.62189	52.0362
L<2>	L<3>	0.29001	24.2665
L<2>	L<1>	0.28321	23.6973

Details Histogram

From	To	Coupling C (fF)	Coupling/Total (%)
L<2>.M1	L<3>.M1	0.2469	85.1359
L<2>.M1	L<3>.M1	0.041629	14.3542
L<2>.M1	L<3>.M1	0.0010215	0.352212
L<2>.M1	L<3>.M1	0.00045706	0.157601

History

Net	Total C (fF)	Total Coupling C (fF)	Total Ground C (fF)	Constraint	Corner Name
L<3>	1.2499	0.61121	0.63865	Pass	N/A
L<2>	1.1951	0.57322	0.62189	Fail	N/A

Load

☒ Zoom to Fit ☒ Export to Console ☐ Export to Error Viewer ☒ Save History ☐ Run in Background ☐ Report RC

Help Defaults Reset Report Export ... Options Close

You can use the `le::reportCapacitance -verbose` command to include additional details in the wire capacitance report, such as the list of paths and vias.

During the reporting operation, you can:

- Interrupt the operation by pressing Ctrl+C.
- Stop the operation using the Job Monitor.

After the report operation is finished, you can check the Job Monitor to confirm that the operation completed successfully and view the extraction log file (`star.log`).

Reporting Resistance and Capacitance

The **Report RC** option reports resistance and capacitance simultaneously.

To generate reports of resistance and capacitance:

1. Set the `leRCReporterByOneExtraction` preference to `true` to enable the **Report RC** option.

Note:

Since **Report RC** runs with StarRC only, make sure that you choose **Options > Resistance > StarRC**.

2. Select **Report**. The results are displayed in the **Details** section of the **Capacitance** tab.

Note:

If you do not select the following options:

- **Depth > Full**
- **Area > Design**
- **Include Parent Design > OFF**

the following warning message appears:

When "Report RC" is ON, the Depth, Area, and Include Parent Designs options in Resistance Reporter impacts the Capacitance result. Set Depth: Full, Area: Design, and Include Parent Designs: OFF to provide the result equivalent to Capacitance Reporter alone.

3. To view the history of RC reports, select the **Save History** option and run the RC report. The results are displayed in the **History** section.

See Also

- [Viewing Capacitance Reports](#)
- [Exporting Capacitance Reports](#)

Viewing Capacitance Reports

The capacitance report in the **Capacitance** tab consists of the following:

- **Nets** table: Lists the nets you selected, the total capacitance, total coupling capacitance, total ground capacitance, and the capacitance constraint value (if it was set).
- **Summary report** table: Contains the names of two nets (From, To), the calculated coupling capacitance value on the net segment, and the ratio of the coupling capacitance against the total capacitance.
- **Details** tab: Displays specific information for individual capacitance values.
- **Histogram** tab: Shows the distribution of the various data in the capacitance report.

Parasitic Reporter

Resistance Capacitance

Net: L<1> L<2> L<3> Draw: ☐ Cycle

Net	Total C (fF)	Total Coupling C (fF)	Total Ground C (fF)	Constraint
L<1>	1.0213	0.28321	0.73808	Fail
L<2>	1.1951	0.57322	0.62189	Fail
L<3>	1.2499	0.61121	0.63865	Pass

Group Results by Net Filter Display by Percentage >= 0 %

From	To	Coupling C (fF)	Coupling/Total (%)
L<2>	Ground	0.62189	52.0362
L<2>	L<3>	0.29001	24.2665
L<2>	L<1>	0.28321	23.6973

Details Histogram

From	To	Coupling C (fF)	Coupling/Total (%)
L<2>.M1	L<3>.M1	0.2469	85.1359
L<2>.M1	L<3>.M1	0.041629	14.3542
L<2>.M1	L<3>.M1	0.0010215	0.352212
L<2>.M1	L<3>.M1	0.00045706	0.157601

History

Net	Total C (fF)	Total Coupling C (fF)	Total Ground C (fF)	Constraint	Corner Name
L<3>	1.2499	0.61121	0.63865	Pass	N/A
L<2>	1.1951	0.57322	0.62189	Fail	N/A

Load

☒ Zoom to Fit ☒ Export to Console ☐ Export to Error Viewer ☒ Save History ☐ Run in Background ☐ Report RC

Help Defaults Reset Report Export ... Options Close

Before you start, if you want to view a saved report, in the **Parasitic Reporter** dialog box:

1. Select **Save History**.
2. In the **History** section that appears, select the reports (rows) you want to view.
3. Click **Load**. The report displays in the tab.

Each report you load is appended to the existing data displayed in the tab.

History

Net	Total C (fF)	Total Coupling C (fF)	Total Ground C (fF)	Constraint	Corner Name
L<3>	1.2499	0.61121	0.63865	Pass	N/A
L<2>	1.1951	0.57322	0.62189	Fail	N/A

Load

- To preview details about a saved report, rest the pointer on the row. A tooltip displays the additional information.

To view the capacitance report, on the **Capacitance** tab:

- Set up the report:
 - Choose **Group Results By: Net, Layer, or All**.
 - See [Generating Capacitance Reports](#) and [Parasitic Reporter - Capacitance Tab Reference](#) for other setup information.

Parasitic Reporter

Resistance Capacitance

Net: ntail Draw: ☐ Cycle

Net	Total C(fF)	Total Coupling C(fF)	Total Ground C(fF)	Constraint(fF)
ntail	1.3693	0.86368	0.50564	≤ 1.5000 (fF)

Group Results By: Net Layer All Filter Display By: Percentage >= 1 %

From	To	Coupling C(fF)	Coupling/Total(%)
ntail	Ground	0.50564	36.9293
ntail	vinp	0.22591	16.4981
ntail	vinm	0.20869	15.2404
ntail	gnd	0.021852	1.5958

- Review the data in the net table.

Column Name	Description
Net	Name of the net that is queried.
Total C (fF)	Total capacitance, in femtofarads. Total C = Total Coupling C + Total Ground C The value is green if it meets the capacitance constraint value. The value is red if it does not meet the constraint value.
Total Coupling C (fF)	Total coupling capacitance, in femtofarads.

Column Name

Description

Total Ground C (fF)

Total ground capacitance, in femtofarads.

Constraint

Capacitance constraint and the capacitance wire matching constraints.

If you run the constraint, the status is displayed as either **Pass** or **Fail**.

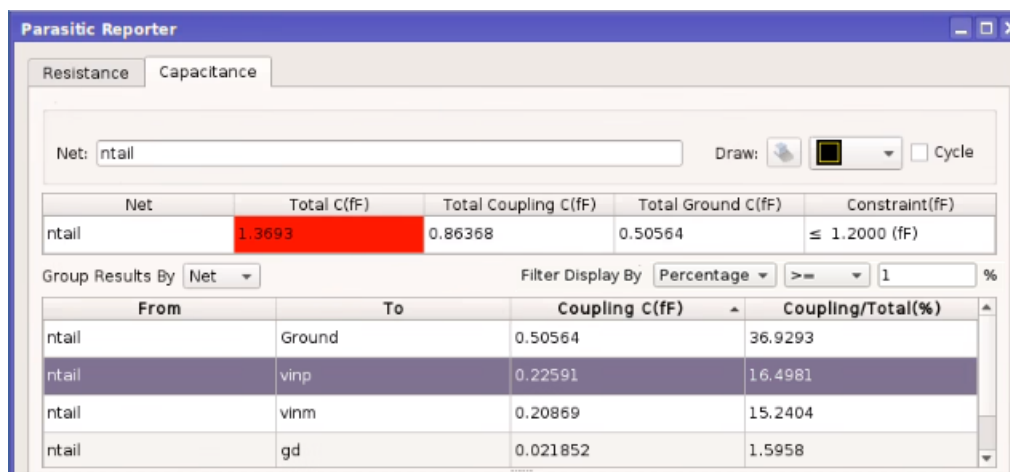
- If the constraint value is **Pass**, the background of the value in the **Net** field turns green.
- If the constraint value is **Fail**, the background of the value in the **Net** field turns red.

The extraction usually fails if the constraints are not defined accurately.

To view the details of the constraint, right-click on the value and select the **Show Detail** option. A table with the constraint values appears, as shown in the following example:

Constraint Type	Constraint Value (fF)	Net1	Net1 Capacitance (fF)	Net2	Net2 Capacitance (fF)	Result
Wire Matching Capacitance	<=1	ntail	1.4211	vn0	3.3153	Fail
Capacitance	<=2	ntail	1.4211	N/A	N/A	Pass

The following example report shows that the total capacitance value does not meet the capacitance constraint. The calculated total capacitance is 1.3693 fF, which is greater than the constraint value of 1.2000 fF.



3. Review the data in the summary report table:

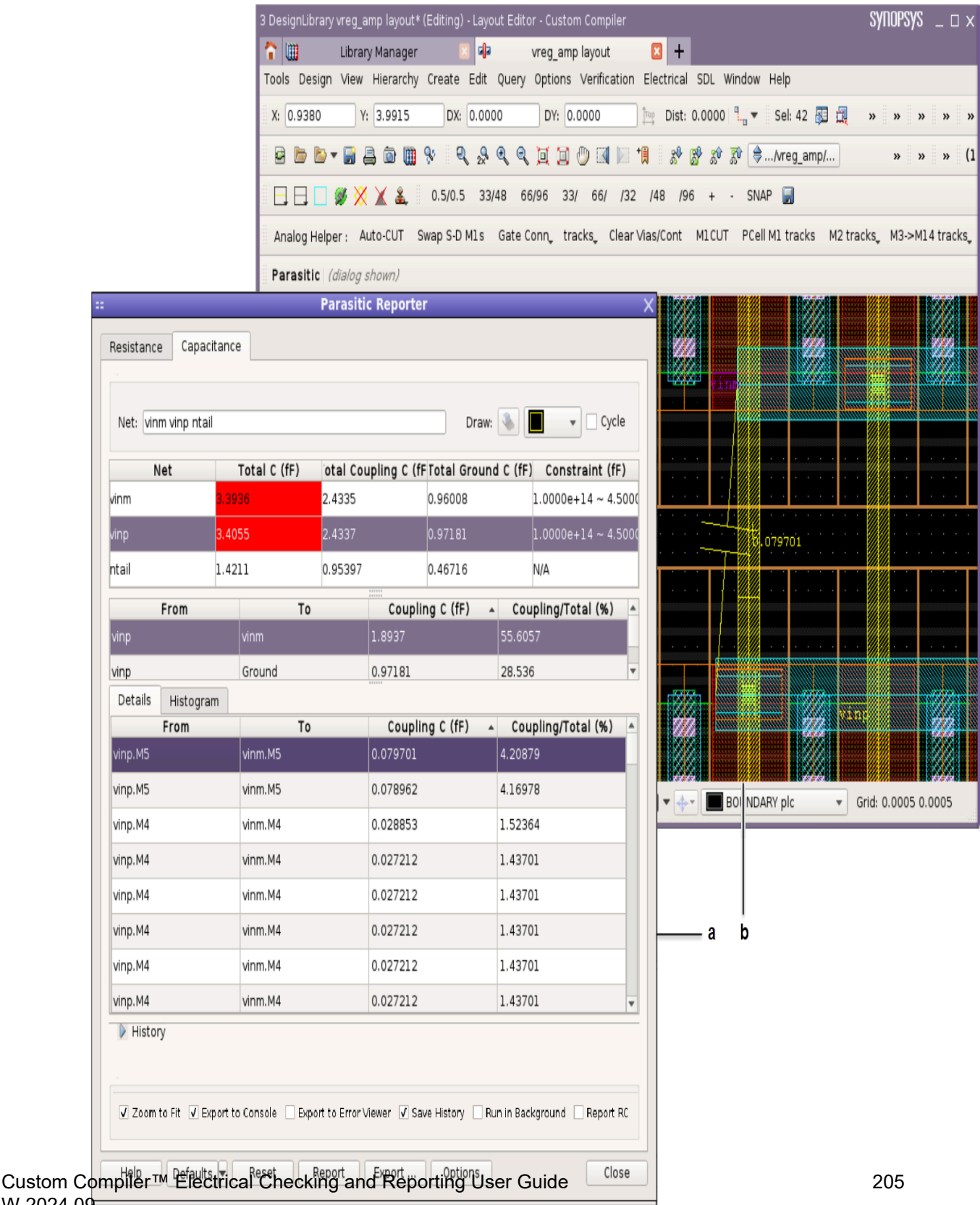
Column Name	Description
From	Name of the first net.
To	Name of the second net.

Column Name	Description
Coupling C (fF)	The coupling capacitance, in femtofarads, calculated between the two nets.
Coupling/Total (%)	The coupling capacitance between the two nets as a percent of the total capacitance.

4. To display specific information for individual capacitance values, click a row in the summary report table. The **Details** tab populates with the following:

Column Name	Description
From	A point on netA.
To	A point on netB.
Coupling C(fF)	The coupling capacitance from the point on netA to the point on netB, in femtofarads.
Coupling/Total (%)	The coupling capacitance from the previous column as a percentage of the net-to-net total.

5. To view the pair of nets on the canvas, select a row in the summary report table. The canvas displays the area of the design that contains the pair of nets.
6. To view an individual capacitance value on the canvas, select a row in the detailed report table (a). The canvas zooms in and displays a magnified view for that capacitance value (b).

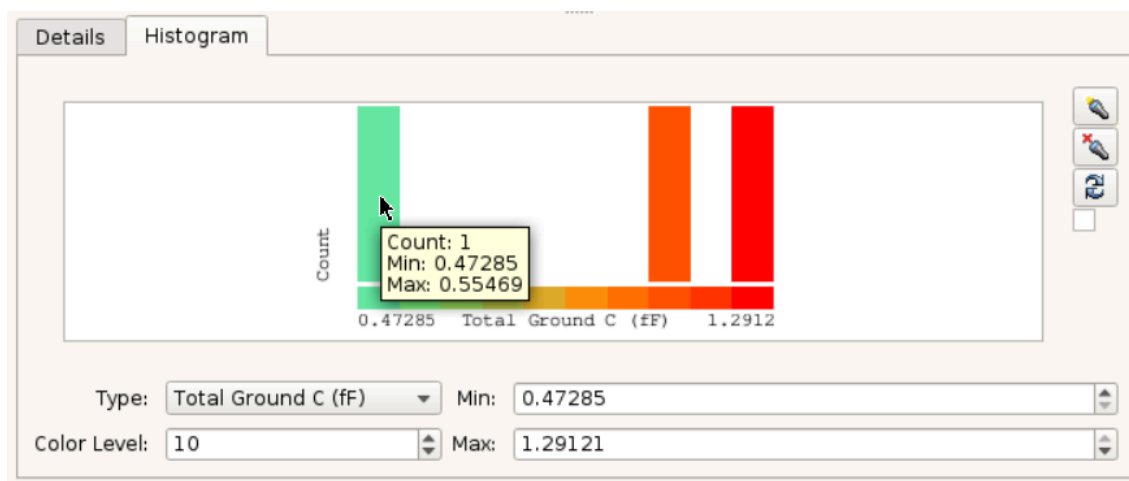


7. To view the histograms of the report data, click the **Histogram** tab, and then:
- Choose the **Type** of data you want to view. Only Total C (fF), Total Coupling C (fF), and Total Ground C (fF) are available in heat map views.

For the descriptions of the data, see steps 2, 3, and 4.
 - Set the **Color Level**, the number of colors you want to use when displaying the histogram and heat map on the canvas.

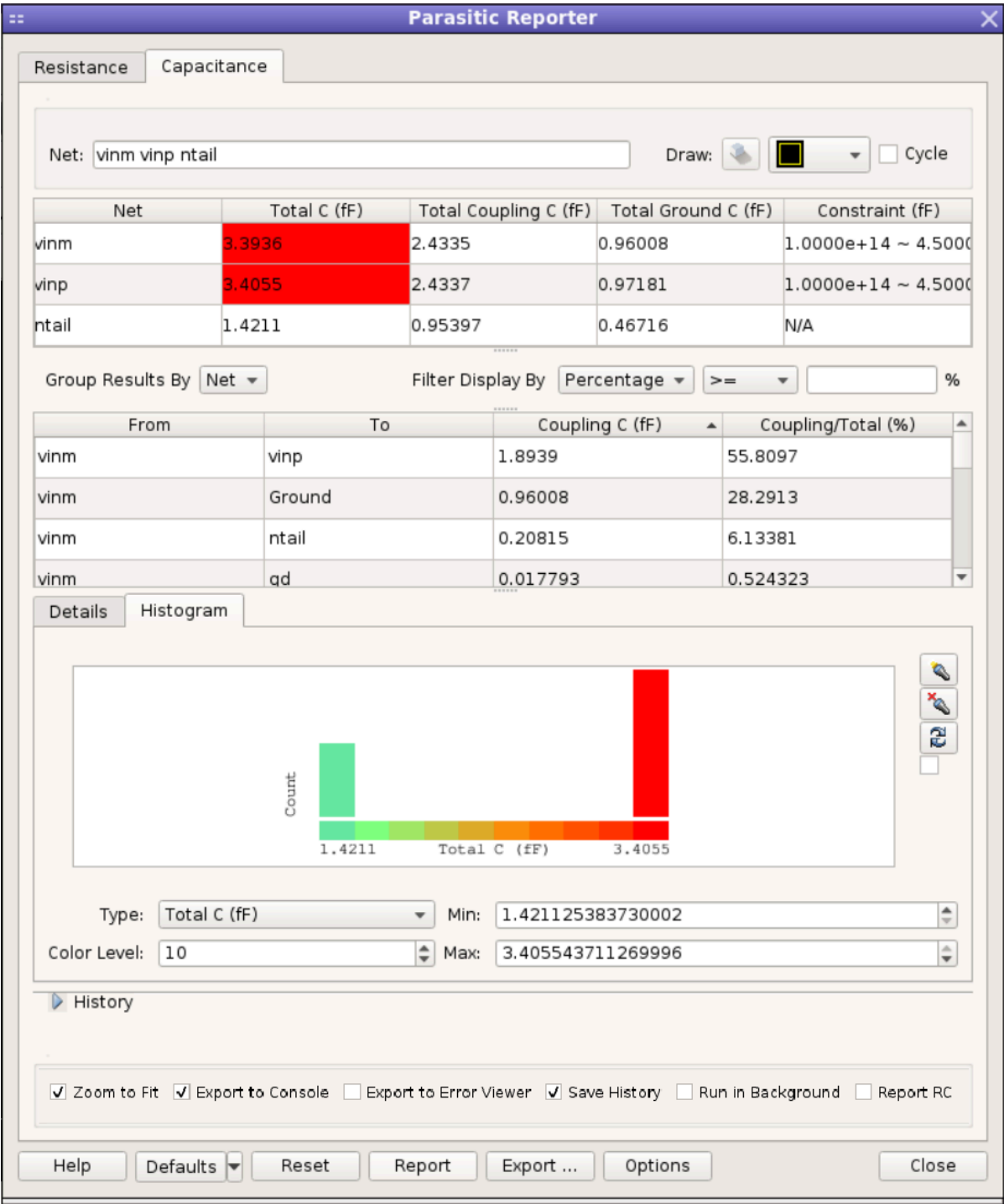
The data between the **Min** and **Max** values are binned into the number of colors you specify.
 - Specify the **Min** and **Max** values for the data type you chose in step a.
 - View the bar chart that shows the distribution of the data type you selected.
 - To view the corresponding heat map on the canvas, click **Show heat map** . To hide the heat map, click **Hide heat map** .

To view the number of values within a bin range, rest the pointer on a colored bar in the histogram.



- f. Repeat steps a through e to view other data types and the corresponding distribution. To refresh the histogram, click **Refresh** .

If you want to automatically refresh the histogram, select the check box below .



8. Adjust the design and generate new reports until the capacitance calculations are satisfactory.

Before generating a new report, click the **Reset** button to clear old results.

Note:

The Export feature is available for Resistance reporting only.

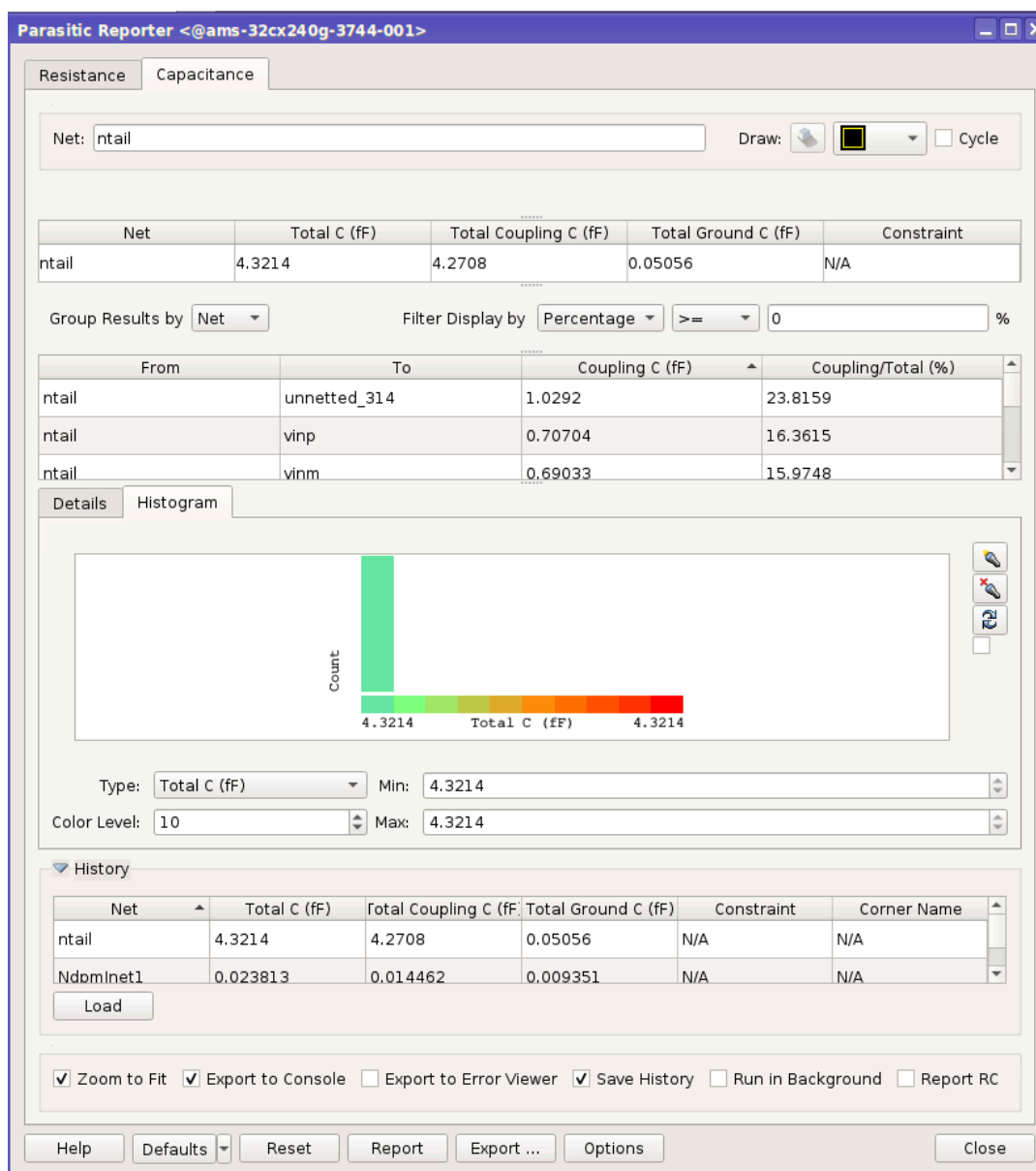
Exporting Capacitance Reports

You can export capacitance reports to comma-separated values (CSV) files, in which the comma character separates each field of text. You can open CSV files in applications such as Microsoft Excel.

To export capacitance reports to the console:

1. In the **Capacitance** tab, select **Export to Console**.
2. If you want to view the violations in the Error Viewer, select **Export To Error Viewer**.

3. Click **Report**. The Parasitic Reporter exports the report to the Console.



To export capacitance reports to CSV files:

1. In the **Capacitance** tab, click **Export**. The **Export Parasitic** dialog box appears.
2. Select the location for the file, type a name in the **File name** box, and click **Save**. The Parasitic Reporter exports the report to a CSV file.

Parasitic Reporter - Capacitance Tab Reference

This section provides a quick reference of the **Capacitance** tab in the **Parasitic Reporter** dialog box. To generate capacitance reports, see [Generating Capacitance Reports](#).

For the quick reference of the **Resistance** tab, see [Parasitic Reporter - Resistance Tab Reference](#).

The **Capacitance** tab contains the following options:

Parasitic Reporter

Resistance Capacitance

Net: L<1> L<2> L<3> Draw:   Cycle ☐

Net	Total C (fF)	Total Coupling C (fF)	Total Ground C (fF)	Constraint
L<1>	1.0213	0.28321	0.73808	Fail
L<2>	1.1951	0.57322	0.62189	Fail
L<3>	1.2499	0.61121	0.63865	Pass

Group Results by Net Filter Display by Percentage >= 0 %

From	To	Coupling C (fF)	Coupling/Total (%)
L<2>	Ground	0.62189	52.0362
L<2>	L<3>	0.29001	24.2665
L<2>	L<1>	0.28321	23.6973

Details Histogram

From	To	Coupling C (fF)	Coupling/Total (%)
L<2>.M1	L<3>.M1	0.2469	85.1359
L<2>.M1	L<3>.M1	0.041629	14.3542
L<2>.M1	L<3>.M1	0.0010215	0.352212
L<2>.M1	L<3>.M1	0.00045706	0.157601

History

Net	Total C (fF)	Total Coupling C (fF)	Total Ground C (fF)	Constraint	Corner Name
L<3>	1.2499	0.61121	0.63865	Pass	N/A
L<2>	1.1951	0.57322	0.62189	Fail	N/A

Load

☒ Zoom to Fit ☒ Export to Console ☐ Export to Error Viewer ☒ Save History ☐ Run in Background ☐ Report RC

Help Defaults Reset Report Export ... Options Close

Net and Display Options

Option	Description
Net	The net that will be queried for the capacitance report. Default: The net you selected in the Design Navigator in Generating Capacitance Reports
Draw	Applies the Shadow (contrast) setting and the selected flight line color to the results on the canvas.
Cycle	Cycles flight line colors as you view different wire segments on the canvas. Disabling this option uses the selected flight line color.

Report Display Options

Option	Description
Group Results by	Organizes the results by: • Net: Lists the results by net name only. • Layer: Lists the results by net and layer name only. • All: Lists the results without specific order. For more information, see Viewing Capacitance Reports .
Filter Display by	Specifies how the results are filtered. Choose the type of values to display: • Percentage: The values in percentage. • Absolute: The femtofarad values. Choose the values to display: • <=: Values that are less than or equal to the percentage or femtofarad value that you specify. • >=: Values that are greater than or equal to the percentage or femtofarad value that you specify. • Range: Values within the range of percent or femtofarad values that you specify. In the text box, type the percent or femtofarad value or values.

Net Table

Column Name	Description
Net	Name of the net that is queried.
Total C (fF)	Total capacitance, in femtofarads. Total C = Total Coupling C + Total Ground C The value is green if it meets the capacitance constraint value. The value is red if it does not meet the constraint value.

Column Name	Description
Total Coupling C (fF)	Total coupling capacitance, in femtofarads.
Total Ground C (fF)	Total ground capacitance, in femtofarads.
Constraint	Capacitance constraint and the capacitance wire matching constraints. If you run the constraint, the status is displayed as either Pass or Fail. - If the constraint value is Pass, the background of the value in the Net field turns green. - If the constraint value is Fail, the background of the value in the Net field turns red. The extraction usually fails if the constraints are not defined accurately. To view the details of the constraint, right-click on the value and select the Show Detail option. A table with the constraint values appears, as shown in the following example:

Constraint Type	Constraint Value (fF)	Net1	Net1 Capacitance (fF)	Net2	Net2 Capacitance (fF)	Result
Wire Matching Capacitance	<=1	ntail	1.4211	vn0	3.3153	Fail
Capacitance	<=2	ntail	1.4211	N/A	N/A	Pass

Summary Report Table

Displays the total capacitance between pairs of nets.

Column Name	Description
From	Name of the first net.
To	Name of the second net.
Coupling C (fF)	The coupling capacitance, in femtofarads, calculated between the two nets.
Coupling/Total (%)	The coupling capacitance between the two nets as a percent of the total capacitance.

Detailed Report Table

Detailed information of the net-pair you selected in the summary report table.

Column Name	Description
From	A point on netA.
To	A point on netB.

Column Name	Description
Coupling C(fF)	The coupling capacitance from the point on netA to the point on netB, in femtofarads.
Coupling/Total (%)	The coupling capacitance from the previous column as a percentage of the net-to-net total.

Histogram Tab

Column Name	Description
Type	Type of data you want to display. See the data descriptions in the Summary Report Table and the Detailed Report Tab sections.
Color Level	Number of colors to use when displaying the histogram and heat map on the canvas. The data between the Min and Max values are binned into the number of colors you specify.
Min	Minimum value of the data.
Max	Maximum value of the data.
 Show heatmap	Displays the heat map based on the histogram on the canvas.
 Hide heatmap	Hides the heat map.
 Refresh histogram	Updates the histogram based on updated settings in the Histogram tab.
☒ Automatically refresh histogram	Automatically updates the histogram as soon as you modify a setting in the Histogram tab.

Global Options

Option	Description
Zoom to fit	(For viewing the report only) Adjusts the viewport to display the design subsection containing the specified wire segment.

Option	Description
Export to Console	Prints the results to the Console.
Export to Error Viewer	Exports the violations to the Error Viewer.
Save History	Saves the generated capacitance report in the History section automatically, so that you can recall it in future design sessions.
Run In Background	Runs the capacitance analysis in the background so that you can continue to work in the Layout Editor.
Report RC	Reports resistance and capacitance simultaneously. See: Reporting Resistance and Capacitance
Report	Generates the capacitance report based on the settings you specified. See Generating Capacitance Reports .
Export	Exports the point-to-point capacitance report to a CSV file. See Exporting Capacitance Reports .
Options	Opens the Electrical Options dialog box. See About the Electrical Options Dialog Box .

Checking Capacitance

You can use the **Check Capacitance** and **Check Intent** commands to determine whether selected nets meet the capacitance constraints in the layout. For a more detailed report, see [Using the Capacitance Reporter](#).

The methods you can use to check capacitance are:

Method	For...
Check Capacitance command in the Design Navigator	Displaying violations in the Error Viewer for further investigation.
Check Intent command in the Constraint Editor	Producing quick pass/fail reports.
<code>le::checkCapacitance</code> Tcl command	Inclusion in scripts for batch processing.

This topic describes the procedures for the **Check Capacitance** and **Check Intent** commands.

Before You Start

If you want to run the checker in the background so that you can continue to work in the Layout Editor:

- Set the `leReportEMRCBackgroundMode` Tcl preference to true. Alternatively, choose **Electrical > Parasitic Reporter**, and select **Run in Background**.

Background mode supports both StarRC shared and non-shared modes.

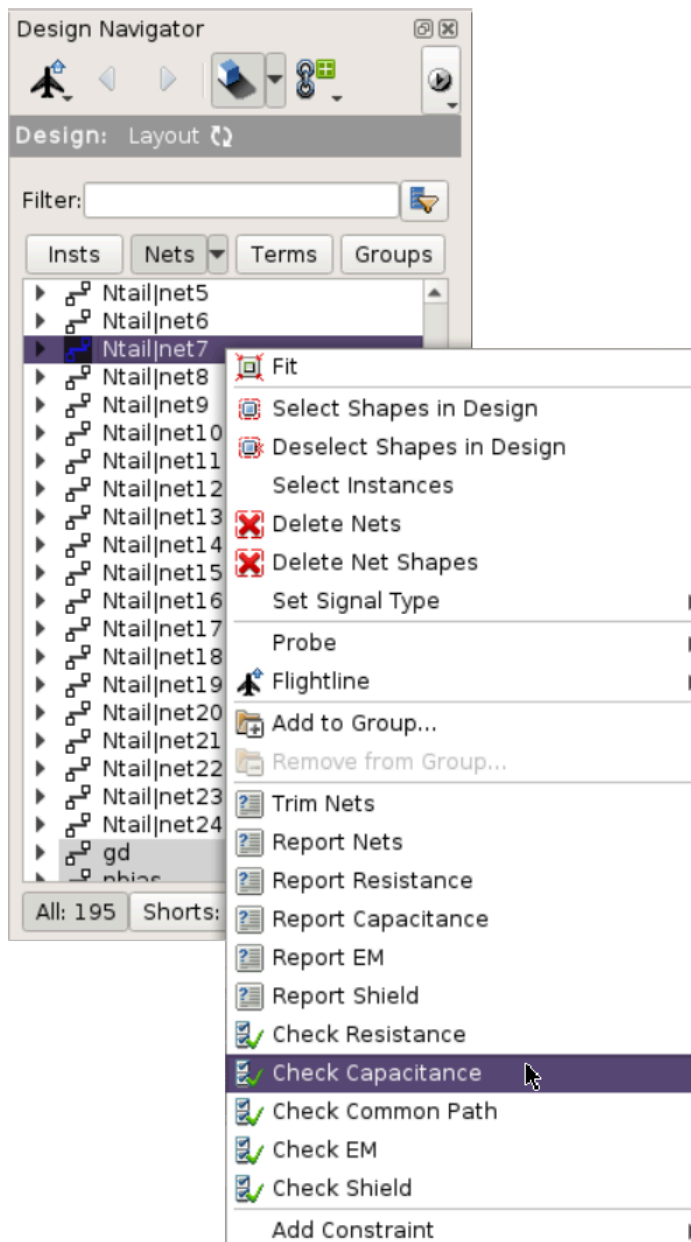
Checking Capacitance From the Design Navigator

To check capacitance from the Design Navigator:

1. In the Layout Editor, open the Design Navigator.
2. Click the **Nets** tab and select the nets. You can do this from either the **Design: Layout** mode or the **Design: Schematic** mode.

You can also select multiple nets and choose **Check Capacitance** to extract parasitics in a single run. If you select more than 10 nets, the **Reporter - Interactive StarRC** dialog box appears. Click **OK** to continue or **Cancel**. If you select **Cancel**, you should select less than 10 nets and continue extraction in the interactive mode.

3. Right-click the selected nets, and then choose **Check Capacitance**.

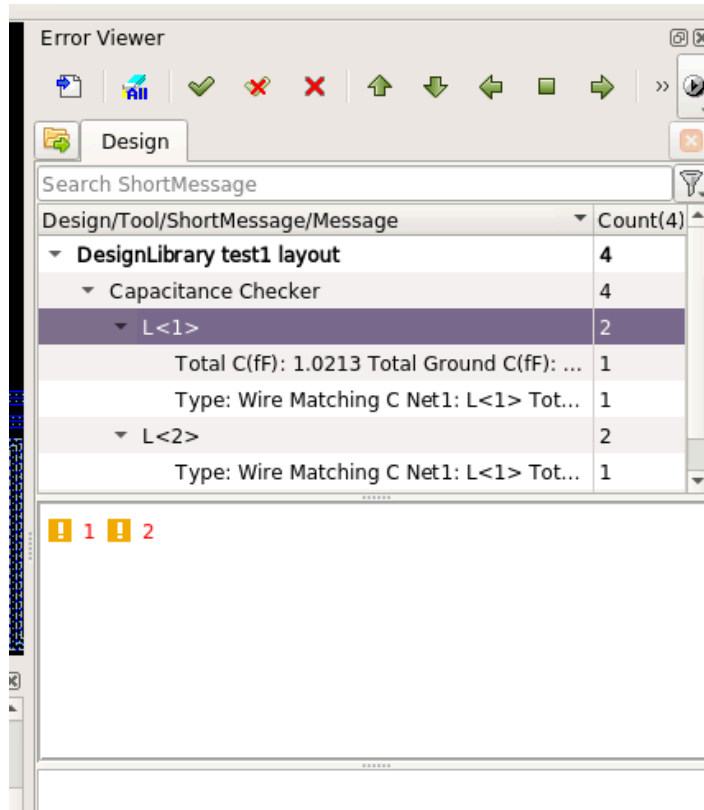


The tool calculates the net capacitance and the wire matching capacitance, compares the calculated values against the capacitance constraint for the net or the wire, and reports any violations to the Error Viewer. If the tool finds violations, the Error Viewer displays automatically and lists the violations.

If there are no violations, you do not need to proceed to the next step.

4. In the Error Viewer, expand the list to show the constraint violations. For a given constraint that failed, the assistant displays details about the error in the lower portion of the assistant.

For a wire matching constraint violation, if the given constraint has failed, the error viewer includes the capacitance values of the net and the wire matching constraint.



Where a measurement is in violation, adjust the design components, and check the capacitance again.

You can also view the pass or fail constraint status in the **Parasitic Reporter** dialog box. See [Using the Capacitance Reporter](#).

7

Setting EM, Resistance, and Capacitance Options

The Electrical Options dialog box contains required settings for in-design electromigration, resistance, and capacitance reporting. You typically access the dialog box from the Parasitic Reporter and EM Reporter dialog boxes.

This chapter contains the following topics:

- [About the Electrical Options Dialog Box](#)
- [Opening the Electrical Options Dialog Box](#)
- [Creating and Managing Defaults](#)
- [Electrical Options Dialog Box Reference](#)
- [Including Additional StarRC Commands](#)

For in-design electrical reporting instructions, see [Reporting and Checking Electromigration \(EM\)](#), [Reporting and Checking Resistance](#), and [Reporting and Checking Capacitance](#).

About the Electrical Options Dialog Box

The **Electrical Options** dialog box allows you to specify general options that are required for electromigration, resistance, and capacitance reporting.

In the **Electrical Options** dialog box, you can specify the following:

- Reporting tool (Custom Compiler built-in reporter, StarRC tool, or StarRC/PrimeSim tools)
- Rule file format (source of the electrical constraints)
- Rule file and mapping file locations
- Extraction method (StarRC device-level or gate-level extraction) and command file to use

- Layout extraction settings
- Analysis temperature

The **Electrical Options** dialog box contains the following tabs. Some tabs are displayed based on the electrical reporter and rule file format selections.

Tab	Content
Main Tab	Electromigration, resistance, and capacitance reporting specifications.
Nxtgrd Tab	Nxtgrd and mapping file locations for resistance and capacitance reporting.
TLU+ Tab	TLUPlus and mapping file locations for resistance reporting. The tab is displayed only if you select the Custom Compiler built-in reporter and the TLUPlus rule file format.
ITF-EM/RATCL/iRCX Tab	ITF-EM, RATCL, or iRCX and mapping file locations for electromigration reporting. The tab is displayed only if you select the corresponding rule file format.

Electrical Options

Main Nxtgrd TLU+ RATCL

Flow Settings

Resistance: ☒ Built-in ☐ StarRC TLUPlus

Capacitance: ☒ StarRC Nxtgrd

Electromigration: ☐ Built-in ☒ StarRC / PrimeSim EMIR RATCL

StarRC Settings

☒ Device Layers

Command File:

Halo Width: um

EM Settings

Current Scaling Factor: %

PrimeSim XA Command File:

PrimeSim EMIR Command File:

☐ Load Report in Results Viewer ☒ Open on Completion

☐ Enable Self-Heating Flow

Analysis Temperature (C)

R/C: EM (Max): EM (Delta):

Layout Extraction Settings

Top Pin Extract Rule: Technology Library

Global: ☒ From layout ☐ Density:

Local: ☒ From layout ☐ Spacing: times of Smin

Width: times of Wmin

Inter Layer Separation: layers

Inter Layer Orientation: Perpend

Job Parameters

Job Class: InDesignEMRC Options...

Help Defaults Close

Opening the Electrical Options Dialog Box

The **Electrical Options** dialog box contains required settings for electromigration, resistance, and capacitance reporting. The dialog box can be accessed directly from the Layout Editor or from within the **EM Reporter** and **Parasitic Reporter** dialog boxes.

You can open the **Electrical Options** dialog box using one of the following methods:

- [From the Layout Editor Electrical Menu](#)
- [From the Parasitic Reporter or the EM Reporter Dialog Box](#)

From the Layout Editor Electrical Menu

You typically open the **Electrical Options** dialog box while already working in the **EM Reporter** or **Parasitic Reporter** dialog box. However, if you want to open the dialog box directly, you can do so within the Layout Editor.

To open the **Electrical Options** dialog box from the **Electrical** menu:

1. In the Layout Editor, make sure the **Electrical** menu is displayed on the menu bar.

If the menu is not displayed, choose **Tools > Electrical**.

2. Choose **Electrical** > **Options**

Electrical Options

Main Nxtgrd TLU+ RATCL

Flow Settings

Resistance: ☒ Built-in ☐ StarRC TLUPlus

Capacitance: ☒ StarRC Nxtgrd

Electromigration: ☐ Built-in ☒ StarRC / PrimeSim EMIR RATCL

StarRC Settings

☒ Device Layers

Command File:

Halo Width: 10 um

EM Settings

Current Scaling Factor: 100 %

PrimeSim XA Command File:

PrimeSim EMIR Command File:

☐ Load Report in Results Viewer ☒ Open on Completion

☐ Enable Self-Heating Flow

Analysis Temperature (C)

R/C: 25 EM (Max): 110 EM (Delta): 5

Layout Extraction Settings

Top Pin Extract Rule: Technology Library

Global: ☒ From layout ☐ Density: 0.35

Local: ☒ From layout ☐ Spacing: 1 times of Smin

Width: 1 times of Wmin

Inter Layer Separation: 1 layers

Inter Layer Orientation: Perpend

Job Parameters

Job Class: InDesignEMRC Options...

Help Defaults Close

From the Parasitic Reporter or the EM Reporter Dialog Box

While working in the **EM Reporter** or **Parasitic Reporter** dialog box, you can open the **Electrical Options** dialog box to set the required options for in-design electrical reporting.

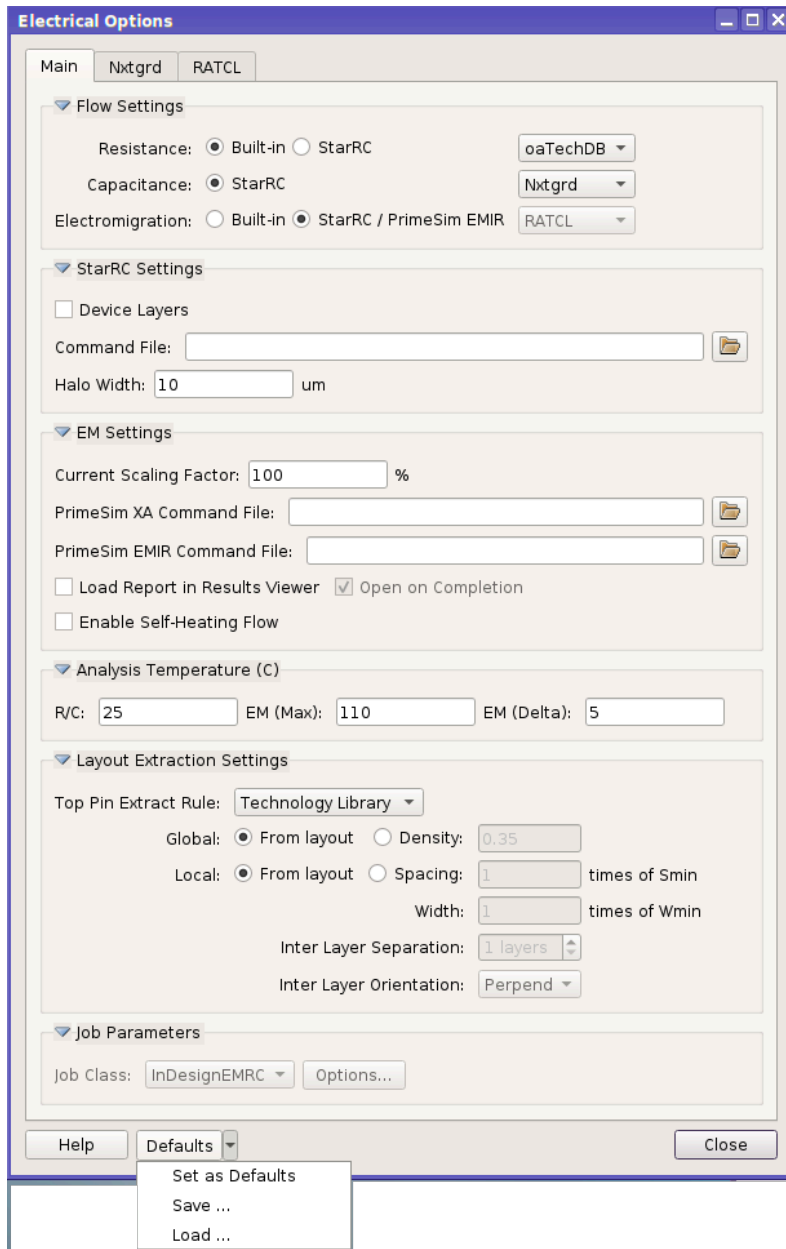
To open the **Electrical Options** dialog box while in the **EM Reporter** or **Parasitic Reporter** dialog box

- ▶ See one of the following topics:
 - [Reporting Electromigration](#)
 - [Reporting and Checking Resistance](#)
 - [Using the Capacitance Reporter](#)

Creating and Managing Defaults

You can save the information in the Electrical Options dialog box to be reloaded in later sessions. Reloading saved settings can save time and prevent inconsistent data entry.

You create and load the defaults using the **Defaults** menu at the bottom of the **Electrical Options** dialog box.



This topic describes the following:

- [Creating Defaults](#)
- [Loading Defaults](#)

Creating Defaults

To create defaults:

1. In the **Electrical Options** dialog box, set the options. See [About the Electrical Options Dialog Box](#).
2. Click , and then choose one of the following:

Command	Description
Set as Defaults	Creates the default based on the current settings in the dialog box.
Save	Saves the current settings in all the Electrical Options dialog box to an XML file.

Loading Defaults

To load settings that were previously saved in the dialog box:

1. Choose one of the following:

Command	Description
Defaults	Loads the saved settings available at the highest scope level. The order of the scope levels is cellview, cell, library, and user. For example, if you did not save any settings at the cellview level, then the cell-level settings are loaded.
Load	Lets you select the file that contains the settings you want to use.

2. Review the loaded settings and make sure they are applicable to the report you want to generate.

Electrical Options Dialog Box Reference

This topic provides a quick reference of the **Electrical Options** dialog box. For instructions on setting up for in-design electrical reporting, see [Reporting and Checking Electromigration \(EM\)](#), [Reporting and Checking Resistance](#), and [Reporting and Checking Capacitance](#).

This topic describes the following tabs in the dialog box:

- [Main Tab](#)
- [Nxtgrd Tab](#)
- [TLU+ Tab](#)
- [ITF-EM/RATCL/iRCX Tab](#)
- [Self-Heating Options Tab](#)

Main Tab

The **Main** tab of the **Electrical Options** dialog box contains the following sections:

- [Flow Settings](#)
- [StarRC Settings](#)
- [EM Settings](#)
- [Analysis Temperature](#)
- [Layout Extraction Settings](#)
- [Job Parameters](#)

Electrical Options

Main Nxtgrd TLU+ RATCL

Flow Settings

Resistance: ☒ Built-in ☐ StarRC TLUPlus

Capacitance: ☒ StarRC Nxtgrd

Electromigration: ☐ Built-in ☒ StarRC / PrimeSim EMIR RATCL

StarRC Settings

☒ Device Layers

Command File:

Halo Width: 10 um

EM Settings

Current Scaling Factor: 100 %

PrimeSim XA Command File:

PrimeSim EMIR Command File:

☐ Load Report in Results Viewer ☒ Open on Completion

☐ Enable Self-Heating Flow

Analysis Temperature (C)

R/C: 25 EM (Max): 110 EM (Delta): 5

Layout Extraction Settings

Top Pin Extract Rule: Technology Library

Global: ☒ From layout ☐ Density: 0.35

Local: ☒ From layout ☐ Spacing: 1 times of Smin

Width: 1 times of Wmin

Inter Layer Separation: 1 layers

Inter Layer Orientation: Perpend

Job Parameters

Job Class: InDesignEMRC Options...

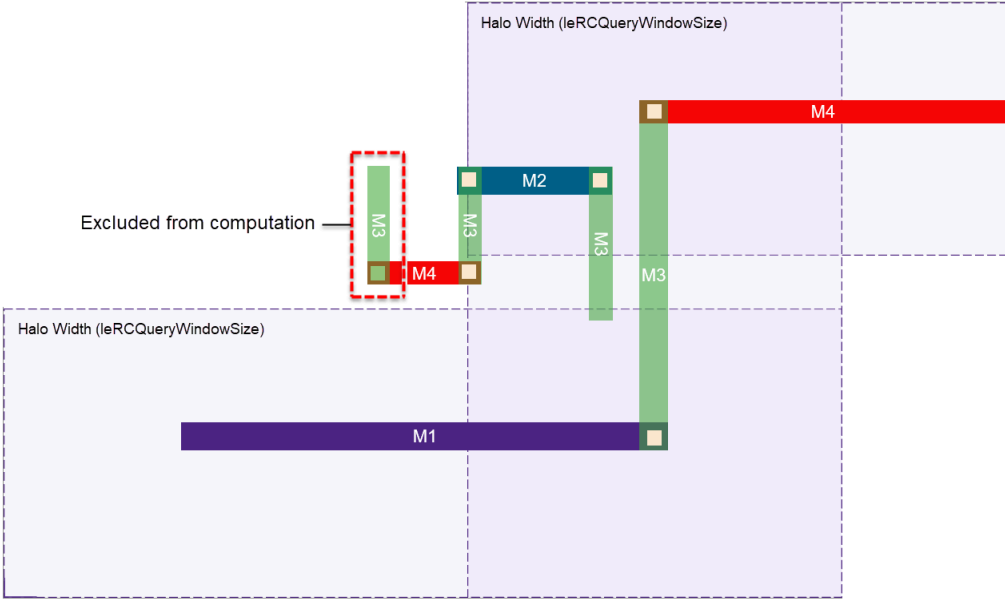
Help Defaults Close

Flow Settings

The **Flow Settings** section allows you to specify the electrical reporter to use and the source of the constraints for the electrical calculations.

Option	Description
Resistance	Specifies the tool to use for resistance calculations and the corresponding format. • Built-in: Uses the Custom Compiler tool to calculate resistance. Choose the oaTechDB or TLUPlus format. See also TLU+ Tab and Custom Compiler Resistance Reporting With TLUPlus Files. • StarRC: Uses the StarRC tool to calculate resistance. The nxtgrd format will be used. See also Nxtgrd Tab and StarRC Resistance Reporting With nxtgrd Files. Tcl Preference: <code>leResistanceModel</code>, <code>leStarRCReportResistance</code> Valid values for <code>leStarRCReportResistance</code> are <code>true</code> (StarRC) and <code>false</code> (Built-in). Default: <code>false</code>
Capacitance	Specifies the capacitance reporter to use: StarRC. The StarRC tool uses the nxtgrd file as the constraint source for capacitance calculations. See also Nxtgrd Tab and Preparing the Design and Technology. Tcl Preference: <code>leCapacitanceModel</code>
Electromigration	Specifies the source of the process-level current density constraints: • Built-in: Uses the Custom Compiler sources. Select Built-in for static EM analysis. • StarRC/PrimeSim EMIR: Uses the StarRC or PrimeSim EMIR source. Select StarRC/PrimeSim EMIR for dynamic EM analysis. If you select Built-in, choose one of the following available constraint sources: • oaTechDB: Uses the <code>maxACCurrentDensity</code> constraint in the technology database. (default) • iRCX: Uses the constraint information in the iRCX file. • ITF-EM: Uses the constraint information in the ITF-EM file. • RATCL: Uses the constraint information in the RA-TCL file. See also ITF-EM/RATCL/iRCX Tab and Defining Current Density Constraints. Tcl Preferences: <code>leStarRCReportEM</code>, <code>leEMConstraintModel</code> Because the EM Reporter uses resistance data to determine electromigration effects, you must also set the resistance options.

StarRC Settings

Option	Description
Device Layers	Runs StarRC device-level extraction, during which metal, device, base, and contactless layers are considered. Clear the check box to run StarRC gate-level extraction, in which only metal layers are considered. Device-level extractions are for advanced-node and analog designs. Be sure to prepare a mapping file containing layer derivations so that the StarRC tool can bridge mapping gaps and produce more accurate results. See Derived Layer Definitions. For co-design flows, clear the check box to run gate-level extraction. Tcl preference: <code>leInteractiveStarRCEngineBias</code>
Command File	Specifies the StarRC command file to use for the extraction process. Tcl preference: <code>leAdditionnalStarCmd</code>
Halo Width	Non-default width of the enclosure around the net of interest. Neighboring shapes within the enclosure are considered during extraction. The minimum $\frac{1}{2}$ halo size is 4 times the wire width. The maximum $\frac{1}{2}$ halo size is 20 μm.  The diagram shows a central horizontal purple wire labeled M1. Above it, a blue wire M2 and a green wire M3 are connected. To the right, a red wire M4 extends horizontally. A dashed purple rectangle, labeled 'Halo Width (leRCQueryWindowSize)', encloses M1 and extends upwards to include M2 and M3. A dashed red rectangle highlights a portion of M3, with an arrow pointing to it from the text 'Excluded from computation'. Another dashed purple rectangle is shown below M1, also labeled 'Halo Width (leRCQueryWindowSize)'. Halo Width can be used with the <code>leNetQueryWindowSize</code> Tcl preference to reduce report runtime if you anticipate heavy computations, especially when reporting total capacitance. See Runtime Reduction. Tcl preference: <code>leRCQueryWindowSize</code>

EM Settings

Option	Description
Analysis Temperature (C)	
Current Scaling Factor	Scales device terminal currents by the specified factor to enable EM what-if analysis. Default: 100% To increase current values, specify a value > 100. To decrease current values, specify a value < 100. For example, to increase the current values by 25%, specify 125%. To decrease current values by 25%, specify 75%. Tcl Preference: <code>leCurrentScalingFactor</code>
PrimeSim XA Command File	Specifies the PrimeSim XA command file.
PrimeSim EMIR Command File	Specifies the PrimeSim EMIR command file.
Load Report in Results Browser	(Dynamic EM analysis only) Automatically opens the PrimeSim EMIR results viewer and displays the EM report. This selection is only applicable if you selected StarRC/PrimeSim EMIR in the Flow Settings section. If you do not select this option, the EM report displays in the EM Reporter dialog box.
Enable Self-Heating Flow	Displays the Self-Heating Options Tab . See also Setting Up for In-Design SHE EM Flow .

Analysis Temperature

Option	Description
Analysis Temperature (C)	
R/C	The temperature (°C) to use when calculating resistance and capacitance. The default temperature is 25 °C. Tcl Preference: <code>leRCAnalysisTemperature</code>
EM (Max)	The maximum temperature (°C) to use when calculating the current density. The default temperature is 110 °C. Tcl Preference: <code>leEMAnalysisTemperature</code>
EM (Delta)	The global self-heating temperature (°C) factor applied to all objects during analysis. The default temperature is 5 °C. Tcl Preference: <code>leDeltaTemperature</code>

Layout Extraction Settings

Option	Description
Top-Pin Extract Rule	Top-level pin location information, which is derived from: • Same LPP: The shapes that have the same layer-purpose pair (LPP) as the overlapping text label. • Technology Library: (default) The pinTextLayers section of the technology library. • Map: The file that defines the mapping of layer-purpose pairs for shapes and text labels. Click , and then select the mapping file you want to use. For information about mapping files, see Preparing the Layer Mapping File. Tcl preference: <code>leGeneratePinLPPMethod</code>
Global	Calculates the metal density From layout, or use a fixed Density value that you supply, when performing resistance and capacitance analysis with TLUPlus and nxtgrd files, respectively. Because electromigration analysis includes net resistance calculations, you need to set this option. For instructions, see Reporting Resistance for Selected Path (Pin-to-Pin, Terminal-to-Terminal, Instance-to-Instance, Point-to-Point). Tcl Preference: <code>leReportResistanceDensityMode</code>
Local	Extracts locally From Layout, or extract using the supplied Spacing, Width, and Inter Layer information. Because electromigration analysis includes net resistance calculations, you need to set this option. For instructions, see Reporting Resistance for Selected Path (Pin-to-Pin, Terminal-to-Terminal, Instance-to-Instance, Point-to-Point).

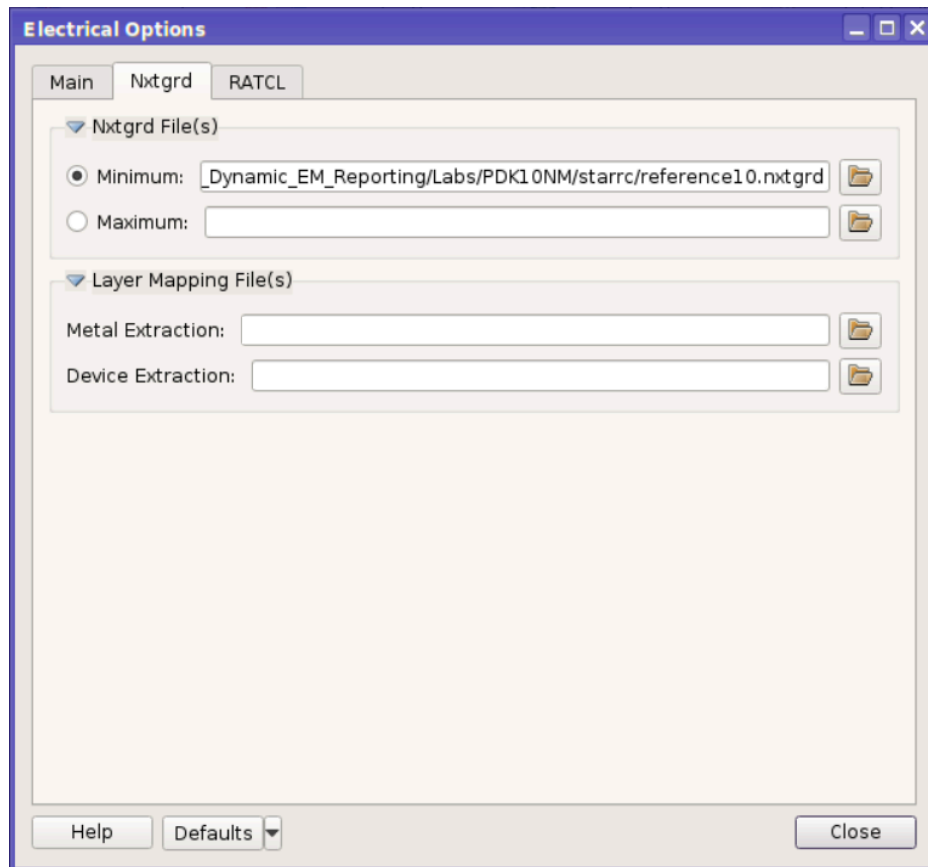
Job Parameters

To enable the **Job Parameters** option, select the **Run In Background** option in the **Parasitics Reporter** dialog box.

Option	Description
Job Class	Combines settings for individual analysis tools. You can configure existing or user-defined job classes. The available option is: InDesignEMRC. For details, see Adding Job Classes With xt::createJobClass Tcl in the <i>Custom Compiler Environment User Guide</i>.
Options	The job settings to control batch processing limits, and to optionally set a queue engine in the Job Management Options dialog box. For details, see in the Custom Compiler Environment User Guide

Nxtgrd Tab

If StarRC process corners are not defined, the **Nxtgrd** tab appears as follows:



If StarRC process corners are defined using the `xtLPEStarCornerDef` Tcl preference, the **Nxtgrd Files** section displays the list of process corners defined by the Tcl preference. The syntax of the preference is as follows:

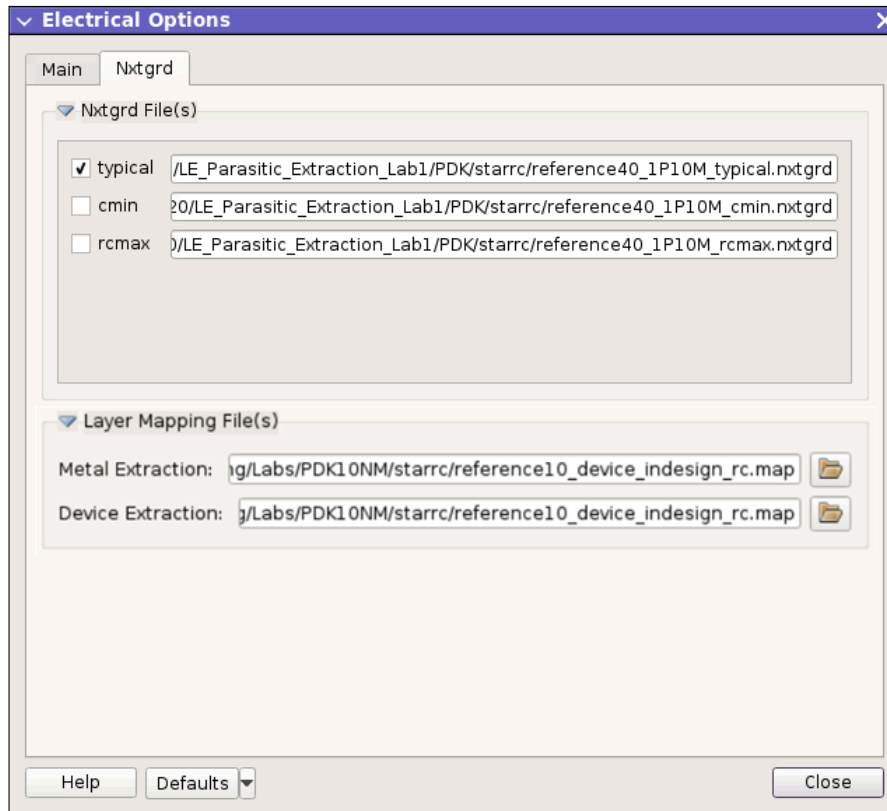
```
xtLPEStarCornerDef -value {  
  corner1 {nxtgrd_file1_path}  
  ...  
  cornern {nxtgrd_filen_path}  
}
```

The order in which you specify the corners in the preference is the order in which they appear in the tab. You have the option of specifying the location of the mapping file.

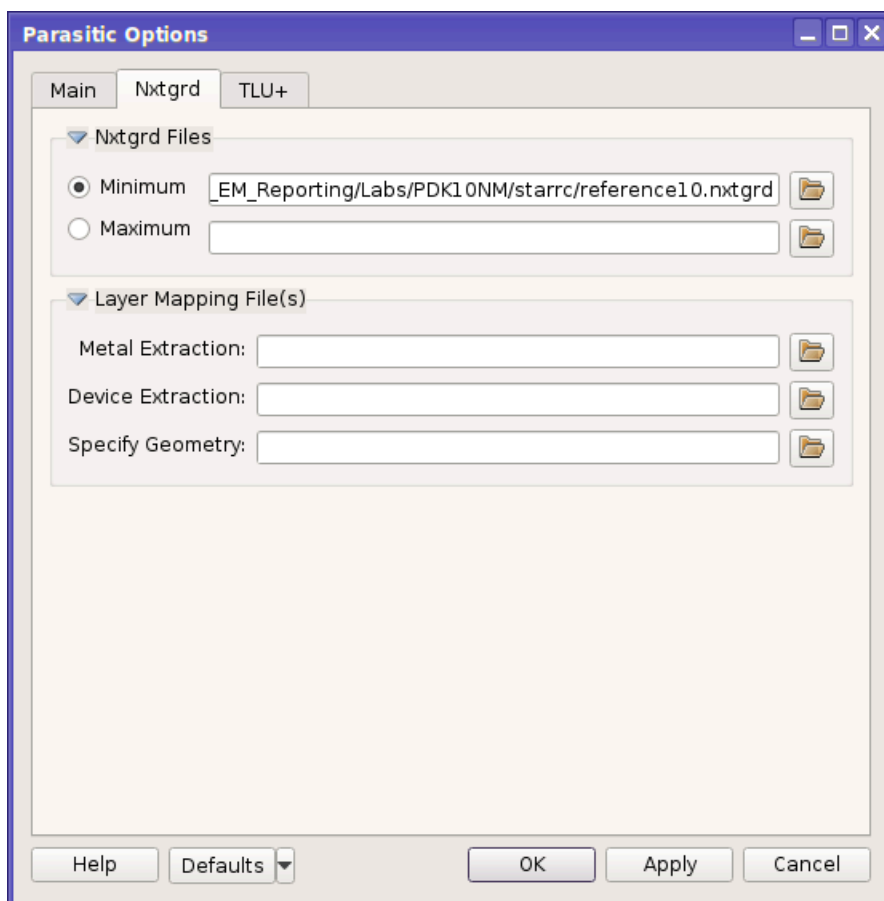
In the following example, the Tcl preference defines three corners: `typical`, `cmin`, and `rcmax`. The order in which they are specified in the preference is reflected in the tab.

Note that the capacitance analyses ignores the optional mapping file defined using the `xtLPEStarCornerDef` Tcl preference.

```
db::setPrefValue "xtLPEStarCornerDef" -value {  
    typical { $PDK_INSTALL_DIR/starrc/reference40_1P10M_typical.nxtgrd }  
    cmin {    $PDK_INSTALL_DIR/starrc/reference40_1P10M_cmin.nxtgrd }  
    rcmax {   $PDK_INSTALL_DIR/starrc/reference40_1P10M_rcmax.nxtgrd }  
}
```



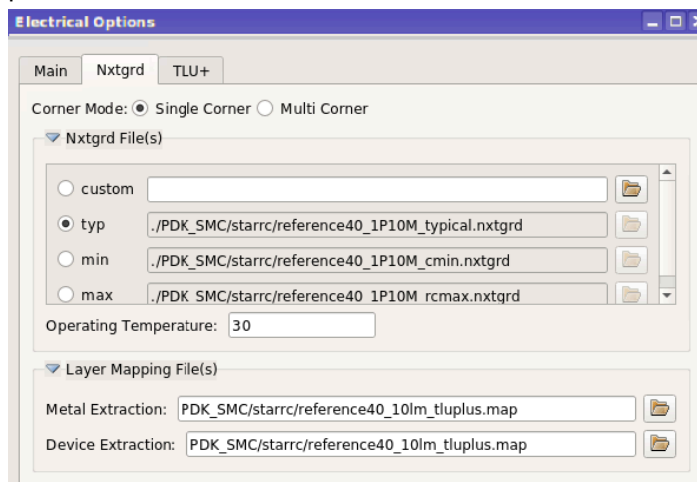
If you opened the dialog box from the Estimated Parasitics assistant, the dialog box is called **Parasitic Options**. The content of the **Parasitic Options** dialog box is nearly identical to the content of the **Electrical Options** dialog box.



Option	Description
Minimum/Maximum	(Process corners not defined) The minimum and maximum operating conditions. Select either Minimum or Maximum, and then click  to select the TLUPlus file that defines either the minimum or maximum conditions. Tcl Preferences: leNxtgrdOperatingPoint, leTCadGRDMinFile, leTCadGRDMaxFile
Corner Mode	

Option	Description
--------	-------------

Single Corner: You can select one corner only using the `xtLPEStarCornerDef` preference.

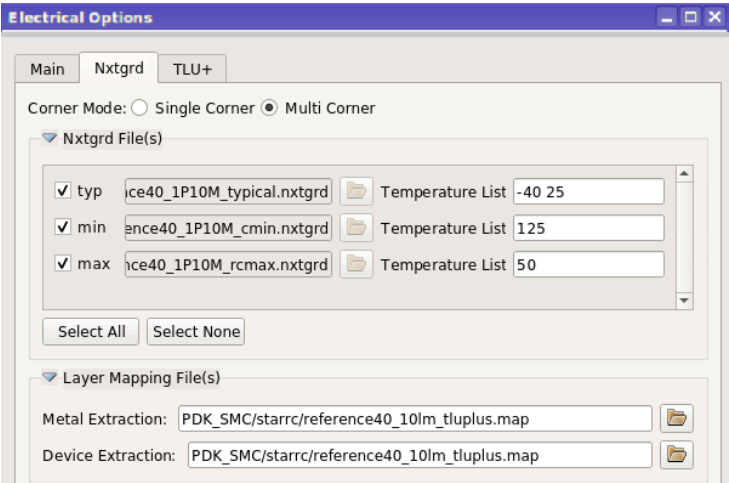


- Displays up to five `nxtgrd` files. To choose the preferred corner, click the option button next to the name of the corresponding `nxtgrd` file. For example: **typ**, **min**, **max**
- To use the corner options of your choice, click the **Custom** option to use your preferred `nxtgrd` file. The other corner options are disabled.
- Specify the value of the **Operating Temperature** field in Celsius (C) only.

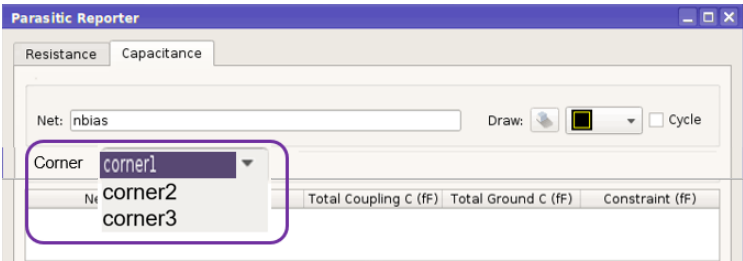
Option	Description
--------	-------------

Multi Corner

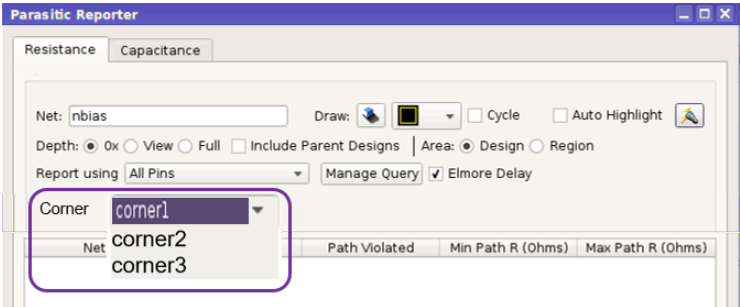
You can select one or more corners using the `xtLPESStarCornerDef` preference.



- Displays up to five `nxtgrd` files. To choose the preferred corner, select the check box next to each of the corresponding `nxtgrd` files. For example: **typ**, **min**, **max**.
The field to enter the name of the selected corners is disabled.
- Specify the value of the **Operating Temperature** field in Celsius (C) only.
- Enable the `leElectricalReporterCornerDisplay` preference to display the result in the **Parasitic Reporter** dialog box.
 - When you select **Multi corner** only, the **Corner** field appears in the **Capacitance** tab:



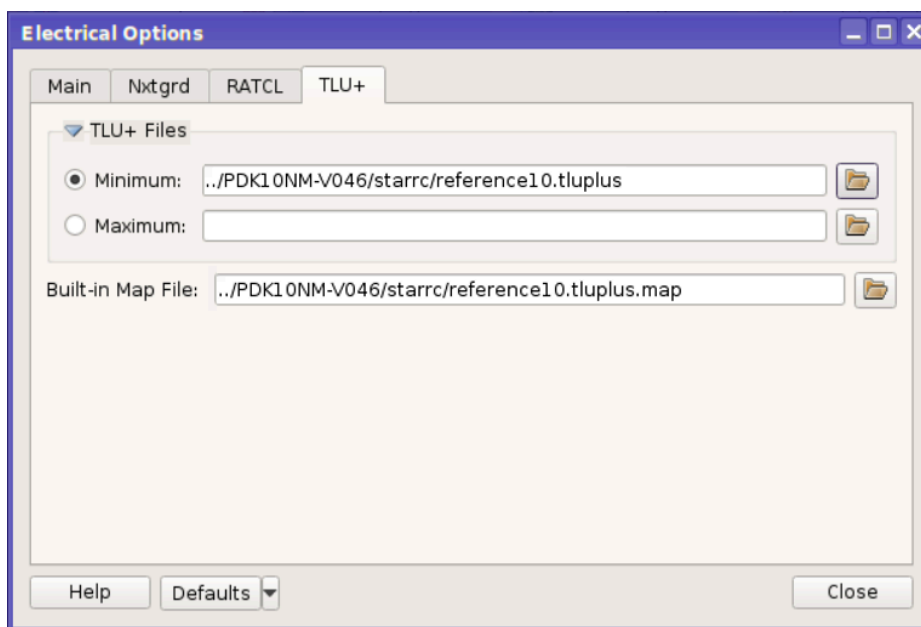
- When you select **Star RC** and **Multi corner** in the **Parasitic Reporter**, the **Corner** field appears in the **Resistance** tab:



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- To view the results in the **Parasitic Reporter** dialog box, select the name of your preferred corner from the drop-down menu. The default is the name of the first corner.
 - To save the results in the **Parasitic Reporter**, select the **Save History** check box at the bottom of the **Parasitic Reporter** dialog box.
- 238

Option	Description
<corner_name>	List of StarRC process corners and corresponding nxtgrd files that are defined using the <code>xtLPESStarCornerDef</code> Tcl preference.
Mapping	Specifies the layer mapping file for one of the following extraction methods: • Metal Extraction • Device Extraction • Specify Geometry (Parasitic Options dialog box only) Click  to select the preferred layer map file. By default, the <code>leNXTGRDMapFile</code> Tcl preference value is used if you do not select a file. Layer mapping files map OA or digital layer names to the corresponding ITF layer names. For more information, see Technology and Display Resource Files. Tcl Preference: <code>leNXTGRDMetalMapFile</code>, <code>leNXTGRDDeviceMapFile</code>, <code>leNXTGRDSpecifyGeometryMapFile</code>, <code>leNXTGRDMapFile</code>

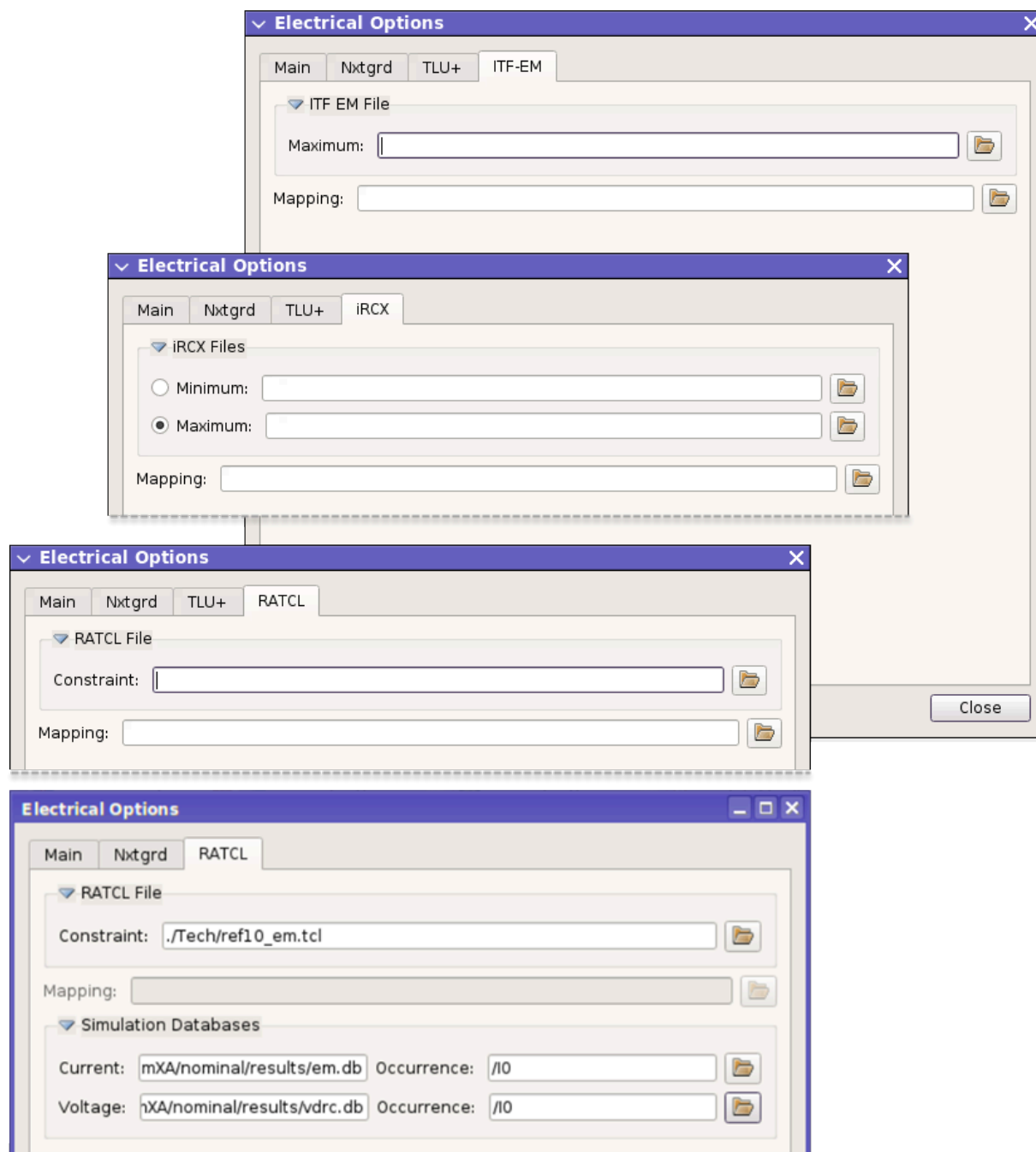
TLU+ Tab



Option	Description
Minimum/Maximum	The minimum and maximum operating conditions. Select either Minimum or Maximum, and then click  to select the TLUPlus or common nxtgrd file that defines either the minimum or maximum conditions. (The Resistance Reporter requires TLUPlus data, and the data can be from TLUPlus or common nxtgrd files.) Tcl Preferences: leTLUPlusOperatingPoint, leICCTLUPlusMaxFile, leICCTLUPlusMinFile
Built-in Map File	Specifies the layer mapping file. Click  to select the lpreferred ayer map file. Layer mapping files map OA or Milkyway layer names to the corresponding ITF layer names. For more information, see Preparing the Layer Mapping File. Tcl Preference: leICCTLUPlusMapFile

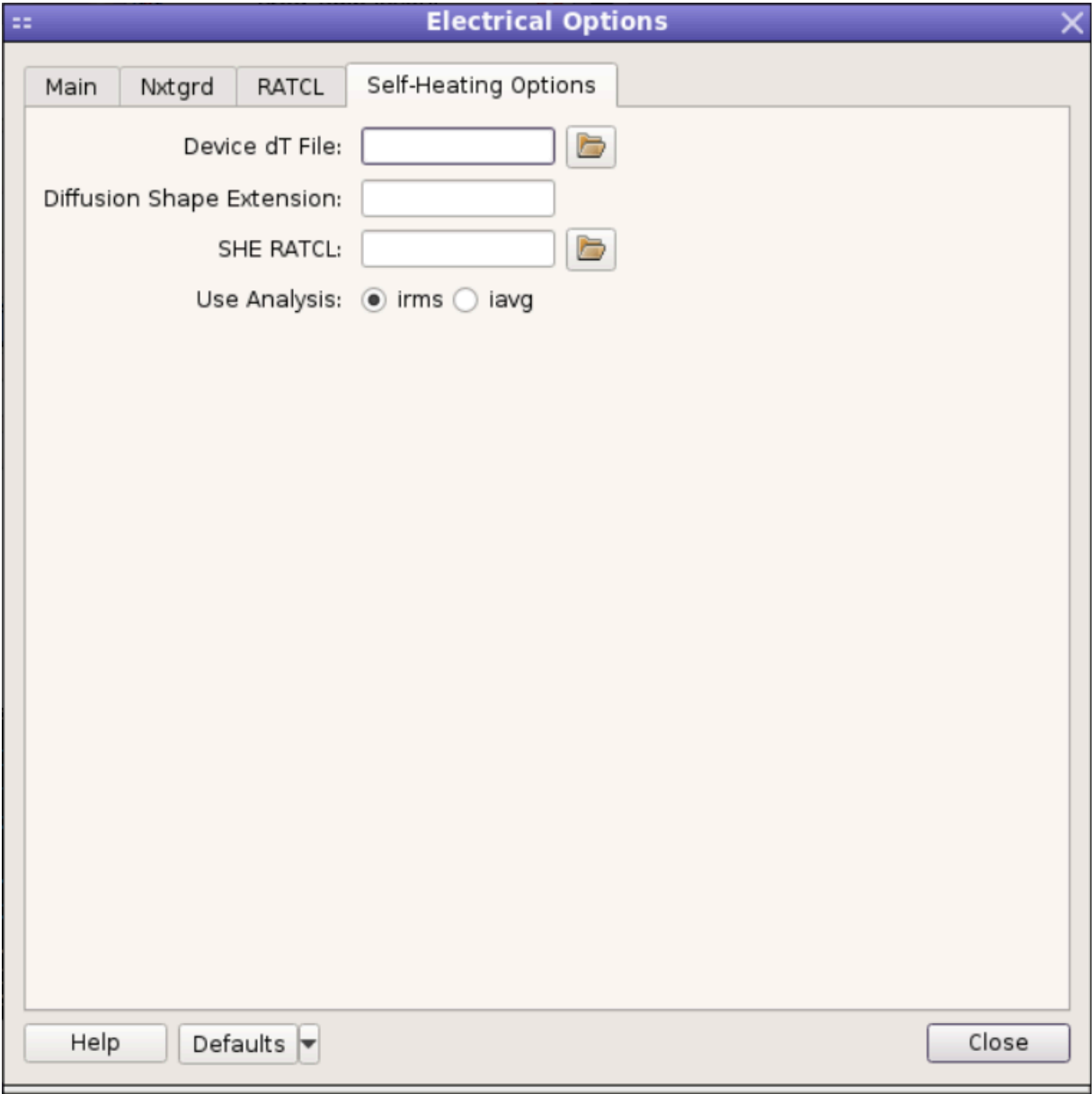
ITF-EM/RATCL/iRCX Tab

The **ITF-EM**, **RATCL**, or **iRCX** tab is displayed depending on the **Electromigration** setting you chose in the [Main Tab](#).



Option	Description
Minimum/Maximum	(iRCX, ITF-EM) The minimum and maximum operating conditions. Click  to select the iRCX or ITF-EM file that defines the minimum and maximum conditions. Tcl Preferences: leIRCXOperatingPoint, leIRCXMinFile , leIRCXMaxFile, leITFEMMaxFile
Constraint	(RATCL) The RA-TCL file containing the constraints. Tcl Preferences: leRATCLMaxFile
Mapping	(iRCX, ITF-EM, RATCL) The layer mapping file. Click  to select the preferred layer map file. Layer mapping files map OA or Milkyway layer names to the corresponding iRCX, ITF, and RATCL layer names. For more information, see Preparing the Layer Mapping File. Tcl Preference: leIRCXMapFile, leITFEMMapFile, leRATCLMapFile
Simulation Databases	(RATCL, Dynamic EM analysis) The <code>em.db</code> and <code>vdrc.db</code> files that you need as input for the PrimeSim EMIR tool, which generates the CCDB file required for results analysis. The <code>em.db</code> and <code>vdrc.db</code> files are generated when you ran simulation to obtain current and voltage data. See Measuring Terminal Current and Voltage Data Using the PrimeWave Design Environment. The Simulation Databases section appears only if you selected StarRC/PrimeSim EMIR as the Electromigration Flow Setting on the Main tab.

Self-Heating Options Tab



Option	Description
Device dT File	The XML file that contains the device delta temperature information. The file is generated from the prelayout simulation.
Diffusion Shape Extension	The amount to extend the diffusion shapes, in μm .

Option	Description
SHE RATCL	Additional RA Tcl file containing the constraints.
Use Analysis	The type of analysis to run: irms or iavg .

Including Additional StarRC Commands

You can include additional StarRC commands for resistance and capacitance reporting in the Custom Compiler environment.

You can include the following StarRC commands in an optional file:

- COUPLE_TO_GROUND
- DENSITY_BASED_THICKNESS
- DENSITY_OUTSIDE_BLOCK
- IGNORE_CAPACITANCE
- MERGE_VIAS_IN_ARRAY
- MULTIGATE_MODELS
- REFERENCE_DIRECTION
- SKIP_CELLS
- SKIP_INSTANCES
- VIA_ARRAY_COUNT_LAYER

For details about the StarRC commands, see the [StarRC User Guide and Command Reference](#) on SolvNetPlus.

The optional file must be in the ASCII format, and contain two columns. The first column contains the command name, the second column contains the values to be processed by the command.

To include the StarRC commands for resistance and capacitance reporting:

- Use the `leAdditionalStarCmd` Tcl preference to specify the location of the StarRC options file.

During capacitance reporting, the Parasitic Reporter will check the Tcl preference and process the StarRC options in the file.

8

Preparing the Layer Mapping File

When you generate resistance and total capacitance reports using external rule sources, you have the option of referencing a layer mapping file that maps OA, Fusion Compiler, IC Compiler II, or IC Compiler layer names to corresponding StarRC layer names in the `nxtgrd` files. The same mapping file is used for both resistance and total capacitance reporting.

This topic describes the following:

- [Layer Mapping File Format and Content](#)
- [Derived Layer Definitions](#)
- [Defining and Mapping Derived Layers](#)
- [Adding Blocking Layers](#)
- [Mapping StarRC Conducting Layers](#)
- [Mapping Via Layers](#)
- [Adding Layers to Ignore During Capacitance Reporting](#)

For information about the StarRC mapping file, see the “Mapping Files” topic in the [StarRC User Guide and Command Reference](#).

Layer Mapping File Format and Content

The mapping files must be in the ASCII format. Each file can consist of the following sections:

Section Content	Description
Derived Layer Definition	(Optional) Defines the following: • Derived layer names, and maps them to OA layer names. See Derived Layer Definitions. • Blocking layers that are listed in the <code>skip_file</code>. See Adding Blocking Layers. This section is only used when StarRC layers do not have direct mappings to OA layers.

Section Content	Description
Conducting Layers	Maps the OA or derived conducting layers to StarRC conducting layers. See Mapping StarRC Conducting Layers .
Via Layers	Maps the OA or derived via layers to StarRC via layers. See Mapping Via Layers . To connect layers that do not require physical vias in the design, you must define fake vias in this section for resistance reporting.
Mappings to Ignore	Tells the Capacitance Reporter to ignore the specified layers. See Adding Layers to Ignore During Capacitance Reporting .

Derived Layer Definitions

When StarRC layers do not have direct mappings to OA layers, you can bridge the gap by defining derived layers in the mapping file.

OA_layerName  Derived_layerName  StarRC_layerName

In FinFET nodes, the derived layers are created by the LVS comparison process for StarRC to produce accurate extractions. You can define these derived layers in the layer mapping file to bridge the gap.

This topic describes the derived layer section of the layer mapping file. The sections are:

- [Syntax](#)
- [Section Demarcation](#)
- [Legacy Derived-Layer Syntax Components](#)
- [Derived-Layer Syntax Components](#)
- [Syntax Support for Silicon-on-Insulator \(SOI\) Processes](#)
- [Example](#)

To map the derived layers, see [Defining and Mapping Derived Layers](#).

Syntax

The derived-layer definitions constitute the first section in the layer mapping file. The syntax for each line in the section is as follows:

```
*BEGIN_LAYER_DERIVATION_SECTION  
  
*CC_IN_DESIGN derived_layerName source_layerName
```

```
[type = n | p]
[region = active | sti]
[recognition_layer = layerName | layerName,purposeName]
[contactless_metal]
[connect = layerName]

*CC_IN_DESIGN derived_layerName =
  [layerName]
  [layerName_1 and layerName_2]
  [layerName_1 or layerName_2]
  [layerName_1 not layerName_2]
  [layerName_1 interact layerName_2]
  [layerName_1 not_interact layerName_2]
  [layerName_1 touch layerName_2]
  [layerName size +/-floatValue[, +/-floatValue]]
  [layerName scale floatValue]
  [cell_extent]

*CC_IN_DESIGN
  [non_buried_oxide_layer layerName | layerName_1, ..., layerName_n]

*END_OF_DERIVATION_SECTION
```

The argument names are not case-sensitive.

Section Demarcation

You can use the following lines to demarcate the beginning and the end of the derived-layer definitions section in the mapping file. Although they are not required, they can facilitate searches in large layer mapping files.

Argument	Description
*BEGIN_OF_LAYER_DERIVATION_SECTION	Indicates the start of the derivative layer section in the mapping file.
*END_OF_DERIVATION_SECTION	Indicates the end of the derivative layer section in the mapping file.

Legacy Derived-Layer Syntax Components

Before S-2021.09, the derived-layer mapping syntax specified the following:

Argument	Description
*CC_IN_DESIGN	Indicates to the Parasitic Reporter that the derived layer is to be replaced with the OA layer in the Layer Mapping section of the file. * is the comment character that tells the signoff tool to ignore this line.
derived_layerName	Name of the derived layer from the LVS process. The net information for the derived layer is obtained from one of the following: • The <code>connect</code> argument. • The first source layer encountered. • Any of the source layers if an OR operator is used to specify multiple source layers. If shapes from two layers touch or overlap, and the net information does not match, the shapes are ignored and not copied to the derived layer. 
source_layerName	Name of the physical layer in the OA database, or name of the existing derived layer.
type	(Optional) Layer type. Valid values are: • <code>n</code>: n-implant • <code>p</code>: p-implant If you do not specify the <code>type</code>, the tool assumes both types.
region	(Optional) Layer region. Valid values are: • <code>active</code>: Regions where poly overlaps with diffusion. • <code>sti</code>: All other regions.
recognition_layer	(Optional) Name of the recognition layer or layer-purpose pair (LPP). Recognition layers are typically used for I/O, memory, low-voltage, and high-voltage components.
contactless_metal	(Optional) Name of the fake-via layers. Fake vias connect contactless metal layers and should be defined in the <code>via_layers</code> section of the mapping file. Although fake vias must be considered for resistance reporting, they should be ignored during capacitance reporting. To ensure that the Capacitance Reporter ignores the fake vias, define an <code>ignore_cap_layers</code> section and list the mock via layers.

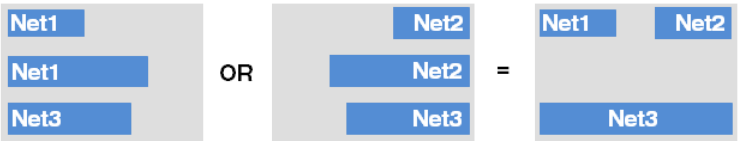
Argument	Description
connect	(Optional) Uses net information from the specified layer for the derived layer. In the following example, the net information from layerD is applied to derivedLayerA. <code>*CC_IN_DESIGN derivedLayerA sourceLayerB and SourceLayerC connect = layerD</code>

Derived-Layer Syntax Components

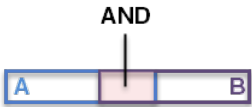
Starting from S-2021.09, the type, region, recognition_layer, and other legacy components are replaced with advanced logical operators so that you no longer have to provide redundant statements and implicit definitions.

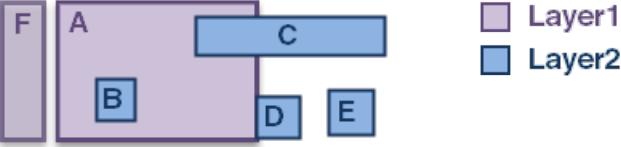
The syntax for the derived-layer section consists of the following:

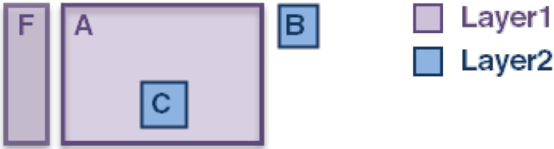
Argument	Description
*CC_IN_DESIGN	Indicates to the Parasitic Reporter that the derived layer is to be replaced with the OA layer in the Layer Mapping section of the file. * is the comment character that tells the signoff tool to ignore this line.
derived_layerName	Name of the derived layer from the LVS process. The net information for the derived layer is obtained from one of the following: • The <code>connect</code> argument. • The first source layer encountered. • Any of the source layers if an OR operator is used to specify multiple source layers. If shapes from two layers touch or overlap, and the net information does not match, the shapes are ignored and not copied to the derived layer.



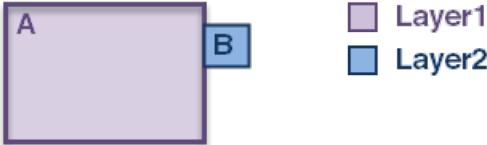
and (Optional) Includes only the intersection of the specified layers.



Argument	Description
or	(Optional) Includes all specified layers. OR 
not	(Optional) Excludes the specified layers. NOT 
interact	(Optional) Includes specified shapes that intersect (or interact). For example: <pre>*CC_IN_DESIGN derivedLayer = layer1 interact layer2</pre> The layer being assessed is layer1, based on its interaction with layer2. Shape A on layer1 interacts with the following shapes: • B on layer2 • C on layer2 • D on layer2 The following shapes do not interact: • A on layer1 and E on layer2 • F on layer1 and B on layer2 • F on layer1 and C on layer2 • F on layer1 and D on layer2 • F on layer1 and E on layer2  The resulting derived layer will consist of shape A on layer1, because shape A interacts with at least one shape on layer2. The derived layer will not contain shape F, because shape F does not interact with any shape on layer2.

Argument	Description
not_interact	(Optional) Includes shapes that do not intersect (or interact). For example: <pre>*CC_IN_DESIGN derivedLayer = layer1 not_interact layer2</pre> The layer being assessed is layer1, based on its lack of interaction with layer2. Shape A on layer1: - Does not interact with shape B on layer2. - Interacts with shape C on layer2. Shape F on layer1 does not interact with any shapes on layer2.  The resulting derived layer will consist of shape F on layer1, because shape F does not interact with any shape on layer2.

Argument	Description
size	(Optional) Increases or decreases the size of a shape. Positive values increase the size, negative values decrease the size. Do not add a character space between the + or - symbol and the value. To increase or decrease the size in all directions by the same amount, specify a single value. For example: <pre>*CC_IN_DESIGN Derived1 layer1 size +0.5</pre> <pre>*CC_IN_DESIGN Derived1 layer1 size -0.5</pre> Before layer1 size +0.5 layer1 size -0.5  To increase or decrease the size in the X and Y directions by different amounts, specify two values (x,y). Do not add a character space after the comma. For example: <pre>*CC_IN_DESIGN Derived1 layer1 size +0.5,0</pre> Before layer1 size +0.5,0  <pre>*CC_IN_DESIGN Derived1 layer1 size +0.5,-0.5</pre> Before layer1 size +0.5,-0.5  To increase or decrease the size in four directions by different amounts, specify four different values (left,right,top,bottom). Do not add a character space after each comma. For example: <pre>*CC_IN_DESIGN NPC PC size +0.5,+0.3,+0.1,+0.4</pre> Before layer1 size +0.5,+0.3,+0.1,+0.4 

Argument	Description
scale	(Optional) Increases or decreases a rectangular shape in all dimensions by a positive multiplier value. Use Scale for blocking layers only. For example: <pre>*CC_IN_DESIGN Derived1 = layer1 scale 2</pre> <pre>*CC_IN_DESIGN Derived1 = layer1 scale 0.5</pre> Before layer1 scale 2 layer1scale 0.5 
touch	(Optional) Includes specified shapes that do not interact, but their edges overlap. For example: <pre>*CC_IN_DESIGN Derived1 = layer1 touch layer2</pre>  The resulting derived layer will consist of shape A on layer1, because the edge of shape A overlaps (or touches) the edge of shape B on layer 2, even though the two shapes do not interact.
cell_extent	(Optional) Specifies a layer that was created by the <code>cell_extent()</code> IC Validator function in the LVS runset. The layer consists of a rectangle in each of the specified cells that is equal to the area of the cell. In the following example, the derivation creates a rectangle shape on the Derived1 layer with the same dimensions of the top cell. <pre>*CC_IN_DESIGN Derived1 = cell_extent</pre>

When using the logical operations (AND, OR, NOT), note the following:

- You can specify multiple logical operators. The order of the operations is from left to right, and you cannot use brackets to prioritize operations. For example:

```
*CC_IN_DESIGN poly_lvt_srm PO not OD and VTL and OD_SRM
```

```
*CC_IN_DESIGN gate_hvt_io PO and OD and VTH and OD_IO
```

- A derived layer can be repeated within the section and specified within the logic. For example:

```
*CC_IN_DESIGN gate PO and OD
```

```
*CC_IN_DESIGN poly PO not gate
```

Syntax Support for Silicon-on-Insulator (SOI) Processes

For non-buried oxide layers in silicon-on-insulator (SOI) processes, the following syntax components are available:

Argument	Description
*CC_IN_DESIGN	Indicates to the Parasitic Reporter that the derived layer is to be replaced with the OA layer in the Layer Mapping section of the file. * is the comment character that tells the signoff tool to ignore this line.
non_buried_oxide_layer	(Optional) Names of non-buried oxide layers that are used for connections to substrates in silicon-on-insulator (SOI) processes, such as fully depleted SOI (FDSOI) processes. In the following example, the NB layer is defined as a non-buried oxide layer. Although the example shows only one layer name, you can specify multiple layer names, using a comma between the names. *CC_IN_DESIGN non_buried_oxide_layer NB

Example

```
*BEGIN_OF_LAYER DERIVATION_SECTION

***** Derived ***** Source **** Rules *****

*CC_IN_DESIGN gate PO and OD
*CC_IN_DESIGN poly PO not gate

*CC_IN_DESIGN n_region NP or VTS_N or VTL_N
*CC_IN_DESIGN p_region PP or VTS_P or VTL_P

*CC_IN_DESIGN n_poly poly and n_region
*CC_IN_DESIGN p_poly poly and p_region
*CC_IN_DESIGN n_gate gate and n_region
*CC_IN_DESIGN p_gate gate and p_region

*CC_IN_DESIGN n_diff OD and n_region
*CC_IN_DESIGN p_diff OD and p_region

*CC_IN_DESIGN md_od MD and OD
*CC_IN_DESIGN md_sti MD not md_od

*CC_IN_DESIGN md_od_n md_od and n_region
*CC_IN_DESIGN md_od_p md_od and p_region

*CC_IN_DESIGN n_od_tap n_diff contactless = MD
*CC_IN_DESIGN p_od_tap p_diff contactless = MD
```

```
*CC_IN_DESIGN    n_poly_tap        n_poly        contactless = MP
*CC_IN_DESIGN    p_poly_tap        p_poly        contactless = MP
*CC_IN_DESIGN    n_gate_tap        n_gate        contactless = MP
*CC_IN_DESIGN    p_gate_tap        p_gate        contactless = MP

*END_OF_DERIVATION_SECTION
```

Defining and Mapping Derived Layers

You define derived layers in the layer mapping file to indirectly map StarRC layers to OA layers. See [Derived Layer Definitions](#).

To define and indirectly map the StarRC layers to OA layers:

1. In the layer mapping file, add the following lines to create a new layer derivation section:

```
*BEGIN_OF_LAYER_DERIVATION_SECTION

*END_OF_DERIVATION_SECTION
```

2. Within the layer derivation section, add the appropriate lines for each StarRC layer that does not have a direct mapping to an OA layer.

For the description of the arguments and a full example of a layer derivation section, see [Derived Layer Definitions](#).

3. When you are finished, save the mapping file.
-

Adding Blocking Layers

If the StarRC command file contains the `SKIP_PCELL_LAYERS_FILE` command to specify layers that block extraction, you must add these blocking layers to the layer mapping file. Doing so ensures proper resistance and total capacitance calculations.

In the layer mapping file, you add the blocking layers in the layer derivation section. The syntax of the blocking layers is as follows:

```
*BEGIN_OF_LAYER_DERIVATION_SECTION
*CC_IN_DESIGN blocking_layerName OA_layerName [additional_keywords]
*END_OF_DERIVATION_SECTION
```

For information about the layer derivation section, see [Derived Layer Definitions](#).

To add the blocking layers to the layer mapping file:

1. Open the `star.cmd` file, locate the `SKIP_PCELL_LAYERS_FILE` command, and check the location of the input file that contains the blocking layer information.

In the following example, `skip.pcells_file` is the input to the `SKIP_PCELL_LAYERS_FILE` command.

star.cmd

```
EXTRACTION: RC
REDUCTION: NO
MERGE_VIAS_IN_ARRAY: NO
IGNORE_CAPACITANCE: ALL
COUPLE_TO_GROUND: NO
AUTO_RUNSET: YES
SHORT_PINS: NO
COUPLE_TO_PCELL_PINS: YES AUTOMATIC_CG_HANDLING
SKIP_PCELL_LAYERS_FILE: ../dir/skip.pcell_file
```

2. Open the input file to view the list of blocking layers.

In the following example, `skip.pcells_file` lists seven blocking layers.

skip.pcell_file

```
BLOCKING_LAYERS
block_MOMCAP size 2.080 MOM_PO MOM_OD AA AB M1 M4 CN M4_NOT_CN
block_MIM_ABM5 MIM_PO MIM_OD
block_ACCMOS ACC_PO ACC_OD
block_DZHV DZHV_PO DZHV_OD
block_LA LA_PO LA_OD
block_NMD NMD_PO NMD_OD
block_RESCON CONT CW
```

3. Add the blocking layer names to the layer mapping file:
 - a. In the layer mapping file, locate the layer derivation section. See [Derived Layer Definitions](#).
 - b. In the layer derivation section, add each blocking layer using the following syntax:

```
*CC_IN_DESIGN blocking_layerName OA_layerName [additional_keywords]
```
 - c. At the bottom of the layer mapping file, add a new `remove_layers` section and list the names of the blocking layers that you added in the layer derivation section.

In the following example, the blocking layers from skip.pcell_file are added in the layer derivation section. So, the names of the blocking layers are added in the `remove_layers` section.

layer.map

```
*BEGIN_OF_LAYER_DERIVATION_SECTION
...

*CC_IN_DESIGN block_MOMCAP      CAPDUMMY or MOM
*CC_IN_DESIGN block_MIM_ABM5    AB and M5 and CAPDUMMY and MIM
*CC_IN_DESIGN block_ACCMOS      CAPDUMMY and ACC
*CC_IN_DESIGN block_DZHV        DDUMMY and DZHV
*CC_IN_DESIGN block_LA          DDUMMY and LA
*CC_IN_DESIGN block_NMD         DDUMMY and NMD
*CC_IN_DESIGN block_RESCON      RESDUMMY or RESCON

*END_OF_LAYER_DERIVATION_SECTION

...

remove_layers

block_MOMCAP
block_MIM_ABM5
block_ACCMOS
block_DZHV
block_LA
block_NMD
block_RESCON
```

4. When you are finished, save the mapping file.

Mapping StarRC Conducting Layers

StarRC conducting layers must be added in the layer mapping file to ensure accurate resistance and total capacitance reporting.

In the layer mapping file, the `conducting_layers` section maps OA or derived conducting layers to StarRC conducting layers. The syntax of the `conducting_layers` section is as follows:

```
conducting_layers
<OA_layer or derived_layer> <StarRC_layer>
```

To map the conducting layers in the layer mapping file:

1. In the layer mapping file, add a `conducting_layers` section.
2. In the `conducting_layers` section, add the following line for each StarRC conducting layer:

```
<OA_layer or derived_layer> <StarRC_layer>
```

3. When you are finished, save the mapping file.

For example:

```
conducting_layers
md_n n_md
md_p p_md
npoly p_poly
```

In the example:

- The `md_n`, `md_p`, and `npoly` are derived layers. They are defined in the [Derived Layer Definitions](#) section.
- The `n_md`, `p_md`, and `p_poly` are StarRC layers.

Mapping Via Layers

Vias layers must be added in the layer mapping file to ensure accurate resistance and total capacitance reporting.

In the layer mapping file, the `via_layers` section maps OA or derived via layers to StarRC via layers. The syntax for the mapping is as follows:

```
via_layers
<OA_layer or derived_layer> <StarRC_layer>
```

To add vias in the layer mapping file:

1. In the layer mapping file, add a `via_layers` section.
2. In the `via_layers` section, add the following line for each via:

```
<OA_layer or derived_layer> <StarRC_layer>
```
3. (Resistance reporting only) If fake vias are used to connect contactless layers, add the fake vias in the `via_layers` section.
4. When you are finished, save the mapping file.

For example:

```
via_layers
V1_A V1
V1_B V1
V1_C V1
```

In the example:

- V1_A, V1_B, and V1_C are OA via layers.
- V1 is the StarRC via layer.
- Because the StarRC via layer has direct mapping to the corresponding OA via layers, derived layers are not needed.

(Resistance reporting only) In the following example, because fake vias are used to connect contactless layers, the fake vias are added in the `via_layers` section.

```
*BEGIN_OF_LAYER_DERIVATION_SECTION
*CC_IN_DESIGN N_ODTAP OD type=n contactless_metal=MD
*CC_IN_DESIGN P_ODTAP OD type=p contactless_metal=MD
*END_OF_DERIVATION_SECTION

via_layers
N_ODTAP n_odtap
P_ODTAP p_pdtap
```

In the example:

- N_ODTAP and P_ODTAP layers are derived layers. They are defined in the layer derivation section.
- n_odtap and p_odtap are StarRC layers.
- OD and OD are OA layers.

Adding Layers to Ignore During Capacitance Reporting

Fake vias must be mapped for resistance reporting and can be ignored for capacitance reporting. To ensure that the Capacitance Report ignores fake via layers, you must add an `ignore_cap_layers` section and list the fake via layers.

The syntax of the `ignore_cap_layers` section is as follows:

```
ignore_cap_layers
<OA_fake_via_layer or derived_fake_via_layer>
```

To map the fake vias in the layer mapping file:

1. In the layer mapping file, add an `ignore_cap_layers` section.
2. In the `ignore_cap_layers` section, add the following line for each fake via:

```
<OA_fake_via_layer or derived_fake_via_layer>
```

3. When you are finished, save the mapping file.

For example:

```
*BEGIN_OF_LAYER_DERIVATION_SECTION
*CC_IN_DESIGN N_ODTAP OD type=n contactless_metal=MD
*CC_IN_DESIGN P_ODTAP OD type=p contactless_metal=MD
*END_OF_DERIVATION_SECTION
```

```
via_layers
N_ODTAP n_odtap
P_ODTAP p_pdtap
```

```
ignore_cap_layers
N_ODTAP
P_ODTAP
```

In the example:

- `N_ODTAP` and `P_ODTAP` are derived layers, and they are mapped to the `OD OA` layer in the layer derivation section.
- `n_odtap` and `p_odtap` are StarRC layers. They are mapped to the corresponding derived layers in the via layers section.
- The `ignore_cap_layers` section lists `N_ODTAP` and `P_ODTAP` so that the Capacitance Reporter will ignore them during the reporting process.

9

Managing StarRC Version and License Use

The Custom Compiler environment provides features that let you optimize StarRC tool version and license use.

This chapter describes the following:

- [Specifying the Version of StarRC Tool](#)
- [Stopping StarRC Jobs](#)

Specifying the Version of StarRC Tool

You can switch to different versions of the StarRC tool for different design phases. For optimal results, you should always run the latest version of the StarRC tool for in-design electrical reporting in the Custom Compiler environment. Later, you can switch to an older, well-validated version of StarRC tool for layout parasitic extraction (LPE).

To use a specific version of the StarRC tool with the Custom Compiler in-design electrical reporters:

- At the Linux prompt, set the following environment variable to define the path of the StarRC executable file: `SYNOPSISYS_CUSTOM_IN_DESIGN_STARRC_PATH`.

For example:

```
% setenv SYNOPSISYS_CUSTOM_IN_DESIGN_STARRC_PATH /global/apps/
starrc_2019.12-SP5-3/bin/:/global/apps/starrc_2020.09/bin
```

The `leStarRCEnvSettingFile` Tcl preference is used to define the path of the file to be sourced before starting the StarRC tool. Do not use it to set the `$PATH`. Instead, use the `SYNOPSISYS_CUSTOM_IN_DESIGN_STARRC_PATH` environment variable.

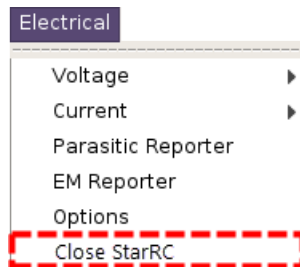
Stopping StarRC Jobs

StarRC licenses are checked out when you use the integrated StarRC engine for in-design resistance and capacitance reporting. The Custom Compiler environment provides

features that let you optimize license use. For example, you can release licenses that are used by sessions that have been inactive for long periods of time.

To stop the StarRC tool and release the licenses, use one of the following methods:

- Close the Custom Compiler session.
- In the Job Monitor, stop the StarRC job that is currently in progress. See the “Job Monitor Tool” topic in the Custom Compiler Environment User Guide.
- In the Layout Editor, choose **Electrical > Close StarRC**.



- Wait for the Custom Compiler tool to time out.

To specify the length of time to wait for input before releasing the licenses, set the `dbIdleLicenseTimeout` Tcl preference.

10

Annotating Voltage Data for VDRC

This chapter explains how to annotate design schematics and layouts with voltage data.

This chapter contains the following topics:

- [About Voltage Data and VDRC](#)
- [Measuring Voltage Data Using the PrimeWave Design Environment](#)
- [Measuring and Saving Voltage Data Using the Schematic Editor](#)
- [Annotating Schematics With Voltage Data](#)
- [Viewing Voltage Annotations in Schematics](#)
- [Editing Voltage Annotations in Schematics](#)
- [Annotating Layouts With Voltage Data](#)
- [Viewing Voltage Data in the Layout](#)
- [Editing Voltage Data in the Layout](#)
- [Verifying Edited Voltage Data](#)
- [Enabling Sync to VDRC Assistant for User-defined Flag](#)
- [Checking for Voltage-Dependent Spacing Violations as You Edit the Layout](#)
- [Creating and Removing Physical Annotations in the Layout Editor](#)
- [Checking Physical Voltage Annotations in the Layout Editor](#)
- [Troubleshooting Physical Annotation Errors](#)

About Voltage Data and VDRC

Voltage-dependent rule checking (VDRC) is a physical verification rule check to ensure that the quality of electrical signals is preserved while the minimum amount of area is used for layout. The rules rely on the voltage differences between nets to determine the minimum space required for the nets.

Traditionally, VDRC is performed at the end of design processes, when layout-versus-schematic (LVS) comparisons map voltages from the design schematic to design layout. However, using Custom Compiler tools and features, you can measure voltages directly from simulations and map the voltage data to design schematics and layout for partially completed designs. Doing so helps you catch violations early in the design phases, thus ensuring an efficient design process while reducing the overall design timeline.

This topic describes the following:

- [Tools for Measuring and Annotating Voltage Data](#)
- [Voltage Data Sources and Annotation Methods](#)
- [Workflow for Annotating Voltage Data \(Simulated Voltage Data and SDL\)](#)
- [Workflow for Annotating Voltage Data \(Manual Entries\)](#)
- [Workflow for Annotating Voltage Data \(Voltage Flags\)](#)

Tools for Measuring and Annotating Voltage Data

You can use the following Custom Compiler tools when annotating designs with voltage data:

Tool	Description
Custom Compiler PrimeWave Design Environment	Measures and saves voltage information from design simulations to a <code>vdrc.db</code> file.
Schematic Editor	Processes voltage data from waveform files and saves them in a <code>vdrc.db</code> file.
Electrical Tools	Annotates design schematics or layouts with voltage data generated in the PrimeWave Design Environment or from waveform files.
VDRC Assistant in the Schematic Editor	Allows you to view and write node voltage data to nets in design schematics.
VDRC Assistant in the Layout Editor	Allows you to view and write node voltage data to nets in design layouts.
Tcl API	Runs the commands for obtaining node voltage data from simulations and saving the data to a VDRC file.

Voltage Data Sources and Annotation Methods

Two key components of the voltage annotation process are:

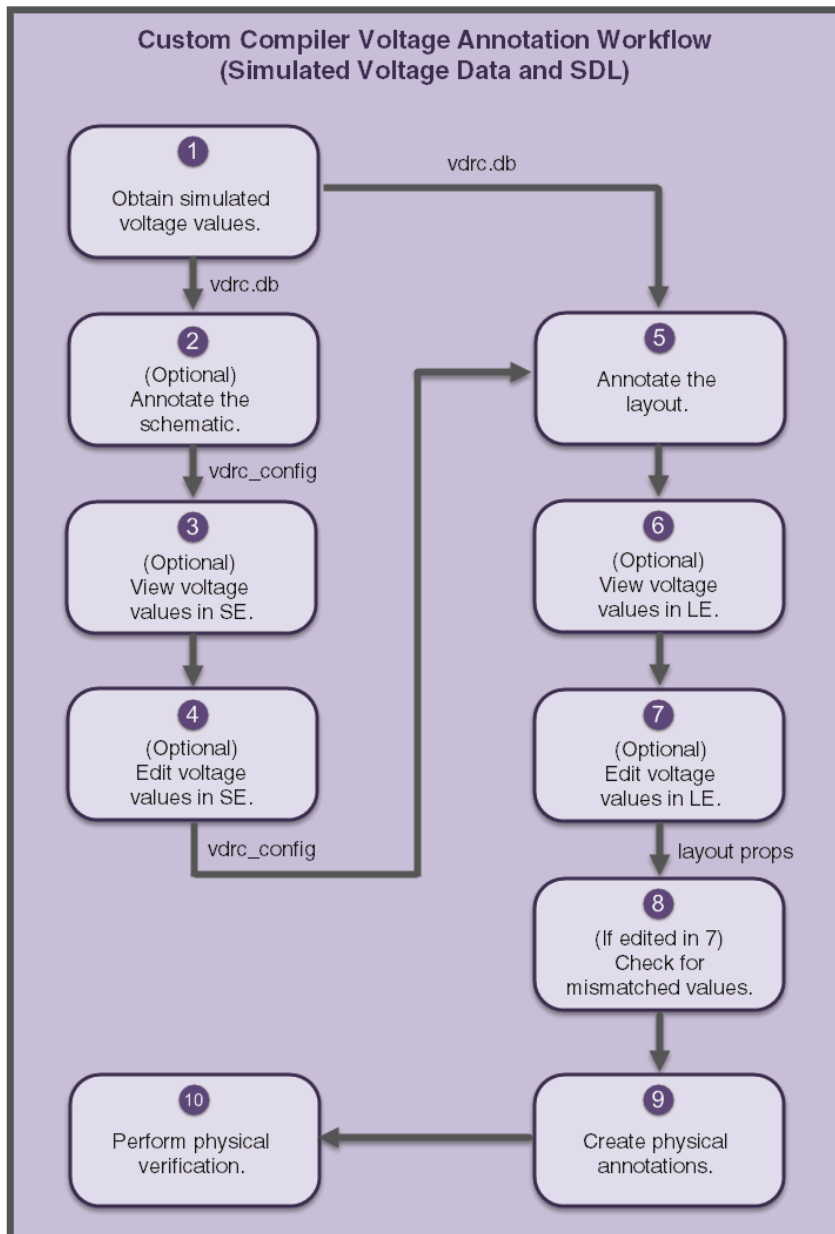
Component	Description
vdrc.db file	Contains voltage data from simulation runs.
vdrc_config cellview	Points to the <code>vdrc.db</code> file, and stores post-simulation modifications, including new entries that you make manually. The Custom Compiler electrical tools use the data in the <code>vdrc_config</code> cellview to annotate design schematics and layouts. The <code>vdrc_config</code> cellview should be treated as the master source of voltage data. Layout designers should annotate layouts using only the <code>vdrc_config</code> cellview from the schematic designers.

The annotation method you use can depend on the source of the voltage data:

Voltage Data Source	Annotation Method
Simulations	You can run simulations that produce estimated voltage values. The simulated voltage values are automatically saved in the <code>vdrc.db</code> file. After the simulation, you can create a <code>vdrc_config</code> cellview, which references the <code>vdrc.db</code> file.
Manual modifications	You can modify (or override) the simulated voltage values to provide more realistic values. You save the modifications in the <code>vdrc_config</code> cellview with the simulated values.
Manual entries only	Instead of obtaining voltage values through simulations, you can enter the voltage values manually using one of the following methods: • Create a new <code>vdrc_config</code> cellview, and type the maximum and minimum voltage values for each net. • Add voltage flags, which contain minimum and maximum voltage information. You can import the voltage data into the <code>vdrc_config</code> cellview. You can also export the voltage data from <code>vdrc_config</code> cellview to voltage flags.

Workflow for Annotating Voltage Data (Simulated Voltage Data and SDL)

The following workflow shows the steps for obtaining voltage data from simulations and manually modifying the voltage annotations. Follow this workflow if you use schematic-driven layout (SDL) to automate the layout tasks.



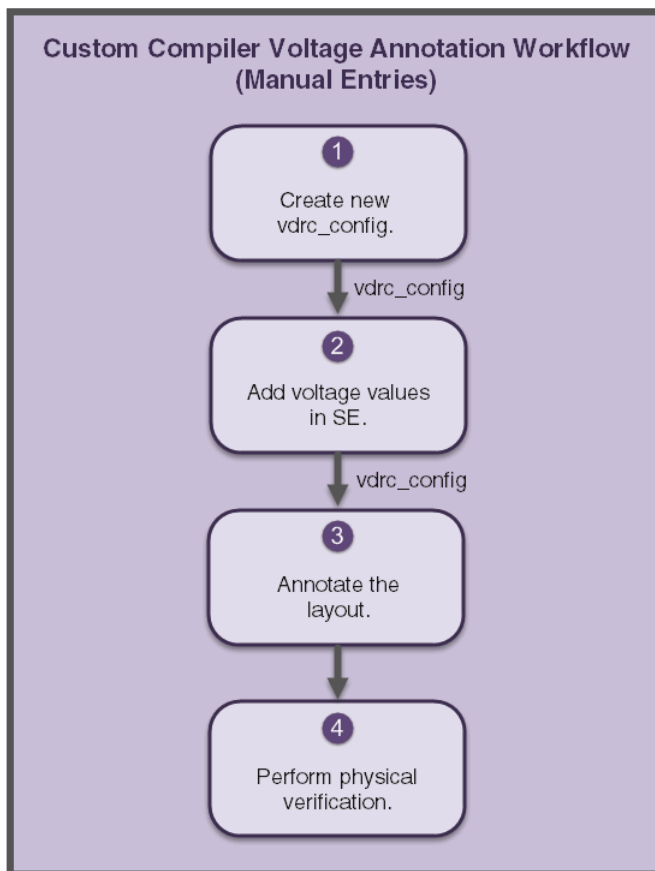
Step		Description
1	Obtain simulated voltage values.	Obtain voltage measurements using one of the following: • <i>PrimeWave Design Environment</i>: Runs simulations to measure the voltages, and then save the data to a <code>vdrc.db</code> file. See Measuring Voltage Data Using the PrimeWave Design Environment. • <i>Schematic Editor</i>: Processes voltage data from existing waveform files, and then creates the voltage database. See Measuring and Saving Voltage Data Using the Schematic Editor. • <i>sa::createVDRCDDB</i>: Post-processes simulation results and saves the voltage data to a file. You can create a Tcl script to include the <code>sa::createVDRCDDB</code> command. See the <i>Custom Compiler Tcl Reference Manual</i>.
2	(Optional) Annotate the schematic.	In the Schematic Editor, use the electrical tools to annotate the design schematic with voltage measurements obtained from step 1. You can also use the Schematic Editor to create an empty <code>vdrc_config</code> cellview in preparation for manual voltage value entries. See Annotating Schematics With Voltage Data . Perform this step only if you want to visualize the voltage data in the design schematic.
3	(Optional) View voltage values in schematics.	In the Schematic Editor (SE), use the VDRC Assistant to view the voltage values or the voltage heat map. See Viewing Voltage Annotations in Schematics .
4	(Optional) Edit voltage values in schematics.	In the Schematic Editor (SE), edit voltage annotations in the design schematic to provide more realistic values. You can also use this method to manually add voltage values in the design schematic instead of automating the process in step 2. See Editing Voltage Annotations in Schematics .
5	Annotate the layout.	In the Layout Editor (LE), use the electrical tools to annotate the design layout with voltage information from steps 1 through 4. See Annotating Layouts With Voltage Data .
6	(Optional) View voltage values in the layout.	In the Layout Editor (LE), view the voltage data on the canvas. In addition, you can use the VDRC Assistant to view the voltage heat map. See Viewing Voltage Data in the Layout .
7	(Optional) Edit voltage values in the layout.	In the Layout Editor (LE), edit the voltage values using either the VDRC Assistant or the Property Editor. See Editing Voltage Data in the Layout .
8	(If edited in 7) Verify the edited values.	In the Layout Editor, verify the edited voltage values. See Verifying Edited Voltage Data . This step is only necessary if you have edited voltage values in step 7.

Step		Description
9	Create physical annotations.	In the Layout Editor, create foundry-specific physical annotations so that physical verification tools can process the voltage information in the design. See Creating and Removing Physical Annotations in the Layout Editor . If changes occur late in the design process, you can use the Check Annotations command to catch discrepancy errors. See Checking Physical Voltage Annotations in the Layout Editor .
10	Perform physical verification.	See Design Checking and Physical Verification.

Workflow for Annotating Voltage Data (Manual Entries)

The following workflow shows the steps for manually annotating voltage information in your designs. Follow this workflow if:

- You perform all layout tasks manually and do not use schematic-driven layout (SDL).
- You do not want to use simulated voltage data.
- You want to create a `vdrc_config` cellview and type the voltage values for the nets manually.

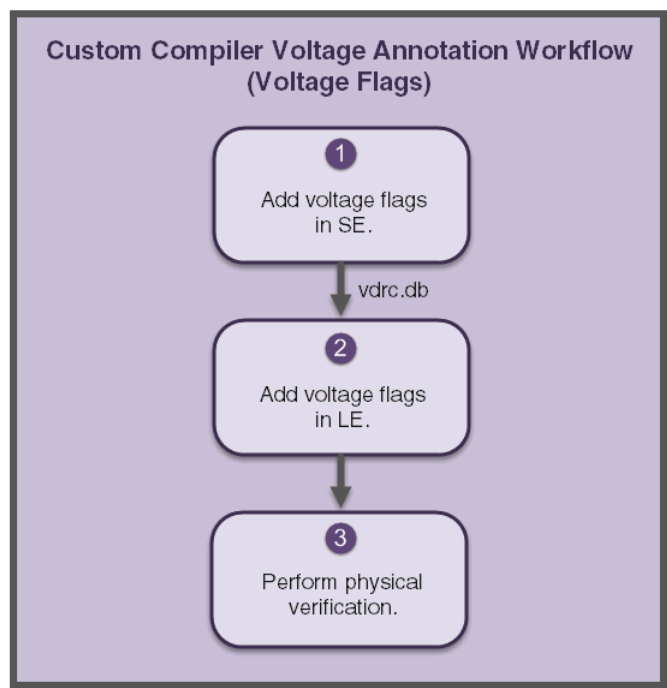


Step	Description	
1	Create new <code>vdrconfig</code> .	Create an empty <code>vdrconfig</code> cellview so that you can manually populate it with voltage data. See Annotating Schematics With Voltage Data .
2	Add voltage values in SE.	In the Schematic Editor (SE) VDRC Assistant, manually add the voltage values using one of the available editing methods described in Editing Voltage Annotations in Schematics . The voltage data is saved in the <code>vdrconfig</code> cellview you created in step 1.
3	Annotate the layout.	In the Layout Editor, annotate the layout using the data in the <code>vdrconfig</code> cellview. See Annotating Layouts With Voltage Data .
4	Perform physical verification.	See Design Checking and Physical Verification.

Workflow for Annotating Voltage Data (Voltage Flags)

The following workflow shows the steps for manually annotating voltage information in your designs using voltage flags only. Follow this workflow if:

- You perform all layout tasks manually and do not use schematic-driven layout (SDL).
- You do not want to use simulated voltage data.
- You want to add voltage values using voltage flags.



Step	Description	
1	Add voltage flags in SE.	In the Schematic Editor (SE), annotate the schematic by adding voltage flags. See Adding Voltage Annotations Using Voltage Flag Symbols .
2	Add voltage flags in LE.	In the Layout Editor (LE), annotate the layout by adding voltage flags. See Adding Voltage Annotations Using Voltage Flags .
3	Perform physical verification.	See Design Checking and Physical Verification.

Measuring Voltage Data Using the PrimeWave Design Environment

To annotate the design schematic and layout with simulated voltage data, you must first use one of the following methods to obtain the voltage data and save the information to a VDRC file (`vdrc.db`):

Method	Description
PrimeWave Design Environment	Runs simulations to measure the voltages, and then saves the voltage data to the VDRC files.
Schematic Editor	Processes voltage data from existing waveform files, and then saves the voltage data to VDRC files. Use this method if you have an existing waveform file that contains all the voltage data for the design.
Tcl API	Runs commands, such as the <code>sa::createVDRCDDB</code> Tcl command, to obtain node voltage data from simulations and saves the data to a VDRC file.

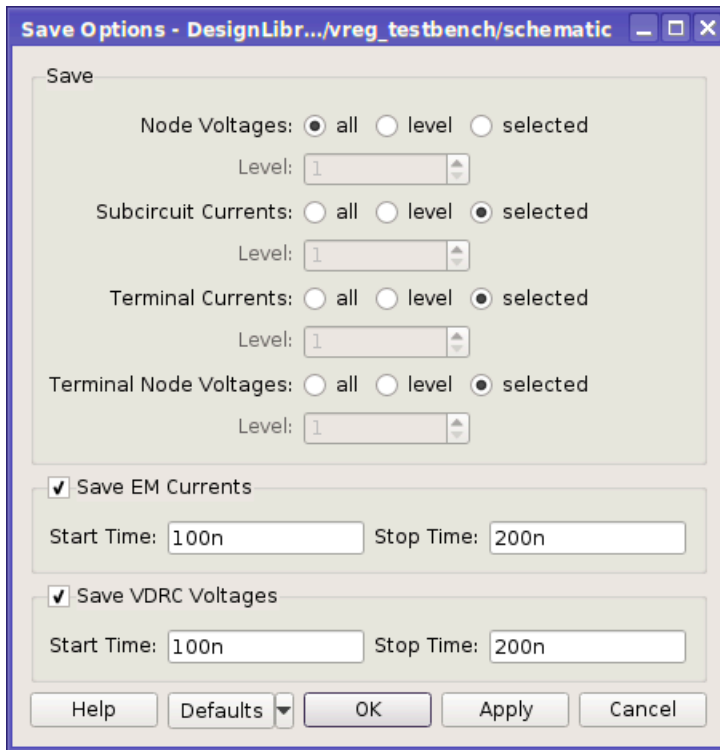
This topic describes how to run the PrimeWave Design Environment to obtain voltage measurements only. For instructions on how to run the simulators for other uses, see [Setting Up Testbenches](#).

For information on using the Schematic Editor to obtain the voltage data, see [Measuring and Saving Voltage Data Using the Schematic Editor](#).

For information about the `sa::createVDRCDDB` Tcl command, see the *Custom Compiler Tcl Reference Manual*.

To run the PrimeWave Design Environment and save the current data:

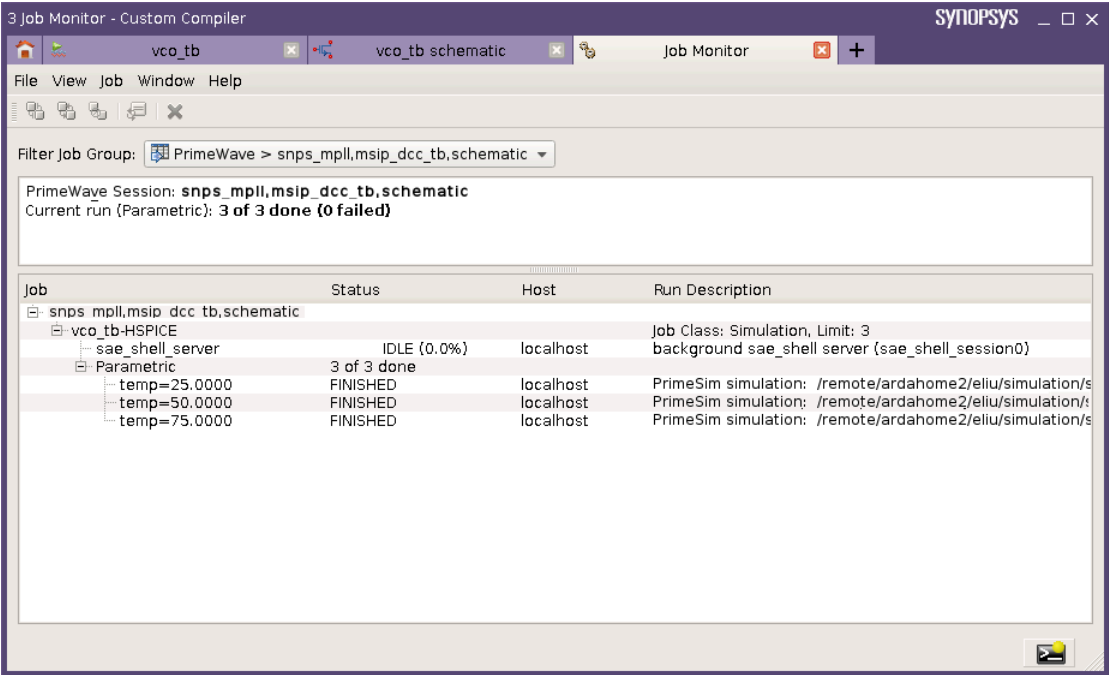
1. Open the design schematic. See [Open Designs](#).
2. In the Schematic Editor, choose **Tools > PrimeWave** to open the PrimeWave Design Environment. The design testbench displays.
3. In the PrimeWave Design Environment, set up a new or load an existing simulation state as you would normally.
4. In the PrimeWave Design Environment, choose **Outputs > Save Options**.



In the **Save Options** dialog box:

- a. Select **Save VDRC Voltages**. In the **Voltage & Independent Source Currents** area, the **Save** selection automatically changes to **all**.
 - b. In the **Start Time** and **Stop Time** boxes, specify the time window within which the PrimeWave Design Environment will obtain the minimum and maximum voltage values while post-processing the waveforms. Make sure you specify the abbreviated unit of time with the values (for example, 1n).
 - c. Select the remaining settings.
 - d. When you are finished, click **OK**. The voltage data settings are saved.
5. In the PrimeWave Design Environment, confirm or set the simulation parameters, then choose **Simulation > Netlist and Run**. The design simulation starts. The Job Monitor opens and displays the simulation status.

When the simulation is finished, the WaveView window opens and displays plotted waveforms. The Job Monitor shows that the simulation has ended and displays run messages.



6. Verify that the PrimeWave Design Environment created a `vdrc.db` file, which contains the voltage measurements, the minimum and maximum values for the time interval you specified, and the waveform information.

Measuring and Saving Voltage Data Using the Schematic Editor

To annotate the schematic and layout with simulated voltage data, you must first use one of the following methods to obtain the voltage data and save the information to a VDRC file (`vdrc.db`):

Method	Description
Custom Compiler PrimeWave Design Environment	Runs simulations to measure the voltage differences, and then saves the voltage data to <code>vdrc.db</code> files.
Schematic Editor	Processes voltage data from existing waveform files, and then saves the voltage data to <code>vdrc.db</code> files. Use this method if you have an existing waveform file that contains all the voltage data for the design.
Tcl API	Runs commands, such as the <code>sa::createVDRCDDB</code> Tcl command, to obtain node voltage data from simulations and saves the data to a VDRC file.

This topic describes how to use the Schematic Editor to obtain voltage data. For instructions on how to use other features, see [Schematic Editor Description](#).

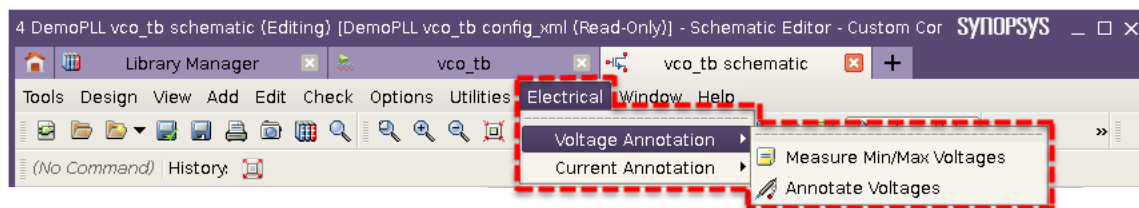
For information on using the PrimeWave Design Environment to obtain the voltage data, see [Measuring Voltage Data Using the PrimeWave Design Environment](#).

For information about the `sa::createVDRCDB` Tcl command, see the *Custom Compiler Tcl Reference Manual*.

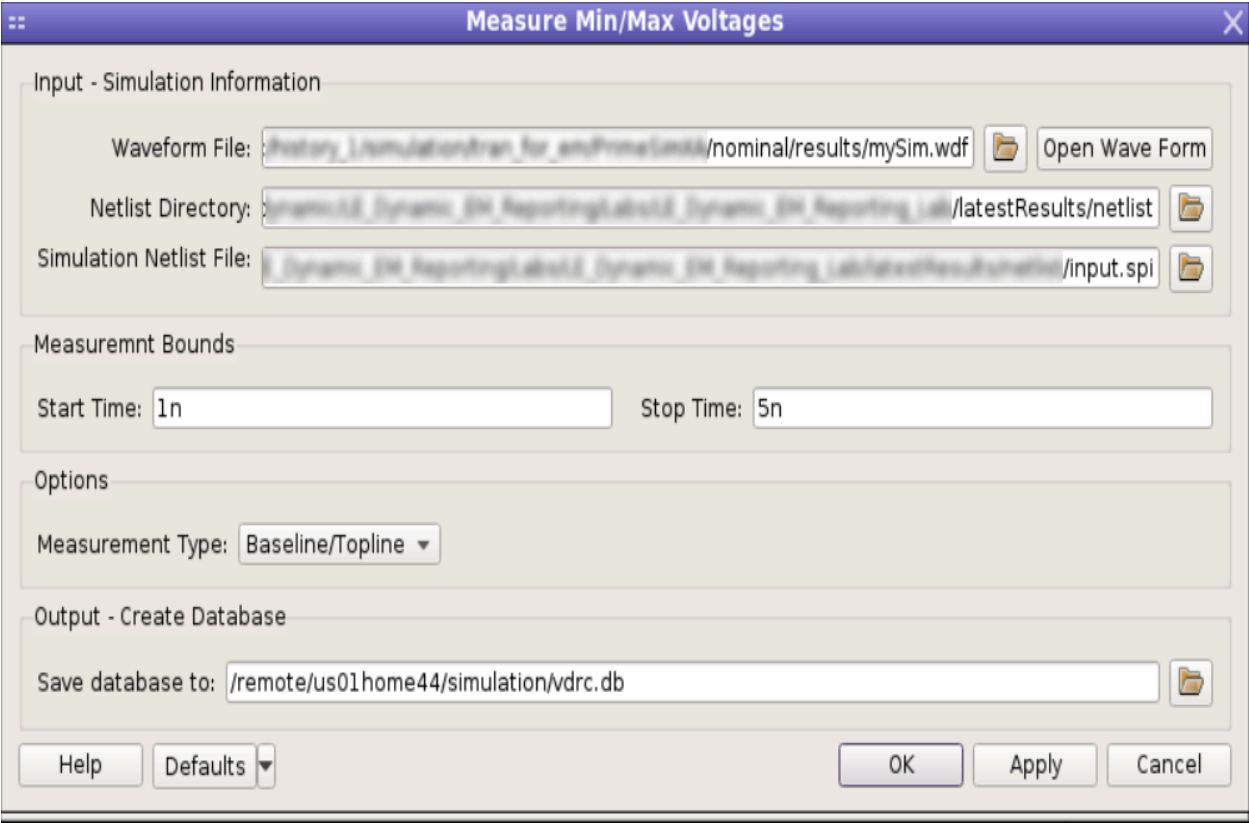
Before you start, make sure you have the waveform file that contains all the voltage data for the design. In addition, you must have the corresponding simulation netlist file and netlist directory ready. The netlist directory should contain name mapping information and the simulation input file (SPICE netlist file). If you do not have these files, run a simulation according to the instructions in [Measuring Voltage Data Using the PrimeWave Design Environment](#).

To use the Schematic Editor to obtain the voltage data:

1. Open the design schematic. See [Open Designs](#).
2. In the Schematic Editor, choose **Tools > Electrical**. The **Electrical** menu appears on the menu bar.



3. Choose **Electrical > Voltage Annotation > Measure Min/Max Voltage**. The **Measure Min/Max Voltage** dialog box opens.



- 4. In the **Input** section:
 - a. Select the **Waveform File**, an HSPICE WDF, PSF, or FSDB file.
 - b. Select the **Netlist Directory**.
 - c. Select the **Simulation Netlist File**, a .spi or .sp file in the netlist folder.
- 5. In the **Options** section, choose the **Measurement Type** for minimum voltage ($V_{\min}$) and maximum voltage ($V_{\max}$):

Argument	Description
Min/Max (default)	In typical wave plots, where: $V_{\min}$ = lowest of the low voltage values. $V_{\max}$ - highest of high voltage values.

Argument	Description
Baseline/Topline	In voltage plots that form square-shaped waves, a histogram is used to determine the low-voltage bin and the high-voltage bin. $V_{\min}$ = baseline level voltage value, or the low-voltage value that occurred the most number of times within the low-voltage bin. $V_{\max}$ = topline level voltage value, or the high-voltage value that occurred the most number of times within the high-voltage bin.

6. In the **Measurement Bounds** section, set the **Start Time** and **Stop Time** to specify the time window within which the PrimeWave Design Environment should obtain the minimum and maximum voltage values while post-processing the waveforms.

Make sure you specify the abbreviated unit of time with the values (for example, 1n).

7. In the **Output - Create Database** section, specify the location for the `vdrc.db` file.
8. When you are finished, click **OK**. The Schematic Editor creates the `vdrc.db` file and saves it in the folder you specified.

Annotating Schematics With Voltage Data

After you obtain the simulated voltage data (vdrc.db), you can annotate the design schematic with the voltage values.

Perform this step if:

- You have simulated voltage data, and you want to visualize the values in the design schematic. Otherwise, proceed to [Annotating Layouts With Voltage Data](#).

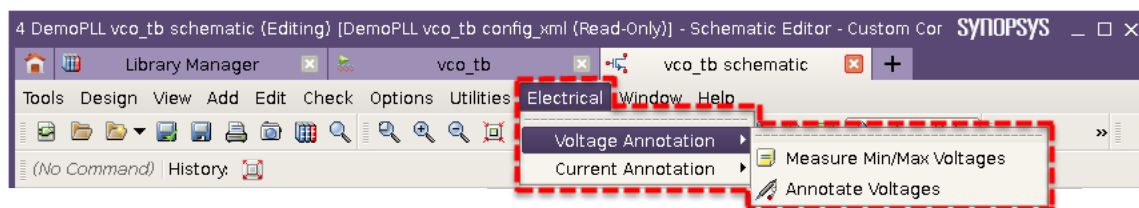
You use the electrical tools to annotate the design schematic. The tools use information in the `vdrc.db` file that was created in [Measuring Voltage Data Using the PrimeWave Design Environment](#) or [Measuring and Saving Voltage Data Using the Schematic Editor](#).

- You do not want to use simulated voltage data. You want to create a new `vdrc_config` cellview that is empty so that you can manually enter voltage values.

After you create the `vdrc_config` cellview, see [Editing Voltage Annotations in Schematics](#) for instructions on how to add voltage values manually.

To annotate the design schematic with voltage data:

1. Open the design schematic. See [Open Designs](#).
2. In the Schematic Editor, choose **Tools > Electrical**. The **Electrical** menu appears on the menu bar.



3. Choose **Electrical > Voltage Annotation > Annotate Voltages**. The **Annotate Voltages** dialog box opens. The **Library**, **Cell**, and **View** fields are prepopulated.

If you are creating a new `vdrc_config` cellview and you want to manually add voltage data, skip to [Step 10](#).

Annotate Voltages

Current CellView

Library: demoPLL_ref10

Cell: LEVELCONV_l2h_reset

View: schematic

Configuration

View: vdrc_config

☒ Copy Simulation Databases ☒ Preserve Overrides

Save Load

Database

Root directory: ./simulation

Get Voltage Data Clear Table

	File Name	Testbench	Simulation Netlist	Occurrence Path
1	...E/nominal/results/vdrc.db	...ef10_tb/L2H_tb/schematic	...E/nominal/netlist/input.spi	/I0

Annotation Type: Annotate One Result(Select one row)

Help OK Apply Cancel

- If you have an existing annotation setup and want to use the existing `vdrc_config` cellview, click **Load** in the **Configuration** area to populate the fields in the dialog box, and then skip to [Step 7](#).
- (If you have simulated voltage data) Expand the **Database** section. In the **Root directory** box, type the directory in which the voltage data are stored. In the **Get Voltage Data** menu, select whether to get the data **From .db Files Only** or **From Any Database File**, and then click **Get Voltage Data**. The list of voltage datasets that are relevant to the opened design appears in the dialog box.

To narrow the search, in the **Get Voltage Data** menu, choose one or more of the following:

Get Voltage Data

- ☒ From .db Files Only
- ☐ From Any Database File
- ☒ For Current Cellview Only
- ☐ For Any Cellview

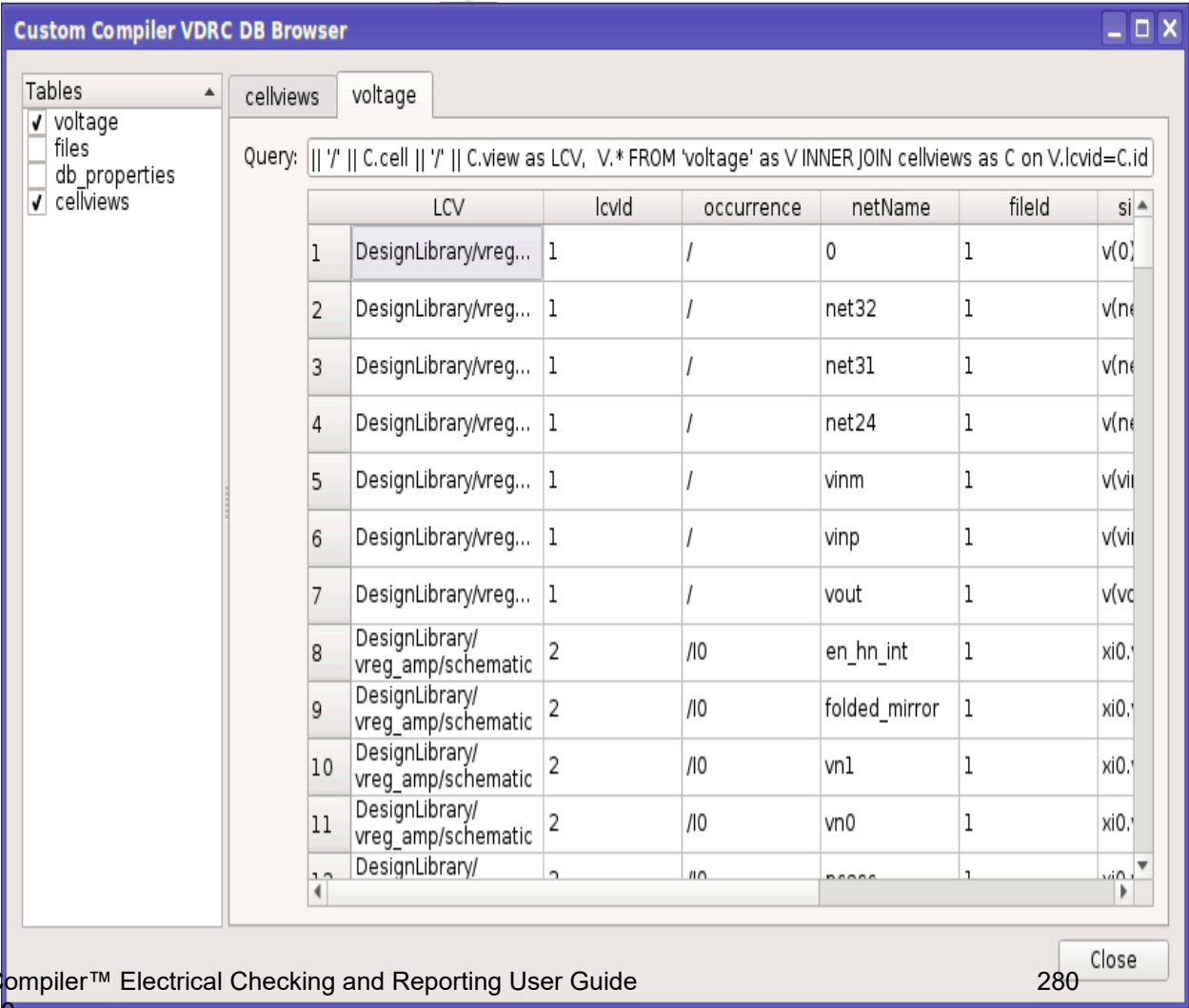
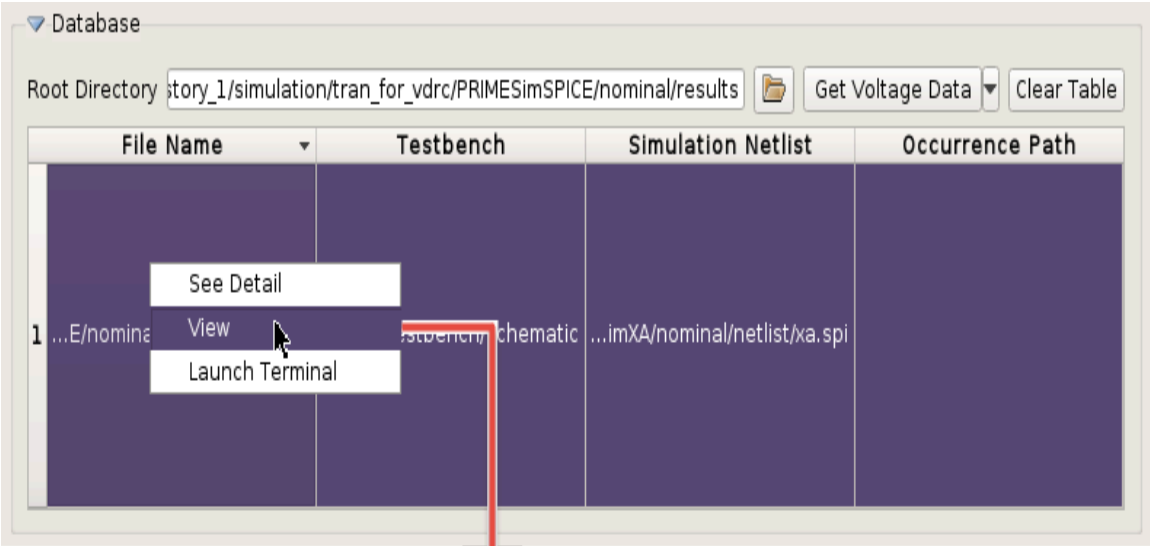
- **From .db Files Only:** Searches for .db files only.
- **From Any Database File:** Searches for all SQLite3 files.

- **For Current Cellview Only:** Searches vdrc.db files that contain data for the current cellview.
 - **For Any Cellview:** Searches all vdrc.db files. When you select this option, the **Occurrence Path** column in the **Database** table remains empty.
6. (If you have simulated voltage data) In the **Annotation Type** list, choose one of the following:

Annotation Type	Description
Annotate One Result (Select One row)	Only the data from one dataset is used for the annotation. You will be permitted to select only one dataset from the list in Step 7 .
Worst Case (Across All)	Only the highest maximum voltage value and the lowest minimum voltage value from across all available voltage datasets for the design are used for the annotation. You will not be permitted to select any dataset in Step 7 .
Worst Case (Across Selected Rows)	Only the highest maximum voltage value and the lowest minimum voltage value from across all the datasets are used for the annotation. You will be permitted to select one or more datasets in the list in Step 7 .

7. (If you have simulated voltage data) Select the voltage datasets you want to use for the annotation process.

If you want to view the contents of a voltage dataset before you start the annotation, right-click the dataset **File Name**, and choose **View**. The **VDRC DB Browser** dialog box opens and displays the dataset contents. When you are finished, close the dialog box.



Your ability to select voltage datasets depends on the **Annotation Type** you chose in [Step 6](#). If you selected **Worst Case (across all)**, you will not be permitted to select any dataset.

8. (Optional, and if you have simulated voltage data) In the **Configuration** area, select **Preserve Overrides** if you want to retain the modified (override) voltage values from [Editing Voltage Annotations in Schematics](#).

Select this option if you plan to annotate the schematic using both the original voltage values from the simulation and the modified (override) voltage values. Clear the check box if you plan to annotate using only the original voltage values from the simulation. (You would not have override values if this is the first iteration in the annotation process.)

9. To save the annotation setup (the voltage datasets from the simulations, the annotation type, and override values), in the **Configuration** area:

- a. In the **View** list, choose **vdrc_config**.
- b. Select **Copy Simulation Databases**, if you want to copy all VDRC datasets listed in the table to a local self-contained `.../databases` directory. Having the simulation datasets locally ensures their availability when you reload the `vdrc_config` cellview.

Tcl Preference: `!eCopyElectricalSimulationToConfig`

- c. Click **Save**.

If you selected **Copy Simulation Databases**, the copy operation deletes any existing local datasets, copies all simulation datasets to the specified config view, and updates the `config.xml` file to reference the local directory.

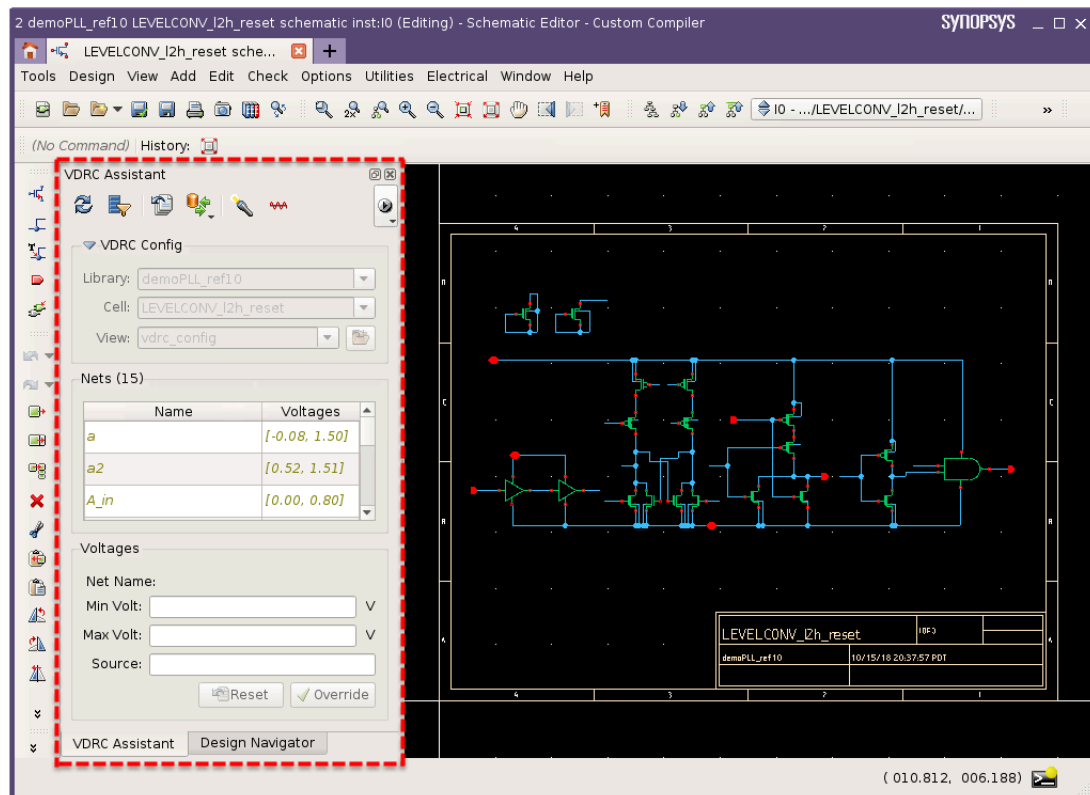
If you did not select **Copy Simulation Databases**, the copy operation deletes any existing local datasets, and updates the `config.xml` file to reference the original simulation directories.

You can reload ([Step 4](#)) and use the same setup later in the Schematic Editor. You can also load this setup in the Layout Editor for annotating the design layout. Doing so allows you to proceed to annotate the layout without having to re-enter the annotation settings.

10. When you are finished, click **OK** or **Apply**.
 - The VDRC Assistant appears on the left side of the window so that you can view and edit the voltage annotations. See [Editing Voltage Annotations in Schematics](#).
 - A `vdrc_config` cellview is created.

(If you have simulated voltage data) The `vdrc_config` cellview contains the link to the `vdrc.db` file, which contains the original voltage values from the simulation. If you selected **Preserve Overrides** in [Step 8](#), the modified (override) values are also stored in the `vdrc_config` cellview.

The annotations are added hierarchically. To view the annotations on lower levels, navigate to those levels.



(If you do not have simulated voltage data) An empty `vdrc_config` cellview is created. You can use the VDRC Assistant to manually add voltage values. See [Editing Voltage Annotations Using the VDRC Assistant](#).

11. To clear the existing annotation and re-annotate using different data selections, click **Reset** in the VDRC Assistant, and then repeat steps 4 through 10. You can re-annotate as many times as desired.

Caution:

Clearing the existing annotation deletes any edited values you provided in the [VDRC Assistant](#).

Viewing Voltage Annotations in Schematics

You use the Schematic Editor VDRC Assistant to view and edit voltage annotations.

This topic describes the following for viewing the schematic voltage data:

- [Before You Start](#)
- [Displaying the Schematic Editor VDRC Assistant](#)
- [Viewing Voltage Data in the VDRC Assistant](#)
- [Viewing the Voltage Heat Map](#)

Before You Start

To optimize the viewing of the schematic voltage annotations, review the prerequisites described in this topic.

If you have voltage annotations stored for different cells at lower levels of the design, and you want to automatically load those `vdrc_config` cellviews (voltage data) as you descend the design:

- Set the `vdrcAutoLoadConfig` Tcl preference to true. (It is false by default.)
- Make sure the voltage annotations were added according to the instructions in [Annotating Schematics With Voltage Data](#).
- The `vdrc_config` cellview names must all be identical. For example:

	Top-Level Design vdrc_config cellview	Level 1 Cell vdrc_config cellview	Level 2 Cell vdrc_config cellview
Library	Library_A	Library_A	Library_A
Cell	TopCell	Level1Cell	Level2Cell
View	xyz	xyz	xyz

See Also

- [Displaying the Schematic Editor VDRC Assistant](#)
- [Viewing Voltage Data in the VDRC Assistant](#)
- [Viewing the Voltage Heat Map](#)

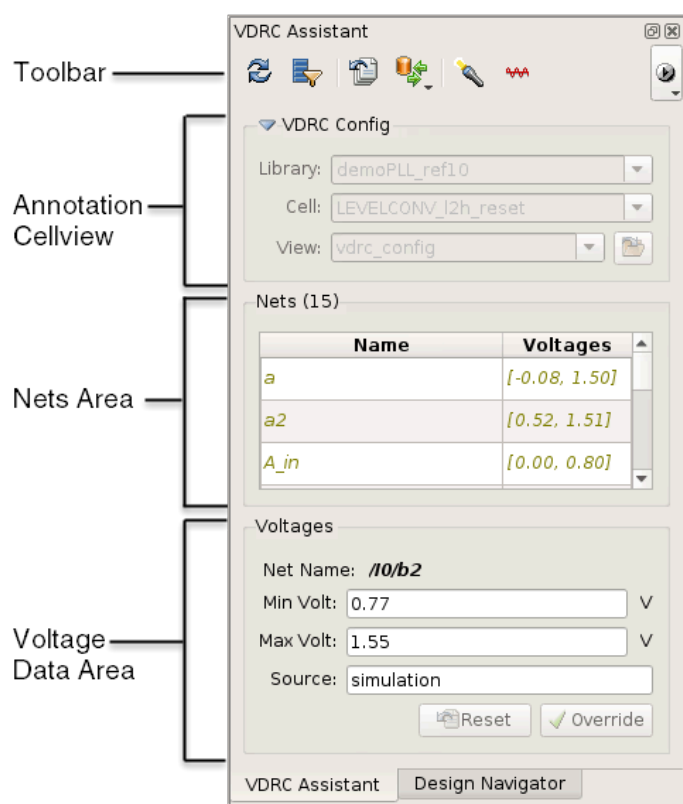
Displaying the Schematic Editor VDRC Assistant

The Schematic Editor VDRC Assistant is a graphical user interface that lets you view voltage annotations in the design schematic. This topic provides a high-level overview of the assistant.

The VDRC Assistant appears when you click **OK** or **Apply** in the **Annotate Voltages** dialog box. Alternatively, you can do the following at any time to display the assistant:

- In the Schematic Editor, choose **Window > Assistants > VDRC Assistant**.

The VDRC Assistant consists of the following:



See Also

- [Before You Start](#)
- [Viewing Voltage Data in the VDRC Assistant](#)
- [Viewing the Voltage Heat Map](#)

Viewing Voltage Data in the VDRC Assistant

The Schematic Editor VDRC Assistant lets you select schematic voltage config views and view the voltage data. This topic describes how to use the assistant.

If you want to automatically load lower-level cell config views as you descend the design, see [Before You Start](#).

To view the voltage data in the VDRC Assistant:

1. Expand the **VDRC Config** area and make sure you have the correct `vdrc_config` cellview open.

The `vdrc_config` cellview contains references to the voltage datasets that you selected for the annotation. The cellview also contains the annotation setup, such as the annotation type information.

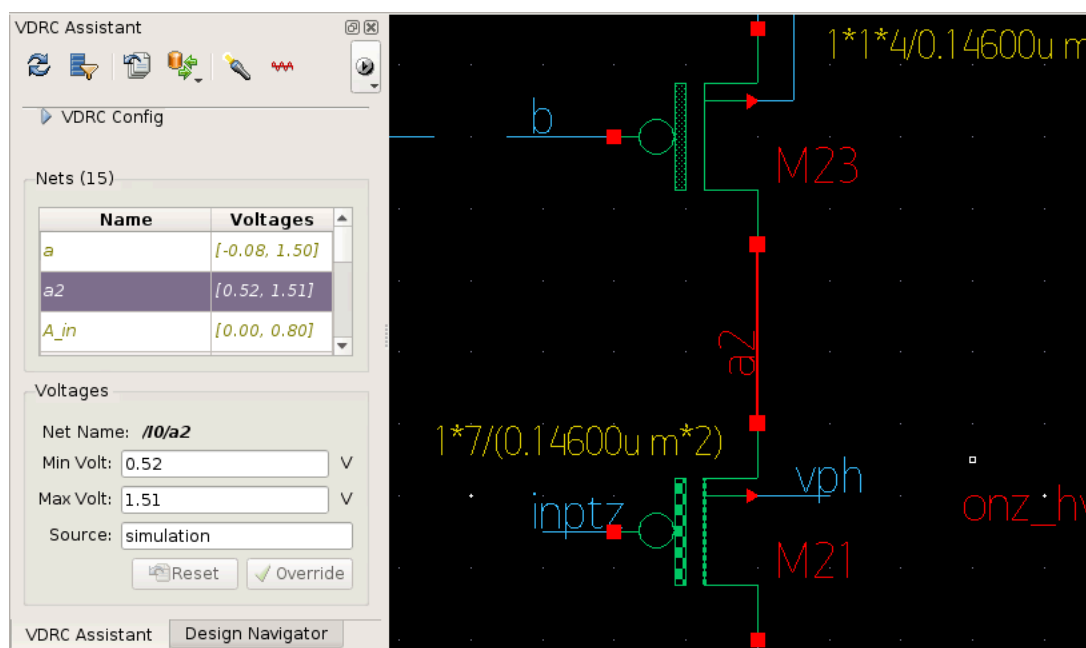
If you are viewing the voltage data immediately after the annotation process ([Annotating Schematics With Voltage Data](#)), the `vdrc_config` cellview is automatically loaded for you.

To view a different `vdrc_config` cellview, choose **Electrical > Voltage Annotation > Annotate Voltages**. See [Annotating Schematics With Voltage Data](#).

2. In the **Nets** area, locate the net of interest. The **Voltage** column displays the corresponding voltage values: (min_voltage, max_voltage). The green font indicates that they are the original values from the simulation run.

If the Nets list is too long, or if you want to view the net on the canvas, you can use one of the following methods to locate the net of interest:

- To filter the **Nets** list, click **Filter Nets by Name**  on the toolbar, and then type the filtering criteria in the filter box that appears.
- Double-click a net name in the **Nets** area to zoom in and display the net in the viewport.
- Select a net on the canvas. The selected net is highlighted on the canvas. The selection is synchronized so that the net name is also highlighted in the VDRC Assistant **Nets** area.



See Also

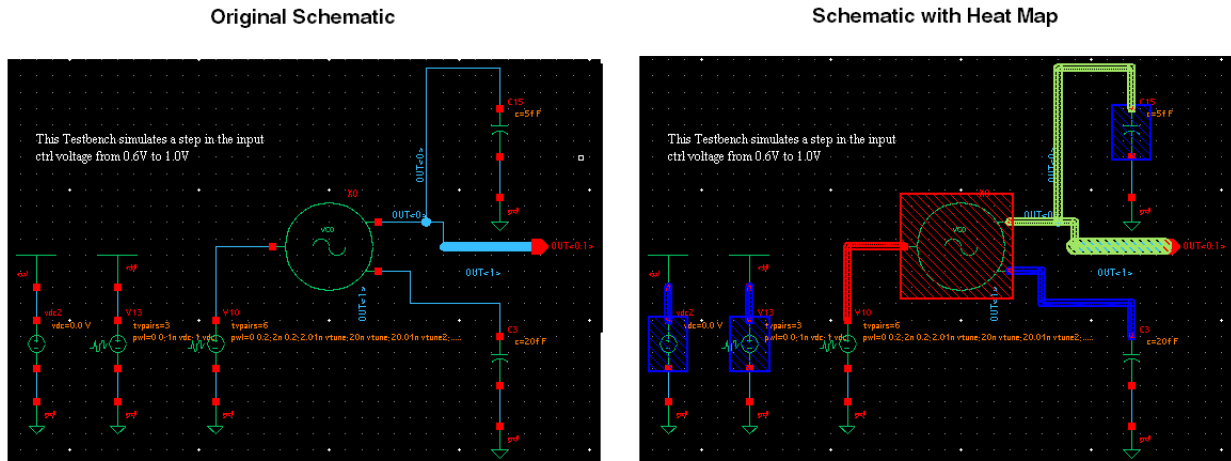
- [Viewing the Voltage Heat Map](#)

Viewing the Voltage Heat Map

The VDRC Assistant heat map feature presents a color-coded overlay of the voltage data to provide a visual summary. This topic describes how to set up and view the heat map view.

You can set up and view the voltage heat map, a graphical representation of the voltage data displayed on the canvas, overlaying the design across the hierarchy. The voltage

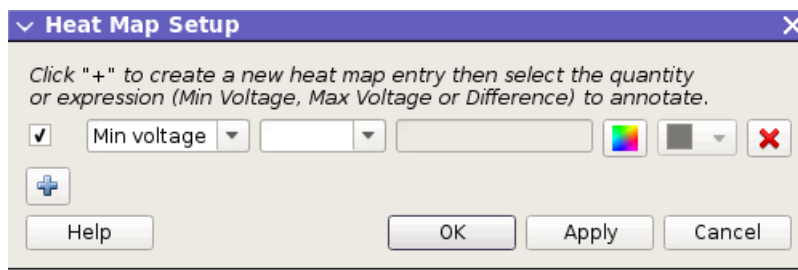
heat map provides a visual summary of all the available voltage values and lets you quickly locate problem areas. In the following example heat map, the red boxes and paths highlight potential electrical problems in the design.



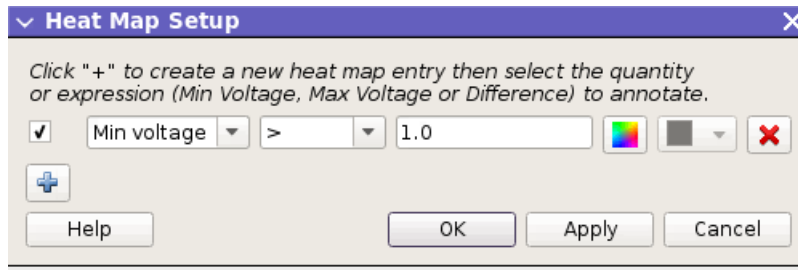
You can use different colors to indicate the voltage range or to highlight different aspects of the data you want to view. For example, you can use red to highlight voltage values that are greater than the maximum permitted, orange to highlight voltage values are near violation, and green to highlight voltage values that are within the permissible range.

To set up and view the heat map:

- On the VDRC Assistant toolbar, click **Show Heat Map Setup** dialog .
- In the **Heat Map Setup** dialog box, choose the data you want to view:
 - Min. voltage:** Minimum voltage
 - Max. voltage:** Maximum voltage
 - Difference:** Voltage difference across the net



- (Optional) Choose a relational operator and type the corresponding value in the adjacent box to specify the condition you want to view.



4. If you chose a relational operator, assign a color to the voltage condition.

For example, if the expression is Max. voltage > 1.0, and you choose the red color, then the heat map will display red highlights wherever the maximum voltage exceeds 1.0.
5. If you did not choose a relational operator, click  to open the **Edit Gradient Settings** dialog box and set up the color gradient for the voltage condition. For instructions, see Step 10 through Step 15 in Editing Heat Map Gradient Settings.
6. To add additional data to the heat map, click , and repeat [Step 2](#) through [Step 5](#).
7. To remove a condition from the heat map, click .
8. When you are finished, click **Apply** or **OK**.
9. To display the heat map, on the VDRC Assistant toolbar, click **Show Heat Map Annotation** . The heat map displays on the canvas, overlaying the schematic.

When viewing the heat map, you can double-click an instance to view the heat map of objects lower in the hierarchy.
10. To hide the heat map, in the VDRC Assistant, click **Hide Heat Map Annotation** .

See Also

- [Viewing Voltage Data in the VDRC Assistant](#)

Editing Voltage Annotations in Schematics

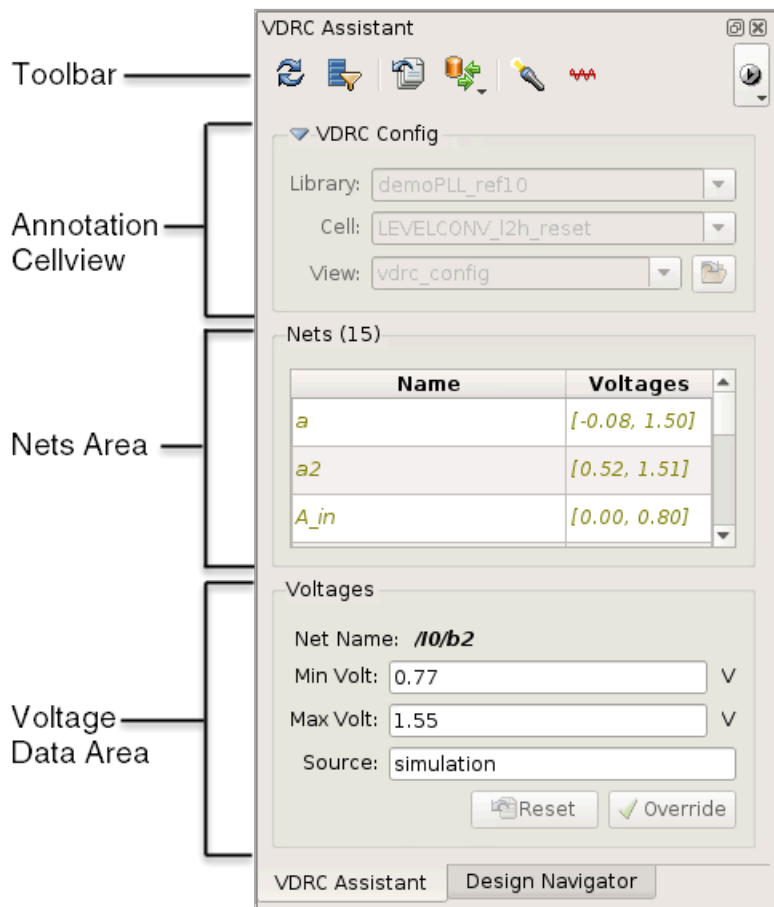
You can edit simulated voltage annotations to provide more realistic values. You can also use editing procedures to add voltage values manually.

This topic describes the following ways to edit voltage annotations in the design schematics:

- [Editing Voltage Annotations Using the VDRC Assistant](#)
- [Updating the Nets List](#)
- [Adding Voltage Annotations Using Voltage Flag Symbols](#)
- [Adding User-Defined Delta Voltage \(UDV\) Instances in Schematics](#)
- [Adding Voltage Flag \(VFlag\) Instances in Schematics](#)
- [Converting Voltage Purposes to Human-Readable Names and Values](#)
- [Adding Vsync Instances in Schematics](#)
- [Overriding vdrc_config with Voltage Flag Symbols](#)
- [Overriding Voltage Symbol Flags with vdrc_config](#)

Editing Voltage Annotations Using the VDRC Assistant

The VDRC Assistant appears when you click **OK** or **Apply** in the **Annotate Voltages** dialog box. Alternatively, you can choose **Window > Assistants > VDRC Assistant** at any time to display the assistant. See [Displaying the Schematic Editor VDRC Assistant](#).



This section describes how to use the VDRC Assistant to edit the voltage data. For information about using the VDRC Assistant to view voltage values or the heat map, see [Viewing Voltage Annotations in Schematics](#).

Note:

The VDRC Assistant lets you edit voltage annotations at the top level only. However, if you edit a voltage value for a net that is connected to lower levels in the hierarchy, the value is propagated to the lower levels. The edits are saved in the `vdrc_config` of the top cell.

You can select and edit multiple nets simultaneously.

To edit voltage annotations in the schematic using the VDRC Assistant:

1. In the VDRC Assistant, check the **VDRC_config** area to make sure you are working with the correct `vdrc_config` cellview.
2. Display the voltage data in the **Voltages** area using one of the following methods:
 - Select one or more nets in the **Nets** area to edit the voltage values simultaneously.

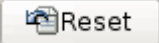
To filter the **Nets** area, click **Filter Nets by Name**  on the toolbar, and then type the filtering criteria in the filter box that appears.

Caution:

If you change any of the nets in the schematic after you open the VDRC Assistant, you must click **Refresh nets from current design**  on the toolbar to synchronize the changes between the design and the VDRC Assistant.

- Double-click a net name in the **Nets** area to zoom in and display the net in the viewport. On the canvas, rest the mouse on a net of interest.
- Select a net on the canvas.

Green net Names and Voltage values indicate that the nets have the original annotated values. The values are not yet modified.

3. To edit the selected net's voltage values, in the **Voltages** area, type new values in the **Min** or **Max** boxes.
4. Click . The annotations are updated across the hierarchy and saved to the `vdrc_config` cellview. The name in the **Nets** list and the values in the Voltages column are in black font, indicating that they have been modified.
5. To verify the edits you made, select the net name in the **Nets** list and check the values in the **Voltages** area.
6. To revert to the original annotation, click one of the following:
 -  (in the Voltages area): Reverts to the original annotation for the selected net.
 -  (on the toolbar): Reverts to original annotations for all nets.

Updating the Nets List

If you added, removed, or renamed a net in the design schematic, you must refresh the VDRC Assistant so that the net name is displayed in or removed from the **Nets** area. To

refresh the VDRC Assistant, click  on the VDRC Assistant toolbar.

If you modify the nets list, you must also rerun the simulation. See [Workflow for Annotating Voltage Data \(Simulated Voltage Data and SDL\)](#).

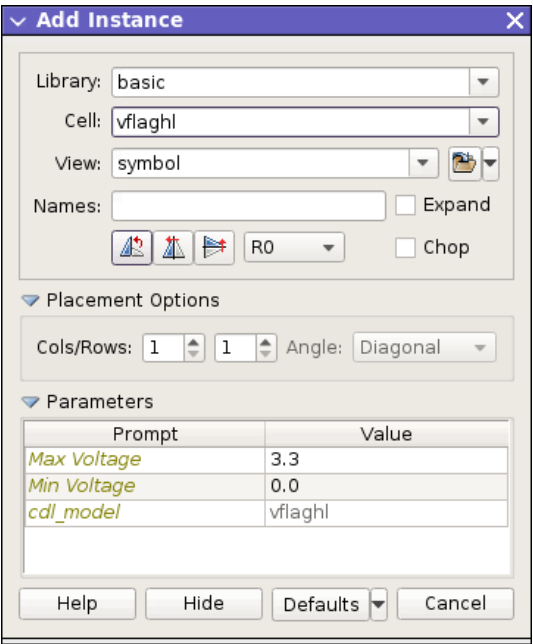
Adding Voltage Annotations Using Voltage Flag Symbols

Voltage flag symbols are symbol instances containing the maximum and minimum voltage information. You can add voltage flag symbols in the design schematic to supplement existing voltage annotations. You can also use this method to manually add voltage data.

A voltage flag symbol cellview (basic/vflaghl/symbol) is supplied in the Custom Compiler environment. The flag symbols that you create are stored in the schematic cellview.

To add voltage flag symbols in the design schematic:

1. In the Schematic Editor, choose **Add > Instance**. The **Add Instance** dialog box opens.



2. Choose the following:
 - **Library:** basic
 - **Cell:** vflaghl
 - **View:** symbol
3. (Optional) Use the **Rotate** or **Flip** options to rotate or flip the voltage flag symbols. See Creating Instances.
4. In the **Parameters** area, set the following:

Option	Description
Max Voltage	Maximum voltage value.
Min Voltage	Minimum voltage value.

5. On the canvas:
 - a. Move the pointer to a net.
 - b. Click to place the voltage flag on the net.



6. Repeat [Step 3](#) through [Step 5](#) to add more voltage flag symbols.
7. When you are finished, press Esc to close the **Add Instance** dialog box.

Adding User-Defined Delta Voltage (UDV) Instances in Schematics

User-defined delta voltage (UDV) symbols are symbol instances you add between two nets in design schematics to overwrite the tool-calculated delta voltage values.

A UDV symbol cellview (basic/udv/symbol) is supplied in the Custom Compiler environment. Instances of the UDV symbol that you create are stored in the schematic cellview and are transferred to the layout during SDL operations.

Before you start, set the following Tcl preferences:

- `leVDRCUdvLayerPurpose`

Specifies the layer purpose of the UDV shapes. For example:

```
db::setPrefValue leVDRCUdvLayerPurpose -value UserDV_m
```

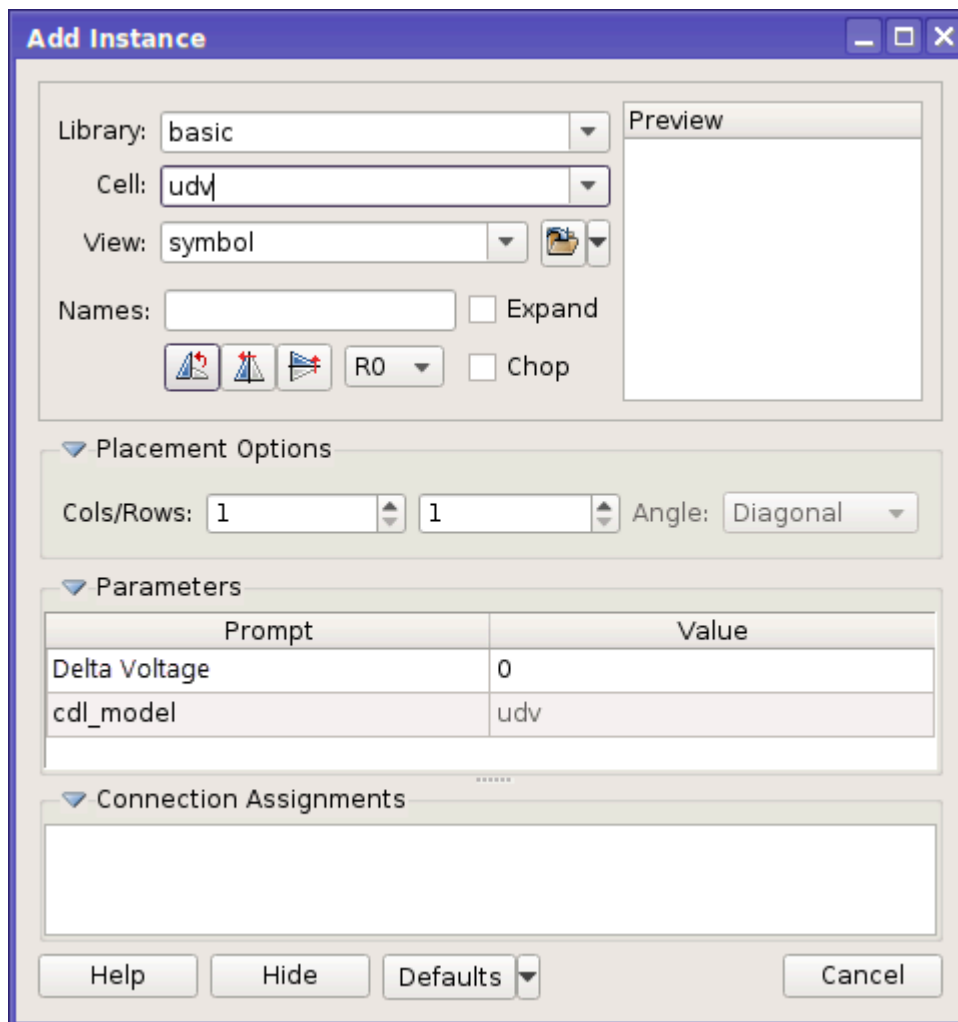
- `leVDRCUdvTextPurpose`

Specifies the layer purpose of the UDV text labels. For example:

```
db::setPrefValue leVDRCUdvTextPurpose -value UserDV
```

To add a UDV symbol in the design schematic:

1. In the Schematic Editor, choose **Add > Instance**. The **Add Instance** dialog box opens.



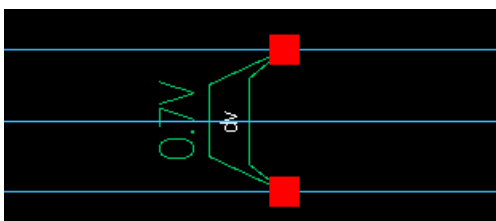
The **Add Instance** dialog box is shown with the following settings:

- Library:** basic
- Cell:** udv
- View:** symbol
- Names:** (empty text field)
- ☐ Expand
- ☐ Chop
- Placement Options:**
 - Cols/Rows:** 1
 - Angle:** Diagonal
- Parameters:**

Prompt	Value
Delta Voltage	0
cdl_model	udv
- Connection Assignments:** (empty text area)

Buttons at the bottom: Help, Hide, Defaults, Cancel.

2. Choose the following:
 - **Library:** basic
 - **Cell:** udv
 - **View:** symbol
3. In the **Parameters** section, type the **Delta Voltage** value you want to use to overwrite the tool-calculated delta voltage.
4. See Creating Instances for setting the remaining options.
5. On the canvas:
 - a. Move the pointer to the two nets of interest.
 - b. Click to place the UDV symbol between the two nets.



6. When you are finished, press Esc to close the **Add Instance** dialog box.

The newly added UDV instance displays in the Design Navigator **Insts** tab. In addition, you can view and edit its properties in the Property Editor. When you use SDL to generate the layout, the UDV instance will appear in the layout.

See Also

- [Adding User-Defined Delta Voltage \(UDV\) Instances in Layouts](#)
- [Design Navigator Assistant](#)
- [Property Editor Assistant](#)
- [Placing Schematic Instances Into Layout](#)

Adding Voltage Flag (VFlag) Instances in Schematics

If you are only interested in the highest voltage value on a net, you can add a VFlag symbol instance to specify the single voltage value for that net. For the lowest voltage value, the DRC feature will use the default value for the device type.

A VFlag symbol cellview (basic/vflag/symbol) is supplied in the Custom Compiler environment. Instances of the VFlag symbol that you create are stored in the schematic cellview and are transferred to the layout during SDL operations.

Before you start, set the following Tcl preferences:

- `leVDRCVflagVoltagePurposes`

Specifies the list of voltage names of the VFlag shapes. The list populates the **Select Voltage** option in the **Create Instance** dialog box.

```
db::setPrefValue leVDRCVflagVoltagePurposes -value "v_1_2 v_1_5 v_1_8"
```

If the voltage purposes are not human-readable, you can convert them to human-readable voltage names and values. See [Converting Voltage Purposes to Human-Readable Names and Values](#).

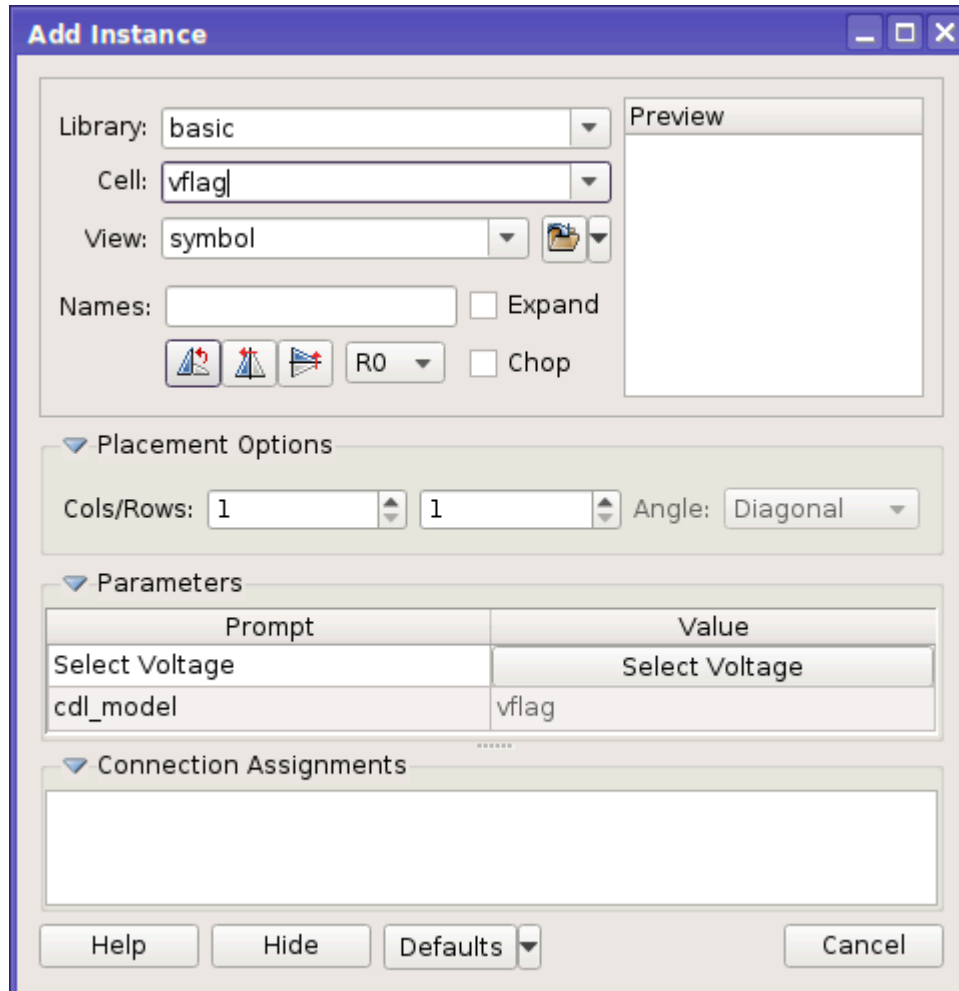
- `leVDRCVflagPinPurpose`

Specifies the layer purpose of VFlag pins. The default is `drawing`, but you can change the value to adjust the results (for example, use the `pin` layer purpose).

If the voltage purposes are not human-readable, you can convert them to human-readable voltage names and values. See [Converting Voltage Purposes to Human-Readable Names and Values](#).

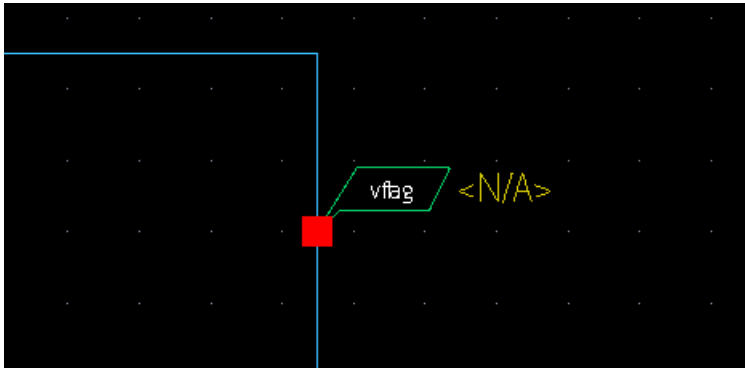
To add a VFlag symbol in the design schematic:

1. In the Schematic Editor, choose **Add > Instance**. The **Add Instance** dialog box opens.



2. Choose the following:
 - **Library:** basic
 - **Cell:** vflag
 - **View:** symbol
3. In the **Parameters** tab, click **Select Voltage** and select the voltage purpose name.
4. See Creating Instances for setting the remaining options.

5. On the canvas:
 - a. Move the pointer to the net.
 - b. Click to place the VFlag symbol on the net.



6. When you are finished, press Esc to close the **Add Instance** dialog box.

The newly added VFlag instance displays in the Design Navigator **Insts** tab. In addition, you can view and edit its properties in the Property Editor. When you use SDL to generate the layout, the VFlag instance will appear in the layout.

See Also

- [Adding Voltage Flag \(VFlag\) Instances in Layouts](#)
- [Design Navigator Assistant](#)
- [Property Editor Assistant](#)
- [Placing Schematic Instances Into Layout](#)

Converting Voltage Purposes to Human-Readable Names and Values

Before adding voltage flags (VFlags) to schematics or layouts, you must define the list of voltage purposes. In case the voltage purposes you want to use are not human-readable, you can convert them to readable pairs of voltage names and values using the `vdrcConvertVoltagePurposeToNameValue` Tcl procedure.

The `vdrcConvertVoltagePurposeToNameValue` Tcl procedure converts voltage purposes specified by the `leVDRCVflagVoltagePurposes` Tcl preference to human readable names and values. For example, you can use the Tcl procedure to convert a voltage purpose (`v_0_1`) to a voltage name (`0V1`) and the corresponding voltage value (`0.1`). The converted voltage names and values are then displayed in the Create Instance dialog box and in the Property Editor.

The following is an example `vdrcConvertVoltagePurposeToNameValue` procedure. You can customize the example to meet the foundry requirements.

```
proc vdrcConvertVoltagePurposeToNameValue { purposeName } {  
    regsub {v_} $purposeName {} vv  
    regsub {_} $vv {.} v  
    set l [string length $v]  
    if { $l < 1 } {  
        return ""  
    } elseif { $l == 1 } {  
        append vn $vv "V0"  
        return [list $vn $v]  
    } elseif { $l == 3 } {  
        regsub {_} $vv {V} vn  
        return [list $vn $v]  
    }  
    return ""  
}
```

To use the Tcl procedure:

1. Customize the example. Be sure to retain the Tcl procedure name, `vdrcConvertVoltagePurposeToNameValue`.
2. Source the Tcl procedure after starting the Custom Compiler tool.

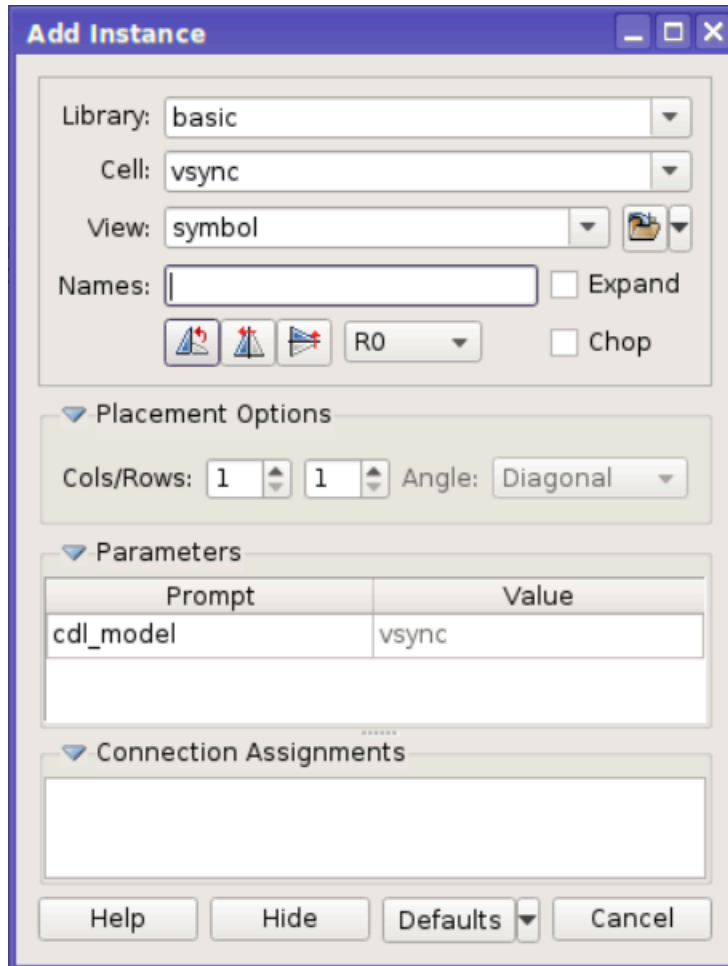
Adding Vsync Instances in Schematics

Vsync symbols are symbol instances you add in schematics to indicate that two nets are synchronous.

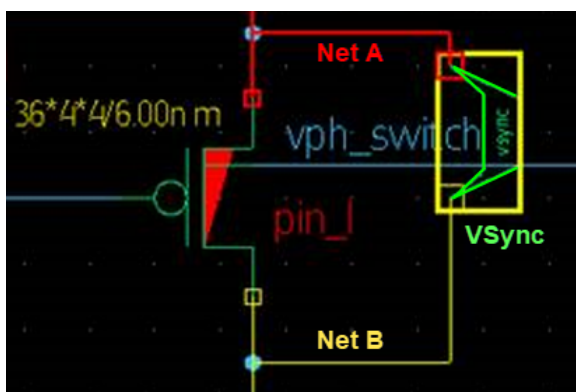
A vsync symbol cellview (`basic/vsync/symbol`) is supplied in the Custom Compiler environment. Instances of the vsync symbols that you create are stored in the schematic cellview and are transferred to the layout during SDL operations.

To add a vsync symbol in the design schematic:

1. In the Schematic Editor, choose **Add > Instance**. The **Add Instance** dialog box opens.



2. Choose the following:
 - **Library:** basic
 - **Cell:** vsync
 - **View:** symbol
3. See Creating Instances for setting the remaining options.
4. On the canvas:
 - a. Move the pointer to a net.
 - b. Click to place the vsync symbol between two nets that are synchronized.



5. When you are finished, press Esc to close the **Add Instance** dialog box.

The newly added vsync instance displays in the Design Navigator **Instance** tab. In addition, you can view its properties in the Property Editor. When you use SDL to generate the layout, the vsync instance appears in the layout.

See Also

- [Adding Vsync Instances in Layouts](#)
- [Design Navigator Assistant](#)
- [Property Editor Assistant](#)
- [Placing Schematic Instances Into Layout](#)

Overriding vdrc_config with Voltage Flag Symbols

You can synchronize the voltage data between the voltage flag symbols and the `vdrc_config` cellview. This section describes how to override the voltage values in the `vdrc_config` cellview with data from the voltage flag symbols. If conflicts occur, you have the option of choosing which values to use.

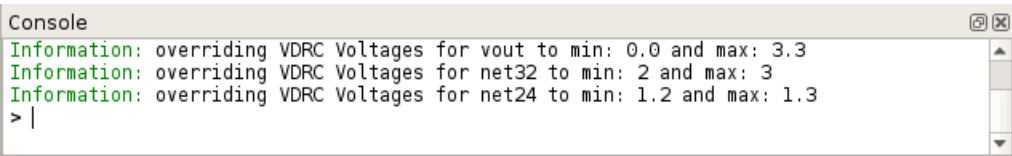
Caution:

Make sure you are at the top level of the design. The synchronization operation is not available if you descend into a lower hierarchy level in a device.

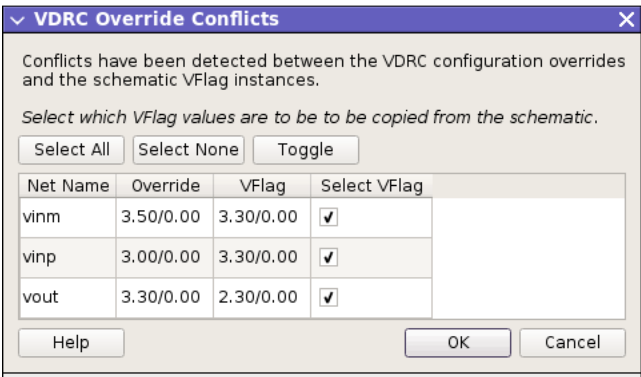
To override the `vdrc_config` cellview with voltage data from voltage flag symbols:

- On the VDRC Assistant toolbar, click  and choose one of the following:
- **Sync vflags with vdrc_config data (overrides only):** Overrides only the voltage values that were modified in the VDRC Assistant.
 - **Sync vflags with vdrc_config data:** Overrides all voltage values in the `vdrc_config` cellview.

The voltage data from voltage flag symbols are synchronized with the `vdrc_config` cellview. The Console reports the overridden voltage data.



If the synchronization process encounters conflicting override values from different sources, the **VDRC Override Conflicts** dialog box displays.



The dialog box lists conflicting values by net. Review each case and select the values you want to use:

Column Name	Description
Override	Voltage values in the <code>vdrc_config</code> view.
VFlag	Voltage values from the vflag symbols.
Select VFlag	Uses the voltage values in the VFlag column, which are from the vflag symbols. To use the Override values, clear the check box.

To use all VFlag values, click **Select All**.

To use all Override values, click **Select None**.

Click **Toggle** to switch between selecting all and selecting none of the VFlag values.

When you are finished, click **OK**. The import process continues based on the selections you made in the dialog box.

Overriding Voltage Symbol Flags with `vdrc_config`

You can synchronize the voltage data between the voltage flag symbols and the `vdrc_config` cellview. This section describes how to create new or override existing voltage values in flag symbols using the data from the `vdrc_config` cellview.

Caution:

Make sure you are at the top level of the design. The synchronization operation is not available if you descend into a lower hierarchy level in a device.

Note:

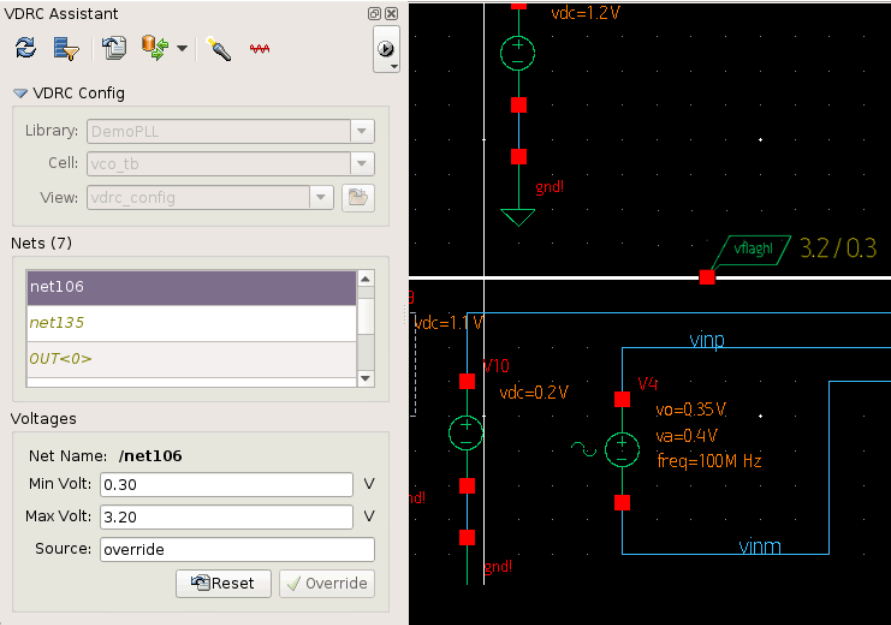
- The new or overriding voltage data are also propagated to connected bus bits if all the bits have the same {vflagh, vflagl} values. (For example, all five bits in a trunk have {0.0, 0.3}, and the overriding values {0.03 , 1.20} will propagate to the five bits.)
- If conflicts occur, you have the option of choosing which values to use.

To create new or override existing voltage values in voltage flag symbols:

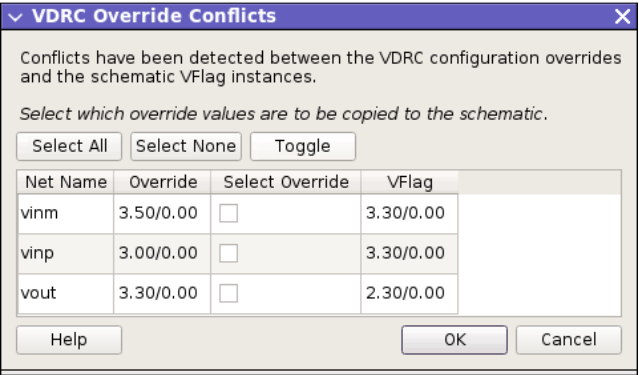
- On the VDRC Assistant toolbar, click  and choose **Sync `vdrc_config` with vflag overrides**.

The voltage data from the `vdrc_config` cellview are exported to the voltage flag symbol cellview. Check the voltage flag symbols on the canvas. You should see the new values from the `vdrc_config` cellview.

In the following example, the voltage values for net106 (3.2 and 0.3) are exported and replace the values in the vflaghl cellview. The voltage flags on the canvas show the new values.



If the synchronization process encounters conflicting override values from different sources, the **VDRC Override Conflicts** dialog box displays.



The dialog box lists conflicting values by net. Review each case and select the values you want to use:

Column Name	Description
Override	Voltage values in the vdr_c_config view.
VFlag	Voltage values from the vflag symbols.

Column Name	Description
Select Override	Uses the voltage values in the Override column. To use the VFlag values, clear the check box.

To use all Override values, click **Select All**.

To use all Vflag values, click **Select None**.

Click **Toggle** to switch between selecting all and selecting none of the Override values.

When you are finished, click **OK**. The export process continues based on the selections you made in the dialog box.

Annotating Layouts With Voltage Data

With the simulated voltage data (vdrc.db) as input, you can annotate the design schematic and layout.

This topic describes how to annotate design layouts and covers the following:

- [Voltage Annotation Method for Design Layouts](#)
- [Before You Start to Annotate Layouts With Voltage Data](#)
- [Using the Existing vdrc_config to Annotate Layouts](#)
- [Creating New vdrc_config for Annotating Layouts](#)

Voltage Annotation Method for Design Layouts

You use the `vdrc_config` cellview to annotate the design layout with voltage data.

The annotation method you use depends on the source of the `vdrc_config` cellview:

Source of <code>vdrc_config</code>	See...
Existing <code>vdrc_config</code> cellview created when Annotating Schematics With Voltage Data (typical scenario)	Using the Existing <code>vdrc_config</code> to Annotate Layouts
New <code>vdrc_config</code> cellview that you create based on the <code>vdrc.db</code> file from Measuring Voltage Data Using the PrimeWave Design Environment	Creating New <code>vdrc_config</code> for Annotating Layouts

With either method, you use the Layout Editor **Electrical > Voltage > Import Voltages** command.

Before You Start to Annotate Layouts With Voltage Data

(If you have simulated voltage data and you want to use SDL to automate your layout tasks) Make sure you:

- Open the design in the SDL mode. For instructions, see the *Custom Compiler SDL and Visually Assisted Automation Guide*.
- Realize the instances in the layout with correct connectivity so that you can automate the voltage annotation in the Layout Editor.

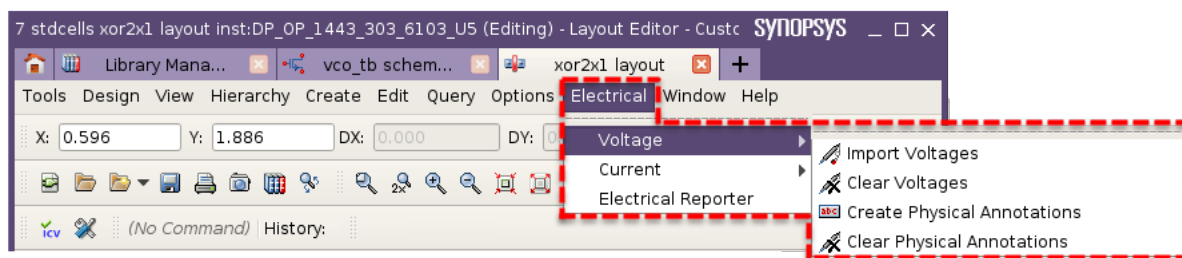
(If you have manually entered voltage data in the Schematic Editor) Make sure you open the design layout.

Using the Existing vdrc_config to Annotate Layouts

The `vdrc_config` cellview that was generated from [Annotating Schematics With Voltage Data](#) is automatically associated with the design. When you open the design layout, the associated `vdrc_config` cellview is readily available for voltage annotation.

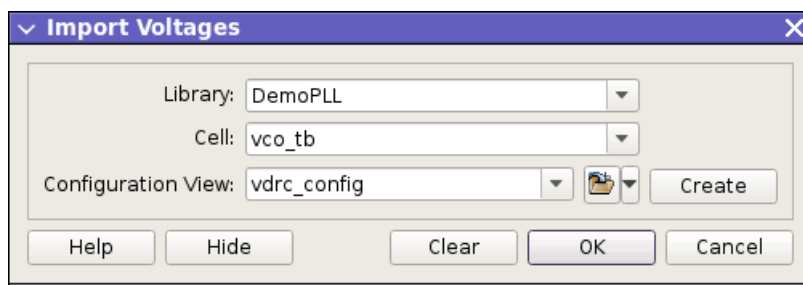
To use an existing `vdrc_config` cellview to annotate the layout:

1. In the Layout Editor, choose **Tools > Electrical**. The **Electrical** menu appears on the menu bar.



2. Choose **Electrical > Voltage > Import Voltages**.

The **Import Voltages** dialog box opens. The **Library**, **Cell**, and **Configuration View** boxes are prepopulated with information from the current design.



3. Click **OK**. The voltage data from the `vdrc_config` cellview is loaded, and the VDRC Assistant appears on the right side of the window. See [Viewing Net Voltage Data in the VDRC Assistant](#).

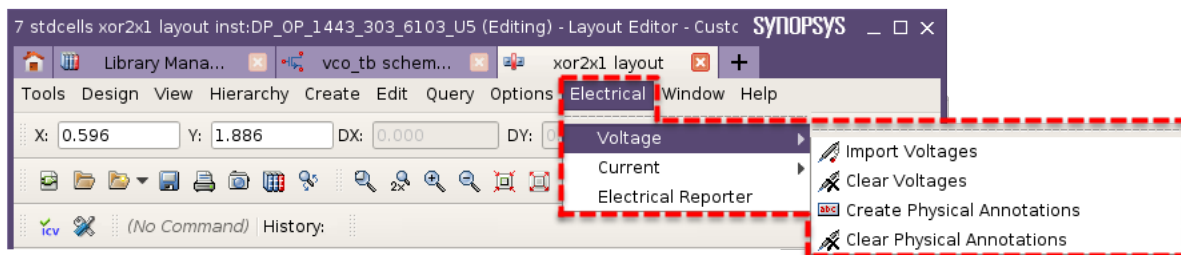
Creating New `vdrc_config` for Annotating Layouts

Typically, you annotate the layout using an existing `vdrc_config` cellview generated from the previous step in the design process. See [Using the Existing `vdrc\_config` to Annotate Layouts](#). If, for some reason, you need to create a new `vdrc_config` cellview for your design layout, you can do so from within the **Import Voltages** dialog box.

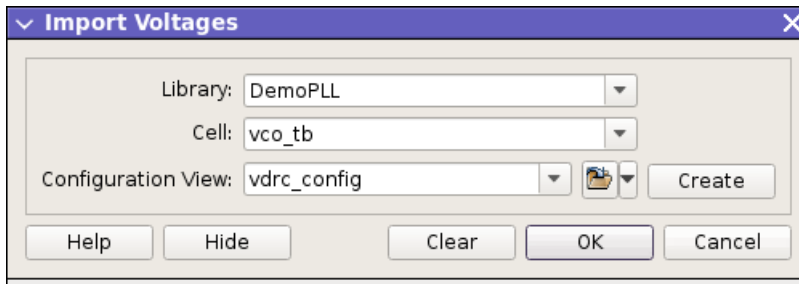
Before you start, make sure you have access to the folder where the voltage data are stored. See [Measuring Voltage Data Using the PrimeWave Design Environment](#).

To create a new `vdrc_config` cellview and annotate the layout with voltage data:

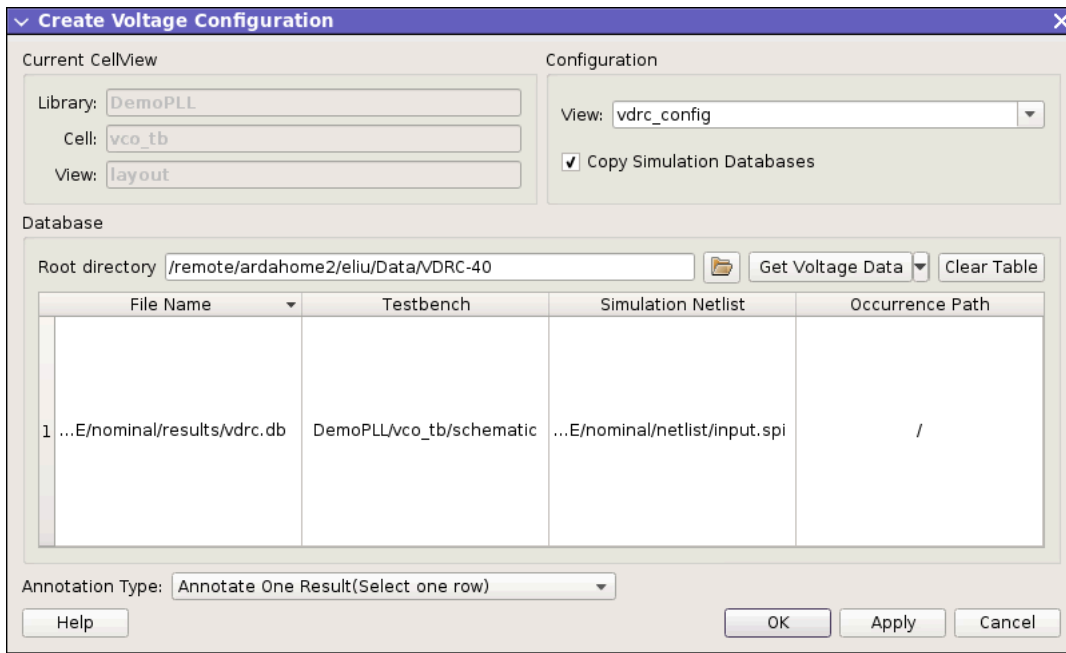
1. In the Layout Editor, choose **Tools > Electrical**. The **Electrical** menu appears in the menu bar.



2. Choose **Electrical > Voltage > Import Voltages**.



3. Click **Create**. The **Create Voltage Configuration** dialog box opens.



4. In the **Configuration** area:

- (Optional) In the **View** box, type a name for the new cellview. Alternatively, you can use the default name, `vdrc_config`.
- Select **Copy Simulation Databases**, if you want to copy all VDRC datasets listed in the table to a local self-contained `.../databases` directory. Having the simulation datasets locally ensures their availability when you reload the `vdrc_config` cellview.

Tcl Preference: `leCopyElectricalSimulationToConfig`

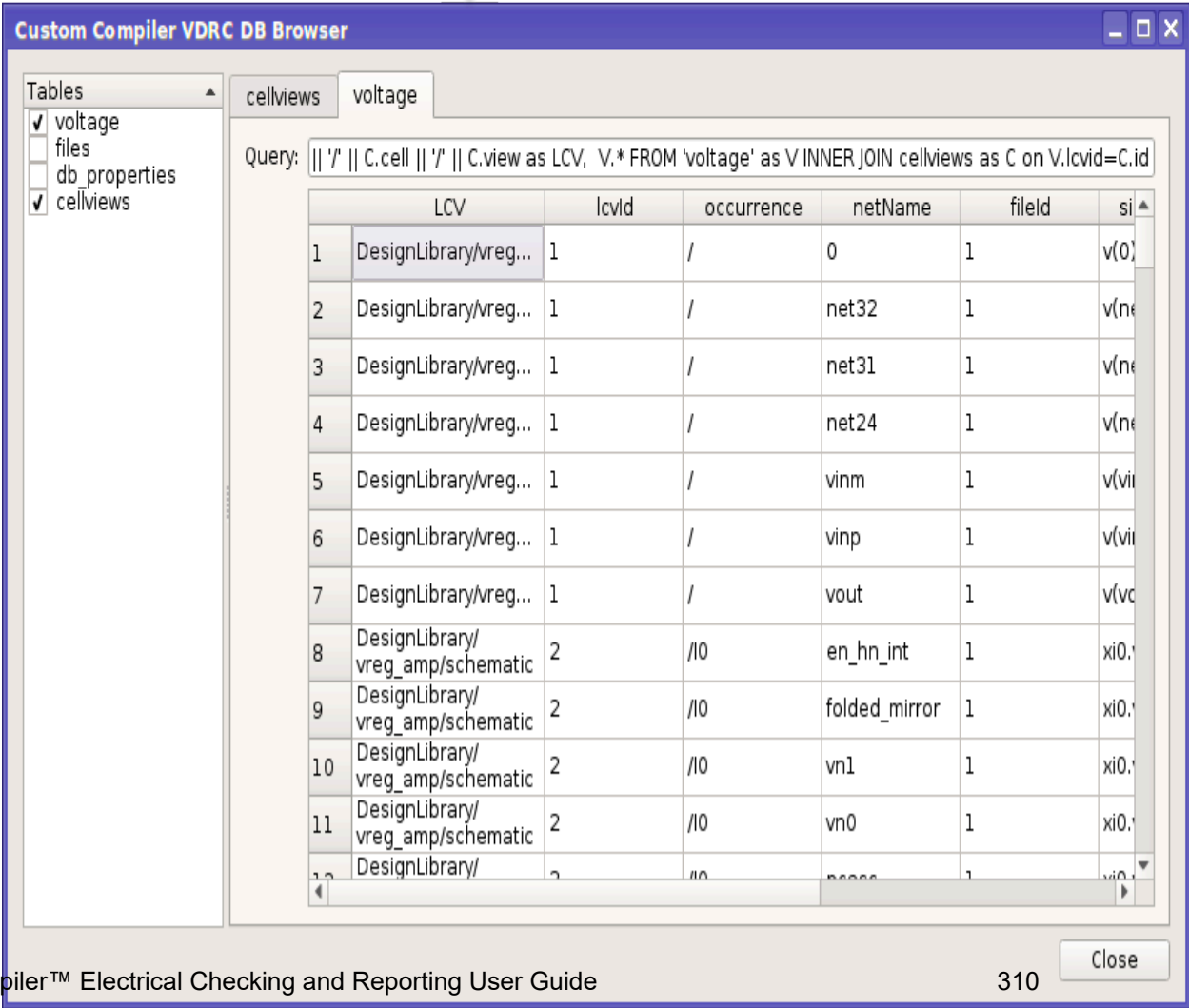
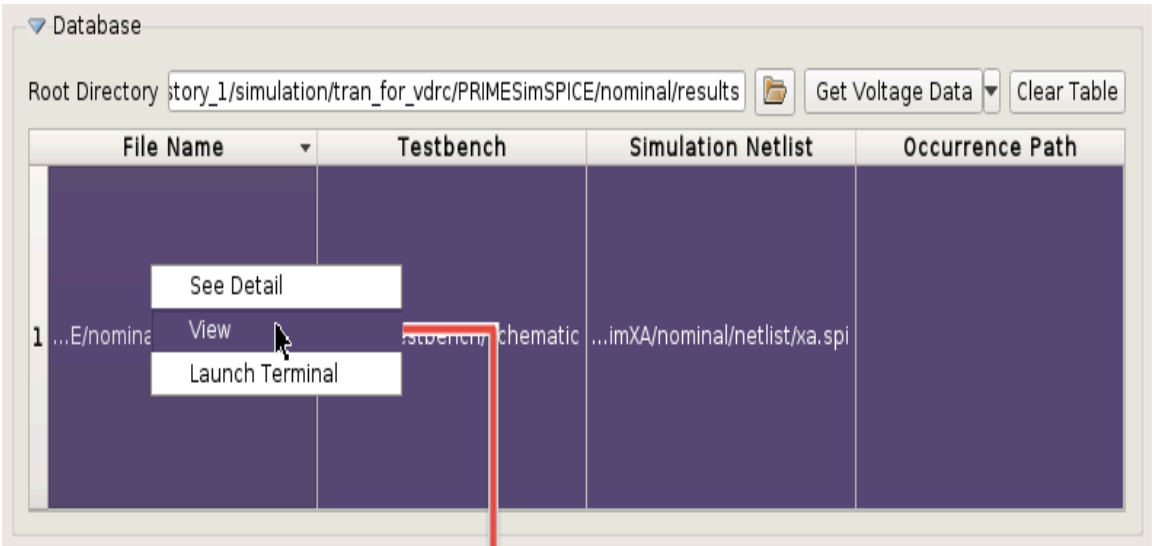
5. In the **Database** area:

- a. In the **Root Directory** box, type the directory in which the `em.db` files are stored. In the **Get Current Data** menu, select whether to get the data **From .db Files Only** or **From Any Database File**, and then click **Get Current Data**. The list of datasets that are relevant to the opened design appears in the dialog box.
- b. In the **Root directory** box, type the directory in which the `vdrc.db` files are stored. In the **Get Voltage Data** menu, select whether to get the data **From .db Files Only** or **From Any Database File**, and then click **Get Voltage Data**. The list of datasets that are relevant to the opened design appears in the dialog box.
- c. In the **Annotation Type** list, choose one of the following:

Annotation Type	Description
Annotate One Result (Select One Row)	Only the data from one dataset is used for annotation. You will be permitted to select only one dataset in the list in step 4.
Worst Case (Across All)	Only the maximum current (worst-case) value from across all available current data sets for the design under test. You will not be permitted to select any dataset in the list in step 4.
Worst Case (Across Selected Rows)	Only the maximum current (worst-case) value from across the selected datasets is used for annotation. You will be permitted to select the datasets in the list in step 4.

- d. In the table, select the datasets you want to use for the annotation process.

If you want to view the contents of a voltage dataset before you start the annotation, right-click the dataset **File Name**, and choose **View**. The **VDRC DB Browser** dialog box opens and displays the dataset contents. When you are finished, close the dialog box.



Your ability to select datasets depends on the Annotation Type you chose in [Step 6](#). If you selected Worst Case (across all), you will not be permitted to select any dataset.

6. Click **Clear Table** to reload different data and repeat step 5.
7. In the **Voltage Annotation** dialog box, click **OK** or **Apply** to close the **Create Voltage Configurations** dialog box.
8. Click **OK** in the **Import Voltages** dialog box. The Layout Editor loads the voltage data into the design layout. The VDRC Assistant appears on the right side of the window. See [Viewing Net Voltage Data in the VDRC Assistant](#).

If you selected **Copy Simulation Databases** in step 4, the copy operation deletes any existing local datasets, copies all simulation datasets to the specified config view, and updates the `config.xml` file to reference the local directory.

If you did not select **Copy Simulation Databases**, the copy operation deletes any existing local datasets, and updates the `config.xml` file to reference the original simulation directories.

Viewing Voltage Data in the Layout

This topic describes the following ways to view the voltage annotations in the layout:

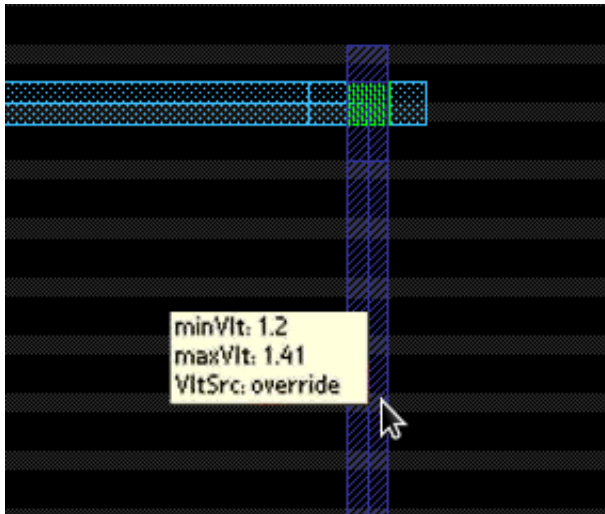
- [Viewing Voltage Data in Tooltips](#)
- [Viewing Net Voltage Data in the VDRC Assistant](#)
- [Setting Up the VDRC Assistant for Heat Map View](#)
- [Viewing the Voltage Heat Map](#)

Viewing Voltage Data in Tooltips

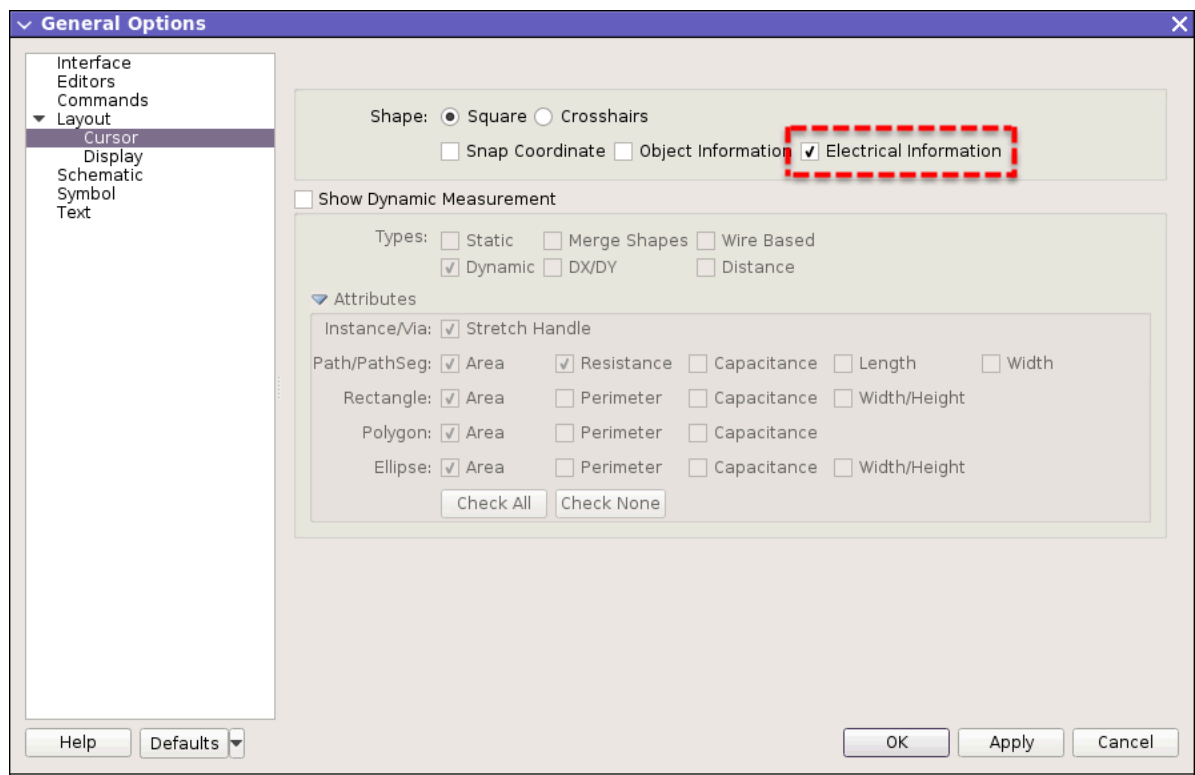
You can rest the pointer on pins or wires and view the voltage data in the tooltips. The tooltip displays the following information:

- Minimum voltage
- Maximum voltage
- Source of the voltage data (simulation, intent, or override)

Intent indicates voltage data that were overridden in the Schematic Editor. Override indicates voltage data that were overridden in the Layout Editor.



If you do not see the voltage data in tooltips, select **Electrical Information** in the **General Options** dialog box. See Setting Layout Display, Cursor, and Selection Options.

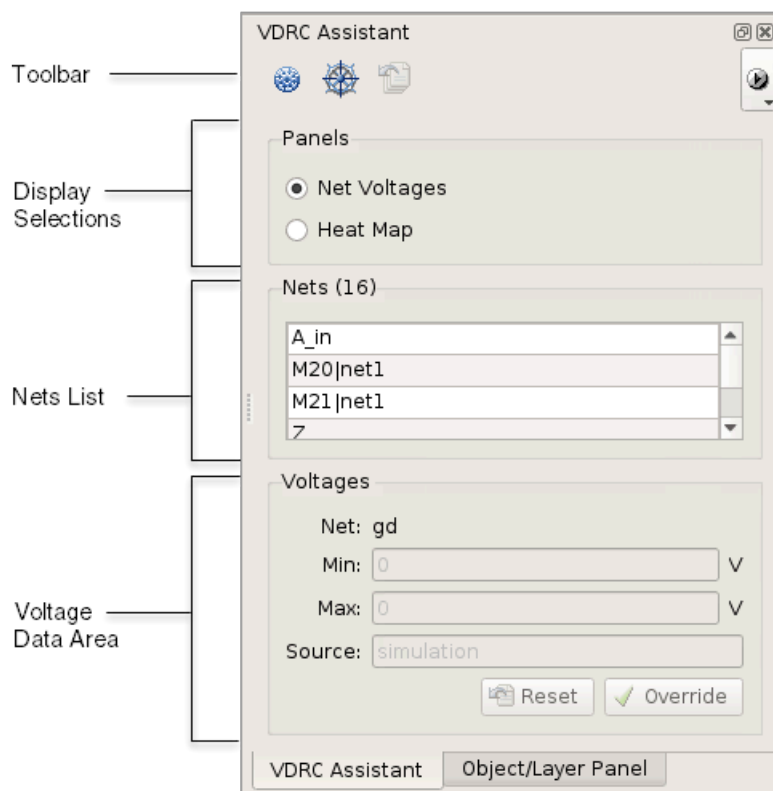


Viewing Net Voltage Data in the VDRC Assistant

The VDRC Assistant appears when you import voltage values from the `vdrc_config` cellview. See [Annotating Layouts With Voltage Data](#). Alternatively, you can choose **Window > Assistants > VDRC Assistant** at any time to display the assistant.

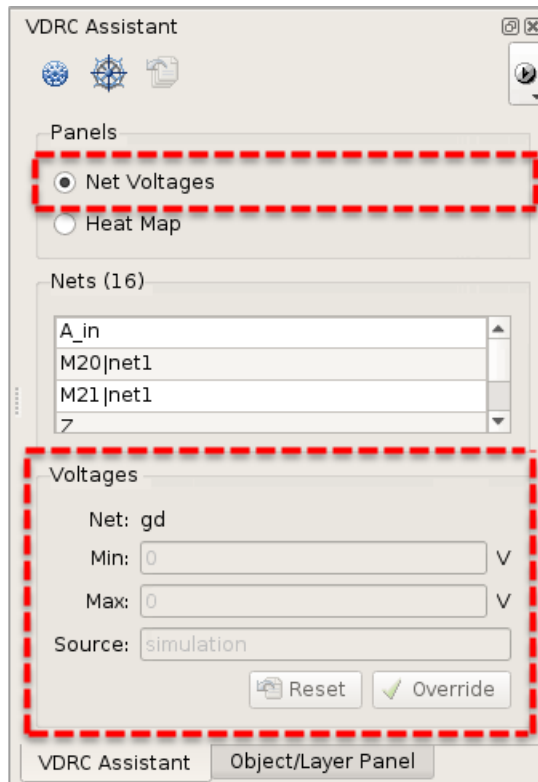
The VDRC Assistant lets you view and edit the simulated voltage data. This section describes how to view the data and display the voltage heat map. To edit the data, see [Editing Voltage Data in the Layout](#).

The VDRC Assistant consists of the following:



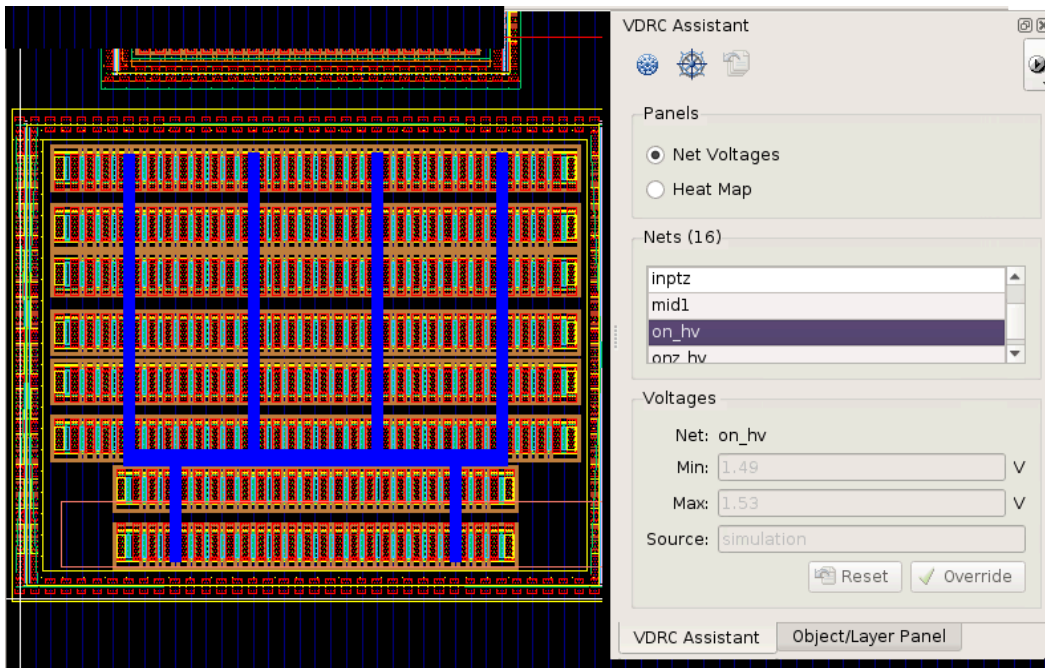
To view the voltage data:

1. In the VDRC Assistant **Panels** area, select **Net Voltages**. The Voltages area appears at the bottom of the assistant, displaying the minimum and maximum voltage values for the selected net.



2. Use one of the following methods to display the voltage data in the **Voltages** area:
 - In the Design Navigator **Nets** tab, select the net.
 - Click a net on the canvas to select it.

When you select a net, the net's name appears in the **Voltages** area.

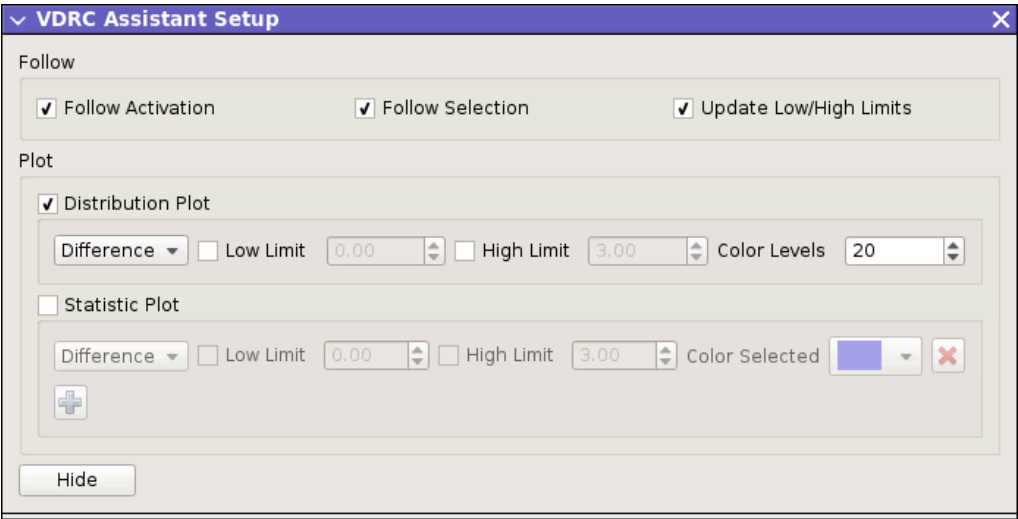


Setting Up the VDR Assistant for Heat Map View

Before you view the voltage heat map and the voltage statistics, you should set the parameters for what you want to view in the heat map.

To set up and view the voltage heat map:

1. On the VDRC Assistant toolbar, click **Open Setup dialog**  icon. The VCRC Assistant Setup dialog box opens.



2. In the **Follow** area, set the following:

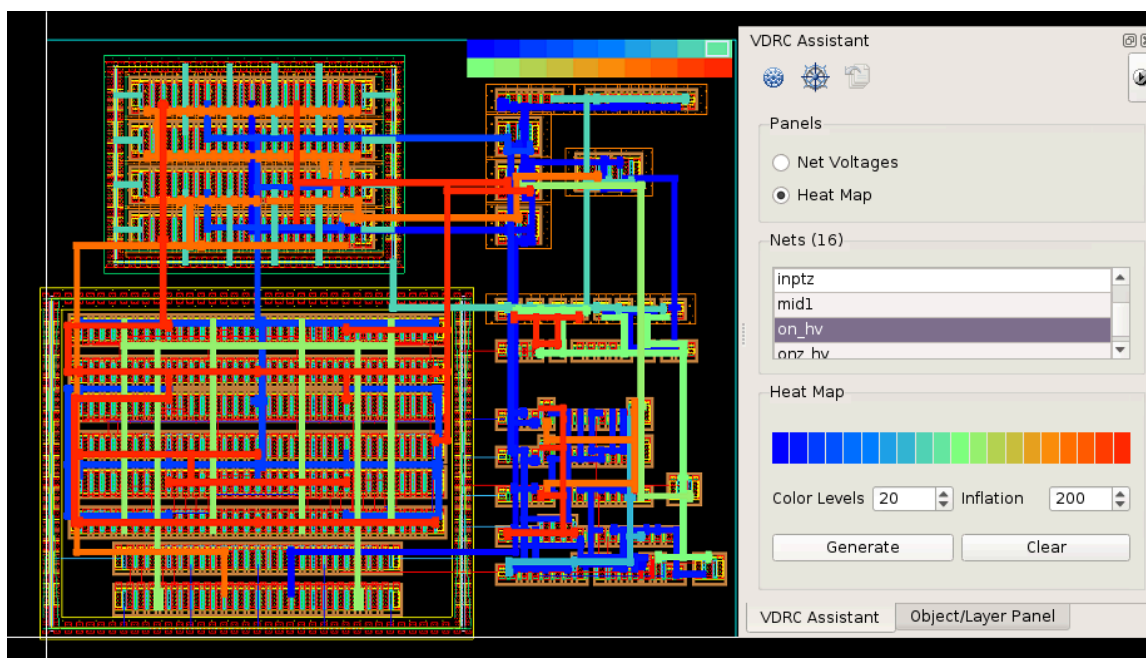
Setting	Description
Follow Activation	Synchronizes with the Net selection in the Design Navigator.
Follow Selection	Synchronizes with the net selection on the canvas.
Update Low/High Limits	Automatically updates the voltage ranges based on the Low Limit and High Limit settings.

3. In the **Plot** area, select **Distribution Plot**, and set up how you want to view the distribution of certain voltage values:
- Select **Minimum**, **Maximum**, or **Difference**.
 - Set the **Low Limit** and **High Limit** values.
 - Set the **Color Levels**, which is the number of colors to use within the low and high voltage values you specify.

4. In the **Statistics Plot** area, select to set up the voltage statistics chart:
 - a. Select **Minimum**, **Maximum**, and **Difference**.
 - b. Set the **Low Limit** and **High Limit** values.
 - c. Select the **Color** to use to represent the specified range.
5. When you are finished, click **Hide** to return to the Layout Editor VDRC Assistant.

Viewing the Voltage Heat Map

You can set up and view the voltage heat map, a graphical representation of the voltage data displayed on the canvas, overlaying the design. The voltage heat map provides a visual summary and lets you quickly locate problem areas. In the following example heat map, the red paths highlight potential electrical problem areas in the design.



You can use different colors to indicate the voltage range or to highlight different aspects of the data you want to view. For example, you can use red to indicate voltage values that are greater than the maximum permitted, orange to indicate voltage values that are near violation, blue to indicate potentially acceptable voltage ranges, and green to indicate voltage values that are within the permissible range.

To set up and view the heat map:

1. In the VDRC Assistant, select **Heat Map** in the **Panels** area. The Heat Map area appears.
2. In the **Heat Map** area:
 - a. Set the **Color Levels**, which is the number of colors to use for the voltage ranges.
 - b. Set the **Inflation** value, which is the amount the heat map overlay overhangs the objects in the design. For example, if Inflation is 200, and the wire size is 1000 x 2000 DBU, then the heat wave overlay is 1200 x 2200 DBU.
3. When you are finished, click **Generate**. The heat map is displayed over the layout on the canvas.

Editing Voltage Data in the Layout

In the typical design workflow, you would re-annotate the design layout using the values (`vdrc_config` cellview) provided by the schematic designer. Although you should not modify the voltage values, you have the option of editing simulated voltage annotations if needed.

Caution:

The `vdrc_config` cellview from schematic designers are considered the master reference for voltage data annotations. Any voltage value updates made in the layout should be communicated to the schematic designers and updated in the master `vdrc_config` cellview to ensure consistency.

This topic describes the following ways to edit the voltage annotations in the layout:

- [Editing Voltage Data Using the VDRC Assistant](#)
- [Adding Voltage Annotations Using Voltage Flags](#)
- [Adding User-Defined Delta Voltage \(UDV\) Instances in Layouts](#)
- [Adding Voltage Flag \(VFlag\) Instances in Layouts](#)
- [Adding Vsync Instances in Layouts](#)
- [Editing Voltage Data Using the Property Editor](#)

Editing Voltage Data Using the VDRC Assistant

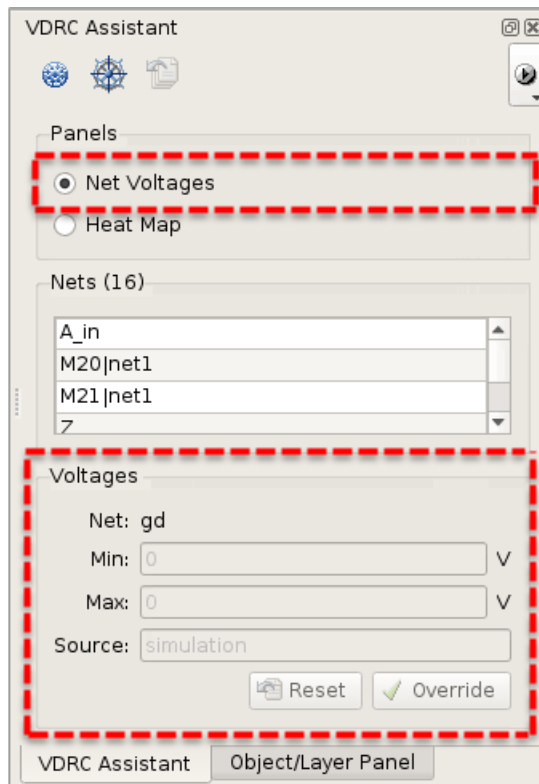
You can modify voltage values in layouts using different methods: VDRC Assistant, voltage flags, or the Property Editor. This topic describes how to use the VDRC Assistant.

Note:

The VDRC Assistant lets you edit voltage annotations at the top level only. You can view voltage annotations at lower hierarchy levels, but you cannot edit the values.

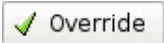
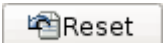
To edit voltage annotations in the layout using the VDRC Assistant:

1. In the VDRC Assistant **Panels** area, select **Net Voltages**. The **Voltages** area appears at the bottom of the assistant, displaying the minimum and maximum voltage values for the selected net.



2. Use one of the following methods to display the voltage data in the **Voltages** area:
 - In the Design Navigator, select a net.
 - Click a net on the canvas to select it.
 - Double-click a net name in the Nets area to zoom in and display the net in the viewport. On the canvas, rest the mouse on a net of interest.

When you select a net, the net's name appears in the Voltages area.

3. To edit the voltage values, in the **Voltages** area, type new values in the **Min** or **Max** boxes.
4. Click  **Override**. The annotations are updated across the hierarchy and saved to the `vdrc_config` cellview.
5. To verify the edits you made, select the net name in the **Nets** list and check the values in the **Voltages** area.
6. To revert to the original annotation, click one of the following:
 -  **Reset** (in the Voltages area): Reverts to the original annotation for the selected net.
 -  (on the toolbar): Reverts to original annotations for all nets.

Adding Voltage Annotations Using Voltage Flags

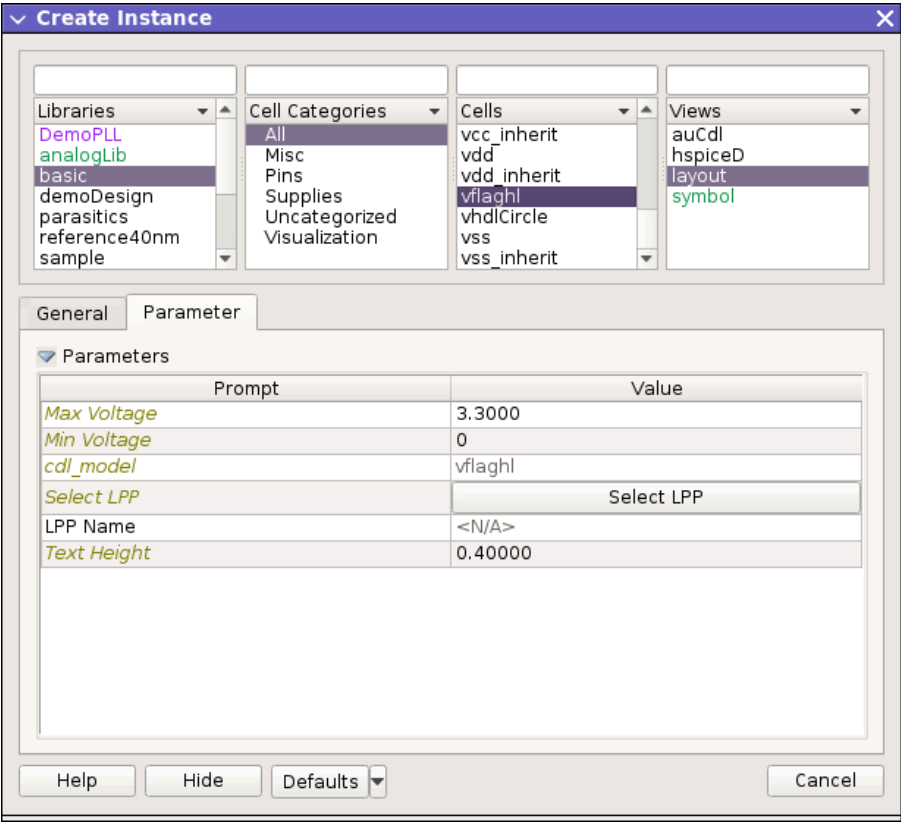
Voltage flag symbols are symbol instances containing the maximum and minimum voltage information. You can add voltage flag symbols in the design layout to supplement existing voltage annotations. You can also use this method to manually add voltage data instead of automating the annotation process using the Layout Editor **Electrical > Voltage Annotation > Annotate Voltages** command.

If you are using this procedure in a non-SDL design workflow, make sure you add a voltage flag per Connectivity Island. Note that you can add voltage PCells even if nets are not fully routed.

A voltage flag layout cellview (`basic/vflaghl/layout`) is supplied in the Custom Compiler environment. The flag PCells that you create are stored in the layout cellview.

To add a voltage flag PCell in the design layout:

1. In the Layout Editor, choose **Create > Instance**. The **Create Instance** dialog box opens.

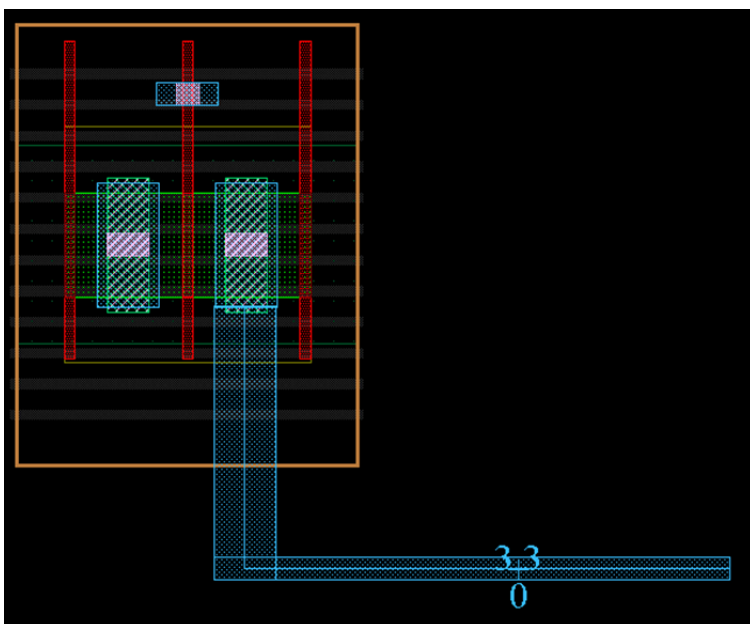


2. Choose the following:
 - Library: basic
 - Cell: vflaghl
 - View: layout
3. (Optional) In the **General** tab, set the desired options. See Instance Toolbar and Create Instance Dialog Box.
4. In the **Parameters** tab, set the following:

Option	Description
Max Voltage	Maximum voltage value.
Min Voltage	Minimum voltage value.

Option	Description
Select LPP	Select the layer-purpose pair specification for the voltage flag PCell.

5. Skip the remaining options.
6. On the canvas:
 - a. Move the pointer to a net.
 - b. Click to place the voltage flag on the net.



7. When you are finished, press Esc to close the **Create Instance** dialog box.

Adding User-Defined Delta Voltage (UDV) Instances in Layouts

You can place a user-defined delta voltage (UDV) instance across two wire shapes in design layouts to overwrite the tool-calculated delta voltage value. If you already added UDV instances in the schematic, the information is transferred to layout during SDL operations. However, if you are using a non-SDL design workflow, you can manually add UDV instances in the layout.

A UDV layout cellview (basic/udv/layout) is supplied in the Custom Compiler environment. The UDV shapes that you create are stored in your layout cellview.

Before you start, set the following Tcl preferences:

- `leVDRCUdvLayerPurpose`

Specifies the layer purpose of the UDV shapes. For example:

```
db::setPrefValue leVDRCUdvLayerPurpose -value UserDV_m
```

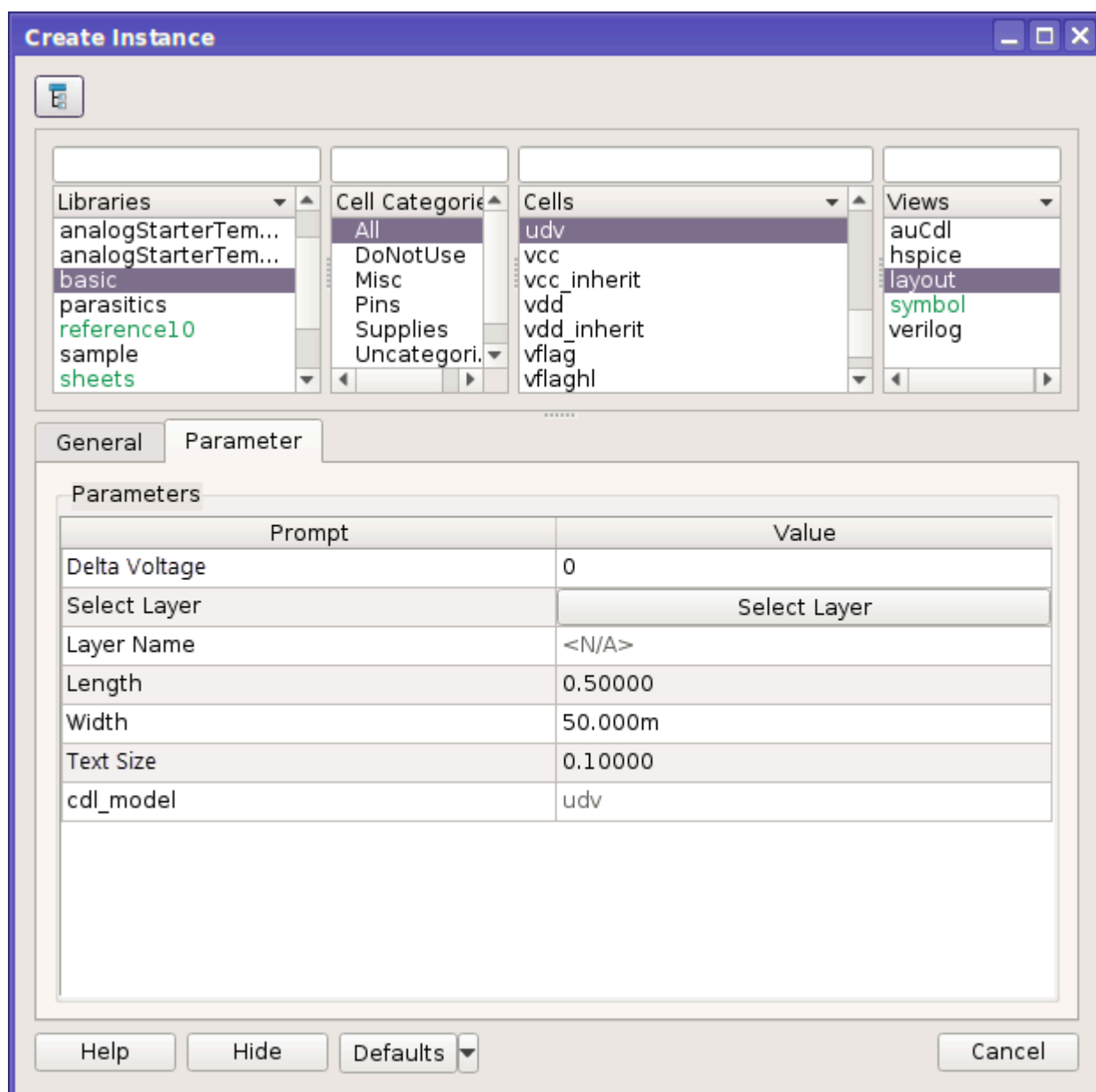
- `leVDRCUdvTextPurpose`

Specifies the layer purpose of the UDV text labels. For example:

```
db::setPrefValue leVDRCUdvTextPurpose -value UserDV
```

To add a UDV instance in the design:

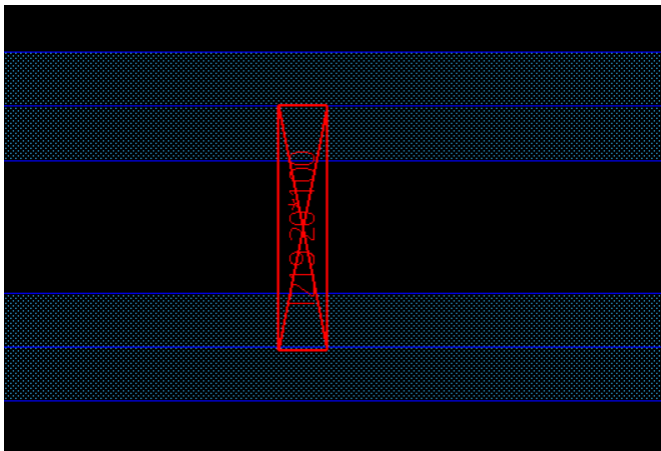
1. In the Object/Layer Panel Assistant, select the layer-purpose pair of the two paths of interest. (You are going to create the UDV instance on the same layer-purpose pair.)
2. Choose **Create > Instance**. The **Create Instance** dialog box opens.



3. Choose the following:

- **Libraries:** basic
- **Cells:** udv
- **Views:** layout

4. (Optional) In the **General** tab, set the options. See Instance Toolbar and Create Instance Dialog Box.
5. In the **Parameters** tab, set the following:
 - **Delta Voltage:** The delta voltage value that will overwrite the tool-calculated value.
 - **Select Layer:** The layer on which you are placing the UDV instance. If you already specified the layer in step 1, confirm that the layer selection is correct. The layer you select appears in the **Layer Name** box.
 - **Length:** The length of the UDV instance.
 - **Width:** The width of the UDV instance.
 - **Text Size:** The size of the text label on the UDV shape.
6. Skip the remaining options.
7. On the canvas, click to place the UDV instance between two nets (paths). Make sure the two UDV pins fully overlap with the two paths.



8. When you are finished, press Esc to close the **Add Instance** dialog box.

The newly added UDV shape displays in the Design Navigator **Insts** tab. In addition, you can view and edit its properties in the Property Editor.

See Also

- [Adding User-Defined Delta Voltage \(UDV\) Instances in Schematics](#)
- [Design Navigator Assistant](#)
- [Property Editor Assistant](#)
- [Placing Schematic Instances Into Layout](#)

Adding Voltage Flag (VFlag) Instances in Layouts

If you are only interested in the highest voltage value on a net, you can add a VFlag instance to specify the single voltage value for that net. If you already added VFlag instances in the schematic, the information is transferred to layout during SDL operations. However, if you are using a non-SDL design workflow, you can manually add VFlag instances in the layout.

A VFlag layout cellview (basic/vflag/layout) is supplied in the Custom Compiler environment. Instances of the VFlag layout that you create are stored in your layout cellview.

Before you start, set the following Tcl preferences:

- `leVDRCVflagVoltagePurposes`

Specifies the list of voltage names of the VFlag shapes. The list populates the **Select Voltage** option in the **Create Instance** dialog box.

```
db::setPrefValue leVDRCVflagVoltagePurposes -value "v_1_2 v_1_5  
v_1_8"
```

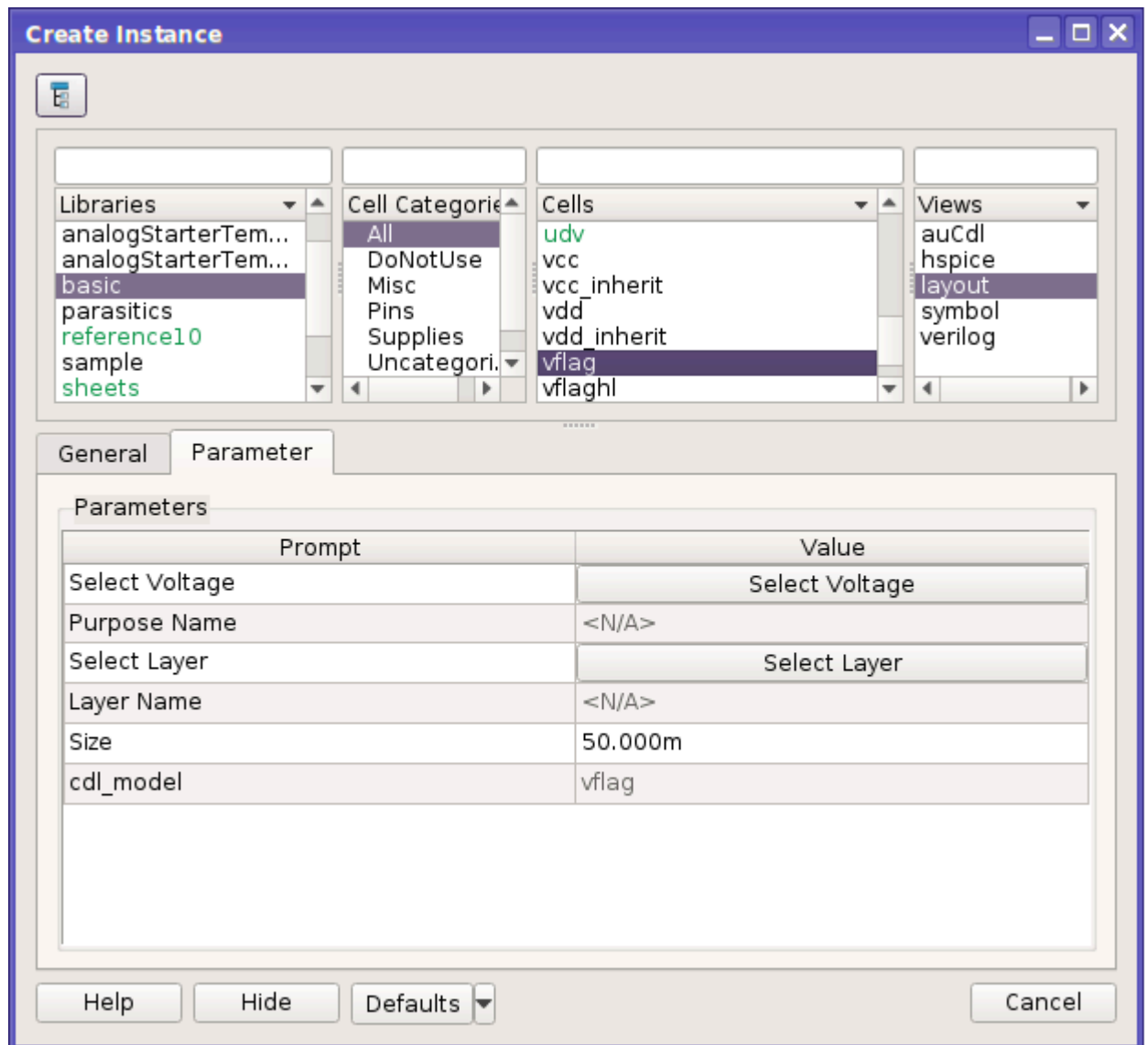
If the voltage purposes are not human-readable, you can convert them to human-readable voltage names and values. See [Converting Voltage Purposes to Human-Readable Names and Values](#).

- `leVDRCVflagPinPurpose`

Specifies the layer purpose of VFlag pins. The default is `drawing`, but you can change the value to adjust the results (for example, use the `pin` layer purpose).

To add a VFlag instance in the design:

1. In the Object/Layer Panel Assistant, select the layer-purpose pair of the two paths. (You are going to create the VFlag instance on the same layer-purpose pair.)
2. Choose **Create > Instance**. The **Create Instance** dialog box opens.



3. Choose the following:
 - **Libraries:** basic
 - **Cells:** vflag
 - **Views:** layout
4. (Optional) In the **General** tab, set the options. See Instance Toolbar and Create Instance Dialog Box.

5. In the **Parameters** tab, set the following:

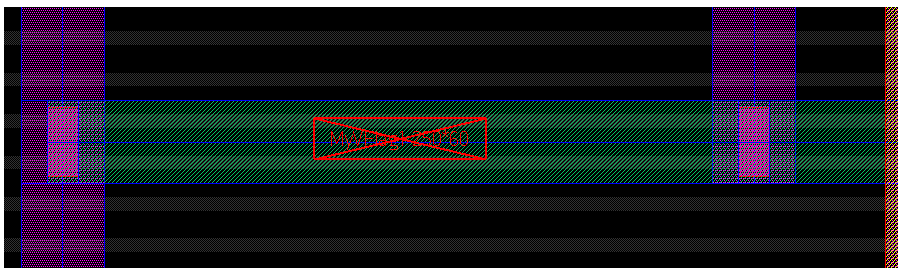
- **Select Voltage:** Click to select the voltage purpose. The selected purpose appears in the **Purpose Name** box.
- **Select Layer:** The layer on which you are placing the VFlag instance. If you already specified the layer in step 1, confirm that the layer selection is correct. The layer you select appears in the **Layer Name** box.

The list of layers that appear are defined in the technology database and have purposes defined by the `leVDRCVflagVoltagePurposes` Tcl preference.

- **Size:** The size of the instance shape.

6. Skip the remaining options.

7. On the canvas, click to place the VFlag instance on the net of interest.



8. When you are finished, press Esc to close the **Add Instance** dialog box.

The newly added VFlag shape displays in the Design Navigator **Insts** tab. In addition, you can view and edit its properties in the Property Editor.

See Also

- [Adding Voltage Flag \(VFlag\) Instances in Schematics](#)
- [Design Navigator Assistant](#)
- [Property Editor Assistant](#)
- [Placing Schematic Instances Into Layout](#)

Adding Vsync Instances in Layouts

To indicate that two different nets are synchronous in a design layout and that special voltage-dependent rules apply, you can add a vsync instance on the two net shapes. If you already added vsync instances in the schematic, the information is transferred to layout during SDL operations. However, if you are using a non-SDL design workflow, you can manually add vsync layout instances in the layout.

A vsync layout cellview (basic/vsync/layout) is supplied in the Custom Compiler environment. The vsync shapes that you create are stored in the layout cellview.

Before you start, set the following Tcl preferences:

- `leVDRCVsyncLayerPurpose`

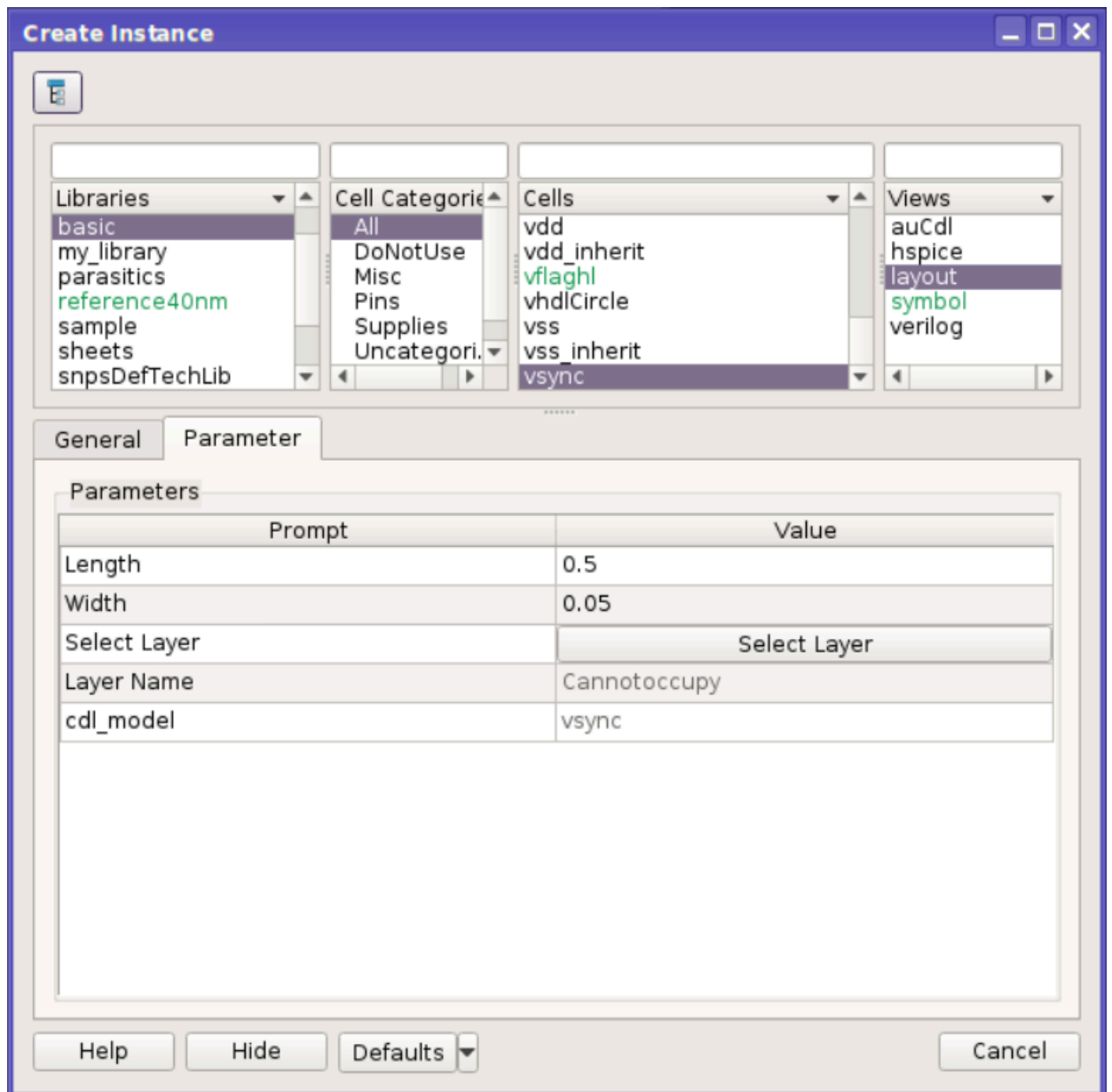
Specifies the layer purpose of the vsync shapes. The default value is `v_sync`, but you can change the value to comply with foundry requirements.

- `leVDRCVsyncPinPurpose`

Specifies the layer purpose of vsync pins. The default is `drawing`, but you can change the value to adjust the results (for example, use the `pin` layer purpose).

To add a vsync instance in the design:

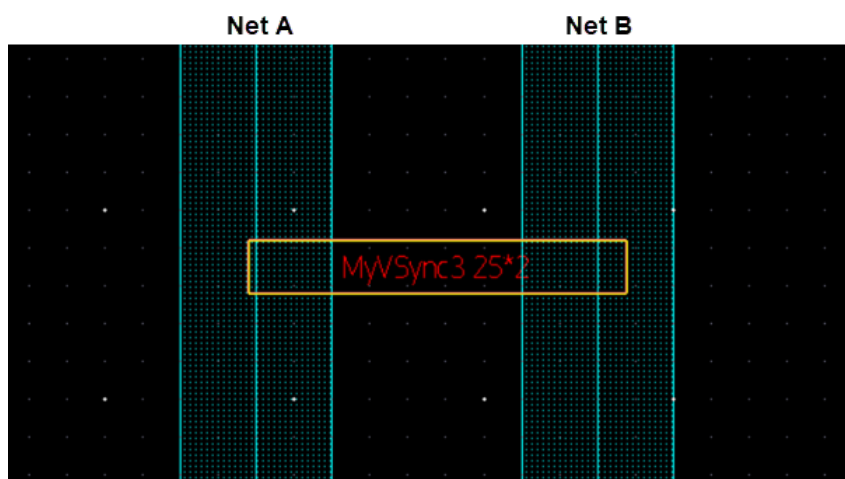
1. In the Object/Layer Panel Assistant, select the layer-purpose pair of the two synchronized paths. (You are going to create the vsync instance on the same layer-purpose pair.)
2. Choose **Create > Instance**. The **Create Instance** dialog box opens.



3. Choose the following:

- **Libraries:** basic
- **Cells:** vsync
- **Views:** layout

4. (Optional) In the **General** tab, set the options. See Instance Toolbar and Create Instance Dialog Box.
5. In the **Parameters** tab, set the following:
 - **Length**: The length of the vsync instance.
 - **Width**: The width of the vsync instance.
 - **Select Layer**: The layer on which you are placing the vsync instance. If you already specified the layer in step 1, confirm that the layer selection is correct.
6. Skip the remaining options.
7. On the canvas, click to place the vsync instance between two synchronized nets (paths). Make sure the two vsync pins fully overlap with the two paths.



8. When you are finished, press Esc to close the **Add Instance** dialog box.

The newly added vsync shape displays in the Design Navigator **Instance** tab. In addition, you can view and edit its properties in the Property Editor.

See Also

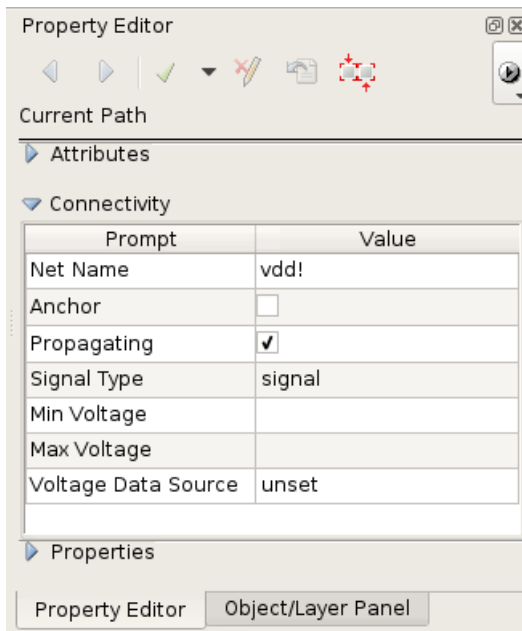
- [Adding Vsync Instances in Schematics](#)
- [Design Navigator Assistant](#)
- [Property Editor Assistant](#)

Editing Voltage Data Using the Property Editor

In design layouts, voltage values are saved as net attributes. You use the Property Editor to view and edit these attributes.

To view and edit the voltage data in design layouts:

1. Display the Property Editor Assistant.
2. In the design, select the net of interest.
3. In the Property Editor, expand the **Connectivity** section, and locate **Min Voltage** and **Max Voltage**.



The screenshot shows the 'Property Editor' dialog box. The 'Connectivity' section is expanded, showing a table with the following data:

Prompt	Value
Net Name	vdd!
Anchor	☐
Propagating	☒
Signal Type	signal
Min Voltage	
Max Voltage	
Voltage Data Source	unset

Below the table, there are tabs for 'Property Editor' and 'Object/Layer Panel'.

4. To edit the values, type new values in the **Min Voltage** and **Max Voltage** boxes.
5. In the **Voltage Data Source** list, choose one of the following to indicate the source of the voltage data: override, intent, simulation.
 - **override**: The voltage value is modified in the Layout Editor.
 - **intent**: The voltage value is modified in the Schematic Editor.
 - **simulation**: The voltage value is obtained from the simulation.
6. When you are finished, click **Apply** .

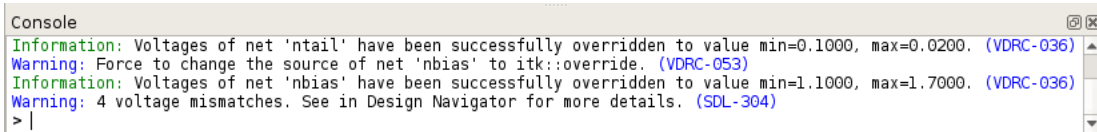
Verifying Edited Voltage Data

After overriding voltage data, you can verify the changes you made before [Creating and Removing Physical Annotations in the Layout Editor](#). Use this procedure if you automate your layout tasks using SDL, and you want to verify that the voltage annotations in the

layout are consistent with those in the `vdrc_config` cellview provided by the schematic designers.

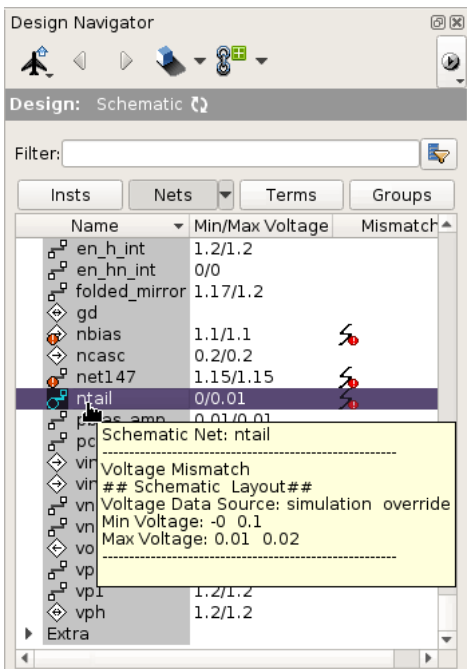
To verify the overrides you made:

1. (Optional) Display the Console.
2. In the Layout Editor, choose **Electrical > Voltage Annotation > Verify Voltages**. The Console displays a message that lets you know whether mismatched voltage values are found. A mismatch occurs when the original voltage value is different from the override or annotated value in the layout.



3. If mismatches are found, display the Design Navigator and search the **Mismatch** column.

 indicates the corresponding net has mismatched voltage values. To find out more about the mismatch, rest the pointer on the net name. A tooltip displays the original value, the source of the original value, and the override value.



4. If you find unintended override values, return to the schematic and revert the value in the `vdrc_config` cellview.

Enabling Sync to VDRC Assistant for User-defined Flag

You can setup the user-device using the following Tcl command:

```
db::defineVflagMaxDevice -deviceLCV demoPLL_ref10/vmax_symbol/symbol  
-targetLib demoPLL_ref10_tb -voltageParam vmax  
db::defineVflagMinDevice -deviceLCV demoPLL_ref10/vmin_symbol/symbol  
-targetLib demoPLL_ref10_tb -voltageParam vmin
```

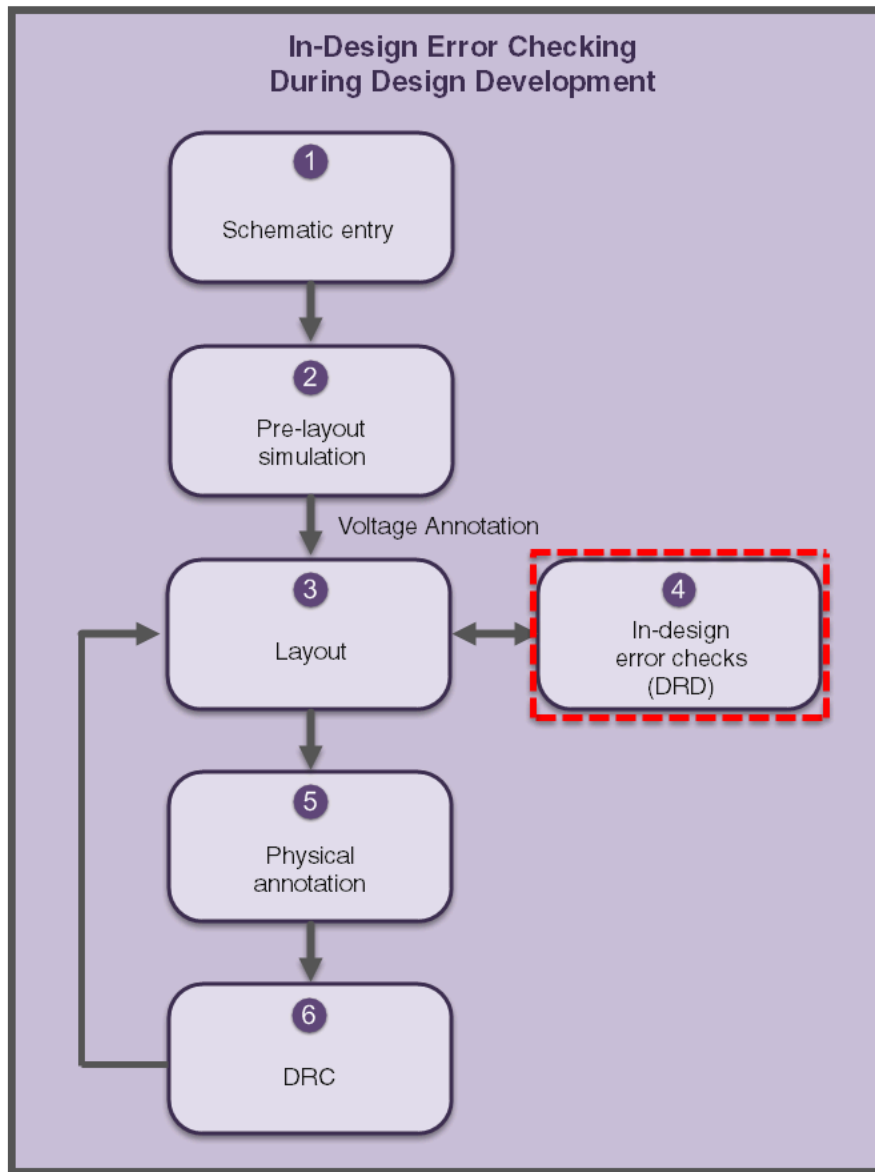
The **VDRC Assistant** refers to the device setting to create or update the voltage symbols.

Checking for Voltage-Dependent Spacing Violations as You Edit the Layout

You can use the Design Rule Driven (DRD) tool to interactively check for voltage-dependent spacing violations as you create and edit interconnects or place routes. Doing so lets you catch errors early in the design process, thus reducing the overall design timeline. The following diagram shows when you would run the DRD tool when developing designs. For information about the DRD tool, see [Checking for Violations as You Edit \(DRD\)](#).

Caution:

Before you use the DRD tool, make sure you define the following constraints in the technology database: `minVoltageSpacing` and `minVoltageClearance`.



Creating and Removing Physical Annotations in the Layout Editor

This topic describes the following:

- [About Physical Annotations](#)
- [Before You Start](#)

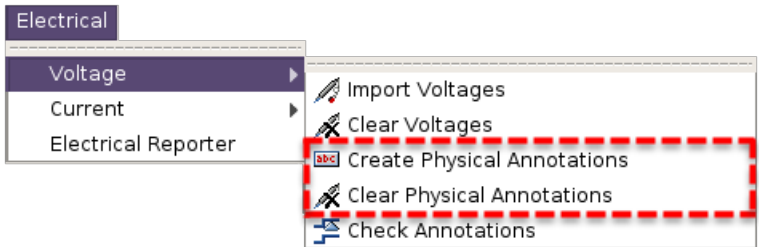
- [Creating Physical Annotations — Text Labels Method](#)
- [Creating Physical Annotations — Custom Method](#)
- [Removing Physical Annotations](#)
- [Create Physical Voltage Annotations Dialog Box Reference](#)

To check and troubleshoot the physical annotations, see [Checking Physical Voltage Annotations in the Layout Editor](#).

About Physical Annotations

Because physical verification tools, such as IC Validator, can only process physical objects in the design, you must convert the voltage annotations to physical annotations before initiating signoff checks.

You can use the following **Electrical > Voltage** commands to create and remove physical annotations:



Command	Description
Create Physical Annotations	Converts voltage annotations to physical annotations. See Creating Physical Annotations — Text Labels Method .
Clear Physical Annotations	Removes physical annotations from the design. See Removing Physical Annotations .

How Physical Annotations Are Added in Hierarchical Designs

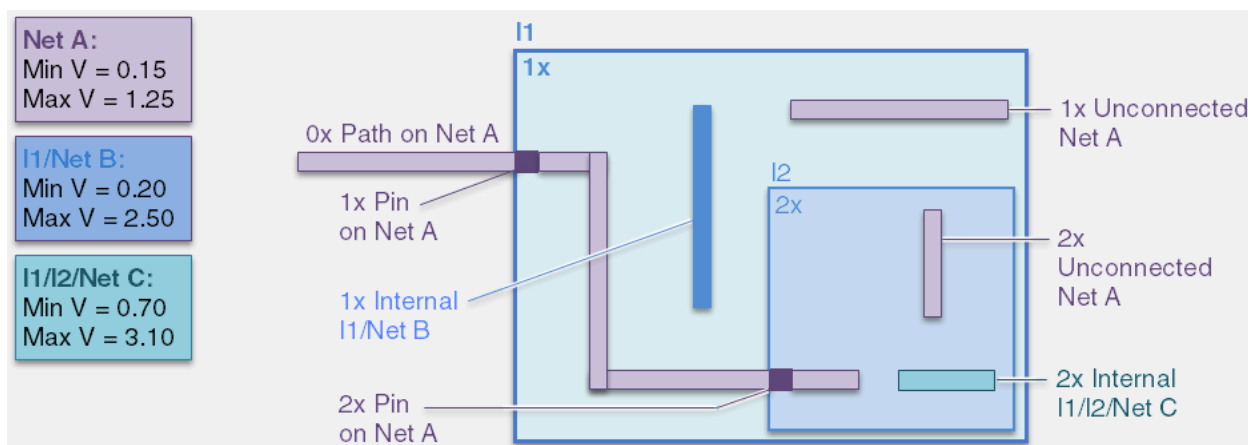
When creating physical annotations:

- You have the option of adding annotations for paths that are at 0x only, up to the view Stop Level, or across all levels of the design hierarchy.
- A pair of voltage values are associated with each fully connected island: minimum voltage value (loV) and maximum voltage value (hiV). The pair of voltage values

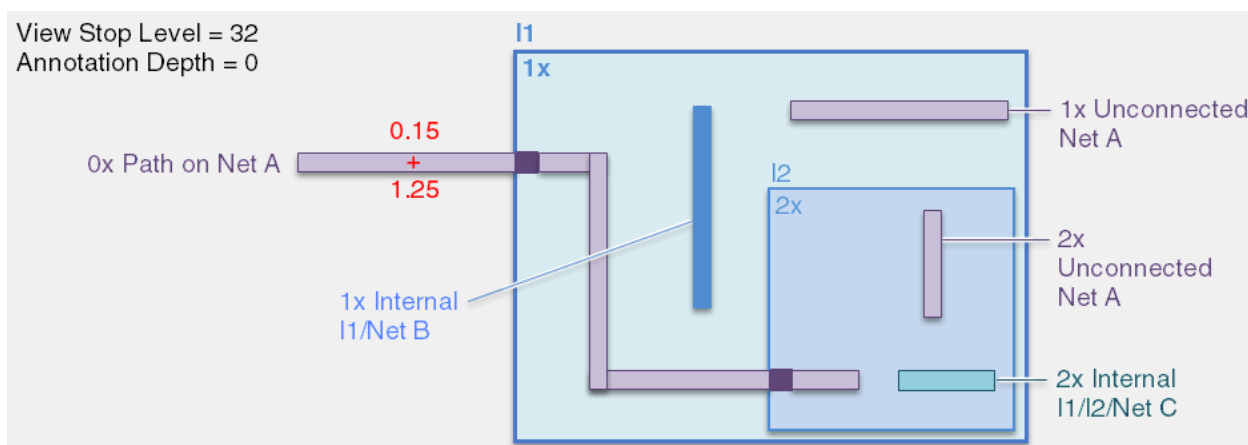
are added for same-island connections that are at each level below 0x, up to the annotation depth that you specify.

- The annotations are placed at 0x only.

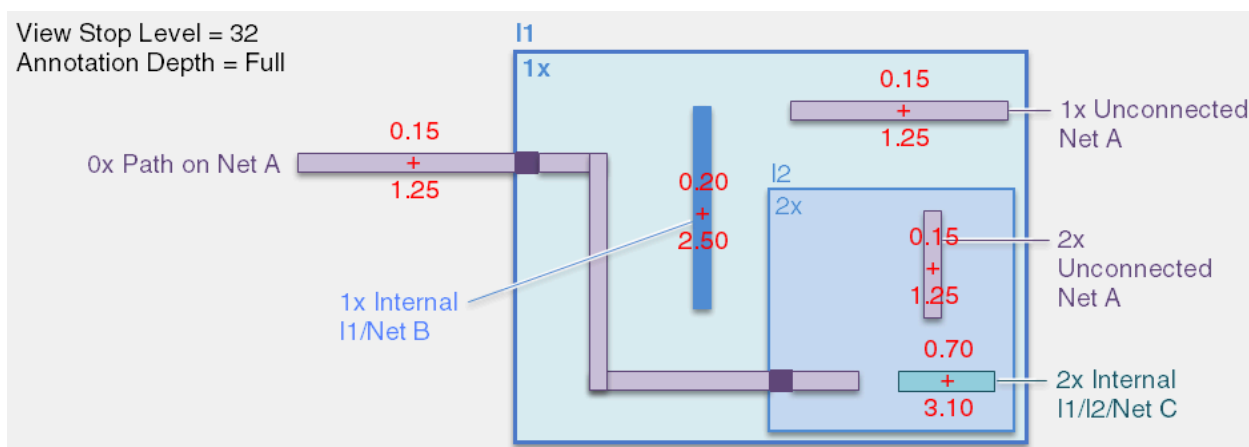
In the following scenario, the design contains two instances (I1 and I2). Shapes are on different nets at the 0x, 1x, and 2x levels.



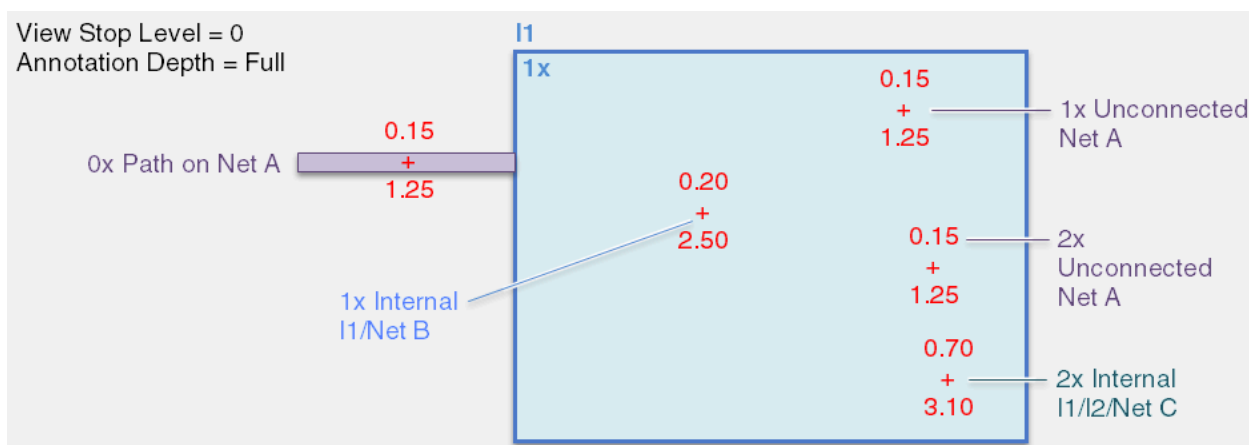
If the annotation depth is 0, annotations are added only to the 0x path segment.



If the annotation depth is Full, annotations are added to each connected island throughout the hierarchy.



When the annotation depth is Full (full hierarchy), but the view Stop Level is 0, you can see the 0x annotations created for all the levels in the hierarchy, but you cannot see the objects below 0x.



Priorities Used During Physical Annotation

The following priority is used during the annotation process:

1. Hierarchy level (0x, 1x, 2x, and so on).
2. Object type within each hierarchy level:
 - a. 0x pins
 - b. 0x paths or path segments
 - c. 0x polygons
 - d. 0x rectangles

- e. 0x vias
- f. 1x pins in 0x instance terminals

3. LPP map order specified using one of the following:

- `leVDRCPhysicalAnnotationLabelTextLPPs` Tcl preference.
- Technology file (`voltageTextLayers` in the `layerRules` section)

See [Before You Start](#).

In the following scenario, both the via and the path are on 0x. However, the annotation will be created on the M2 path, because paths have a higher priority than vias.

Hierarchy Level	Object Type	Text LPPs
0x	Via	M1
0x	Path	M2

In the following scenario, both path1 and path2 are on 0x. The annotation will be created on path1, because M1 has a higher priority than M2 in the LPP Map table.

Hierarchy Level	Object Type	Text LPPs
0x	path1	M1
0x	path2	M2

LPP Map Priority

Typically, a net is composed of multiple shapes, and the different shapes can be on different LPPs. When the shape LPPs match more than one LPP mapping, the first-encountered LPP is used.

In the following example, if the design has shapes on both the M1 drawing and M2 drawing LPPs, then annotation will be created on M1 drawing. However, if the M1 drawing LPP does not have any shapes, M2 drawing will be used.

Object	Vmin	Vmax
M1 drawing	M1 loV	M1 hiV
M2 drawing	M2 loV	M2 hiV

Before You Start

Make sure you:

- Specify the layer-purpose pair mappings.
- (Optional) Specify the viewport halo distance.

To specify the layer-purpose pair (LPP) mappings, use one of the following methods:

- Technology database (voltageTextLayers in the layerRules section)

Example:

```
voltageTextLayers(  
  ( (M1 drawing) (M1 loV) (M1 hiV) )  
  ( (M2 drawing) (M2 loV) (M2 hiV) )  
  ...  
)
```

- leVDRCPhysicalAnnotationLabelTextLPPs Tcl preference.

Example:

```
db::setPrefValue leVDRCPhysicalAnnotationLabelTextLPPs -value \  
{ { (M1 drawing) (M1 loV) (M1 hiV) } \  
  { (M2 drawing) (M2 loV) (M2 hiV) } }
```

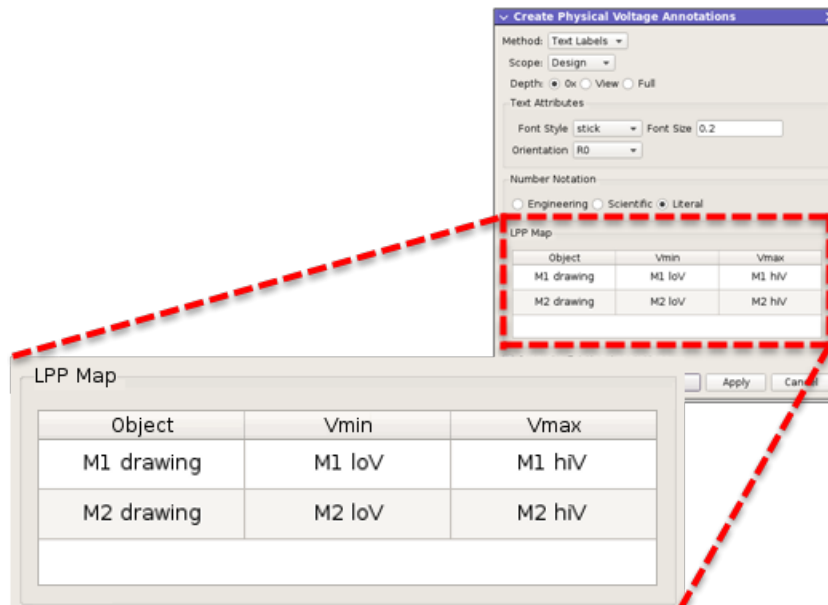
The physical annotation operation always checks the technology database first. If the information is not available in the technology database, the Tcl preference is checked.

If the LPP mapping information is specified using both the Tcl preference and the voltageTextLayers section, the information from the Tcl preference will be used.

The LPP mapping information from the Tcl preference or the technology database will be displayed in the **Create Physical Voltage Annotations** dialog box.

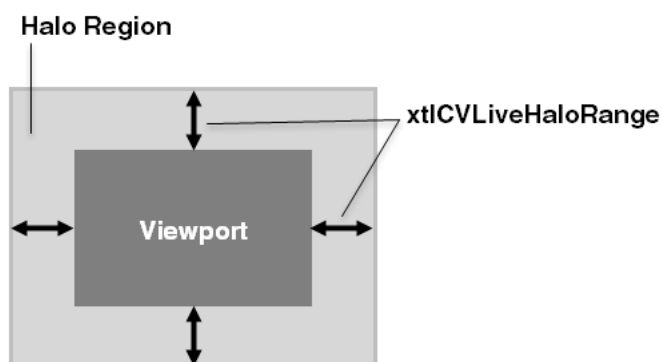
Chapter 10: Annotating Voltage Data for VDRC

Creating and Removing Physical Annotations in the Layout Editor



When you set up for physical voltage annotation, you can specify the scope of the operation: within the design, within the viewport, or selected nets. If you choose to annotate within the viewport, you might want to define a halo region surrounding viewport. The **Create Physical Annotations** command will consider objects within the halo region to improve the annotation results.

To define the halo region, use the `xtICVLiveHaloRange` Tcl preference to specify the distance from the viewport boundary.



Creating Physical Annotations — Text Labels Method

Caution:

Make sure you verify all override values before you create the physical annotations. If you use SDL in your design flow, see [Verifying Edited Voltage Data](#).

Caution:

Because of varying manufacturing tolerances, physical annotations are foundry-specific. Before you create the physical annotations, be sure to check with the foundry and make sure the annotations will meet the criteria.

You can use one of the following methods to convert voltage annotations to physical annotations in the Layout Editor:

Method	Description
Text Labels	Converts annotations to text labels. A pair of text labels, minimum voltage and maximum voltage, will be created for each path in a net. The minimum voltage value will appear above the maximum voltage value.
Custom	Runs the specified custom Tcl procedure that addresses special annotation requirements. The Custom method lets you create new physical annotations incrementally and does not overwrite existing annotations. You have the option of removing existing annotations that were created by a specific Tcl procedure before adding new annotations.

This section describes how to use the **Text Labels** method. To use the Custom method, see [Creating Physical Annotations — Custom Method](#).

To convert the voltage annotations to physical annotations using the **Text Labels** method:

1. In the Layout Editor, choose **Electrical > Voltage > Create Voltage Annotations**. The **Create Physical Voltage Annotation** dialog box opens.
2. In the **Method** list, choose **Text Labels**. The content of the dialog box changes to display the Text Labels options.

▼ Create Physical Voltage Annotations

Method: Text Labels

Scope: Design

Depth: ☒ 0x ☐ View ☐ Full

Text Attributes

Font Style: stick Font Size: 0.2

Orientation: R0

Number Notation

☐ Engineering ☐ Scientific ☒ Literal

LPP Map

Object	Vmin	Vmax
M1 drawing	M1 loV	M1 hiV
M2 drawing	M2 loV	M2 hiV

☒ Overwrite Existing Annotations

Help

Hide

OK

Apply

Cancel

3. In the **Scope** list, choose the extent of the physical annotation operation:

Selection	Description
Design	Adds physical annotations to the entire design.
Viewport	Adds physical annotations to the components within the viewport. If you want the command to consider objects in the halo region just beyond the viewport, see Checking Physical Voltage Annotations in the Layout Editor .
Nets	Adds physical annotations to the specified nets. The command considers the shapes of the nets in the entire design. To specify the nets, type one or more net names in the box, or click  to select the nets in the Design Navigator.

4. For the **Depth** of the routing shapes to consider, select one of the following:

Selection	Description
0x	Adds annotations for connections at the top level only.
View	Adds annotations for connections that are at 0x through the Stop Level that you specify.
Full	Adds annotations for connections at all levels of the design hierarchy.

5. Set the **Text Attributes**:

Option	Description
Font Style	Style of the text. Choose from the list of available styles. Tcl preference: <code>leVDRRCPhysicalAnnotationLabelTextFontStyle</code>
Font Size	Size of the text. Type the size in the box. Tcl preference: <code>leVDRRCPhysicalAnnotationLabelTextFontSize</code>
Orientation	Text orientation: ◦ R0: Does not rotate the text. ◦ R90: Rotates the text 90° counterclockwise. ◦ R180: Rotates the text 180° counterclockwise. ◦ R270: Rotates the text 270° counterclockwise. ◦ MX: Flips the text along the x-axis. ◦ MY: Flips the text along the y-axis. ◦ MXR90: Flips the text along the x-axis and rotates it 90° counterclockwise. ◦ MYR90: Flips the text along the y-axis and rotates it 90° counterclockwise. Tcl preference: <code>leVDRRCPhysicalAnnotationLabelTextFontOrient</code>

6. Select the **Number Notation**.

Option	Example of how 100 is expressed
Engineering	0.10e+03
Scientific	1.00e+02
Literal	100.00

7. Review the **LPP Map** table.

The LPP Map table contains information from the `leVDRCPhysicalAnnotationLabelTextLPPs` Tcl preference or the `voltageTextLayers` subsection of the technology database. See [Before You Start](#).

Column	Description
Object	LPP of the wires.
Vmin	LPP on which the minimum voltage annotation objects will be placed.
Vmax	LPP on which the maximum voltage annotation objects will be placed.

8. Select **Overwrite Existing Annotations** if you want to overwrite existing voltage annotations within the **Scope** you chose in [Step 3](#). Clear the check box if you want to add voltage annotations incrementally.
9. When you are finished, click **OK** or **Apply**. The voltage annotations are converted to physical objects.

Creating Physical Annotations — Custom Method

Caution:

Make sure you verify all override values before you create the physical annotations. If you use SDL in your design flow, see [Verifying Edited Voltage Data](#).

Caution:

Because of varying manufacturing tolerances, physical annotations are foundry-specific. Before you create the physical annotations, be sure to check with the foundry and make sure the annotations will meet the criteria.

You can use one of the following methods to convert voltage annotations to physical annotations in the Layout Editor:

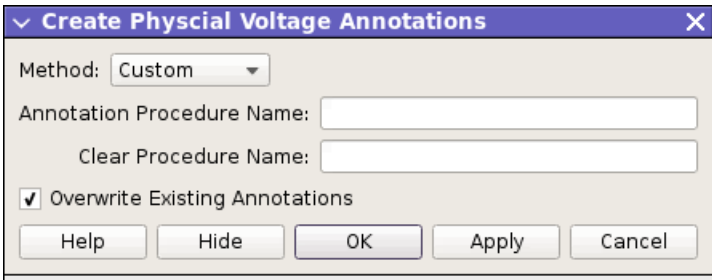
Method	Description
Text Labels	Converts annotations to text labels. A pair of text labels, minimum voltage and maximum voltage, will be created for each path in a net. The minimum voltage value will appear above the maximum voltage value.

Method	Description
Custom	Runs the specified custom Tcl procedure that addresses special annotation requirements. The Custom method lets you create new physical annotations incrementally and does not overwrite existing annotations. You have the option of removing existing annotations that were created by a specific Tcl procedure before adding new annotations.

This section describes how to use the **Custom** method. To use the Text Labels method, see [Creating Physical Annotations — Text Labels Method](#).

To convert the voltage annotations to physical annotations using the Custom method:

1. In the Layout Editor, choose **Electrical > Voltage > Create Voltage Annotations**. The **Create Physical Voltage Annotation** dialog box opens.
2. In the **Method** list, choose **Custom**. The content of the dialog box changes to display the Custom options.



3. Set the following options:

Setting	Description
Annotation Procedure Name	Name of the Tcl procedure that will create the custom physical annotations. The procedure is run when you click OK or Apply in this dialog box. Tcl preference: <code>leVDRCPhysicalAnnotationCustomProcedureName</code>
Clear Procedure Name	Name of the Tcl procedure that will remove physical annotations previously added using the create physical annotation process. The clear procedure is run when you select Overwrite Existing Annotations and click Apply or OK in this dialog box. (Alternatively, you can choose Electrical > Voltage > Clear Physical Annotations in the Layout Editor. See Removing Physical Annotations .) Tcl preference: <code>leVDRCPhysicalAnnotationCustomClearProcedureName</code>

Setting	Description
Overwrite Existing Annotations	Overwrites existing voltage annotations in the design. Clear the check box if you want to add voltage annotations incrementally. Tcl preference: <code>leVDRCPhysicalAnnotationOverwrite</code>

4. When you are finished, click **OK** or **Apply**. The voltage annotations are converted to physical objects.

Removing Physical Annotations

To remove all existing physical annotations in the design:

- In the Layout Editor, choose **Electrical > Voltage > Clear Physical Annotations**.

Create Physical Voltage Annotations Dialog Box Reference

This section presents a quick reference of the options in the **Create Physical Voltage Annotations** dialog box. For instructions on creating physical voltage annotations, see [Creating Physical Annotations — Text Labels Method](#) and [Creating Physical Annotations — Custom Method](#).

The content of the **Create Physical Voltage Annotations** dialog box changes depending on the **Method** you choose:

- [Method: Text Labels](#)
- [Method: Custom](#)
- [Overwrite Existing Annotations](#)

Tcl preference: `leVDRCPhysicalAnnotationMethod`

Regardless of the method, you can specify whether you want to [Overwrite Existing Annotations](#).

Method: Text Labels

▼ Create Physical Voltage Annotations

Method: Text Labels

Scope: Design

Depth: ☒ 0x ☐ View ☐ Full

Text Attributes

Font Style: stick Font Size: 0.2

Orientation: R0

Number Notation

☐ Engineering ☐ Scientific ☒ Literal

LPP Map

Object	Vmin	Vmax
M1 drawing	M1 loV	M1 hiV
M2 drawing	M2 loV	M2 hiV

☒ Overwrite Existing Annotations

Help Hide OK Apply Cancel

Scope:

Option	Description
Design	Adds physical annotations to the entire design.
Viewport	Adds physical annotations to the components within the viewport.
Nets	Adds physical annotations to the specified nets. The command considers the shapes of the nets in the entire design. To specify the nets, type one or more net names in the box, or click  to select the nets in the Design Navigator.

Tcl preference: leVDRCAnnotationScope

Depth:

Option	Description
0x	Adds annotations for connections at the top level only.
View	Adds annotations for connections that are at 0x through the Stop Level that you specify.
Full	Adds annotations for connections at all levels of the design hierarchy.

Tcl preference: `leVDRCAnnotationDepth`

Text Attributes:

Option	Description
Font Style	Style of the text. Choose from the list of available styles. Tcl preference: <code>leVDRCPhysicalAnnotationLabelTextFontStyle</code>
Font Size	Size of the text. Type the size in the box. Tcl preference: <code>leVDRCPhysicalAnnotationLabelTextFontSize</code>
Orientation	Text orientation: • R0: Does not rotate the text. • R90: Rotates the text 90° counterclockwise. • R180: Rotates the text 180° counterclockwise. • R270: Rotates the text 270° counterclockwise. • MX: Flips the text along the x-axis. • MY: Flips the text along the y-axis. • MXR90: Flips the text along the x-axis and rotates it 90° counterclockwise. • MYR90: Flips the text along the y-axis and rotates it 90° counterclockwise. Tcl preference: <code>leVDRCPhysicalAnnotationLabelTextFontOrient</code>

Number Notation:

Option	Example of how 100 is expressed
Engineering	0.10e+03
Scientific	1.00e+02
Literal	100.00

Tcl preference: `leVDRCPhysicalAnnotationNumberNotation`

LPP Map:

The LPP Map table displays the layer-purpose pair (LPP) mappings for the physical annotations.

Column	Description
Object	LPP of the wires.
Vmin	LPP on which the minimum voltage annotation objects will be placed.
Vmax	LPP on which the maximum voltage annotation objects will be placed.

The LPP mapping information is derived from the `leVDRCPhysicalAnnotationLabelTextLPPs` Tcl preference or the `layerRules` section (`voltageTextLayers` subsection) of the technology database. See [Before You Start](#).

Method: Custom

The Custom method has the following options:

▼ Create Physical Voltage Annotations

Method: Custom

Annotation Procedure Name:

Clear Procedure Name:

☒ Overwrite Existing Annotations

Help Hide OK Apply Cancel

Setting	Description
Method	See Creating Physical Annotations — Text Labels Method .
Annotation Procedure Name	Name of the Tcl procedure that will create the custom physical annotations. The procedure is run when you click OK or Apply in this dialog box. Tcl preference: <code>leVDRCPhysicalAnnotationCustomProcedureName</code>

Setting	Description
Clear Procedure Name	Name of the Tcl procedure that will remove physical annotations previously added using the create physical annotation process. The clear procedure is run when you select Overwrite Existing Annotations and click Apply or OK in this dialog box. (Alternatively, you can choose Electrical > Voltage > Clear Physical Annotations in the Layout Editor. See Removing Physical Annotations .) Tcl preference: leVDRCPhysicalAnnotationCustomClearProcedureName
Overwrite Existing Annotations	Overwrites existing voltage annotations in the design. Clear the check box if you want to add voltage annotations incrementally. Tcl preference: leVDRCPhysicalAnnotationOverwrite

Overwrite Existing Annotations

The Overwrite Existing Annotations option overwrites the existing voltage annotations in the design. By default, the option is selected. If you clear the check box, the voltage annotations are added incrementally.

Tcl preference: leVDRCPhysicalAnnotationOverwrite

Checking Physical Voltage Annotations in the Layout Editor

Although you can use the in-design tools to catch errors and make modifications early in the design process, some changes might be necessary in later design phases, such as during ECO.

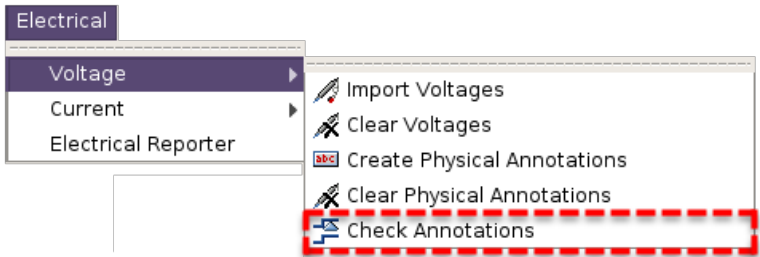
Modifying voltage values or changing net shapes in later design phases might cause discrepancies between the net information and the physical annotations. To locate these differences and check for other errors, you can use the **Electrical > Voltage > Check Annotations** command:

This topic describes the following:

- [About the Check Annotations Command](#)
- [Before You Start Checking Physical Voltage Annotations in the Layout Editor](#)
- [Checking for Physical Annotation Errors](#)
- [Check Voltage Annotations Toolbar and Dialog Box Reference](#)

About the Check Annotations Command

Modifying voltage values or changing net shapes in later design phases might cause discrepancies between the net information and the physical annotations. To locate these differences and check for other errors, you can use the **Electrical > Voltage > Check Annotations** command:

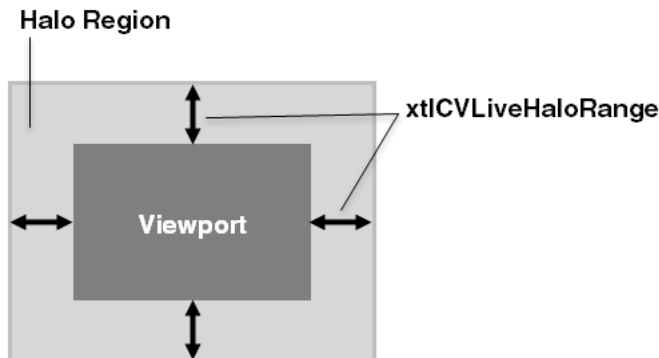


Command	Description
Check Annotations	Checks for: • Missing voltage values on nets • Missing text-label pairs on connection islands • Excess text labels on connection islands • Incorrect text-label positions • Incorrect text-label voltage values • Incorrect text-label layer-purpose pairs

Before You Start Checking Physical Voltage Annotations in the Layout Editor

When you set up for voltage annotation checking, you can specify the scope of the checks: within the design, within the viewport, or selected nets. This topic describes how to prepare for viewport scope voltage annotation checking.

If you choose to check within the viewport, you might want to define a halo region surrounding viewport. The **Check Annotations** command considers objects within the halo region to improve the error checking results. To define the halo region, use the `xtICVLiveHaloRange` Tcl preference to specify the distance from the viewport boundary.

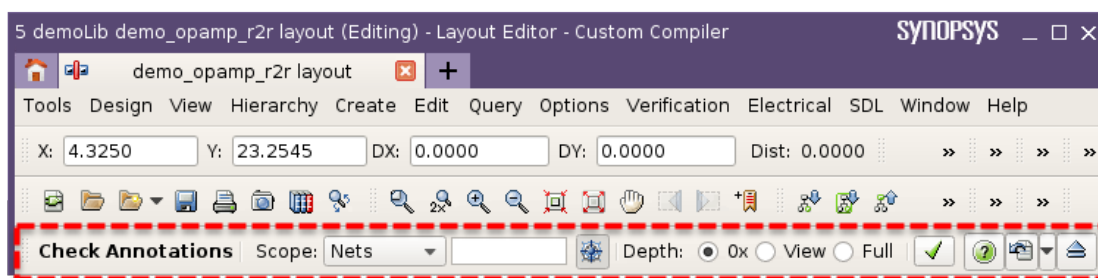


Checking for Physical Annotation Errors

The **Check Annotations** command lets you check for errors introduced during late design phase voltage value modifications. Checking and resolving the errors ensures successful ECO processes.

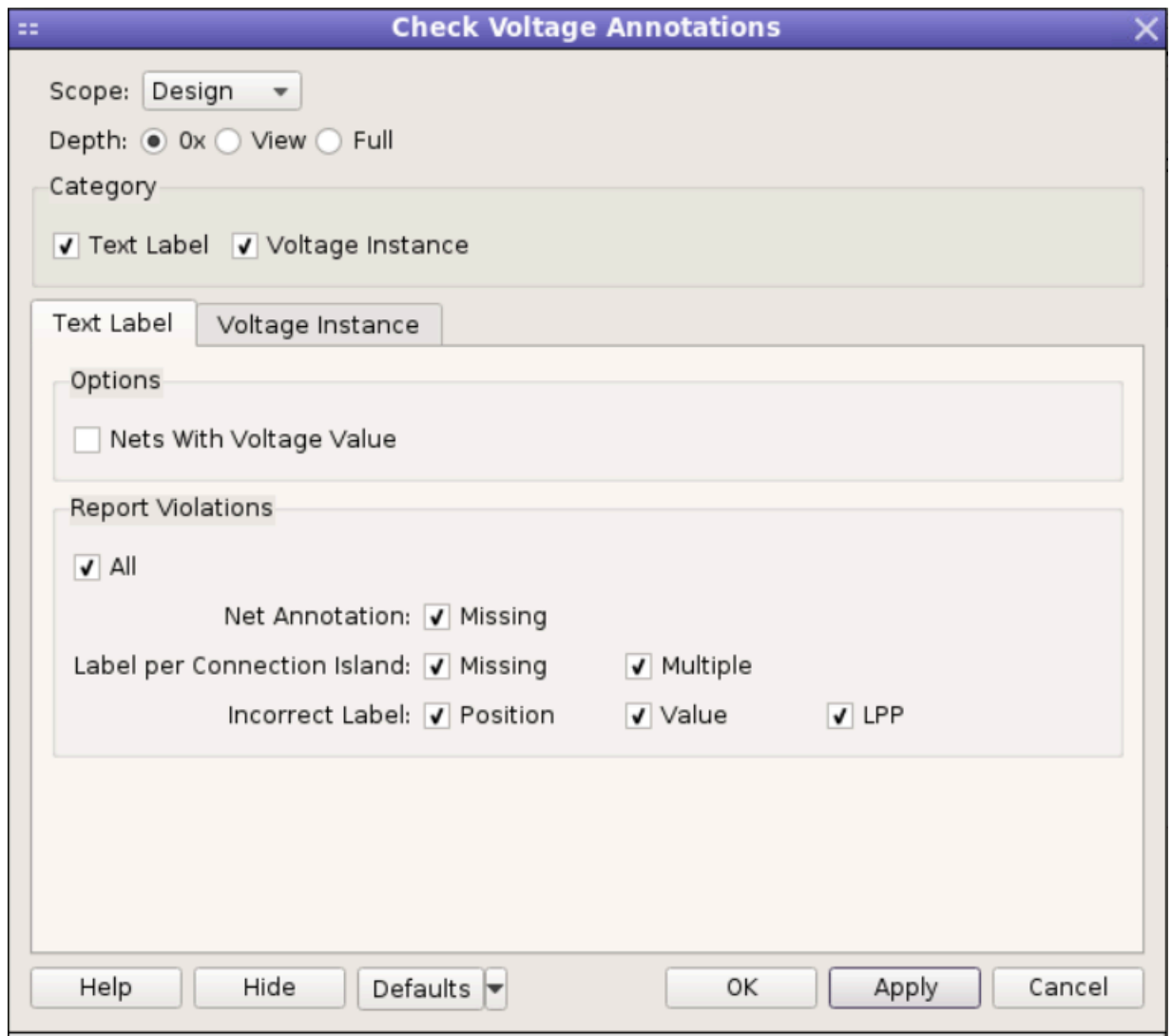
To check for physical annotation errors:

1. In the Layout Editor, choose **Electrical > Voltage > Check Annotations**. The **Check Annotations** toolbar appears.



2. (Optional) On the toolbar, click Eject  to display the **Check Voltage Annotations** dialog box. The dialog box contains additional options.

To hide the dialog box and return to the toolbar, click **Hide** in the dialog box.



The "Check Voltage Annotations" dialog box is shown with the following settings:

- Scope: Design
- Depth: 0x (selected), View, Full
- Category: ☒ Text Label, ☒ Voltage Instance
- Text Label tab is selected.
- Options: ☐ Nets With Voltage Value
- Report Violations: ☒ All
 - Net Annotation: ☒ Missing
 - Label per Connection Island: ☒ Missing, ☒ Multiple
 - Incorrect Label: ☒ Position, ☒ Value, ☒ LPP
- Buttons: Help, Hide, Defaults (dropdown), OK, Apply, Cancel

3. At the top of the dialog box, set the following options:

Option Name	Description
Scope	Size of the area to consider when checking annotations: ◦ Design: Checks the entire design. ◦ Viewport: Checks the components within the viewport. If you want the command to consider objects in the halo region just beyond the viewport, see Before You Start Checking Physical Voltage Annotations in the Layout Editor. ◦ Nets: Checks the specified nets in the entire design. To specify the nets, type one or more net names in the box, or click  to select the nets in the Design Navigator. Tcl preference: <code>leVDRCAnnotationScope</code>
Depth	Depth of the hierarchy to consider: ◦ 0x: Checks at the top level only. ◦ View: Checks at 0x through the Stop Level that you specify. ◦ Full: Checks all levels of the design hierarchy. Tcl preference: <code>leVDRCAnnotationDepth</code>
Category (Dialog box only)	Type of objects to check: ◦ Text Label: Enables the Text Label tab and lets you check text labels created by you or the tool. Tcl preference: <code>leVDRCAnnotationCheckTextLabel</code> ◦ Voltage Instance: Enables the Voltage Instance tab and lets you check the Vsync, Vflag, and Udv voltage instances. Tcl preference: <code>leVDRCAnnotationCheckVoltageInstance</code>

4. (Dialog box only) If you selected **Category: Text Label**, set the following options on the **Text Label** tab:

- In the **Options** section:

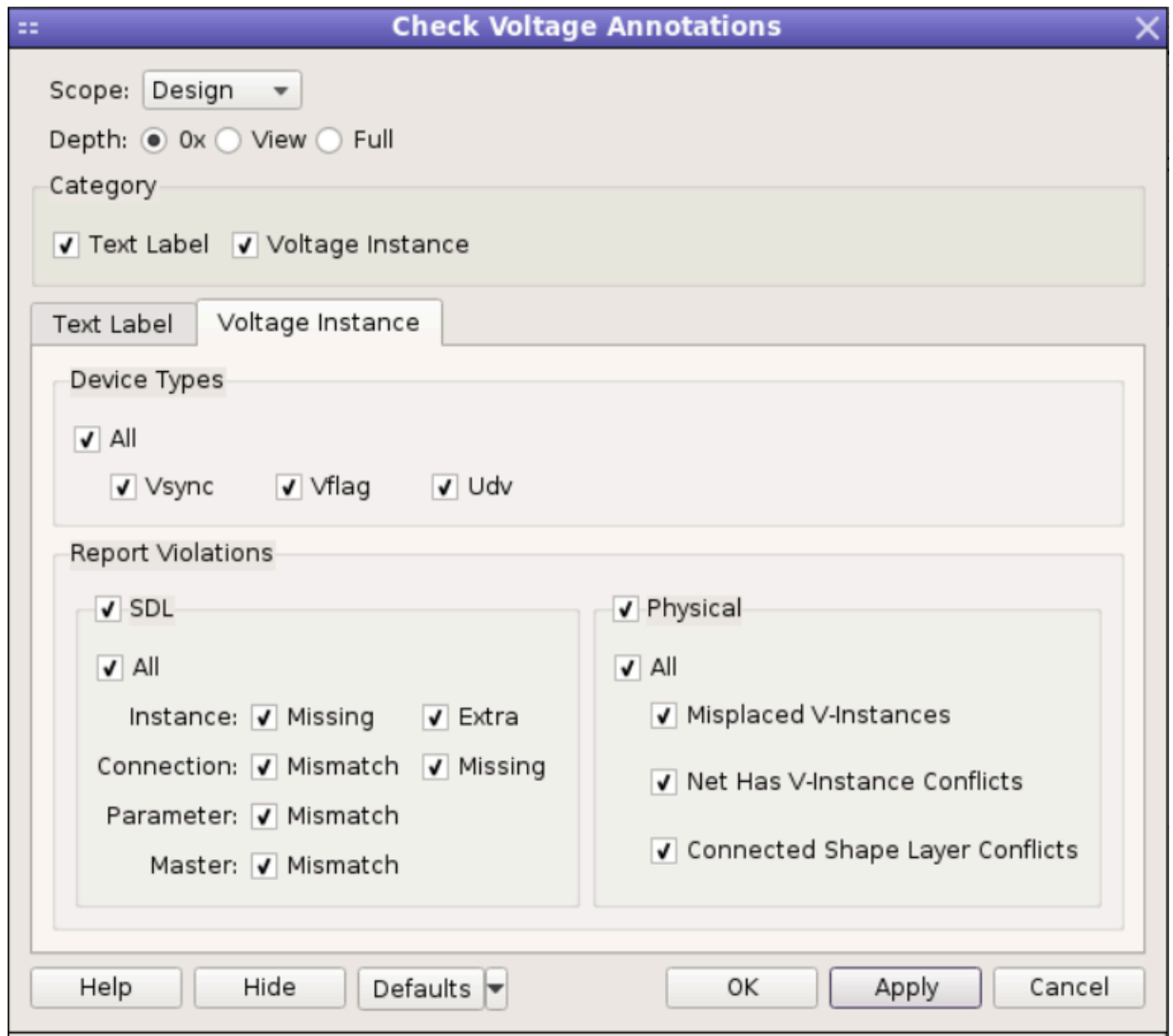
Option	Description
Nets With Voltage Value	Considers nets only if they have voltage values and ignore nets that do not have voltage values. Tcl preference: <code>leVDRCAnnotationNetsWithVoltageValue</code>

- In the **Report Violations** section:

Option	Description
All	(Default) Checks for all error types.

Option	Description
Net Annotation	Checks for missing voltage values on the nets. Tcl preference: <code>leVDRCAnnotationMissingNetAnnotation</code>
Label per Connection Island	Checks for: ▪ Missing: Missing pairs of text labels (min and max voltage values) on connection islands. Tcl preference: <code>leVDRCAnnotationMissingTextLabels</code> ▪ Multiple: Excess pairs of text labels on connection islands. Tcl preference: <code>leVDRCAnnotationMultipleTextLabels</code> (Each connection island should only have one pair of text labels. See How Physical Annotations Are Added in Hierarchical Designs.)
Incorrect Label	Checks for: ▪ Position: Text labels whose origins are not on connection islands. Tcl preference: <code>leVDRCAnnotationIncorrectTextLabelPosition</code> ▪ Value: Text labels that have incorrect voltage values. Tcl preference: <code>leVDRCAnnotationIncorrectTextLabelValue</code> ▪ LPP: Text labels that are on incorrect layer-purpose pairs (LPPs). Tcl preference: <code>leVDRCAnnotationIncorrectTextLabelLPP</code>

5. (Dialog box only) If you selected **Category: Voltage Instance**, set the following options on the **Voltage Instance** tab:



The image shows the "Check Voltage Annotations" dialog box. It has a title bar with a close button. The "Scope" is set to "Design". The "Depth" has three radio buttons: "0x" (selected), "View", and "Full". The "Category" section has two checkboxes: "Text Label" and "Voltage Instance", both of which are checked. Below this are two tabs: "Text Label" and "Voltage Instance". The "Voltage Instance" tab is active. It contains a "Device Types" section with checkboxes for "All", "Vsync", "Vflag", and "Udv", all of which are checked. Below that is a "Report Violations" section with two columns. The left column is for "SDL" and the right column is for "Physical". Both columns have an "All" checkbox checked. Under "SDL", there are checkboxes for "Instance" (Missing, Extra), "Connection" (Mismatch, Missing), "Parameter" (Mismatch), and "Master" (Mismatch), all of which are checked. Under "Physical", there are checkboxes for "Misplaced V-Instances", "Net Has V-Instance Conflicts", and "Connected Shape Layer Conflicts", all of which are checked. At the bottom of the dialog are buttons for "Help", "Hide", "Defaults" (with a dropdown arrow), "OK", "Apply", and "Cancel".

Check Voltage Annotations

Scope: Design

Depth: ☒ 0x ☐ View ☐ Full

Category

☒ Text Label ☒ Voltage Instance

Text Label Voltage Instance

Device Types

☒ All

☒ Vsync ☒ Vflag ☒ Udv

Report Violations

☒ SDL

☒ All

Instance: ☒ Missing ☒ Extra

Connection: ☒ Mismatch ☒ Missing

Parameter: ☒ Mismatch

Master: ☒ Mismatch

☒ Physical

☒ All

☒ Misplaced V-Instances

☒ Net Has V-Instance Conflicts

☒ Connected Shape Layer Conflicts

Help Hide Defaults OK Apply Cancel

- In the **Devices Types** section:

Option	Description
Device Types	▪ All: All device types (Vsync, Vflag, and Udv). ▪ Vsync: Vsync devices only. Tcl preference: <code>leVDRCAnnotationCheckVsync</code> ▪ Vflag: Vflag devices only. Tcl preference: <code>leVDRCAnnotationCheckVflag</code> ▪ Udv: Udv devices only. Tcl preference: <code>leVDRCAnnotationCheckUdv</code>

- In the **Report Violations: SDL** section:

Option	Description
SDL	Checks the consistency of voltage instances between the design schematic the corresponding schematic-driven layout. This section is available only if the layout was generated from a schematic through the SDL operation. Tcl preference: <code>leVDRCAnnotationSDLErrorGroup</code>
All	Checks for all SDL error types.
Instance	Checks for: ▪ Missing: Missing voltage instances. Tcl preference: <code>leVDRCAnnotationMissingVInst</code> ▪ Extra: Excess voltage instances. Tcl preference: <code>leVDRCAnnotationExtraVInst</code>
Connection	Checks for: ▪ Mismatch: Instance terminal and voltage instance connections that are mismatched between the schematic and layout. Tcl preference: <code>leVDRCAnnotationMismatchedVInstConn</code> ▪ Missing: Missing instance terminal and voltage instance connections. Tcl preference: <code>leVDRCAnnotationMissingVInstConn</code>
Parameter	▪ Mismatch: Voltage instance parameter values that are mismatched between the schematic and layout. Tcl preference: <code>leVDRCAnnotationMismatchedVInstParams</code>
Master	▪ Mismatch: Masters of the voltage instances are mismatched between the schematic and layout. Tcl preference: <code>leVDRCAnnotationMismatchedVInstParams</code>

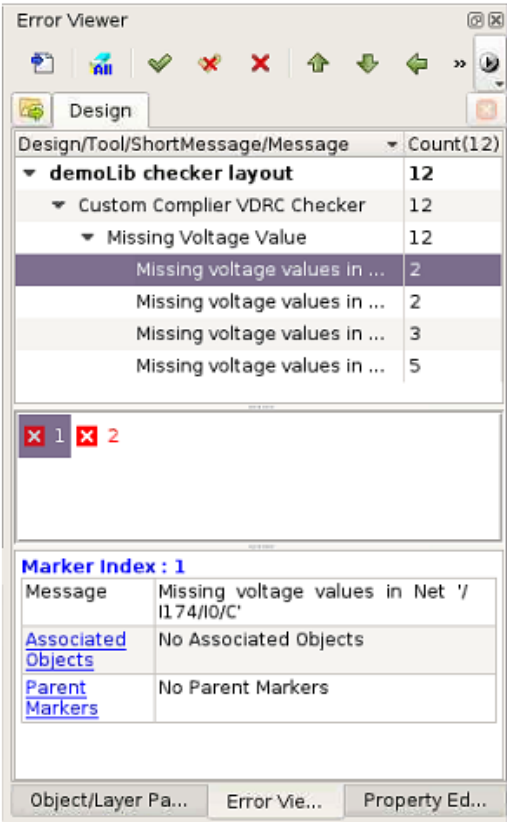
- In the **Report Violations: Physical** section:

Option	Description
Physical	Checks that the voltage instances are placed and connected correctly in the design layout. Tcl preference: <code>leVDRCAnnotationPhysicalErrorGroup</code>
All	Checks for all error types.
Misplaced V-Instances	Checks for incorrectly placed or connected voltage instances. Tcl preference: <code>leVDRCAnnotationMisplacedVInst</code>
Net Has V-Instance Conflicts	Check for nets that have multiple same-type voltage instances with different parameter values. Tcl preference: <code>leVDRCAnnotationNetHasVInstConflicts</code>
Connected Shape Layer Conflicts	Check whether shapes that are connected to voltage instances have layer conflicts. Tcl preference: <code>leVDRCAnnotationConnectedShapeLayerConflicts</code>

6. When you are finished, click **OK** or **Apply**.

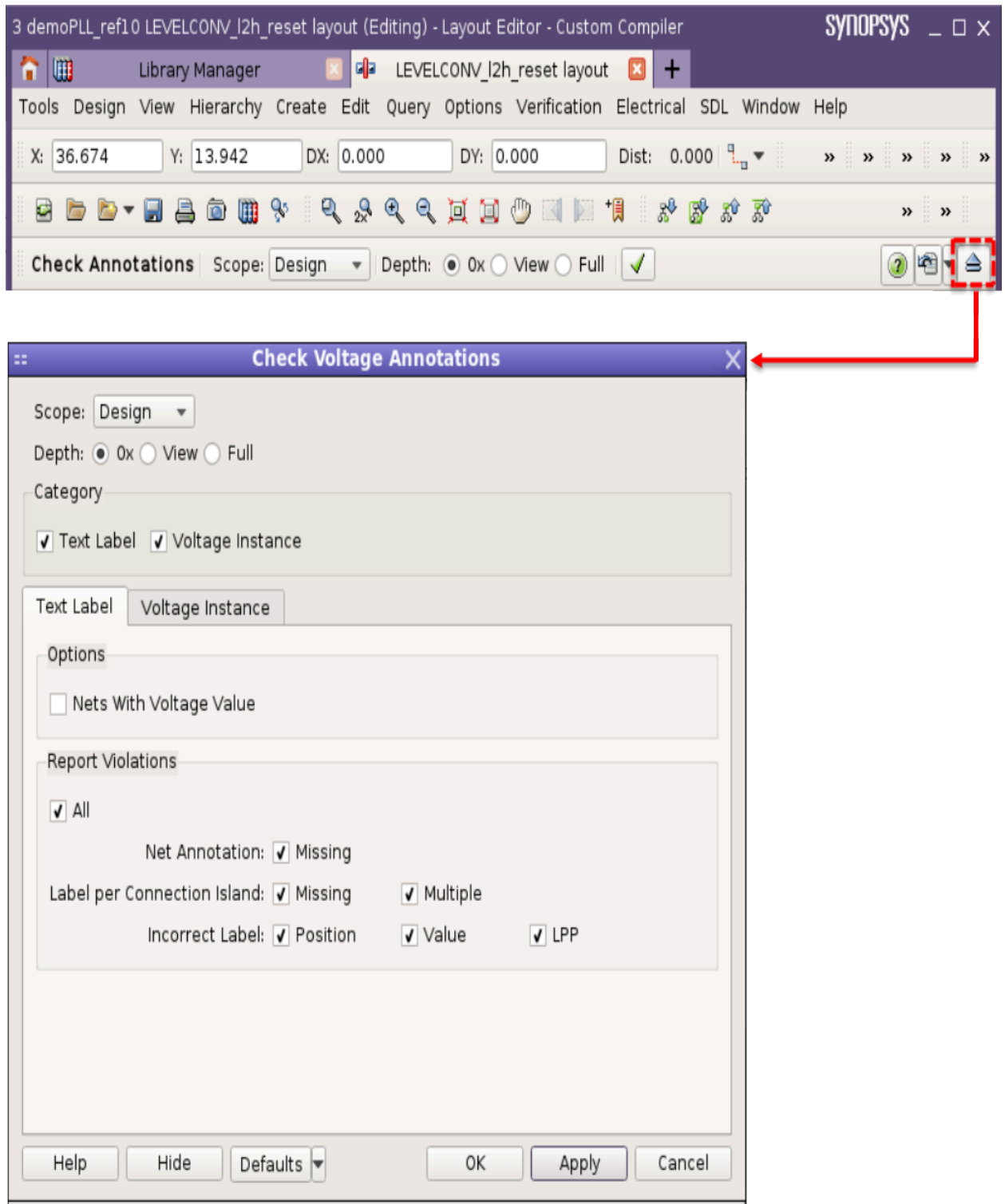
The command checks for the types of errors you specified in the dialog box. If errors are found, markers are added in the design. To view information about the errors,

display the Error Viewer and select errors of interest. To troubleshoot the errors, see [Troubleshooting Physical Annotation Errors](#).



Check Voltage Annotations Toolbar and Dialog Box Reference

This topic provides a quick reference of the **Check Voltage Annotations** dialog box. For instructions on how to locate and resolve voltage annotation errors, see [Checking for Physical Annotation Errors](#).



General Options

Option Name	Description
Scope	Size of the area to consider when checking annotations: • Design: Checks the entire design. • Viewport: Checks the components within the viewport. If you want the command to consider objects in the halo region just beyond the viewport, see Before You Start Checking Physical Voltage Annotations in the Layout Editor. • Nets: Checks the specified nets in the entire design. To specify the nets, type one or more net names in the box, or click  to select the nets in the Design Navigator. Tcl preference: <code>leVDRCAnnotationScope</code>
Depth	Depth of the hierarchy to consider: • 0x: Checks at the top level only. • View: Checks at 0x through the Stop Level that you specify. • Full: Checks all levels of the design hierarchy. Tcl preference: <code>leVDRCAnnotationDepth</code>
Category (Dialog box only)	Type of objects to check: • Text Label: Enables the Text Label tab and lets you check text labels created by you or the tool. Tcl preference: <code>leVDRCAnnotationCheckTextLabel</code> • Voltage Instance: Enables the Voltage Instance tab and lets you check the Vsync, Vflag, and Udv voltage instances. Tcl preference: <code>leVDRCAnnotationCheckVoltageInstance</code>

Text Label Tab

Options Section:

Option	Description
Nets With Voltage Value	Considers nets only if they have voltage values and ignore nets that do not have voltage values. Tcl preference: <code>leVDRCAnnotationNetsWithVoltageValue</code>

Report Violations Section:

Option	Description
All	(Default) Checks for all error types.
Net Annotation	Checks for missing voltage values on the nets. Tcl preference: <code>leVDRCAnnotationMissingNetAnnotation</code>

Option	Description
Label per Connection Island	Checks for: • Missing: Missing pairs of text labels (min and max voltage values) on connection islands. Tcl preference: <code>leVDRCAnnotationMissingTextLabels</code> • Multiple: Excess pairs of text labels on connection islands. Tcl preference: <code>leVDRCAnnotationMultipleTextLabels</code> (Each connection island should only have one pair of text labels. See How Physical Annotations Are Added in Hierarchical Designs.)
Incorrect Label	Checks for: • Position: Text labels whose origins are not on connection islands. Tcl preference: <code>leVDRCAnnotationIncorrectTextLabelPosition</code> • Value: Text labels that have incorrect voltage values. Tcl preference: <code>leVDRCAnnotationIncorrectTextLabelValue</code> • LPP: Text labels that are on incorrect layer-purpose pairs (LPPs). Tcl preference: <code>leVDRCAnnotationIncorrectTextLabelLPP</code>

Voltage Instance Tab

(Scope: Design only, with Depth: 0x or Depth: Full)

Device Types:

Option	Description
Device Types	• All: All device types (Vsync, Vflag, and Udv). • Vsync: Vsync devices only. Tcl preference: <code>leVDRCAnnotationCheckVsync</code> • Vflag: Vflag devices only. Tcl preference: <code>leVDRCAnnotationCheckVflag</code> • Udv: Udv devices only. Tcl preference: <code>leVDRCAnnotationCheckUdv</code>

Report Violations: SDL Section

Option	Description
SDL	Checks the consistency of voltage instances between the design schematic the corresponding schematic-driven layout. This section is available only if the layout was generated from a schematic through the SDL operation. Tcl preference: <code>leVDRCAnnotationSDLErrorGroup</code>
All	Checks for all SDL error types.
Instance	Checks for: • Missing: Missing voltage instances. Tcl preference: <code>leVDRCAnnotationMissingVInst</code> • Extra: Excess voltage instances. Tcl preference: <code>leVDRCAnnotationExtraVInst</code>
Connection	Checks for: • Mismatch: Instance terminal and voltage instance connections that are mismatched between the schematic and layout. Tcl preference: <code>leVDRCAnnotationMismatchedVInstConn</code> • Missing: Missing instance terminal and voltage instance connections. Tcl preference: <code>leVDRCAnnotationMissingVInstConn</code>
Parameter	• Mismatch: Voltage instance parameter values that are mismatched between the schematic and layout. Tcl preference: <code>leVDRCAnnotationMismatchedVInstParams</code>
Master	• Mismatch: Masters of the voltage instances are mismatched between the schematic and layout. Tcl preference: <code>leVDRCAnnotationMismatchedVInstParams</code>

Report Violations: Physical Section

Option	Description
Physical	Checks that the voltage instances are placed and connected correctly in the design layout. Tcl preference: <code>leVDRCAnnotationPhysicalErrorGroup</code>
All	Checks for all error types.
Misplaced V-Instances	Checks for incorrectly placed or connected voltage instances. Tcl preference: <code>leVDRCAnnotationMisplacedVInst</code>
Net Has V-Instance Conflicts	Check for nets that have multiple same-type voltage instances with different parameter values. Tcl preference: <code>leVDRCAnnotationNetHasVInstConflicts</code>
Connected Shape Layer Conflicts	Check whether shapes that are connected to voltage instances have layer conflicts. Tcl preference: <code>leVDRCAnnotationConnectedShapeLayerConflicts</code>

Troubleshooting Physical Annotation Errors

Understanding physical annotation error messages can help you troubleshoot and resolve the issues.

The following table lists:

- The error messages that you might encounter during physical voltage annotation checking. The messages are listed alphanumerically by message ID.
- The possible causes of the errors.
- Ways to resolve the errors.

If you are still experiencing problems after trying the solutions, or if an error not on the list is displayed, contact Synopsys Support. See [Contacting Customer Support](#) for contact details.

ID	Error Message	Causes	Solutions
VDRC-201	Missing net annotation	• One or more nets were not annotated with voltage data, because the <code>vdrc_config</code> cellview does not contain all the nets. • The design was not annotated with voltage data.	Depending on the cause and whether you intentionally left some nets without annotations, try annotating the design with voltage data, or manually add voltage data to the nets. See Editing Voltage Data in the Layout .
VDRC-202	Missing label per connected island	One or more nets were not fully routed, physical annotations were not added, or the text label was manually deleted.	Make sure each net, or each island that is not a fully routed net, has one text-label pair.
VDRC-203	Multiple labels per connection island	Two text-labeled connection islands were merged into a single connection island.	Review the design and manually check the annotations. Delete one of text-label pairs.
VDRC-204	Incorrect label position	The text-label pair was manually moved outside of the associated shapes.	Move the text-label pair within the associated shape.
VDRC-205	Incorrect label value	The text label voltage values do not match the nets' voltage values.	Modify the text label voltage value for the net. Check the Schematic Editor Design Navigator or the Property Editor for the correct value.

ID	Error Message	Causes	Solutions
VDRC-206	Incorrect label LPP	The old LPP text labels for the connection islands were overwritten or modified manually.	Add new LPP text labels by adding physical annotations, re-import the technology file, or use Tcl preferences.
VDRC-207	SDL: Missing Layout Instance	One or more voltage instances were not placed (or realized) in the layout from the schematic.	Place the voltage instances using the SDL Pick & Place command. See Placing Schematic Instances Into Layout.
VDRC-208	SDL: Extra Layout Instance	One or more voltage instances added in the layout do not match instances in the schematic.	Delete the excess layout voltage instances.
VDRC-209	SDL: Instance Terminal Connectivity Missing	One or more voltage instances added in the layout are not assigned any instance terminal nets.	Assign instance terminal nets to the voltage instances. Make sure the net assignments match the voltage instance connections in the schematic.
VDRC-210	SDL: Instance Terminal Connectivity Mismatch	One or more voltage instances added in the layout have different instance terminal connections than the corresponding voltage instances in the schematic.	Correct the connections in the layout to ensure they match the corresponding voltage instance connections in the schematic.
VDRC-211	SDL: Parameter Mismatch	One or more layout voltage instance parameters were modified so that they do not match the corresponding voltage instance parameters in the schematic.	Correct the layout voltage instance parameter values to ensure they match the corresponding schematic voltage instance parameters.
VDRC-212	SDL: Instance Master Mismatch Incorrect label LPP	The master of the layout voltage instance was modified so that it is different from the corresponding schematic voltage instance.	Correct the master layout voltage instance to ensure it matches the corresponding schematic voltage instance.
VDRC-213	Physical: Misplaced Instances	One or more voltage instances were placed on the wrong net shapes.	Move these voltage instances onto the correct net shapes.
VDRC-214	Physical: Misplaced Instances	One or more voltage instances were placed on the same type of voltage instance so that they overlap.	Move the overlapping voltage instances.

ID	Error Message	Causes	Solutions
VDRC-215	Physical: Misplaced Instances	One or more voltage instances were placed such that they are floating (not connected).	Connect the pin shapes of these voltage instances to net shapes.
VDRC-216	Physical: Shape Layer Conflicts	One or more voltage instances are connected to voltage instances on different layers.	Modify the voltage instance layer parameters so that they are on the same layer as the connected net shapes.
VDRC-217	Physical: Voltage Instance Conflicts	Multiple voltage instances of the same type but different voltage values were placed on the nets.	Remove extra voltage instances and make sure each net has only one voltage instance.

11

Reporting and Checking Shield Coverage

This chapter explains how to generate reports about shield coverage.

This chapter contains the following topics:

- [About the Shield Reporter and Shield Checker](#)
- [Preparing for Shield Reporting](#)
- [Workflow for Checking and Reporting Shield Coverage](#)
- [Reporting Shield Coverage](#)
- [Checking Shield Coverage](#)

Checking and reporting shield coverage is part of in-design electrical checking, which includes [Reporting and Checking Electromigration \(EM\)](#), [Reporting and Checking Resistance](#).

About the Shield Reporter and Shield Checker

To reduce signal noise and interference, you can add shielding to isolate certain circuits from other parts of the design. You can add shielding to the wires during or after routing. For information about how to add shielding, see [Net Shielding Constraint](#), [Adding Post-Route Shielding](#), and [Shielding Styles and Behavior](#).

To ensure optimal shield coverage in the design, you can use the following to automate shield checking:

- [Shield Reporter](#)
- [Shield Checker](#)

When using the Shield Reporter and Shield Checker, you should be aware of their [Limitations](#).

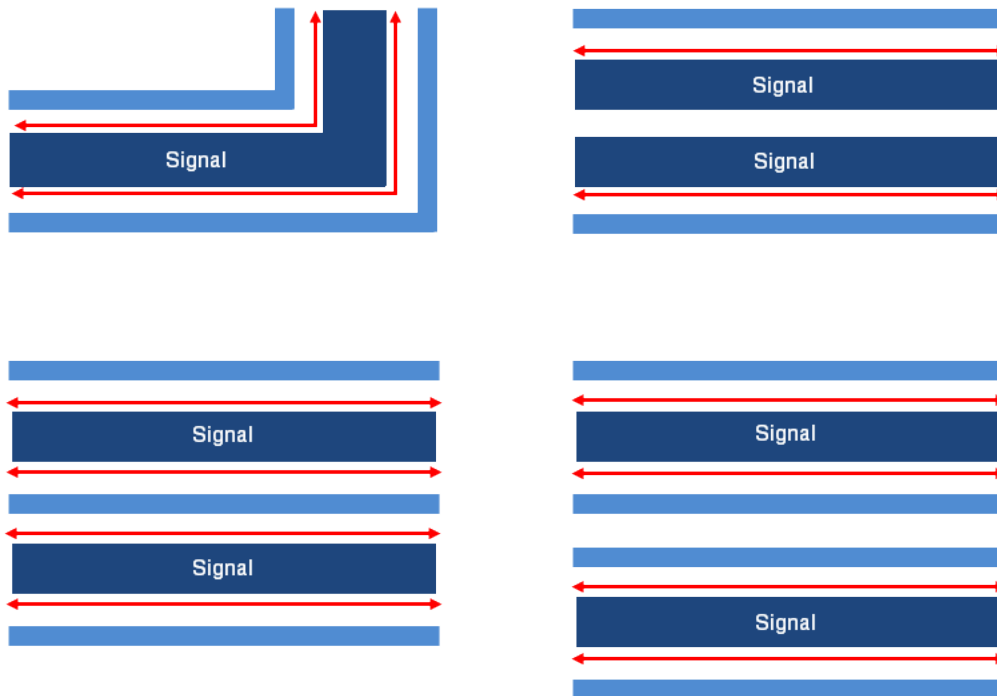
Shield Reporter

The Shield Reporter calculates and displays the coverage ratio for each layer. The ratio is calculated as follows:

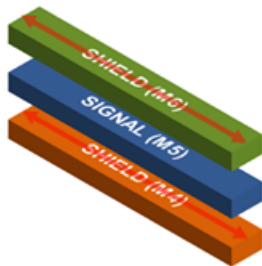
$$\text{Ratio} = \frac{\alpha * \sum \text{Length}_{\text{covered by parallel}} + \beta * \sum \text{Length}_{\text{covered by tandem}}}{\sum \text{Length}_{\text{signal}}}$$

where:

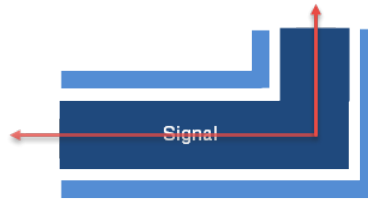
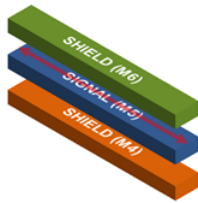
- Length_{covered by parallel} is the length of the two sides of the wire that are covered by parallel shielding



- Length_{covered by tandem} is the length of wires that are covered by tandem shielding



- Length_{signal} is the length of the signal wires that are shielded



- a and b values depend on the shield type:

Shield Type	a	b
parallel	1/2	0
tandem	0	1/2
coaxial	1/4	1/4
split coaxial	1/4	1/2
center coaxial	1/4	1/4
below	0	1
above	0	1
offset below	0	1
offset above	0	1
offset tandem	0	1/2
offset coaxial	1/4	1/4
below split coaxial	1/4	1/6
above split coaxial	1/4	1/6
below center coaxial	1/4	1/2
above center coaxial	1/4	1/2
above solid coaxial	1/4	1/2
below solid coaxial	1/4	1/2
offset above solid coaxial	1/4	1/2
offset below solid coaxial	1/4	1/2
offset split coaxial	1/4	1/12

Shield Type	a	b
offset above split coaxial	1/4	1/6
offset below split coaxial	1/4	1/6
offset center coaxial	1/4	1/4
offset above center coaxial	1/4	1/2
offset below center coaxial	1/4	1/2
parallel multilayer	1/4	1/8

Note:

Because the **Shield Placement** parameter for diff-pair and bus bit nets only works for the parallel style, for other styles such as tandem and coaxial, the Shield Reporter and Shield Checker will take these bit nets as common nets regardless of the value of the **Shield Placement** parameter.

For differential pair (diff-pair) and bus shielding, the shield coverage ratio is reported using the same shield coverage ratio formula given in [Shield Reporter](#). The values of a and b depend on the value of the **Shield Placement** parameter.

Shield Placement	a	b
Outside	1	0
Interleave	1/2	0
Double Interleave	1/2	0

For group shielding, the a and b values depend on the shielding style and layers used:

Shield Placement	a	b
Parallel / Single Layer	1	0
Parallel / Multiple Layers	1/2	1/4
Split Axial / Both Layers (Above and Below)	1/2	1/8
Split Axial / Above or Below Layer	1/2	1/4

Shield Checker

The Shield Checker reports the following types of violations:

Violation	Description
Shield width	The width of the shield wires exceeds the Default Width specified in the Net Shielding Constraint.
Overhang	The shields do not align with the wires. If the Shield Overhang Constraint is set, the overhanging distances do not comply with the specifications in the constraint.
Gap	The spacing between wires and corresponding shields do not comply with the gap settings in the Net Shielding Constraint.
Sharing	For differential-pair and bus shielding, Shield Placement is Double Interleave, but only one shield is found between two adjacent bits. (Two shields are expected between adjacent bits.)
Aggressor	Additional wires are found between the signal wires and the corresponding shields. The area between the wires and the corresponding shields should be clear. If your design requires you to insert partial coaxial shields in these areas to supplement parallel shielding, you can use the <code>leCheckTandemAggressorForParallelShielding</code> Tcl preference to turn on or off the aggressor checks in these particular cases. Set the preference before running the checker.
Shield coverage	The actual shield coverage ratio is less than the expected ratio. The Shield Checker compares the minimum percent shield coverage you specified using the <code>leShieldMinimumPercentage</code> Tcl preference with the calculated (actual) shielding coverage ratio. If the actual ratio is less than the minimum percent, a violation is reported in the Error Viewer. Note that the default minimum shield coverage is 80% for all layers.
Floating shields	Shields that are not connected to pins. To check for floating shields, set the <code>leCheckFloatingShield</code> Tcl preference to <code>true</code> before you run the checker.
Via coverage	Vias are not fully enclosed by shields, as required by the Enclose Via setting in the Net Shielding Constraint. Set the <code>leCheckEncloseViaMode</code> Tcl preference to <code>on</code> to always check via enclosures regardless of the Net Shielding constraint setting. Set the preference to <code>auto</code> to check the via enclosures according to the Net Shielding constraint Via Enclosure setting. Set the <code>leCheckViaArrayForCoaxialMode</code> Tcl preference to <code>on</code> to report via coverage for solid coaxial shield that are in tandem. Set the Tcl preferences before running the checker.

Violation	Description
Maximum length of wire that is not shielded	To report the maximum wire length that is unshielded, you can set the <code>leReportMaxUnshieldedLength</code> Tcl preference to <code>true</code> . The default setting is <code>false</code> . Set the preference before running the checker.

The Shield Checker also can tailor its violation reports on signal wires that are not shielded, either limiting reports to signal wires that have no aggressor, or reporting on the entire signal wire. The violations are based on the value of preference `leIgnoreNotFullyShieldedViolationWithoutAggressor`. If this preference is set to `true`, only the part of the signal wire that is not shielded and that has an aggressor will report a Not-Fully Shielded violation. If the preference is set to `false`, all of the signal wire that is not shielded will report this violation.

When checking tandem and center-only coaxial shields, the Shield Checker will consider the overhanging widths in the upper and lower shields.



For instructions on running the Shield Checker, see [Checking Shield Coverage](#).

Limitations

The Shield Checker and Shield Reporter:

- Report and check top-level (0x) nets and 1x pins only. Extended shapes that are on the 1x level are not included during the reporting and checking operation.
- Focus only on interconnect objects (paths, path segments, and vias).

Workflow for Checking and Reporting Shield Coverage

[Table 3](#) shows the high-level workflow for reporting and checking shield coverage in your design. Note that checking shield coverage is an iterative process that involves reporting, checking, viewing violations, adjusting shield constraints, and rechecking until violations are fixed and optimal coverage is achieved.

Table 3 *Checking and Reporting Shield Coverage Workflow*

Step	Description
1	Prepare for shield reporting and coverage. See Preparing for Shield Reporting .
2	Report the shield coverage in the design. Generate shield coverage reports to view the percent of shield coverage in the design by layer. See Reporting Shield Coverage .
3	Check the shield coverage. Check for shield coverage violations in your design. For information, see Checking Shield Coverage .
4	Adjust the shielding to meet the minimum shielding requirement. Adjust shielding to optimize shield coverage. For example, you can manually edit the shield shapes, such as move or stretch the shapes, to ensure improved shield coverage. To edit shapes, see Reporting and Checking Shield Coverage .

Preparing for Shield Reporting

Before you run the Shield Reporter and Shield Checker, make sure you:

- Set the minimum percent shield coverage using the `leShieldMinimumPercentage` Tcl preference.
- Set the Net Shielding constraint.
- Set the Shield Overhang constraint, if overhanging shielding is desired.
- Create routes and add shielding. For information, see the Custom Router Description.

Also consider the following:

- For differential pair and bus shielding, you do not need to set the Differential Group and Bus Routing constraints before starting to run the Shield Reporter and Shield Checker. The Shield Reporter and Shield Checker will check these constraints, however, to determine whether the reported/checked signal nets are bit nets of differential group or bus.
- For checking shield coverage only, you can set `leIgnoreNotFullyShieldedViolationWithoutAggressor` to `true` if you want to see not-fully-shielded violations for signal wires that have no aggressor. The default value is `false`.

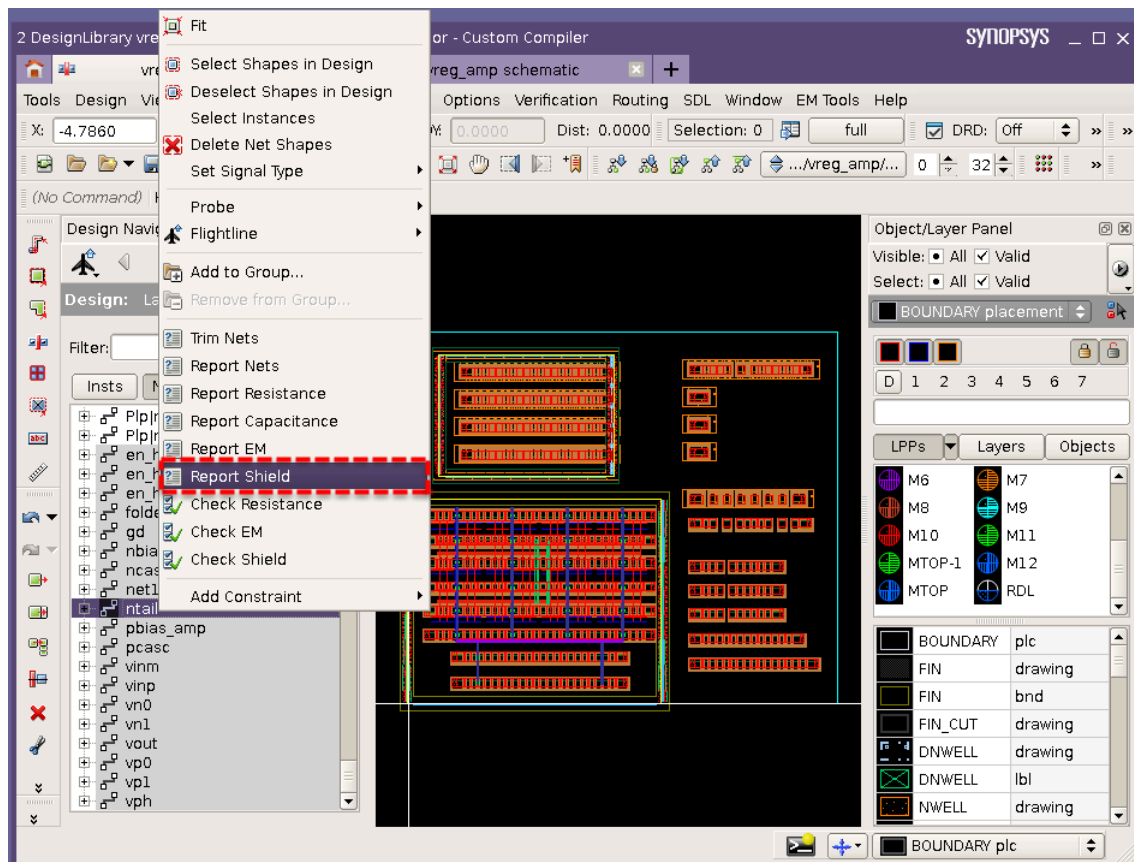
- If your design requires that you insert partial coaxial shields to supplement parallel shielding, set the `leCheckTandemAggressorForParallelShielding` Tcl preference to `false`.
- For coaxial shields, if you want to report the top and bottom shields separately, set the `leShieldReportAboveBelowTandemSeparately` Tcl preference to `true`.

Reporting Shield Coverage

You use the Design Navigator to report shield coverage.

To report shield coverage:

1. Open the design.
2. Display the Console.
3. Display the Design Navigator.
4. In the Design Navigator, select the nets of interest.
5. In the Design Navigator, right-click the net name, and then choose **Report Shield**.

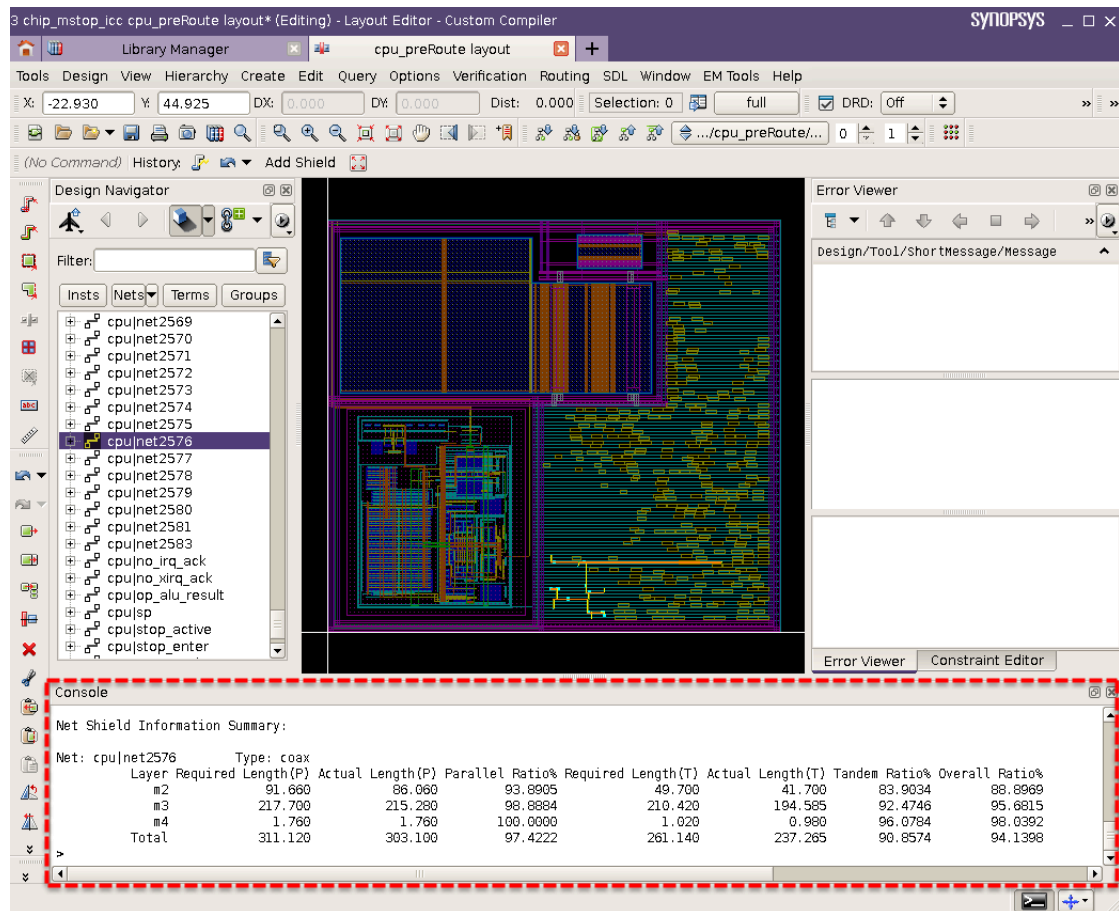


Alternatively, you can enter the following Tcl command in the Console:
`le::reportShield.`

The Shield Reporter calculates the shield coverage ratio for each layer, and then displays the results in the Console. The report shows the following information:

- The names of the nets you selected for the reporting
- The types of shielding found in the selected nets
- The layers that have shielding
- The percent shielding per layer, by shielding type
- The percent shielding per shielding type

- The total shielding coverage



In the example shown, the shield coverage report is for a net called `cpu|net2576`. The types of shielding found on this net are parallel and tandem. The layers that have shielding are m2, m3, and m4. The total parallel shielding coverage for all three layers is 97.4222. The total tandem shielding coverage for all three layers is 90.8574. The total shielding coverage for all layers and all shielding types is 94.1398.

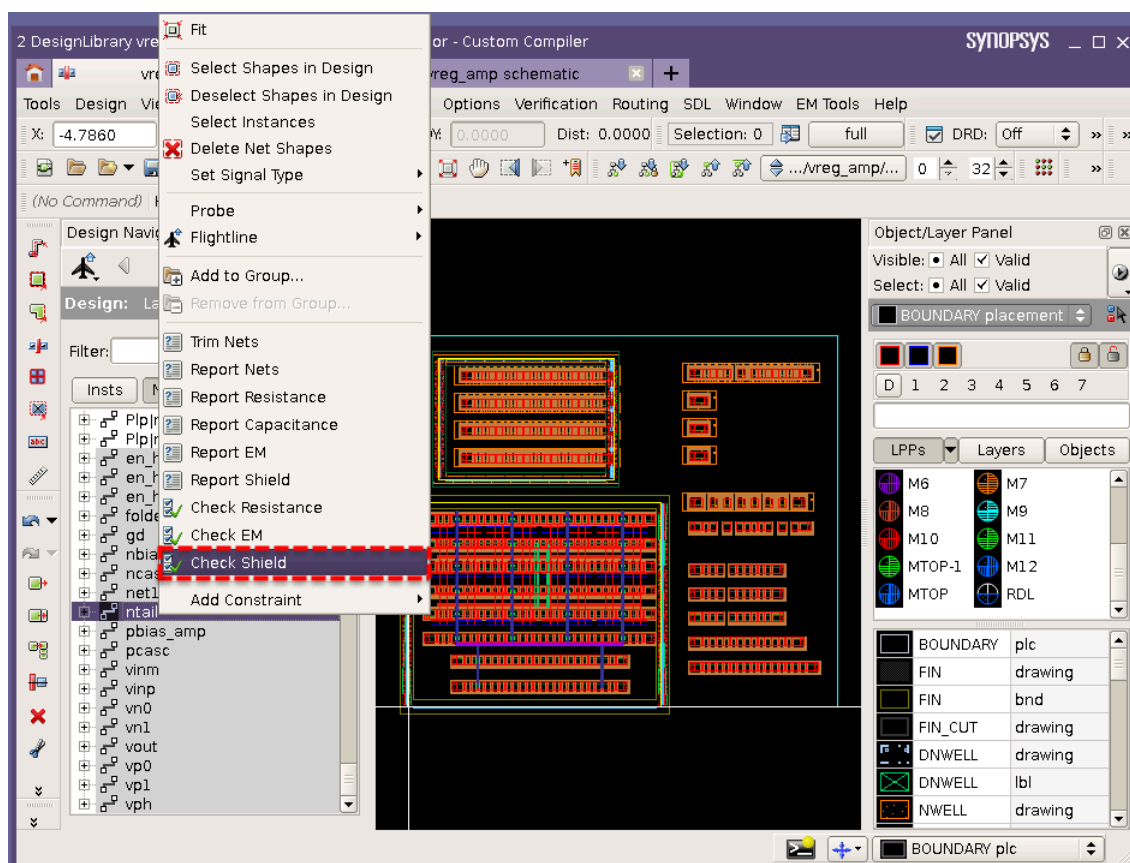
Checking Shield Coverage

You use the Design Navigator to check shield coverage.

To check shield coverage:

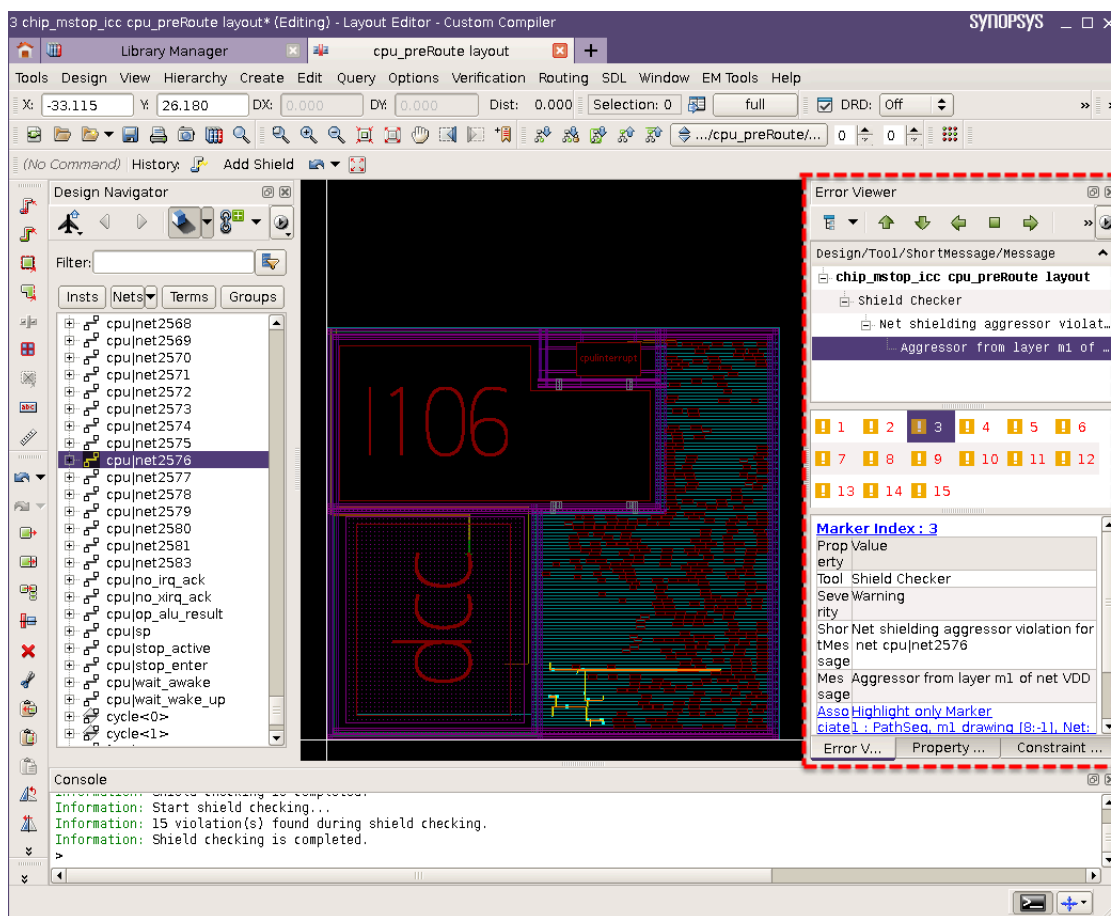
1. Open the design.
2. Open the Design Navigator.

3. Open the Error Viewer.
4. In the Design Navigator, select the nets of interest.
5. In the Design Navigator, right-click the net name, and then choose **Check Shield**.



Alternatively, you can enter the following Tcl command in the Console:
`le::checkShield.`

The Shield Checker compares the calculated (actual) shield coverage with the default minimum shield coverage or the minimum you specified. If violations are discovered, the Shield Checker displays them in the Error Viewer. For information about the types of violations reported, see [About the Shield Reporter and Shield Checker](#).



6. In the Error Viewer, expand the list to display the full message of each violation.

To address the violations, you can adjust the shielding options. For information about editing shielding options, see Adding Post-Route Shielding and Shield Tab.